Prime-Representing Constants and Related Structures

Encodings, Arithmetic, Geometry, and Modular Obstructions

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Abstract

We study constants whose prescribed powers have prime integer parts, together with related algebraic power sequences. Exact prime cells yield certified minima, effective independent perfect sets, and reproducible large-prime certificates. Geometric arguments give full Hausdorff dimension for unbounded power schedules and iterated-exponential prime orbits. Modular return and local minimality give transcendence criteria for perturbed exponential and factorial schedules. Further results concern polynomial growth constants, ceiling endpoints, optimal finite prime covers, and local computability. We extend support-sensitive independence to all left boundary minima, obtain density-one cubic compositeness results at fixed norm, and lower a Type-C transcendence threshold. The final arithmetic analysis proves infinitely many composite outputs for non-totally-real cubic Pisot numbers, exact parity-dependent 3-adic trace depths in a cubic-unit branch, and a short-interval prime deficit forced by a hypothetical rational square-clock minimum. An exact finite-state counterexample limits a proposed prime-divisor return mechanism. Related sections treat polynomial patterns, palindromes, quotient sums, and dimension profiles. Proofs and standalone integer verification programs are included in one self-contained source. The unconditional transcendence of Mills' constant, the irrationality of $\xi(2^k)$, the universal totally real cubic Pisot problem, and the twin-prime conjecture are not resolved.

Keywords: prime-representing constants; Pisot numbers; transcendence; Hausdorff dimension; modular dynamics; finite prime covers.

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1 Objects, principal results, and logical scope

For a sequence $C = (C_k)$ of positive real exponents, write

$$W(C) = \{A > 1 : \lfloor A^{C_k} \rfloor \text{ is prime for every } k \geq 1\}, \qquad \xi(C) = \min W(C)$$

whenever the set is nonempty. Write $c_1 = C_1$ and $c_{k+1} = C_{k+1}/C_k$. The classical Mills choice is $C_k = 3^k$ [26]; the arithmetic and geometry of more general choices are developed in Saito and Saito–Takeda [33, 38, 36, 39]. This manuscript combines exact constructions with extensions of those methods and with related problems from the same research programme.

1.1 The objects that must be distinguished

The set $W(C)$, its least element, a selected branch inside it, and the least element after deleting initial conditions are different objects. The following conventions remain in force throughout.

Object	Meaning and scope
$W(C)$	All permitted bases for one fixed exponent sequence. For successive ratios at least two its Hausdorff dimension is one, by Theorem 14.3 and Corollary 14.5.
$\xi(C)$	The global least permitted base. Statements about this specific number require a minimality argument.
Local left minimum	A point $A \in W(C)$ with an interval immediately to its left containing no other point of $W(C)$. The localized transcendence criteria concern these points.
Selected perfect set K	A constructed subset of $W(C)$. Some such sets have upper box dimension zero and algebraically independent points. Their dimension need not equal the dimension of $W(C)$.
Tail minimum ξ_N	The least base subject only to the conditions with index at least N. It generally differs from $\xi(C)$.
Adaptive exponent sequence	A sequence chosen during a construction. Its flexibility does not imply the same result for a previously fixed sequence or its global minimum.

The ordinary factorial clock $C_k = k!$, the perturbations $k! + k$ and $k! + k^2$, and the sparse recursion $E_{n+1} = E_n E_n!$ are also distinct. The enormous decimal-prefix estimates in Part I belong to the explicitly defined sparse schedules there. They are not asserted for every factorial perturbation.

1.2 Main conclusions and their dependencies

Part I begins with the finite exponent list $(4, 6, 7, 12)$ and proves its exact minimum $29^{1/12}$. A certified prime seed extends this finite problem to explicit infinite schedules. The construction supplies algebraically independent perfect sets, preserves Turing degrees, and gives deterministic certificates for a 1966-digit prime and two finite twin-prime pairs. The twin-output window equivalence is kept separate from a proof that its windows are nonempty infinitely often. Spectral applications transfer the constructed algebraic independence to finite matrices.

Part II proves full dimension for arbitrary unbounded multiplier sequences with successive ratios at least two, and for Wright prime orbits in every real base $b > 1$. It then establishes local dimension, support-sensitive separation, and relative independence of logarithms and finite exponential towers. The complete independent binary tree in Part I contains its greedy

minimum. A later analytic selection need not retain that minimum; in particular it cannot retain it when independence is required over a field already containing it. Likewise, the field generated by a selected thin set is distinct from the ring generated by the full spectrum.

Part III develops modular return and local minimality. In particular, the full minima for

$$3^k - k, \quad 3^k - 2k, \quad 3^k - 2k + 1, \quad k! + k, \quad k! + k^2 \qquad (k \geq 1)$$

are transcendental; these are consequences of Saito's criteria and the arithmetic verifications given here, not a new proof of those external criteria. The more general family $P(a^n) + Q(n)$ is treated with its exact growth and nonvanishing assumptions; its general conclusion concerns sufficiently late minima. For the ordinary Mills constant A, the degree conclusion is conditional:

$$A \in \overline{\mathbb{Q}} \quad \Longrightarrow \quad [\mathbb{Q}(A) : \mathbb{Q}] = 3^{t+1}, \qquad \mathrm{MinPoly}_A(X) = f(X^{3^t})$$

for some integer $t \geq 0$ and a monic irreducible cubic $f \in \mathbb{Z}[X]$. It is a necessary condition, not a contradiction. The same distinction applies to the cubic prime-divisor sieve and bounded-degree certificates.

Part IV treats selected related questions in additive combinatorics and digital structure. Its statements are collected with their exact scopes in the following table.

Source question or extension	Conclusion proved here
Saito, dimension Questions 1.4–1.5 in arXiv:2102.04038v1	Full dimension for unbounded power schedules and for Wright prime orbits; Theorems 14.3 and 15.1.
Pongsriiam–Saito, palindrome Question 27	The coloring and positive-relative-density assertions hold exactly for integer bases $b \geq 3$; Theorem 32.1.
Saito–Yoshida, polynomial-pattern Question 6.1	The stated order of magnitude holds under positive lower density; Theorem 33.2.
Saito–Yoshida, three-term waiting-time conjecture	$L_{\alpha,3,r}(x) = O_{\alpha,r}(x^{1-\alpha/2})$ for $1 < \alpha < 2$ and fixed r; Theorem 34.1.
Positive balanced equations	An exceptional-dimension upper bound and almost-everywhere finiteness; Theorem 35.1. No exact dimension is claimed.
Normality and lattice dimension profiles	A sparse analytic equivalence criterion and all permissible numerical lattice profiles. No specified algebraic irrational is proved normal.

1.3 Attribution, notation, and verification

Question numbers always refer to the source version explicitly identified at the point of use. Some consequences settle the literal statements of questions in those versions; historical priority has not been established by a comprehensive literature search. The paper is a research manuscript, not a claim of completed external peer review. Existing theorems used as inputs are attributed where needed. The conditional results are not counted as unconditional resolutions.

All logarithms are natural unless a base is displayed. A prime is a positive rational prime. Algebraic independence means that every nonzero polynomial in finitely many distinct specified entries is nonzero. Hausdorff dimension is denoted by $\dim_H$; computability is understood in the Cauchy-name sense and carries no efficiency assertion. Arithmetic progressions are nonconstant unless stated otherwise. In the final pattern sections, "almost every" means Lebesgue almost every. Appendix A gives executable exact integer checks for the finite numerical certificates; they were rerun during consolidation. Proofs of infinite assertions remain separate from these finite checks.

1.4 Extensions incorporated in this edition

Alkauskas–Dubickas constructed transcendental prime-representing constants for suitable rapidly growing schedules [1]. Saito–Takeda's results already include transcendence of a least factorial constant [38]. These antecedents distinguish existence of transcendental members from arithmetic conclusions about a specified least member. Parts V and VI incorporate the two subsequent research manuscripts, preserving their substantive proofs and finite certificates while unifying the notation and bibliography. Part V treats polynomial growth, factorial return, ceiling geometry, finite covers, and computability under integer successive multipliers at least two with infinitely many at least three. Part VI develops left-boundary independence, fixed-norm cubic sieves, square-clock criteria, non-reversible windows, the refined Type-C cutoff, and the two density applications. Part VII incorporates the subsequent focused analysis of the three principal remaining arithmetic problems. In particular, the non-totally-real cubic theorem is included with its full proof; no irrational-angle equidistribution hypothesis is used.

Target	Status of this manuscript
Unconditional transcendence of Mills' least constant	Necessary arithmetic conditions and finite exclusions; not proved.
Irrationality of $\xi(2^k)$	Exact conditional prime-deficit theorem; not proved.
Infinitely many composite cubic Pisot outputs at 3^n	Non-totally-real case proved; individual totally real exceptions remain.
Infinite twin primes	Exact output windows and finite certificates only.

The word "final" identifies the consolidated edition of this research programme. It does not certify external peer review or historical priority. All conclusions retain the hypotheses stated in their individual results.

Part I
Exact prime encodings and arithmetic certificates

This part constructs specified constants and selected sets from exact prime cells. It establishes their minimality or branch status before deriving arithmetic and computational consequences.

2 Prime cells and the exact seed minimum

The constructions below distinguish the global least admissible constant for a fixed exponent sequence, a selected branch inside an admissible cell, and a constant for a modified exponent sequence. They build on the prime-representing framework of Mills [26] and the algebraic independence results of Saito and Takeda [38]. The explicit constructions and the consequences proved here do not imply infinitely many twin primes or a general primality-testing speedup.

Write $\mathbb{P}$ for the positive primes and $\mathrm{nextprime}(x)$ for the least prime strictly greater than the real number x. For a positive integer E and a prime s, the half-open interval

$$J_E(s) = [s^{1/E}, (s+1)^{1/E})$$

is exactly the set of positive bases whose Eth power has integer part s. We also use its closed hull $I_E(s) = [s^{1/E}, (s+1)^{1/E}]$. The upper endpoint of $I_E(s)$ must not be treated as admissible without an additional argument.

Lemma 2.1 (Strict nesting). *Let $E_{n+1} = E_n R_n$, where R_n is an integer greater than one. Suppose primes s_n satisfy*

$$s_n^{R_n} < s_{n+1}, \qquad s_{n+1} + 1 < (s_n + 1)^{R_n}.$$

Then $I_{E_{n+1}}(s_{n+1})$ is contained in the interior of $I_{E_n}(s_n)$. If $E_n \to \infty$, the intervals have a unique common point α, and $\lfloor \alpha^{E_n} \rfloor = s_n$ for every n.

Proof. Taking positive E_{n+1}th roots proves strict containment. The mean value theorem gives $|I_E(s)| \le 1/E$. Hence the nested compact intervals have a unique intersection. The next interval is strictly inside each given parent, so the intersection point avoids both endpoints of every parent. This proves the floor identities. □

For a finite exponent list $A = (a_1, \ldots, a_t)$, let

$$\eta(A) = \min\{x > 1 : \lfloor x^{a_i} \rfloor \in \mathbb{P} \text{ for all } i\},$$

whenever the set is nonempty. A finite lower-bound certificate can prove such a minimum exactly. If every admissible x is at least u and $\lfloor u^c \rfloor = m$ is composite, then every admissible x must be at least $\mathrm{nextprime}(m)^{1/c}$. The jump is forced because no intervening integer is prime.

Proposition 2.2 (An exact four-condition minimum). *For $A = (4, 6, 7, 12)$,*

$$\eta(A) = 29^{1/12}, \qquad (\lfloor \eta^4 \rfloor, \lfloor \eta^6 \rfloor, \lfloor \eta^7 \rfloor, \lfloor \eta^{12} \rfloor) = (3, 5, 7, 29).$$

Every $x \in [29^{1/12}, 30^{1/12})$ has these four outputs.

Proof. The first condition forces $x \ge 2^{1/4}$. Table 1 applies the preceding forced-jump principle. Each entry is an exact comparison of integer powers, not a floating-point floor computation. At the final lower bound all four outputs are prime. Monotonicity, together with the values at the two endpoints, verifies the entire half-open interval. □

Table 1: An exact certificate of the finite global minimum.

Current lower bound	Outputs at $(4,6,7,12)$	Next forced lower bound
$2^{1/4}$	$(2,2,3,8)$	$11^{1/12}$
$11^{1/12}$	$(2,3,4,11)$	$5^{1/7}$
$5^{1/7}$	$(2,3,5,15)$	$17^{1/12}$
$17^{1/12}$	$(2,4,5,17)$	$5^{1/6}$
$5^{1/6}$	$(2,5,6,25)$	$7^{1/7}$
$7^{1/7}$	$(3,5,7,28)$	$29^{1/12}$
$29^{1/12}$	$(3,5,7,29)$	all conditions satisfied

Define the seed prime

$$p_0 = 513284270838218284273581257165 3 = 29^{21} + 24 = \text{nextprime}(29^{21}). \tag{2.1}$$

The exact certificates in Section 8 and Appendix A prove both primality and the absence of an intervening prime. In particular this is not an assumed probable prime. Put

$$b = p_0^{1/252}.$$

Since $p_0 + 1 < 2 \cdot 29^{21} < 30^{21}$,

$$I_{252}(p_0) \subset (29^{1/12}, 30^{1/12}) \subset (1, 4/3). \tag{2.2}$$

3 The least constant for the factorial tail

The exponent sequence used first is

$$C = (4,6,7,12,E_0,E_1,\ldots), \qquad E_0 = 252, \quad R_n = (E_n)!, \quad E_{n+1} = E_n R_n. \tag{3.1}$$

Define the greedy prime chain by

$$p_{n+1} = \text{nextprime}(p_n^{R_n}), \qquad n \geq 0. \tag{3.2}$$

Theorem 3.1 (Existence and exact global minimality)**.** *The set of $x > 1$ for which $\lfloor x^c \rfloor$ is prime for every exponent c in (3.1) has a least element ξ. It is*

$$\xi = \lim_{n\to\infty} p_n^{1/E_n},$$

and its first five outputs are $3, 5, 7, 29, p_0$. In particular $\xi > b > 29^{1/12}$.

Proof. Bertrand's postulate supplies a prime between each integer $N > 1$ and $2N$. Consequently

$$p_n^{R_n} < p_{n+1} < 2p_n^{R_n}, \qquad p_{n+1} + 1 \leq 2p_n^{R_n}.$$

Inductively all root cells remain below 2, so $p_n < 2^{E_n} < (E_n)! = R_n$. Bernoulli's inequality gives

$$(p_n + 1)^{R_n} \geq p_n^{R_n}(1 + R_n/p_n) > 2p_n^{R_n}.$$

Lemma 2.1 and (2.2) therefore construct an admissible limit.

Any admissible x is at least $29^{1/12}$ by Proposition 2.2. Its 252nd power is at least the composite integer 29^{21}, so its prime floor must be at least p_0 and $x \geq p_0^{1/252}$. Inductively, $x \geq p_n^{1/E_n}$ implies $x^{E_{n+1}} \geq p_n^{R_n}$. The latter is composite, so the next prime floor is at least p_{n+1}. Hence every admissible x is at least every greedy lower endpoint, and therefore at least their limit. Strict nesting proves $\xi > b$. □

Corollary 3.2 (No upper size bound)**.** *The prime outputs p_n increase without bound. More generally, any finite compatible prime-cell construction below 2 ending at an exponent $E \geq 252$ extends indefinitely by the rule $E \mapsto EE!$ and greedy prime selection.*

Proof. The same nesting proof applies. Since the limiting base is greater than one and the exponents tend to infinity, its prime floors tend to infinity. □

4 Algebraic approximation and certified decimal agreement

Lemma 4.1 (A one-sided decimal-boundary estimate)**.** *Let $b = m^{1/d} > 0$ with integers $m, d \geq 1$. For an integer $D \geq 1$, set*

$$t = \frac{\lfloor 10^D b \rfloor + 1}{10^D}.$$

If $t \leq B$, then

$$t - b \geq \frac{10^{-dD}}{dB^{d-1}}.$$

Thus $b \leq x < b + 10^{-dD}/(dB^{d-1})$ implies $\lfloor 10^D x \rfloor = \lfloor 10^D b \rfloor$.

Proof. By definition $t > b$. The positive rational $t^d - m$ has denominator dividing 10^{dD}, so it is at least 10^{-dD}. On $[b, t]$ the derivative of u^d is at most dB^{d-1}. The mean value theorem proves the estimate. No computation of the D digits is required. □

Proposition 4.2 (A Bertrand-only estimate for the factorial minimum)**.** *With $E_1 = 252 \cdot 252!$,*

$$0 < \xi - b < b(2^{1/E_1} - 1) < \frac{1}{E_1} < 2 \cdot 10^{-500}.$$

Proof. The first child interval has upper endpoint at most $b2^{1/E_1}$. For the sharper numerical constant, use $b < 4/3$, $\log 2 < 7/10$, and $e^u - 1 < u/(1-u)$ for $0 < u < 1$. The desired inequality follows for $E_1 > 11$. Finally, the integer E_1 has 500 decimal digits and exceeds $5 \cdot 10^{499}$. □

A short-prime-interval theorem drastically strengthens this estimate. Cully-Hugill proves, in the discussion of consecutive cubes in [11, Section 4], that

$$n \geq \exp(\exp(32.892)) \quad \Longrightarrow \quad \mathbb{P} \cap (n^3, (n+1)^3) \neq \varnothing. \tag{4.1}$$

The cited passage is on page 14 of the arXiv version. This is an unconditional input; no Riemann-hypothesis assumption is used below.

Theorem 4.3 (A long common decimal prefix)**.** *For the original least constant ξ,*

$$0 < \xi - p_0^{1/252} < 10^{-10^{498}}.$$

Moreover, with $D = 10^{495}$,

$$\lfloor 10^D \xi \rfloor = \lfloor 10^D p_0^{1/252} \rfloor.$$

The second assertion concerns exact common digits, not merely an absolute-error convention.

Proof. Write $R = 252!$, $E = 252R$, and $M = p_0^{R/3}$. Since $3 \mid R$, M is an integer. Exact integer comparisons give $R > 10^{497}$ and $p_0 > 10^{30}$, whence

$$M > 10^{10R} > 10^{10^{498}}.$$

This also verifies the threshold in (4.1); for example $e^{32.892} < 3^{33} < 10^{16}$, whereas $\log M > 10^{498}$. Thus

$$M^3 < p_1 < (M+1)^3, \qquad p_1 + 1 \le (M+1)^3.$$

Strict containment of later cells and concavity yield

$$0 < \xi - b < b((1+1/M)^{3/E} - 1) < \frac{3b}{EM} < \frac{1}{M} < 10^{-10^{498}}.$$

For decimal agreement apply Lemma 4.1 with $d = 252$ and $B = 2$. The exact integer inequalities

$$252 \cdot 2^{251} < 10^{79}, \qquad 252 \cdot 10^{495} + 79 < 10^{498}$$

show that the next D-digit boundary is farther from b than the displayed error. Thus no decimal boundary is crossed. □

Remark 4.4 (Precision is not the computation of a huge prime). Put $N = p_0^{252!}$. Substituting the algebraic approximation b into the next exponent gives $b^{E_1} = N$, a composite integer, not p_1. The value p_1 has on the order of 10^{498} decimal digits. More explicitly, elementary bounds on $252!$ and $\log_{10} p_0$, together with $N < p_1 < 2N$, place its digit count strictly between $6 \cdot 10^{498}$ and $7 \cdot 10^{498}$. Printing such a number is a different task from proving the approximation theorem.

If a d-digit prime q is to be identified at exponent E using one whole decimal interval for the base, that interval must fit inside its prime cell. For a cell below 2,

$$(q+1)^{1/E} - q^{1/E} < \frac{2}{Eq}.$$

For $E = E_1 > 10^{499}$ this is less than 10^{-d-497}. Therefore this particular interval-certification method requires at least $d + 498$ decimal places. This is not a lower bound for all algorithms for finding or testing primes.

Proposition 4.5 (A certified composite block). *For $N = p_0^{252!}$, all of $N+1, \dots, N+561$ are composite. In particular $p_1 \ge N + 562$.*

Proof. For every prime $\ell < 563$, the integer $\ell - 1$ divides $252!$. This finite assertion is checked by the exact verifier in Appendix A. Since $p_0 > \ell$, Fermat's theorem gives $N \equiv 1 \pmod{\ell}$. For $1 \le h \le 561$, a prime divisor ℓ of $h+1$ is below 563, and divides $N + h$. It is a proper divisor because $N + h > \ell$. □

Corollary 4.6 (A necessary local twin filter). *If $0 \le h \le 563^2 - 4$ and both $N+h$ and $N+h+2$ are prime, then $h+1$ and $h+3$ are primes at least 563.*

Proof. The case $h = 0$ is excluded by the compositeness of N. A composite positive integer below 563^2 has a prime divisor below 563. Applied to $h+1$ or $h+3$, this would give a proper divisor of one of the asserted large primes. Small prime values below 563 are excluded by the same congruence. This is a necessary sieve condition, not a sufficient construction of twin primes. □

5 A quantitative criterion for algebraic independence

The arithmetic argument below separates two inputs: independence of finite radical approximants, and sufficiently rapid approximation of the limiting constants. This separation makes it possible to replace strong prime-gap estimates by faster growth of the exponent sequence. Algebraic independence of prime-representing constants has an established literature; in particular, see Saito and Takeda [38]. No claim of priority for the general approximation strategy is needed here.

Lemma 5.1 (Prime radical basis)**.** *Let $E \geq 2$ be an integer, and let $s_1, \dots, s_r$ be distinct primes. Put $b_i = s_i^{1/E}$, taking positive real roots. The E^r monomials*

$$b_1^{j_1} \cdots b_r^{j_r}, \qquad 0 \leq j_i < E,$$

are linearly independent over $\mathbb{Q}$. Consequently,

$$[\mathbb{Q}(b_1, \dots, b_r) : \mathbb{Q}] = E^r,$$

and $P(b_1, \dots, b_r) \neq 0$ whenever P is a nonzero rational polynomial with $\deg_{X_i} P < E$ for every i.

Proof. Every displayed monomial has its Eth power in $\mathbb{Q}$. If the quotient of two such monomials were rational, taking Eth powers and then valuations at s_i would imply that each difference of exponents is divisible by E. Those differences lie strictly between $-E$ and E, so all are zero. The monomials are therefore pairwise linearly independent over $\mathbb{Q}$. The real-radical independence theorem of Carr and O'Sullivan [9, Theorem 1.1] makes them linearly independent. They also span the generated field, by reduction using $b_i^E = s_i$. This proves all the assertions. In particular, E need not be prime or squarefree. □

For a nonzero polynomial $P = \sum_\nu a_\nu X_1^{\nu_1} \cdots X_r^{\nu_r} \in \mathbb{Z}[X_1, \dots, X_r]$, write $|\nu| = \nu_1 + \cdots + \nu_r$ and set

$$B(P) = \max\left\{2, \sum_\nu |a_\nu| 2^{|\nu|}\right\}, \qquad L(P) = \sum_{|\nu| \geq 1} |a_\nu|\,|\nu|\,2^{|\nu|-1}. \tag{5.1}$$

The second constant is a Lipschitz bound on $[0,2]^r$ for the supremum norm.

Proposition 5.2 (Explicit polynomial separation)**.** *Suppose that $s_1, \dots, s_r$ are distinct primes, $b_i = s_i^{1/E} < 2$, $\alpha_i \in [0,2]$, and $|\alpha_i - b_i| \leq \varepsilon$. If*

$$E > \max_i \deg_{X_i} P, \qquad L(P)\varepsilon \leq \tfrac{1}{2} B(P)^{1-E^r}, \tag{5.2}$$

then

$$|P(\alpha_1, \dots, \alpha_r)| \geq \tfrac{1}{2} B(P)^{1-E^r} > 0. \tag{5.3}$$

Moreover, $P(\alpha_1, \dots, \alpha_r)$ has the same sign as $P(b_1, \dots, b_r)$.

Proof. Let $F = \mathbb{Q}(b_1, \dots, b_r)$ and $\beta = P(b_1, \dots, b_r)$. Lemma 5.1 gives $\beta \neq 0$ and $[F : \mathbb{Q}] = E^r$. Each b_i, and hence β, is an algebraic integer. Every embedding of F into $\mathbb{C}$ sends b_i to an Eth root of s_i, of absolute value $b_i < 2$. Thus $|\sigma(\beta)| \leq B(P)$ for every embedding. The nonzero integer norm then yields

$$1 \leq |N_{F/\mathbb{Q}}(\beta)| \leq |\beta| B(P)^{E^r - 1}, \qquad |\beta| \geq B(P)^{1-E^r}.$$

The mean value inequality gives

$$|P(\alpha_1, \dots, \alpha_r) - \beta| \leq L(P)\varepsilon.$$

Condition (5.2) proves both the lower bound and the assertion about the sign. □

Theorem 5.3 (Rapid prime-root approximation criterion)**.** *Let $K \subset (1,2)$, let $E_n \to \infty$ be integers, and let $\varepsilon_n > 0$. Suppose that, for every $\alpha \in K$, there are primes $s_n(\alpha)$ such that*

$$\left|\alpha - s_n(\alpha)^{1/E_n}\right| \leq \varepsilon_n, \qquad s_n(\alpha)^{1/E_n} < 2.$$

Assume also the following two properties.

1. *For any distinct $\alpha_1, \dots, \alpha_r \in K$, the primes $s_n(\alpha_1), \dots, s_n(\alpha_r)$ are pairwise distinct for every sufficiently large n.*
2. *For every fixed positive integer r,*

$$\frac{\log(1/\varepsilon_n)}{E_n^r} \longrightarrow \infty.$$

Then K is algebraically independent over $\mathbb{Q}$. For any specified nonzero polynomial, every sufficiently late stage also provides the explicit lower bound in Proposition 5.2.

Proof. Fix distinct $\alpha_1, \dots, \alpha_r$ and a nonzero integer polynomial P. At all sufficiently late stages the prime labels are distinct and $E_n > \max_i \deg_{X_i} P$. If $L(P) > 0$, the second assumption gives

$$\log(2L(P)\varepsilon_n) + (E_n^r - 1)\log B(P) \longrightarrow -\infty,$$

which is precisely the eventual validity of $L(P)\varepsilon_n \le \frac{1}{2}B(P)^{1-E_n^r}$. For $L(P) = 0$ the polynomial is a nonzero constant and the assertion is immediate. Proposition 5.2 completes the proof. □

6 Binary families with a common prime prefix

Write

$$p_* = 513284270838218284273581257165 3, \qquad b_* = p_*^{1/252}.$$

The finite computation and the certificate in Section 8 establish that $p_* = \mathrm{nextprime}(29^{21})$ and

$$[b_*, (p_*+1)^{1/252}] \subset (29^{1/12}, 30^{1/12}) \subset (1, 2). \tag{6.1}$$

All constants constructed in this section therefore have the same five initial outputs $3, 5, 7, 29, p_*$ at exponents $4, 6, 7, 12, 252$.

Let $E_0 = 252$ and $E_{n+1} = E_n R_n$, where R_n will be specified below. For each finite binary word w, define a prime s_w recursively. The empty word has label $s_\varnothing = p_*$. If $|w| = n$, let

$$s_{w0} = \mathrm{nextprime}(s_w^{R_n}), \qquad s_{w1} = \mathrm{nextprime}(s_{w0}).$$

Associate to w the closed interval

$$J_w = [s_w^{1/E_n}, (s_w+1)^{1/E_n}]. \tag{6.2}$$

We use closed intervals only to take compact intersections. The strict nesting proved below ensures that no limiting constant occupies an upper endpoint, where the floor condition would fail.

Lemma 6.1 (Nested prime cells)**.** *Suppose that*

$$J_{w0} \cup J_{w1} \subset \mathrm{int}(J_w)$$

for every w, that the two child cells are disjoint, and that $E_n \to \infty$. Then every binary sequence $\omega \in \{0,1\}^{\mathbb{N}}$ determines exactly one real number

$$\alpha_\omega \in \bigcap_{n \ge 0} J_{\omega|n}.$$

The map $\omega \mapsto \alpha_\omega$ is a homeomorphism onto a compact perfect set K. Its points satisfy

$$\lfloor \alpha_\omega^{E_n} \rfloor = s_{\omega|n} \quad (n \ge 0).$$

Different points have different prime labels at every sufficiently late level. If $\log E_n / n \to \infty$, then K has upper box dimension zero, and hence $\dim_{\mathrm{H}} K = 0$.

Proof. The mean value theorem gives

$$|J_w| = (s_w + 1)^{1/E_n} - s_w^{1/E_n} \le E_n^{-1}.$$

Nested compactness and this bound give a unique intersection point. It lies in the interior of every ancestor cell because it belongs to a child cell, proving the floor identity. Distinct binary sequences separate into disjoint cells at their first different bit and never meet again. Agreement on the first n bits gives real distance at most E_n^{-1}, so the map is continuous. A continuous injection from compact Cantor space into $\mathbb{R}$ is a homeomorphism onto its image. Alternatively, perfectness follows directly by changing a sufficiently late bit.

Let $N_\delta(K)$ be the least number of intervals of length at most δ covering K. There are 2^n level-n cells, each of diameter at most E_n^{-1}. For $E_n^{-1} \le \delta < E_{n-1}^{-1}$, therefore,

$$\frac{\log N_\delta(K)}{\log(1/\delta)} \le \frac{n \log 2}{\log E_{n-1}} \longrightarrow 0.$$

The limsup defining the upper box dimension is zero. The Hausdorff dimension is bounded by the upper box dimension, so it too is zero; in particular K has zero Lebesgue measure. □

Proposition 6.2 (The all-zero branch is the global minimum)**.** *For either of the two constructions below, let $\xi_0 = \alpha_{000\ldots}$. Then ξ_0 is the least $x > 1$ for which every exponent in*

$$(4, 6, 7, 12, E_0, E_1, \ldots)$$

has a prime floor output. Thus minimality holds in the full admissible set, not only in the binary subset K.

Proof. The finite-prefix argument forces every admissible x to satisfy $x \ge 29^{1/12}$. At exponent $E_0 = 252$, the value at that lower endpoint is the composite integer 29^{21}, so prime output forces $x \ge p_*^{1/E_0}$. Let $p_n = s_{0^n}$. Inductively, if $x \ge p_n^{1/E_n}$, then $x^{E_{n+1}} \ge p_n^{R_n}$, a composite integer. Its floor can be prime only if it is at least $p_{n+1} = \mathrm{nextprime}(p_n^{R_n})$. Therefore $x \ge p_{n+1}^{1/E_{n+1}}$. Taking the limit gives $x \ge \xi_0$. The all-zero branch itself is admissible by Lemma 6.1, proving attainment and minimality. □

6.1 The factorial schedule

Theorem 6.3 (A thin algebraically independent prime family)**.** *Take $R_n = E_n!$, so that $E_{n+1} = E_n E_n!$. The binary construction is well-defined and gives a compact perfect algebraically independent set $K_{\mathrm{fac}} \subset (1, 2)$ of upper box dimension zero. Its minimum is the original minimum constant ξ. Every member has prime outputs at all the displayed exponents and the common initial outputs $3, 5, 7, 29, p_*$. If $|w| = n$, $s = s_w$, $E = E_n$, $R = E!$, and $b = s^{1/E}$, then every descendant α of w satisfies*

$$0 < \alpha - b < \frac{6b}{ERs^{R/3}} < \frac{12}{ER}\, 2^{-R/3}. \tag{6.3}$$

Proof. We use the following explicit consequence of Cully-Hugill [11, Section 4]: each interval $(m^3, (m+1)^3)$ contains a prime whenever the integer m satisfies $m \ge \exp(\exp(32.892))$. Here $M = s^{R/3}$ is an integer because $3 \mid E!$. The threshold holds already for $E = 252$ and $s = 2$, hence at every node. For example, $252! \ge 126^{126}$ gives

$$\log\log(2^{252!/3}) \ge 126 \log 126 + \log(\log 2/3) > 32.892.$$

There is consequently a prime in each of $(M^3, (M+1)^3)$ and $((M+1)^3, (M+2)^3)$. The first two primes above $s^R = M^3$ therefore satisfy

$$s^R < s_{w0} < s_{w1} < (M+2)^3, \qquad s_{w1} + 1 \le (M+2)^3.$$

For the integer $t = R/3 \geq 2$, the binomial theorem gives $(s+1)^t - s^t > 2$. Thus

$$s^R < s_{wj} < s_{wj} + 1 < (s+1)^R \qquad (j = 0, 1),$$

which proves strict containment of both child cells. Their prime labels are odd and different, so $s_{w0} + 1 < s_{w1}$ and the cells are disjoint.

Every descendant is contained in one of these child cells and hence

$$b < \alpha \leq b(1 + 2/M)^{3/(ER)}.$$

Concavity of $(1+u)^{3/(ER)}$ gives $(1+2/M)^{3/(ER)} - 1 < 6/(ERM)$, proving (6.3). In particular, a uniform error bound is $\varepsilon_n = 12\, 2^{-E_n!/3}$. For every fixed r,

$$\frac{\log(1/\varepsilon_n)}{E_n^r} = \frac{(\log 2)E_n!/3 - \log 12}{E_n^r} \longrightarrow \infty.$$

Lemma 6.1 and Theorem 5.3 now give algebraic independence. Finally, $E_{n+1} \geq E_n^2$, so $\log E_n \geq 2^n \log E_0$ and the Hausdorff dimension is zero. Minimality follows from Proposition 6.2. □

Corollary 6.4 (A common decimal block of prescribed enormous length)**.** *Every $\alpha \in K_{\mathrm{fac}}$ satisfies*

$$0 < \alpha - b_* < 10^{-10^{498}}, \qquad \left\lfloor 10^{\,10^{495}} \alpha \right\rfloor = \left\lfloor 10^{\,10^{495}} b_* \right\rfloor.$$

Thus the entire uncountable algebraically independent set has the same first 10^{495} digits after the decimal point.

Proof. At the initial node, put $R = 252!$ and $M = p_*^{R/3}$. The elementary integer inequalities $p_* > 10^{30}$ and $252! > 10^{497}$ imply $M > 10^{10^{498}}$. Since $6b_* < 12 < 252R$, the error in (6.3) is less than M^{-1}.

To certify the decimal block, apply Lemma 4.1 with $D = 10^{495}$, degree 252, and upper bound 2. If t is the next D-place decimal boundary above b_*, the lemma and $252 \cdot 2^{251} < 10^{79}$ give

$$t - b_* > 10^{-252D-79} > 10^{-10^{498}}.$$

No α reaches the next boundary. This proves the floor identity. □

6.2 An alternative requiring only Bertrand's theorem

The factorial schedule uses a short-prime-interval theorem to obtain its rapid error decay. A different schedule obtains the same arithmetic conclusions from Bertrand's theorem alone. Its exponents are substantially larger, so the following is a trade-off in assumptions and exponent growth, not a faster algorithm for listing prime outputs.

Theorem 6.5 (Bertrand-only construction)**.** *Define*

$$F_0 = 252, \qquad R(F) = 2^{F^F}, \qquad F_{n+1} = F_n R(F_n).$$

Use F_n in place of E_n in the same first-two-primes binary rule, starting from the same seed p_*. The resulting set K_{B} is compact, perfect, algebraically independent, and of upper box dimension zero. All its elements have the initial outputs $3, 5, 7, 29, p_*$, and the all-zero branch is the global minimum for this new exponent sequence. At a node with label s and exponent F, every descendant satisfies

$$0 < \alpha - s^{1/F} < \frac{6}{F\, 2^{F^F}}. \tag{6.4}$$

The only theorem on the distribution of primes used in this construction is Bertrand's theorem.

Proof. Let $R = 2^{F^F}$. Two applications of Bertrand's theorem [4] give, for the first two primes $q_0 < q_1$ above s^R,

$$s^R < q_0 < 2s^R, \qquad q_1 < 2q_0 < 4s^R, \qquad q_1 + 1 \le 4s^R.$$

Inductively the parent cell is below 2, hence $s < 2^F$. For $F \ge 2$, $R \ge 4 \cdot 2^F > 3s$, so Bernoulli's inequality implies

$$(s+1)^R = s^R(1 + 1/s)^R \ge s^R(1 + R/s) > 4s^R.$$

Both child cells therefore lie strictly inside their parent, and they are disjoint. Put $b = s^{1/F} < 2$. Any descendant belongs to one of the two children, so

$$b < \alpha \le b\,4^{1/(FR)}, \qquad \alpha - b < b\frac{3}{FR} < \frac{6}{FR},$$

where concavity gives $4^u - 1 < 3u$ for $0 < u < 1$. For $\varepsilon_n = 6/(F_n 2^{F_n^{F_n}})$,

$$\frac{\log(1/\varepsilon_n)}{F_n^r} = \frac{(\log 2)F_n^{F_n} + \log F_n - \log 6}{F_n^r} \longrightarrow \infty$$

for each fixed r. The general criterion proves algebraic independence. The estimates $F_{n+1} \ge F_n^2$ and Lemma 6.1 prove the topological and dimensional claims. The minimality proof is unchanged. □

Corollary 6.6 (An elementary shared block of 10^{601} digits)**.** *For every $\alpha \in K_{\mathrm{B}}$,*

$$0 < \alpha - b_* < 10^{-10^{604}}, \qquad \left\lfloor 10^{\,10^{601}}\alpha \right\rfloor = \left\lfloor 10^{\,10^{601}} b_* \right\rfloor.$$

Proof. Let $A = 252^{252}$. The exact integer comparison $A > 10^{605}$ and $2^{10} > 10^3$ imply

$$\frac{6}{252\,2^A} < 2^{-A} < 10^{-3A/10} < 10^{-10^{604}}.$$

The comparison $A > 10^{605}$ can also be obtained without writing A: $252^5 > 1.016 \cdot 10^{12}$ implies

$$252^{252} > 252^2\,10^{600}(1.016)^{50} \ge 63504 \cdot 1.8 \cdot 10^{600} > 10^{605}.$$

Apply Lemma 4.1 as in Corollary 6.4, now with $D = 10^{601}$. Since $252D + 79 < 10^{604}$, no member crosses the next boundary. □

7 Effective coding and consequences

Theorem 7.1 (Preservation of Turing degrees)**.** *For either K_{fac} or K_{B}, the homeomorphism $\omega \mapsto \alpha_\omega$ and its inverse are computable in the usual representations of binary sequences and real numbers. In particular,*

$$\deg_T(\alpha_\omega) = \deg_T(\omega).$$

Each of these zero-dimensional algebraically independent sets therefore contains a real of every Turing degree. Its computable elements form a dense subset, and its minimum is computable.

Proof. All exponent schedules are computable integer sequences. Given a finite binary word, its prime labels can be computed by exhaustive deterministic primality testing; termination follows from the existence proofs. The endpoints of its cell are real algebraic numbers, and can be approximated with certified rational errors by bisection and integer powering. Given a requested accuracy 2^{-m} and oracle access to ω, compute enough bits to obtain a cell of diameter

less than 2^{-m-2}, then approximate its endpoints. This gives a rational approximation to α_ω with error less than 2^{-m}.

Conversely, after a finite prefix has been recovered, compute the two child cells. Their positive separating gap contains an effectively obtainable rational number t. A Cauchy oracle for α_ω will eventually determine whether $\alpha_\omega < t$ or $\alpha_\omega > t$, because the point belongs to one child cell and cannot equal t. This determines the next bit and gives the inverse reduction. Since every point is transcendental, the conventional binary expansion has no dyadic ambiguity and is equivalent to its real-number oracle. Thus the two Turing reductions give equality of degrees. Eventually-zero binary sequences are computable and dense, and include the all-zero sequence representing the minimum. □

Remark 7.2 (Computability versus feasible calculation). The algorithms in Theorem 7.1 need not be efficient. The exponent growth is extreme, and finding a prescribed next prime can require enormous computation. Nor does the theorem say that every constant in the construction is computable: it explicitly includes noncomputable binary branches of every noncomputable degree.

Corollary 7.3 (Transcendental combinations and uniqueness of finite sums)**.** *Let K be either of the two constructed sets.*

1. *If $\alpha, \beta \in K$ are distinct, then $\alpha + \beta$, $\alpha - \beta$, $\alpha\beta$, and α/β are transcendental. More generally, a nonconstant rational function over $\mathbb{Q}$ evaluated at a tuple of distinct elements of K is transcendental.*
2. *A finite sum of elements of K, counted with repetitions, determines its multiset of summands uniquely. The analogous assertion holds for finite products. In particular, equal sums or products of distinct multisets are impossible, even when their cardinalities differ.*
3. *For every $\delta > 0$ there is $\beta \in K$ with $0 < \beta - \min K < \delta$, and this difference is transcendental.*

Proof. For the first assertion, algebraic independence ensures that the denominator of any nonzero rational function does not vanish. If a nonconstant rational function had algebraic value, substituting it into the minimal polynomial of that value and clearing denominators would give a nonzero rational polynomial relation among the tuple. Equal finite sums or products from different multisets likewise give nonzero polynomial relations. Finally, change a sufficiently late zero bit of the minimum branch to one. The resulting point is larger than the minimum, remains within the same arbitrarily small ancestor cell, and has transcendental difference from the minimum by the first part. □

For computable distinct branches, the quantitative certificate (5.2) can itself be found effectively. First compute enough bits that their labels differ, then increase the level until the rational error bound satisfies that inequality. The search terminates by Theorem 5.3; it yields both the explicit nonzero lower bound and, by certified evaluation of the radical approximants, the sign of any prescribed nonzero polynomial. This is an effective statement about the constructed constants, not an irrationality test for arbitrary named constants.

Corollary 7.4 (A thin field containing every Turing degree)**.** *For either constructed set K, the real field $F = \mathbb{Q}(K)$ satisfies*

$$\dim_{\mathrm{H}} F = 0, \qquad \operatorname{trdeg}_{\mathbb{Q}} F = 2^{\aleph_0}, \qquad F \cap \overline{\mathbb{Q}} = \mathbb{Q}.$$

It is a dense, meager, proper F_σ subfield of $\mathbb{R}$, and it contains a real of every Turing degree.

Proof. The Cartesian-power cover obtained from covers of K shows that each K^r also has upper box dimension zero. Every element of F is a value of a rational function P/Q over $\mathbb{Q}$ at a finite tuple from K, where the denominator is nonzero. On

$$\{x \in K^r : |Q(x)| \geq 1/m\}, \qquad m = 1, 2, \ldots,$$

the rational function is Lipschitz: polynomial values and derivatives are bounded on $[1, 2]^r$, and the denominator is bounded away from zero. Its image therefore has Hausdorff dimension zero. There are only countably many rational functions and integers m, so their union F has Hausdorff dimension zero. Each displayed domain is compact and its image is compact. These images have empty interior because their dimension is zero. Thus F is also a countable union of closed nowhere dense sets, which proves the F_σ and meagerness assertions. Since K is algebraically independent and has cardinality $2^{\aleph_0}$, it is a transcendence basis for F of that cardinality. After repeated entries in a tuple are identified, its rational function is either constant in $\mathbb{Q}$ or nonconstant in distinct generators. The rational-function assertion in Corollary 7.3 therefore gives $F \cap \overline{\mathbb{Q}} = \mathbb{Q}$. The field is dense because it contains $\mathbb{Q}$, and proper because its Hausdorff dimension is less than that of $\mathbb{R}$. Finally, $K \subset F$ already contains every Turing degree. □

The dimension assertion here is Hausdorff dimension. A dense field does not have upper box dimension zero on bounded intervals; the upper box statement applies to the compact generating set K and its finite Cartesian powers.

8 Certified large primes and finite twin-prime branches

The sparse exponent construction does not by itself make primality testing unnecessary. It does, however, accommodate candidates whose neighboring integers have large, completely factored divisors. This section records seven explicit certified primes, including two twin-prime pairs, and proves that their root cells preserve the preceding outputs. These are finite examples, not a theorem that the displayed polynomial family contains infinitely many primes or twin primes.

8.1 The seed prime and its exact certificate

Put

$$p = 513284270838218284273581257165 3 = 29^{21} + 24. \tag{8.1}$$

This is the least prime strictly exceeding 29^{21}. A complete recursive order certificate is included in the executable appendix. There are 28 nodes, counting the terminal prime 2. Its top-level factorization is

$$p - 1 = 2^2 \cdot 157 \cdot 4457 \cdot 12541 \cdot 355452619 \cdot 411378690803,$$

and the top-level witness is 2. Each prime factor is certified in the same way until the recursion reaches 2.

For clarity, the elementary criterion used in this recursion is as follows. If the complete prime factorization of $n - 1$ is known and

$$a^{n-1} \equiv 1 \pmod{n}, \qquad \gcd(a^{(n-1)/q} - 1, n) = 1 \quad (q \mid n - 1,\ q \text{ prime}),$$

then every prime divisor ℓ of n has an element of order $n - 1$ modulo ℓ. Hence $n - 1 \mid \ell - 1$, so $\ell \geq n$ and n is prime. All the factorizations and witnesses needed for this argument are given explicitly in the appendix; no probable-prime predicate is invoked.

For the intervening integers $29^{21}+i$, $1 \le i \le 23$, proper divisors in increasing order of i are

$$2, 31, 2, 3, 2, 5, 2, 571, 2, 3, 2, 13,$$
$$2, 17, 2, 3, 2, 72673, 2, 7, 2, 3, 2.$$

Thus the minimality of the seed is certified as well as its primality.

8.2 Two complementary primality certificates

Proposition 8.1 (A specialized $n-1$ certificate)**.** *Let s be a certified odd prime and let $N > 1$ satisfy $F = 2s^2 \mid N-1$ and $F > \sqrt{N}$. Suppose an integer a satisfies*

$$a^{N-1} \equiv 1 \pmod{N}, \qquad \gcd(a^{(N-1)/2}-1, N) = \gcd(a^{(N-1)/s}-1, N) = 1. \tag{8.2}$$

Then N is prime.

Proof. For any prime divisor ℓ of N, the order h of a modulo ℓ divides $N-1$. The two gcd conditions force $v_2(h) = v_2(N-1)$ and $v_s(h) = v_s(N-1)$. Consequently $F \mid h$. Since $h \mid \ell - 1$, every prime divisor satisfies $\ell \ge F+1 > \sqrt{N}$. This is impossible for a composite N. □

For the other side of a twin pair, use the Lucas sequence

$$U_0 = 0, \qquad U_1 = 1, \qquad U_{m+2} = PU_{m+1} - QU_m,$$

with discriminant $\Delta = P^2 - 4Q$.

Proposition 8.2 (A specialized $n+1$ certificate)**.** *Let s be a certified odd prime and let $N > 1$ satisfy $F = 2s^2 \mid N+1$ and $F - 1 > \sqrt{N}$. Suppose $\gcd(N, 2Q\Delta) = 1$ and*

$$U_{N+1} \equiv 0 \pmod{N}, \qquad \gcd(U_{(N+1)/2}, N) = \gcd(U_{(N+1)/s}, N) = 1. \tag{8.3}$$

Then N is prime.

Proof. Fix a prime divisor ℓ of N. The assumptions ensure that ℓ is odd and does not divide $Q\Delta$. In $\mathbb{F}_{\ell^2}$, let α, β be the distinct, nonzero roots of $X^2 - PX + Q$. The identity

$$U_m = \frac{\alpha^m - \beta^m}{\alpha - \beta}$$

shows that the least positive z with $U_z \equiv 0 \pmod{\ell}$ is the multiplicative order of α/β. Frobenius either fixes both roots or interchanges them, according as Δ is a quadratic residue or nonresidue modulo ℓ. Therefore

$$z \mid \ell - \left(\frac{\Delta}{\ell}\right), \qquad z \le \ell + 1.$$

The congruence in (8.3) gives $z \mid N+1$. The two gcd conditions force the full 2-adic and s-adic valuations of $N+1$ to occur in z, whence $F \mid z$. It follows that $\ell \ge F - 1 > \sqrt{N}$. Again this excludes a composite N. □

Both criteria are standard order-based primality certificates. Their use here is to provide short, reproducible proofs for the selected branches, not to assert a new general primality test.

8.3 Quartic lifts and preservation of the coding intervals

For an odd prime $s \geq 3$ and an odd integer j with $1 \leq j < s$, define

$$N_{\pm}(s,j) = s^4 + js^2 \pm 1, \qquad F = 2s^2.$$

The conditions on s and j imply

$$s^4 < N_-(s,j) < N_+(s,j) + 1 < (s+1)^4. \tag{8.4}$$

They also imply

$$F \mid N_+(s,j) - 1, \quad F^2 > N_+(s,j),$$

and

$$F \mid N_-(s,j) + 1, \quad (F-1)^2 > N_-(s,j).$$

For example, the last strict inequality follows from

$$(F-1)^2 - N_-(s,j) = 3s^4 - (j+4)s^2 + 2 > 0.$$

All the coefficients used below satisfy $1 \leq j < s$.

Suppose s is the desired prime output at exponent E. If either $N = N_{\pm}(s,j)$ is prime, then (8.4) gives

$$\left[N^{1/(4E)}, (N+1)^{1/(4E)}\right] \subset \left(s^{1/E}, (s+1)^{1/E}\right). \tag{8.5}$$

Thus the quartic lift simultaneously supplies an exactly certified prime and a child cell lying strictly inside the previous prime cell. Subsequent nested extensions preserve both outputs.

Starting with (8.1), set

$$\begin{aligned} P_1 &= p^4 + 33p^2 + 1, \\ P_2 &= P_1^4 + 765P_1^2 + 1, \\ P_3 &= P_2^4 + 4583P_2^2 + 1. \end{aligned}$$

The exact certificates are summarized in Table 2. The displayed witnesses satisfy every congruence and gcd condition in Proposition 8.1.

Table 2: Certified quartic lifts.

Prime	Coefficient j	Decimal digits	Witness a
P_1	33	123	2
P_2	765	492	5
P_3	4583	1966	13

Consequently, the outputs

$$3, 5, 7, 29, p, P_1, P_2, P_3$$

can be realized at the exponents

$$4, 6, 7, 12, 252, 1008, 4032, 16128,$$

respectively, and then extended indefinitely by a factorial tail. These are deliberately selected branches of the coding tree. No claim is made that they are the globally minimal constants for the corresponding exponent sequence, or that they lie on the original greedy factorial branch.

8.4 Two certified twin-prime pairs

Define

$$T_\pm = p^4 + 20303p^2 \pm 1,$$
$$W_\pm = T_+^4 + 447479T_+^2 \pm 1.$$

Both $T_\pm$ have 123 decimal digits, and both $W_\pm$ have 492 decimal digits. Their differences are exactly

$$T_+ - T_- = W_+ - W_- = 2.$$

For the plus signs, Proposition 8.1 applies with witness 17 for T_+ and witness 7 for W_+. For the minus signs, Proposition 8.2 applies with

$$(P, Q) = (1, -1), \quad \Delta = 5 \qquad \text{for } T_-,$$

and

$$(P, Q) = (1, -3), \quad \Delta = 13 \qquad \text{for } W_-.$$

The first Lucas sequence is the Fibonacci sequence; the second satisfies $U_{m+2} = U_{m+1} + 3U_m$. The exact verifier checks the discriminant gcds, all rank congruences, all gcd exclusions, and all size inequalities.

Each of the two cells for $T_\pm$ at exponent 1008 lies strictly inside the cell for p at exponent 252. Each of the two cells for $W_\pm$ at exponent 4032 lies strictly inside the cell for T_+ at exponent 1008. Attaching the same sufficiently sparse factorial tail to the two siblings produces two prime-generating constants whose outputs at the selected exponent differ by exactly 2 and whose earlier outputs agree.

This verifies finite instances of coupled prime-generating constants. The existence of suitable coefficients j at every stage, or at infinitely many stages, has not been proved. In particular, the examples do not establish twin-prime infinitude and impose no upper bound on the size of all primes.

8.5 Reproducibility and scope of the computation

The executable appendix is a self-contained Python 3 program using only integer arithmetic and the standard library. Modular powers use exact binary exponentiation. Lucas terms are evaluated by the identities

$$U_{2m} = U_m(2U_{m+1} - PU_m), \qquad U_{2m+1} = U_{m+1}^2 - QU_m^2.$$

Thus no enormous Lucas number needs to be constructed: all operations are performed modulo the candidate. The program certifies the seed, its minimality above 29^{21}, all seven later primes, their stated digit counts, both gaps of 2, and the strict inclusion of all seven child cells. Its successful execution produces

```
Seed:  28 certificate nodes; all 23 intervening integers composite.
Prime tower:  123, 492, 1966 digits; all certificates passed.
Twin pairs:  123 and 492 digits; all certificates passed.
All seven selected child cells are strictly nested.
```

9 Finite prime cells and certified minimum searches

For a finite set $A \subset \mathbb{N}$ of positive exponents, write

$$\mathcal{W}(A) = \{x > 1 : \lfloor x^c \rfloor \in \mathbb{P} \text{ for every } c \in A\}.$$

For a prescribed tuple $(p_c)_{c\in A}$ of primes, its parameter cell is

$$\bigcap_{c\in A}[p_c^{1/c},(p_c+1)^{1/c}) = \left[\max_{c\in A} p_c^{1/c}, \min_{c\in A}(p_c+1)^{1/c}\right). \tag{9.1}$$

It is nonempty precisely when its left endpoint is strictly smaller than its right endpoint. All comparisons between endpoints are decidable by integer powers: $u^{1/a} < v^{1/b}$ if and only if $u^b < v^a$.

Proposition 9.1 (Certified finite search)**.** *Suppose that $\mathcal{W}(A)$ is nonempty. Then it has a least element. Starting at $x = 2^{1/\min A}$, repeatedly apply the following rule: if $n = \lfloor x^c \rfloor$ is not prime for some $c \in A$, replace x by $q^{1/c}$, where q is the least prime strictly larger than n. This procedure terminates at $\min \mathcal{W}(A)$. If a feasible upper bound B is supplied, all its comparisons and its termination can be certified by a finite calculation with integers.*

Proof. The initial value is a lower bound for every feasible parameter. If x is such a lower bound and $\lfloor x^c \rfloor = n$ is not prime, every feasible $y \geq x$ must satisfy $y^c \geq q$. Thus every replacement preserves the lower-bound property and strictly increases x. Below any feasible B, there are only finitely many breakpoints $m^{1/c}$, with $c \in A$ and $m \leq B^c$. Consequently the procedure terminates. Its terminal point is both feasible and a lower bound for all feasible points. □

This argument is a verification procedure, not an unconditional assertion that every finite exponent set is feasible, and not a polynomial-time complexity assertion. The certificate consists of the forced jumps and the exact prime checks at the terminal point.

Proposition 9.2 (Two exponents cannot force a large first prime)**.** *If $1 \leq a < b$ are integers, then $\mathcal{W}(\{a,b\})$ is nonempty and*

$$\left\lfloor (\min \mathcal{W}(\{a,b\}))^a \right\rfloor = 2.$$

Proof. Put $r = b/a > 1$. If $r < 2$, then $x = 2^{1/a}$ gives first output 2 and second output $\lfloor 2^r \rfloor \in \{2,3\}$. If $r \geq 2$, put $n = \lfloor 2^r \rfloor$. Bertrand's postulate supplies a prime q with $n < q < 2n$. Hence

$$2^r < q < 2^{r+1} < 3^r,$$

so $x = q^{1/b}$ is feasible and has first output 2. The minimum is no larger than this witness and no smaller than $2^{1/a}$; its first output must therefore also be 2. □

There is a sharp version relative to the initial exponent pair $(4,6)$.

Proposition 9.3 (Exactly two extra constraints are needed after $(4,6)$)**.** *For every integer $c > 6$, there exists an x such that*

$$(\lfloor x^4 \rfloor, \lfloor x^6 \rfloor) = (2,3), \qquad \lfloor x^c \rfloor \in \mathbb{P}.$$

In particular, one additional exponent cannot force the first output of the minimum above 2. In contrast, the two additional exponents $7, 12$ give

$$\min \mathcal{W}(\{4,6,7,12\}) = 29^{1/12}, \qquad (\lfloor x^4 \rfloor, \lfloor x^6 \rfloor, \lfloor x^7 \rfloor, \lfloor x^{12} \rfloor) = (3,5,7,29)$$

at that minimum.

Proof. Every $x \in [3^{1/6}, 4^{1/6})$ has first two outputs $(2, 3)$, since $2 < 3^{2/3} < 4^{2/3} < 3$. For $c \geq 15$, $(4/3)^{c/6} > 2$. Apply Bertrand's postulate to $\lfloor 3^{c/6} \rfloor$ to find a prime $q \in (3^{c/6}, 4^{c/6})$. Then $x = q^{1/c}$ is a witness. The remaining cases are certified by

c	7	8	9	10	11	12	13	14
q	5	5	7	7	11	11	11	13

using the exact inequalities $3^c \leq q^6 < 4^c$. The four-exponent assertion is exactly Proposition 2.2, whose forced-jump certificate is given in Table 1. □

The qualification "after $(4, 6)$" is essential: this proposition does not settle whether some other three-exponent prefix can force the first output above 2.

10 Twin-output geometry and an exact moving-window reformulation

Proposition 10.1 (Exact two-cell criterion)**.** *Let $p \geq 2$, $a \geq 1$, and $r \geq 1$ be integers. There exists an $x > 1$ with*

$$\lfloor x^a \rfloor = p, \qquad \lfloor x^{a+r} \rfloor = p + 2$$

if and only if

$$\frac{\log p}{\log(1 + 3/p)} < \frac{a}{r} < \frac{\log(p + 1)}{\log(1 + 1/(p + 1))}. \tag{10.1}$$

Primality of $p, p + 2$ is not needed for this geometric statement.

Proof. The two cells are $[p^{1/a}, (p + 1)^{1/a})$ and $[(p + 2)^{1/(a+r)}, (p + 3)^{1/(a+r)})$. Their intersection is nonempty exactly when

$$p^{1/a} < (p + 3)^{1/(a+r)}, \qquad (p + 2)^{1/(a+r)} < (p + 1)^{1/a}.$$

Taking logarithms and rearranging gives (10.1). □

The two bounds in (10.1) are asymptotic to $p \log p/3$ and $p \log p$, respectively. Thus, along successful twin encodings, $p \to \infty$ if and only if $a/r \to \infty$. The geometric criterion says when an already specified pair can be encoded; it does not assert primality or the existence of new twin pairs.

The next result chooses its exponent from the scale N, without first choosing a twin pair. It also gives an exact topological counting interpretation.

Theorem 10.2 (A target-free moving window)**.** *For an integer $N \geq 16$, put*

$$a_N = \lfloor N \log N \rfloor, \qquad I_N = [N^{1/a_N}, (2N)^{1/a_N}),$$

and let

$$\mathcal{T}_N = \{x \in I_N : \lfloor x^{a_N} \rfloor, \lfloor x^{a_N+1} \rfloor \in \mathbb{P}\}.$$

Every $x \in \mathcal{T}_N$ produces a twin-prime pair. Moreover, $\mathcal{T}_N$ is nonempty if and only if a twin pair $p, p + 2$ exists with $N \leq p < 2N$. Its connected components are in one-to-one correspondence with those twin pairs, and the component for p has length

$$\min\{(p + 1)^{1/a_N}, (p + 3)^{1/(a_N+1)}\} - \max\{p^{1/a_N}, (p + 2)^{1/(a_N+1)}\} > 0. \tag{10.2}$$

Proof. Write $a = a_N$ and $y = x^a \in [N, 2N)$. Then

$$\Delta(y) = x^{a+1} - x^a = y(y^{1/a} - 1)$$

is strictly increasing in $y > 1$. Since $e^t - 1 \geq t$,

$$\Delta(y) \geq \frac{N \log N}{a} \geq 1.$$

Using $e^t - 1 < t/(1-t)$ for $0 < t < 1$ and $a > N \log N - 1$ gives

$$\Delta(y) < \frac{2N \log(2N)}{N \log N - 1 - \log(2N)} < 3.$$

For the last inequality, the denominator is positive and the difference between three times the denominator and the numerator is

$$h(N) = N \log N - 2N \log 2 - 3 \log(2N) - 3.$$

Here $h(16) = 17 \log 2 - 3 > 0$ and $h'(N) = \log N + 1 - 2 \log 2 - 3/N > 0$ for $N \geq 16$. Thus $1 \leq x^{a+1} - x^a < 3$. If both floors are prime, they are odd and distinct; their difference is at most 3, and hence is exactly 2.

Conversely, let $N \leq p < 2N$ be a twin lower prime and set $f(y) = y^{1+1/a}$. The preceding bound gives $f(p) < p + 3$. Strict monotonicity of Δ gives $f(p+1) > p + 2$, including when $p + 1 = 2N$. If $f(p) \geq p + 2$, take $y = p$; otherwise the intermediate value theorem gives a $y \in (p, p+1)$ with $f(y) = p + 2$. This proves existence, and continuity gives a cell of positive length. Its endpoints are exactly those in (10.2). Distinct lower primes give separated cells, since the disjoint intervals $[p, p+1)$ in the y coordinate have positive gaps between them. □

The theorem is an exact reformulation, including a counting identity; it is not an unconditional nonemptiness theorem for all or infinitely many windows. In particular, the twin-prime conjecture is equivalent to $\mathcal{T}_{2^m} \neq \varnothing$ for infinitely many integers m. The minimum taken within I_N must not be confused with the unrestricted minimum of $\mathcal{W}(\{a_N, a_N + 1\})$, whose first output is always 2 by Proposition 9.2.

11 Rigorous obstructions for fixed power spectra

Proposition 11.1 (A fixed-base bound)**.** *Let $x > 1$, let $a < b$ be positive integers, and suppose $\lfloor x^a \rfloor = p$ and $\lfloor x^b \rfloor = p + d$ with $d \geq 1$. Then*

$$p < \frac{d+1}{x-1}. \tag{11.1}$$

Consequently a fixed base has only finitely many twin pairs anywhere in its entire integer-exponent output spectrum.

Proof. Since $b \geq a + 1$, $xp \leq x^{a+1} \leq x^b < p + d + 1$. Rearranging proves the bound. Only finitely many integer exponents have outputs below a fixed bound. □

For the constant in the main construction, $\xi > 29^{1/12} > 13/10$, where the latter comparison follows from $29 \cdot 10^{12} > 13^{12}$. Hence $3/(\xi - 1) < 10$. Therefore its complete integer-power spectrum can contain only the twin pairs $(3, 5)$ and $(5, 7)$: the only possible prime lower members below 10 are $2, 3, 5, 7$, and $2 + 2$ and $7 + 2$ are composite. Both pairs occur at exponents $4, 6, 7$, so the spectrum contains exactly these two twin pairs. This assertion concerns pairs of values in the spectrum, not adjacency of their exponents.

Proposition 11.2 (Scaling does not enlarge the prime outputs)**.** *Suppose a positive integer exponent sequence $C = (C_k)$ has a least prime-generating parameter $\xi(C)$. For an integer $m \geq 1$, put $mC = (mC_k)$. Then*

$$\xi(mC) = \xi(C)^{1/m}, \qquad \lfloor \xi(mC)^{mC_k} \rfloor = \lfloor \xi(C)^{C_k} \rfloor.$$

Thus a family of bases tending to 1 need not yield any new prime values or any larger twin pairs.

Proof. The increasing bijection $x \mapsto x^m$ identifies the two feasible sets and preserves their order. □

A second obstruction applies even when two different fixed bases and two independently chosen exponent sequences are allowed.

Theorem 11.3 (Exponential approximation forced by bounded output gaps)**.** *Let $x, y > 1$ and $D \geq 1$ be fixed. If infinitely many distinct pairs $(a, b) \in \mathbb{N}^2$ satisfy*

$$1 \leq |\lfloor x^a \rfloor - \lfloor y^b \rfloor| \leq D, \tag{11.2}$$

then $\rho = \log x / \log y$ is a Liouville number. More precisely, along an unbounded sequence of such exponents,

$$0 < \left| \rho - \frac{b}{a} \right| < \frac{2(D+1)}{a \log y}\, x^{-a}. \tag{11.3}$$

Proof. Infinitely many distinct exponent pairs satisfying the bounded-gap condition force $a \to \infty$ along a subsequence. Put $A = x^a$ and $B = y^b$. Then $|A - B| < D + 1$. For $A > 2(D+1)$, both A and B exceed $A/2$, so the mean value theorem for the logarithm gives

$$|a \log x - b \log y| < 2(D+1)x^{-a}.$$

The two powers cannot be equal because their floors differ. This proves (11.3).

If $\rho = u/v$ were rational, every nonzero difference $|\rho - b/a|$ would be at least $1/(va)$, contradicting the displayed exponential upper bound for sufficiently large a. Thus ρ is irrational. For every positive integer L, the upper bound is eventually smaller than a^{-L}. Reducing b/a to a fraction with denominator $q \leq a$ only weakens the bound $a^{-L} \leq q^{-L}$; the reduced denominators must be unbounded because ρ is irrational. This is the Liouville property. □

The exponential estimate is stronger than the bare Liouville property, and neither is asserted to be sufficient for bounded nonzero gaps or for primality. In particular, if $\log x / \log y$ is rational or is an algebraic irrational number, only finitely many bounded nonzero gaps can occur. In the algebraic irrational case this follows already from the elementary Liouville approximation inequality, without needing Roth's theorem.

Corollary 11.4 (A common exponent cannot sustain fixed nonzero gaps)**.** *Two fixed bases $x, y > 1$ cannot satisfy $1 \leq |\lfloor x^{E_n} \rfloor - \lfloor y^{E_n} \rfloor| \leq D$ for an unbounded integer sequence E_n. If $x \neq y$, the absolute differences tend to infinity; if $x = y$, they vanish identically.*

Proof. For $x > y$, $x^E - y^E = x^E(1 - (y/x)^E) \to \infty$, and taking floors changes the difference by less than 1. The other cases follow by symmetry. □

Thus finite constructions of two constants with twin outputs at a common exponent do not automatically iterate to infinitely many twin outputs for those same two constants.

12 Adaptive insertion and prescribed residue classes

The freedom to choose new exponents gives a general extension principle that needs no estimate for short prime gaps.

Lemma 12.1 (Insertion of a prescribed large integer)**.** *Let $1 < u < v$, let $m \geq 1$, and let $H > 0$. Every sufficiently large integer q admits an exponent $E \in m\mathbb{N}$ with $E > H$ such that*

$$[q^{1/E}, (q+1)^{1/E}] \subset (u, v).$$

Proof. The required strict inequalities are equivalent to

$$\frac{\log(q+1)}{\log v} < E < \frac{\log q}{\log u}.$$

The interval length tends to infinity as $q \to \infty$, and its lower endpoint also tends to infinity. Eventually its length exceeds m and it lies above H, so it contains a multiple of m strictly between its endpoints. □

Theorem 12.2 (Adaptive coding in arbitrary infinite prime sets)**.** *Let J be a nonempty bounded open interval with $\overline{J} \subset (1, \infty)$. Let $S_1, S_2, \ldots$ be arbitrary infinite subsets of the primes, and let $G_1, G_2, \ldots$ be positive real thresholds. Then there exist an $\alpha \in J$, strictly increasing positive integer exponents E_n, and primes $p_n \in S_n$, such that*

$$E_n \mid E_{n+1}, \qquad p_n > G_n, \qquad \lfloor \alpha^{E_n} \rfloor = p_n \quad (n \geq 1).$$

The exponents can additionally be required at each stage to exceed any specified finite threshold depending on the preceding choices. If all points of J satisfy a given finite prime-output prefix, that prefix is preserved.

Proof. Choose $p_1 \in S_1$ large enough to apply Lemma 12.1 inside J, imposing $p_1 > G_1$. This produces a closed prime cell J_1 in J. Given J_n, apply the lemma in its interior with $m = E_n$, imposing the next exponent threshold and choosing $p_{n+1} \in S_{n+1}$ sufficiently large, in particular above G_{n+1}. This gives $J_{n+1} \subset \mathrm{int}(J_n)$ and $E_n \mid E_{n+1}$. Choose the exponents to tend to infinity. If $\overline{J} \subset [u, v]$ with $u > 1$, the mean value theorem bounds

$$|J_n| \leq \frac{1}{E_n u^{E_n - 1}} \longrightarrow 0.$$

Nested compact intervals therefore have a unique common point α. For every n, this point lies in $J_{n+1} \subset \mathrm{int}(J_n)$, so the floor at exponent E_n is exactly p_n. □

Corollary 12.3 (Arbitrarily prescribed reduced residue classes)**.** *Given positive integers m_n and integers a_n satisfying $\gcd(a_n, m_n) = 1$, one can make the construction in Theorem 12.2 satisfy*

$$\lfloor \alpha^{E_n} \rfloor \equiv a_n \pmod{m_n} \qquad (n \geq 1).$$

Proof. Dirichlet's theorem says that each target set $S_n = \{p \in \mathbb{P} : p \equiv a_n \pmod{m_n}\}$ is infinite. Apply the theorem. □

This extension permits arbitrary prescribed growth thresholds and residue classes while preserving any already established open prefix cell. It constructs an admissible constant, not necessarily the global minimum for the resulting exponent sequence. The exponents are chosen adaptively and need not be the factorial-tail exponents of the main construction. Using a target set of twin lower primes here would require its infinitude as an input; the insertion argument cannot supply that missing arithmetic assertion.

13 An exact spectral encoding from the prime-generating constants

The conclusions in this section are elementary consequences of algebraic independence. They do not solve an open graph-theoretic problem, and the use of generic diagonal entries is standard. Their role here is to show that the prime-generating construction supplies explicit choices of the otherwise generic parameters. In particular, one may use distinct members of the set K constructed above, even when all the chosen members have the exceptionally long common decimal prefix established there.

Fix algebraically independent real numbers $\alpha_1, \dots, \alpha_n$ over $\mathbb{Q}$, and write

$$D = \operatorname{diag}(\alpha_1, \dots, \alpha_n).$$

All matrix entries called rational below belong to $\mathbb{Q}$.

Theorem 13.1 (A quartic trace determines a labeled weighted graph)**.** *For a symmetric rational matrix $A = (a_{ij})$ with zero diagonal, put*

$$\Phi_\alpha(A) = \operatorname{tr}(A + D)^4.$$

If $\Phi_\alpha(A) = \Phi_\alpha(B)$ for two such matrices, then $a_{ij}^2 = b_{ij}^2$ for every i, j. Consequently Φ_α is injective on the class of nonnegative rational symmetric zero-diagonal matrices, and in particular on the adjacency matrices of labeled simple graphs. The exponent 4 is the smallest positive integer with this universal injectivity property for labeled simple graphs of arbitrary order.

Proof. Cyclic invariance of the trace gives

$$\begin{aligned}\operatorname{tr}(A + D)^4 = {} & \sum_i \alpha_i^4 + 4\sum_i (A^2)_{ii}\alpha_i^2 + 4\sum_{i<j} a_{ij}^2 \alpha_i \alpha_j \\ & + 4\sum_i (A^3)_{ii}\alpha_i + \operatorname{tr} A^4. \end{aligned} \tag{13.1}$$

Indeed, the six words containing two occurrences each of A and D contribute $4\operatorname{tr}(A^2D^2) + 2\operatorname{tr}(ADAD)$. The terms with three copies of D vanish because $a_{ii} = 0$. Algebraic independence says that equality of the two evaluated polynomials in (13.1) implies equality of their coefficients. Comparing the coefficient of $\alpha_i\alpha_j$ gives $a_{ij}^2 = b_{ij}^2$.

For a simple graph with m edges, labeled degrees d_i, and t triangles, the first three traces are

$$\begin{aligned}\operatorname{tr}(A + D) &= \sum_i \alpha_i, \\ \operatorname{tr}(A + D)^2 &= \sum_i \alpha_i^2 + 2m, \\ \operatorname{tr}(A + D)^3 &= \sum_i \alpha_i^3 + 3\sum_i d_i\alpha_i + 6t.\end{aligned}$$

The Petersen graph and the pentagonal prism both have ten vertices, are 3-regular, and contain no triangles. For any labeling their first three traces therefore agree. They are different graphs: the prism has 4-cycles, whereas the Petersen graph has none. Thus no exponent 1, 2, or 3 is universally injective. The quartic trace is injective by the preceding argument. □

Remark 13.2. The nonnegativity qualification is necessary for recovering signed weights. If S is a diagonal matrix with entries in $\{1, -1\}$, then $SDS = D$ and

$$SAS + D = S(A + D)S.$$

Thus A and SAS have the same trace at every power, although their edge signs can differ. The theorem recovers squared weights without this qualification and the weights themselves when they are nonnegative.

The invariant is an exact real-number encoding. Algebraic independence alone does not give a useful uniform precision bound for decoding it. For fixed n and fixed computable labels, however, the finitely many values arising from labeled simple graphs are distinct computable reals. Successively refining their certified approximations eventually produces disjoint intervals. This is a terminating finite-precision reconstruction procedure, without a claim of computational efficiency.

Theorem 13.3 (Independent eigenvalues)**.** *Let A be any rational symmetric $n \times n$ matrix. The n eigenvalues of $A + D$ are algebraically independent over $\mathbb{Q}$. In particular, they are simple and are individually transcendental.*

Proof. Replace the α_i temporarily by indeterminates x_i, and write

$$\det(tI - A - \operatorname{diag}(x_1, \ldots, x_n)) = t^n + c_1(x)t^{n-1} + \cdots + c_n(x).$$

The total degree of c_k is k, and its homogeneous part of degree k is $(-1)^k e_k(x)$, where e_k is the kth elementary symmetric polynomial. The c_k are algebraically independent: otherwise, take a nonzero relation $P(c_1, \ldots, c_n) = 0$, give the kth argument of P weight k, and pass to the highest weighted homogeneous part P_{top}. The highest ordinary homogeneous part after substitution would be

$$P_{\mathrm{top}}(-e_1, e_2, \ldots, (-1)^n e_n) = 0,$$

contrary to the algebraic independence of $e_1, \ldots, e_n$. Since evaluation at the algebraically independent tuple α is injective on $\mathbb{Q}[x_1, \ldots, x_n]$, the numbers $c_1(\alpha), \ldots, c_n(\alpha)$ are algebraically independent.

Let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues, counted with multiplicity. Their elementary symmetric polynomials are the $c_k(\alpha)$ up to signs. Hence the field $\mathbb{Q}(\lambda_1, \ldots, \lambda_n)$ has transcendence degree at least n. It is generated by n elements, so these elements must be algebraically independent. This also rules out repeated eigenvalues. □

Theorem 13.4 (Every vertex is cyclic)**.** *Let A be a rational symmetric matrix whose support graph, with edges ij whenever $i \neq j$ and $a_{ij} \neq 0$, is connected. Put $H = A + D$. For every coordinate vector e_v, the matrix*

$$[\, e_v, He_v, \ldots, H^{n-1}e_v \,]$$

is invertible. Equivalently, the linear system $\dot{z} = Hz + e_v u$ is controllable from any single coordinate.

Proof. Absorb the rational diagonal entries of A into the α_i; this preserves algebraic independence. We may therefore assume $a_{ii} = 0$. Consider

$$F(x) = \det(\operatorname{diag}(x_1, \ldots, x_n) - A).$$

We first prove that F is irreducible over $\mathbb{Q}$. It is multiaffine and its coefficient of $x_1 \cdots x_n$ is 1. In any factorization $F = GH$, the degree in each variable is additive, so each variable occurs in at most one factor. A nontrivial factorization therefore partitions the variables into two nonempty sets S, T. Normalize the top monomials of the factors to have coefficient 1. The coefficient of the monomial missing only x_i is $-a_{ii} = 0$, so each corresponding coefficient in its factor is zero. For $i \in S$ and $j \in T$, the coefficient missing both x_i, x_j would then be zero. In F, however, this coefficient is $-a_{ij}^2$. Thus every edge between S and T would vanish, contrary to connectedness.

Adjoin an independent variable t and make the invertible substitution $x_i = t - d_i$. It follows that

$$f(t, d) = \det(tI - A - \operatorname{diag}(d_1, \ldots, d_n))$$

is irreducible in $\mathbb{Q}[t, d]$. Since it is monic in t, it is also irreducible over $\mathbb{Q}(d)[t]$. Let $g_v(t, d)$ be the characteristic polynomial of the principal submatrix with row and column v deleted. It is monic of degree $n-1$, so f and g_v are relatively prime over $\mathbb{Q}(d)$. Their resultant is therefore a nonzero polynomial in $\mathbb{Q}[d]$. Evaluation at $d = \alpha$ cannot annihilate that polynomial. Thus H and the principal submatrix obtained by deleting v have no common eigenvalue.

If an eigenvector of H had zero vth coordinate, deleting that coordinate would give a nonzero eigenvector of the principal submatrix with the same eigenvalue. This is impossible. Every eigenvector therefore has nonzero vth coordinate. Together with simplicity of the spectrum, the spectral decomposition of the Krylov matrix gives a product of an invertible orthogonal matrix, an invertible diagonal matrix, and a Vandermonde matrix on distinct eigenvalues. The Krylov matrix is invertible. □

Corollary 13.5 (No exact vertex revival)**.** *Under the hypotheses of Theorem 13.4, suppose $n \geq 3$. There are no real $t \neq 0$ and complex γ with $|\gamma| = 1$ such that*

$$e^{-itH} e_v = \gamma e_v.$$

There is also no perfect state transfer between distinct coordinate vectors.

Proof. Every eigenvector has nonzero overlap with e_v. A revival would therefore require $e^{-it\lambda_j} = \gamma$ for every j. For three distinct eigenvalues this gives nonzero integers k_2, k_3 such that

$$t(\lambda_2 - \lambda_1) = 2\pi k_2, \qquad t(\lambda_3 - \lambda_1) = 2\pi k_3.$$

Eliminating t produces the nontrivial rational linear relation

$$k_3(\lambda_2 - \lambda_1) - k_2(\lambda_3 - \lambda_1) = 0,$$

contradicting Theorem 13.3. For a real symmetric H, perfect state transfer from v to w would imply revival at v at twice the transfer time: in a real orthonormal eigenbasis the two coordinate overlaps must agree up to sign, so the squared phase is common to every eigencomponent. The already proved absence of revival rules this out. □

These applications are stable under replacing the labels by any other algebraically independent tuple. What the prime-generating framework adds is the simultaneous arithmetic specification of the labels: each label retains the prescribed prime outputs and, in the set K, the shared long decimal prefix. No unbounded twin-prime conclusion is used in, or follows from, these spectral transfers.

Part II
Geometry, sparse relations, and exponential independence

The full spectrum and selected thin subsets are treated separately. Dimension results concern all admissible constants; the later independence results use explicitly refined subsets.

For an increasing positive exponent sequence $E = (E_k)$, write

$$\mathcal{P}(E) = \{x > 1 : \lfloor x^{E_k} \rfloor \in \mathbb{P} \text{ for every } k\}.$$

This part distinguishes the full admissible spectrum from a prescribed compact prime-coding subset. A dimension statement about the first and an independence statement about the second do not imply either statement for the unrestricted least admissible base. We first treat dimension, then quantitative separation, and finally selection within a fixed schedule.

Throughout this part, $\dim_H$ denotes Hausdorff dimension and $\overline{\mathbb{Q}}$ denotes the field of complex algebraic numbers. A perfect set is nonempty, closed, and has no isolated points. Algebraic independence of an indexed family means nonvanishing of every nonzero polynomial in finitely many distinct indexed entries. Computability refers to ordinary real or complex Cauchy names, with no complexity bound asserted.

14 Removing the boundedness hypothesis in the dimension theorem

For a sequence $\mathbf{c} = (c_k)_{k\ge1}$ of positive real numbers, put

$$C_k = \prod_{i=1}^{k} c_i, \qquad \mathcal{W}(\mathbf{c}) = \{x > 1 : \lfloor x^{C_k} \rfloor \text{ is prime for every } k \ge 1\}.$$

The distinction between multipliers c_k and cumulative exponents C_k is essential in this section. The intermediate multipliers introduced below are real numbers, even when all prescribed cumulative exponents are integers.

Theorem 14.1 (Saito's bounded-multiplier theorem)**.** *If $2 \le c_k \le R < \infty$ for every k, then* $\dim_H \mathcal{W}(\mathbf{c}) = 1$.

This is [33, Theorem 1.2]. The same paper asks in Question 1.4 of its arXiv version whether the Hausdorff dimension can be determined when the upper bound on c_k is removed. A bounded refinement settles that question.

Lemma 14.2 (Bounded multiplicative refinement)**.** *Let $a > 1$ and let $c_k \ge a$ be real numbers. There are real numbers $d_j \in [a, a^2)$ and strictly increasing integers N_k such that, with $D_j = \prod_{i=1}^{j} d_i$, one has*

$$D_{N_k} = C_k \qquad (k \ge 1).$$

Consequently, $\mathcal{W}(\mathbf{d}) \subseteq \mathcal{W}(\mathbf{c})$. More generally, for any fixed set $T \subseteq \mathbb{Z}$, satisfying $\lfloor x^{D_j} \rfloor \in T$ for every j implies $\lfloor x^{C_k} \rfloor \in T$ for every k.

Proof. Set $m_k = \lfloor \log c_k / \log a \rfloor \ge 1$ and $N_0 = 0$, $N_k = \sum_{i=1}^{k} m_i$. For $1 \le j \le m_k$, define

$$d_{N_{k-1}+j} = c_k^{1/m_k}.$$

Since $a^{m_k} \le c_k < a^{m_k+1}$,

$$a \le c_k^{1/m_k} < a^{1+1/m_k} \le a^2.$$

Each block has product c_k, so $D_{N_k} = C_k$ exactly. If all outputs at the refined exponents D_j are prime, then in particular the outputs at their subsequence $D_{N_k} = C_k$ are prime. □

Theorem 14.3 (Full dimension without an upper multiplier bound)**.** *For every real sequence $c_k \ge 2$, bounded or unbounded,*

$$\boxed{\dim_H \mathcal{W}(\mathbf{c}) = 1.}$$

Proof. Apply Lemma 14.2 with $a = 2$. The refined multipliers belong to $[2, 4)$, so Theorem 14.1 gives $\dim_H \mathcal{W}(\mathbf{d}) = 1$. Monotonicity of Hausdorff dimension yields

$$1 = \dim_H \mathcal{W}(\mathbf{d}) \le \dim_H \mathcal{W}(\mathbf{c}) \le \dim_H \mathbb{R} = 1.$$

□

Remark 14.4 (Scope and attribution)*.* Theorem 14.3 answers Question 1.4 as stated in [33, arXiv version]. Its proof is an elementary consequence of Saito's substantial bounded-multiplier theorem, rather than a new estimate for primes in short intervals. This observation does not assert historical priority for the refinement argument. This refinement argument alone does not resolve Question 1.5, which concerns iterated exponentials; the separate argument in Section 15 does so. Nor does the refinement imply full dimension for a branch-restricted Cantor subset or for an arbitrarily prescribed finite prefix cell.

Corollary 14.5 (Arbitrary initial scale)**.** *Let $E_k > 0$ satisfy $E_{k+1}/E_k \ge 2$ for all $k \ge 1$. Then*

$$\dim_H\{x > 1 : \lfloor x^{E_k} \rfloor \text{ is prime for all } k \ge 1\} = 1.$$

Proof. Set $c_1 = 2$ and $c_{k+1} = E_{k+1}/E_k$. Then $C_k = 2E_k/E_1$. Under the change of variable $x = y^{2/E_1}$, one has $x^{E_k} = y^{C_k}$. The change of variable is bi-Lipschitz on each compact subinterval of $(1, \infty)$, and countable stability of Hausdorff dimension proves the assertion. □

For example, the conclusion applies to $E_k = k! + k$, beginning with $k = 1$: the first two ratios are 2 and $9/4$, and for $k \ge 2$,

$$(k+1)! + (k+1) - 2(k! + k) = (k-1)(k! - 1) \ge 0.$$

It also applies to $E_k = 3^k - k$ and $E_k = 3^k - 2k$, because respectively

$$E_{k+1} - 2E_k = 3^k + k - 1 > 0, \qquad E_{k+1} - 2E_k = 3^k + 2k - 2 > 0.$$

The second sequence starts at $E_1 = 1$, which is covered by Corollary 14.5. These are statements about the full sets of permissible bases, not about the arithmetic nature of their least elements.

The endpoint 4 in the refinement interval $[2, 4)$ is best possible for this universal factorization device. Indeed, a number $c \in (2, 4)$ cannot be a product of two or more factors each at least 2; if one demanded a universal interval $[2, R]$ with $R < 4$, any $c \in (R, 4)$ would be impossible to refine. This optimality concerns the interpolation lemma alone.

15 Full dimension for Wright prime orbits

Let $b > 1$, put $f_b(x) = b^x$, and write $f_b^{\circ n}$ for its n-fold composition. Define

$$\mathcal{W}_b^{\mathrm{it}} = \{x > 1 : \lfloor f_b^{\circ n}(x) \rfloor \text{ is prime for every } n \geq 1\}.$$

Question 1.5 in the arXiv version of [33] asks for the Hausdorff dimension of $\mathcal{W}_2^{\mathrm{it}}$. The argument below answers this question and allows every fixed real base $b > 1$. It uses ordinary prime counting in fixed-ratio intervals and a uniform upper sieve estimate; it does not use a lower bound for primes in short intervals.

Theorem 15.1 (Dimension of prime iterated-exponential orbits)**.** *For every real $b > 1$,*

$$\boxed{\dim_H \mathcal{W}_b^{\mathrm{it}} = 1.}$$

More precisely, for every sufficiently large prime p_0, there is a compact perfect set $K \subset [p_0 + 1/4, p_0 + 3/4]$ of Hausdorff dimension 1 such that $\lfloor f_b^{\circ n}(x) \rfloor$ is prime for every $x \in K$ and every $n \geq 0$.

We first specify the two external prime-counting inputs. The prime number theorem implies that, for fixed $0 < a < d$,

$$\#(\mathbb{P} \cap [aX + O(1), dX + O(1)]) \sim \frac{(d-a)X}{\log X} \qquad (X \longrightarrow \infty). \tag{15.1}$$

The Montgomery–Vaughan form of the Brun–Titchmarsh inequality [27, Theorem 2], specialized to modulus 1, gives

$$\pi(u+h) - \pi(u) < \frac{2h}{\log h} \qquad (u \geq 1,\ h > 1).$$

In particular, after allowing either convention for interval endpoints and enlarging an absolute constant C, every interval $A \subset [1, \infty)$ of length $h \geq 0$ satisfies

$$\#(A \cap \mathbb{P}) \leq C \frac{h+1}{\log(h+2)}. \tag{15.2}$$

For $0 \leq h \leq 2$ this last assertion also follows directly by counting integers. The uniformity in the location of A is essential.

15.1 Closed prime cells and uniform inverse distortion

Fix $b > 1$, write $f = f_b$ and $g(x) = \log x / \log b$, and let

$$I_p = [p + 1/4, p + 3/4] \qquad (p \in \mathbb{P}).$$

Choose $P > 1$ sufficiently large that

$$\rho := \frac{1}{P \log b} < 1, \qquad b^{p+1/4} - 1/4 \geq p + 1 \quad (p \geq P),$$

and that all counting and contraction estimates below hold for $p \geq P$. Starting with a prime $p_0 \geq P$, call a finite sequence $\omega = (p_0, \ldots, p_n)$ admissible if all its entries are prime and

$$I_{p_{j+1}} \subset f(I_{p_j}) \qquad (0 \leq j < n).$$

Its cylinder in the original parameter axis is

$$J_\omega = g^{\circ n}(I_{p_n}).$$

For every $x \in J_\omega$ and $0 \leq j \leq n$ one has $f^{\circ j}(x) \in I_{p_j}$. In particular, all those integer parts are prime, including at both endpoints of J_ω.

Lemma 15.2 (Distortion and child geometry)**.** *There exist positive constants c_b, C_b, independent of the admissible word and its length, with the following properties. If $J = J_\omega$ has final prime label p, its children are exactly $J_{\omega q} = g^{\circ(n+1)}(I_q)$ for primes q with $I_q \subset f(I_p)$. Their number N_p satisfies*

$$c_b \frac{b^p}{p} \le N_p \le C_b \frac{b^p}{p}, \qquad N_p \ge 2. \tag{15.3}$$

Every child satisfies

$$c_b b^{-p}|J| \le |J_{\omega q}| \le C_b b^{-p}|J|, \tag{15.4}$$

and distinct children are separated by a gap of length at least $c_b b^{-p}|J|$. Moreover,

$$\sup_{x\in J} (f^{\circ(n+1)})'(x) \le C_b \frac{b^p}{|J|}. \tag{15.5}$$

Here and below, constants with the same subscript may be enlarged or decreased when necessary.

Proof. Put $h = g^{\circ n}$. For $u, v \in I_p$, all the intermediate points $g^{\circ j}(u)$ and $g^{\circ j}(v)$ belong to $[P, \infty)$. On that half-line g is ρ-Lipschitz. The derivative formula

$$h'(u) = (\log b)^{-n} \prod_{j=0}^{n-1} \left(g^{\circ j}(u)\right)^{-1}$$

therefore gives

$$\begin{aligned} \left|\log \frac{h'(u)}{h'(v)}\right| &\le \sum_{j=0}^{n-1} \frac{|g^{\circ j}(u) - g^{\circ j}(v)|}{P} \\ &\le \frac{|u-v|}{P(1-\rho)} \le \frac{1}{2P(1-\rho)}. \end{aligned}$$

Thus the distortion of h on I_p is at most $D = \exp(1/(2P(1-\rho)))$, uniformly in n and ω. This includes $n = 0$, when h is the identity. Since $|I_p| = 1/2$,

$$\frac{2|J|}{D} \le h'(u) \le 2D|J| \qquad (u \in I_p). \tag{15.6}$$

Write $a = b^{1/4}$ and $d = b^{3/4}$. The permitted child primes are exactly

$$q \in \mathbb{P} \cap [ab^p - 1/4, db^p - 3/4].$$

Equation (15.1) gives $N_p \sim (d-a)b^p/(p \log b)$, proving (15.3) after increasing P. On $f(I_p) = [ab^p, db^p]$, the derivative $g'(t) = 1/(t \log b)$ is bounded above and below by positive constants depending only on b times b^{-p}. Combining this fact with (15.6) gives (15.4). Distinct integer labels give a gap of at least $1/2$ between their trimmed cells I_q, so the same derivative lower bound proves the asserted gap estimate. Finally, on J the derivative of $f^{\circ n} = h^{-1}$ is at most $D/(2|J|)$. Since $f^{\circ n}(J) = I_p$ and $f'(u) = (\log b)b^u$, the chain rule proves (15.5). □

Increasing P once more, we may suppose that $C_b b^{-p} \le 1/2$ for $p \ge P$ in (15.4). Let K_n be the finite union of all depth-n cylinders, starting with $K_0 = I_{p_0}$, and put

$$K = \bigcap_{n\ge 0} K_n.$$

The cylinders are closed, nested, and disjoint at each level, every cylinder has at least two children, and their maximum length tends to zero. Hence K is a nonempty compact perfect set, and each of its points has one infinite admissible itinerary. The floor condition above shows $K \subset \mathcal{W}_b^{\mathrm{it}}$.

15.2 An interval mass estimate

Give I_{p_0} mass 1 and divide each cylinder's mass equally among its N_p children. The consistent cylinder masses define a Borel probability measure μ on K. Since $N_p \geq 2$, the measure has no atoms. By (15.3),

$$\mu(J_{\omega q}) \leq C_b p b^{-p} \mu(J_\omega). \tag{15.7}$$

Fix $0 < s < 1$. Equations (15.4) and (15.7) imply

$$\frac{\mu(J_{\omega q})}{|J_{\omega q}|^s} \leq C_b c_b^{-s} p b^{-(1-s)p} \frac{\mu(J_\omega)}{|J_\omega|^s}. \tag{15.8}$$

The coefficient on the right is at most 1 for all sufficiently large p, depending on s and b. By the choice of P, every child label is at least its parent label plus 1. Thus the minimum label at depth n tends to infinity. Only finitely many cylinders occur before all labels reach the required threshold. Taking the maximum of $\mu(J)/|J|^s$ over those finitely many cylinders and then using (15.8) gives a finite constant A_s such that

$$\mu(J) \leq A_s |J|^s \qquad \text{for every cylinder } J. \tag{15.9}$$

This estimate alone does not yet give a Hausdorff-dimension bound: the number of children is unbounded, so intervals between cylinder scales must also be controlled. The uniform upper sieve supplies this missing estimate. Let $B \subset \mathbb{R}$ be an interval of positive finite length. If $B \cap K$ contains at most one point, then $\mu(B) = 0$. Otherwise, the unique itineraries of the points of $B \cap K$ have a longest common finite prefix. Let J be its cylinder and p its final label. All points of $B \cap K$ lie in J, and at least two different children of J contain points of $B \cap K$. Put $t = |B|/|J|$. If $t \geq 1$, then

$$\mu(B) \leq \mu(J) \leq A_s |J|^s \leq A_s |B|^s.$$

Suppose henceforth that $0 < t < 1$. The sibling gap estimate implies

$$y := b^p t \geq c_b > 0. \tag{15.10}$$

Even if B is not contained in J, the set $B \cap J$ is an interval. By (15.5), its image under $f^{\circ(n+1)}$ has length at most $C_b b^p t = C_b y$, where n is the depth of J. If a child $J_{\omega q}$ meets B, then I_q meets this image. Therefore all the corresponding prime labels q belong to one interval of length at most $C_b y + 1$. The sieve bound (15.2), together with (15.10), yields

$$\#\{q : J_{\omega q} \cap B \neq \varnothing\} \leq C_b \frac{y+1}{\log(y+2)} \leq C_b \frac{y}{\log(y+2)}.$$

Using the equal-child masses and (15.3), we consequently obtain

$$\begin{aligned} \mu(B) &\leq C_b \mu(J) \frac{p b^{-p} y}{\log(y+2)} \\ &= C_b \mu(J) \frac{pt}{\log(b^p t + 2)} \\ &\leq C_b A_s |B|^s \frac{p t^{1-s}}{\log(b^p t + 2)}. \end{aligned} \tag{15.11}$$

The final factor is bounded uniformly for $p \geq P$ and $0 < t < 1$. Indeed, if $t \geq b^{-p/2}$, then

$$\frac{p t^{1-s}}{\log(b^p t + 2)} \leq \frac{2}{\log b}.$$

If $t < b^{-p/2}$, then it is at most

$$\frac{pb^{-(1-s)p/2}}{\log 2},$$

which is bounded as p varies. Hence there is a finite constant M_s such that

$$\mu(B) \le M_s |B|^s \qquad \text{for every interval } B \subset \mathbb{R}.$$

The mass distribution principle gives $\dim_H K \ge s$. This holds for every $s < 1$ for the same compact set K and the same measure μ. Therefore $\dim_H K = 1$, and Theorem 15.1 follows by $K \subset \mathcal{W}_b^{\mathrm{it}} \subset \mathbb{R}$.

Remark 15.3 (Scope and attribution). The case $b = 2$ answers the exact question stated as Question 1.5 in [33, arXiv version]. The proof combines classical prime-counting and sieve inputs with a bounded-distortion Cantor construction; no historical priority assertion is made here. It is a statement about the full set of permitted starting values and the compact sets explicitly constructed above. It gives no arithmetic classification of a separately specified least Wright constant.

16 Local dimension after a finite prime prefix

The global dimension theorem does not by itself localize the admissible set inside a prescribed prime root cell. This matters for a finite prefix whose successive exponent ratios can be smaller than 2. A quantitative result already established by Saito supplies the required bridge.

Proposition 16.1 (Saito's quantitative bounded-schedule theorem)**.** *For every $R \ge 2$ there is $M_2(R) > 0$ with the following property. For every real sequence (c_j) satisfying $2 \le c_j \le R$, and every $M \ge M_2(R)$, there is a prime $q \in [M, 2M]$ such that, with $D_j = c_1 \cdots c_j$,*

$$\dim_H \left\{ y \in [q^{1/c_1}, (q+1)^{1/c_1}) : \lfloor y^{D_j} \rfloor \text{ is prime for every } j \right\} \ge \left(1 + \frac{R}{q \log q}\right)^{-1}.$$

This is [33, Theorem 6.1]. Its threshold is uniform over all bounded schedules with the stated bound R.

Lemma 16.2 (A local bridge across a large exponent ratio)**.** *Suppose a prime-admissibility problem contains an exponent $E > 0$, followed by $E' = rE$, and all successive ratios after E' are at least 2. Let p be prime. Suppose every point of*

$$J = (p^{1/E}, (p+1)^{1/E})$$

satisfies all the required conditions up to exponent E. If

$$M \ge M_2(4), \qquad p^r < M, \qquad 2M + 1 < (p+1)^r, \tag{16.1}$$

then the set satisfying all required conditions and lying in J has Hausdorff dimension at least

$$\left(1 + \frac{4}{M \log M}\right)^{-1}. \tag{16.2}$$

Proof. Refine every ratio after E' into a finite block of real factors in $[2, 4)$, and prepend the initial factor 2. The products of this bounded schedule, at its block boundaries, are

$$2, \quad 2E''/E', \quad 2E'''/E', \quad \dots,$$

where $E'', E''', \dots$ are the later original exponents. Proposition 16.1 gives a prime $q \in [M, 2M]$ and a set of admissible y in $[q^{1/2}, (q+1)^{1/2})$ of dimension at least $(1 + 4/(q \log q))^{-1}$. Apply the map

$$x = y^{2/E'}.$$

On this bounded interval the map and its inverse are Lipschitz, so Hausdorff dimension is preserved. Its image lies in $[q^{1/E'}, (q+1)^{1/E'})$, which is strictly inside J by (16.1). Thus all earlier conditions hold, the output at E' is q, and all later original outputs occur at refined block boundaries and are prime. Since $q \geq M$, the dimension is at least (16.2). □

The seed condition in this lemma concerns the entire open prime root cell. A nonempty intersection of finitely many prime conditions need not contain such a cell; we do not silently replace one condition by the other.

Lemma 16.3 (Strict continuation of every factorial seed)**.** *Let $E \geq 252$ be an integer and p a prime. Put $r = E!$. There is a prime q satisfying*

$$p^r < q < q + 1 < (p+1)^r.$$

Consequently the closed prime root cell $[q^{1/(Er)}, (q+1)^{1/(Er)}]$ is strictly inside $(p^{1/E}, (p+1)^{1/E})$.

Proof. The consecutive-cubes result in [11, Section 4] gives a prime strictly between m^3 and $(m+1)^3$ whenever $m \geq \exp(\exp(32.892))$. Here set $t = r/3$ and $m = p^t$. The integer r is divisible by 3, and already

$$\log m \geq \frac{252!}{3} \log 2 > \exp(32.892),$$

so the threshold holds. For example, the elementary bound $252! > 10^{90}$ is more than sufficient for the last inequality. Since $q < (m+1)^3$ and both sides are integers, $q + 1 \leq (m+1)^3$. As $t \geq 2$, the binomial theorem gives $p^t + 1 < (p+1)^t$. It follows that

$$p^r < q < q + 1 \leq (p^t + 1)^3 < (p+1)^r.$$

Taking positive roots proves the last assertion. □

Theorem 16.4 (Full local dimension for a factorial continuation)**.** *Let $0 < C_1 < \cdots < C_t = E_0$ be a finite list of real exponents, where $E_0 \geq 252$ is an integer. Let p_0 be prime, and assume that every x in*

$$J_0 = (p_0^{1/E_0}, (p_0+1)^{1/E_0})$$

satisfies $\lfloor x^{C_i} \rfloor \in \mathbb{P}$ for every $i \leq t$. Define

$$E_{n+1} = E_n E_n! \qquad (n \geq 0).$$

Then

$$\dim_H \left\{ x \in J_0 : \lfloor x^{C_i} \rfloor \in \mathbb{P} \ (1 \leq i \leq t), \quad \lfloor x^{E_n} \rfloor \in \mathbb{P} \ (n \geq 1) \right\} = 1. \tag{16.3}$$

No lower bound of 2 is imposed on the ratios inside the finite prefix.

Proof. By Lemma 16.3, construct primes p_n such that

$$I_n = [p_n^{1/E_n}, (p_n+1)^{1/E_n}], \qquad I_{n+1} \subset \operatorname{int}(I_n).$$

Every point of the interior of I_n satisfies the finite prefix and all factorial-tail conditions up to E_n. Choose $B > 1$ with $I_0 \subset (1, B)$. Then $p_n < B^{E_n}$. Writing $r_n = E_n!$, we have

$$\frac{r_n}{p_n} \geq \frac{E_n!}{B^{E_n}} \longrightarrow \infty.$$

Indeed, $E! \geq \lfloor E/2 \rfloor^{\lceil E/2 \rceil}$, whose logarithm minus $E \log B$ tends to infinity. Since $\log(1+1/p) \geq 1/(2p)$ for $p \geq 1$,

$$\frac{(p_n+1)^{r_n}}{p_n^{r_n}} = \exp(r_n \log(1+1/p_n)) \geq \exp\left(\frac{r_n}{2p_n}\right) \longrightarrow \infty.$$

Thus, for all sufficiently large n, the choice $M_n = \lceil 2p_n^{r_n} \rceil$ satisfies

$$M_n \geq M_2(4), \qquad p_n^{r_n} < M_n, \qquad 2M_n + 1 < (p_n+1)^{r_n}.$$

Apply Lemma 16.2 to the seed cell I_n and the jump to E_{n+1}. The set in (16.3) consequently has Hausdorff dimension at least $(1 + 4/(M_n \log M_n))^{-1}$ for every sufficiently large n. As $M_n \to \infty$, these lower bounds tend to 1. The ambient upper bound is 1. □

Corollary 16.5 (The certified common prime prefix)**.** *Take*

$$(C_1, C_2, C_3, C_4, C_5) = (4, 6, 7, 12, 252),$$
$$p_0 = 513284270838218284273581257165 3 = 29^{21} + 24.$$

The exact primality certificate and root-cell comparisons in Section 8, with the verifier in Appendix A, show that the complete factorial-tail admissible set inside $(p_0^{1/252}, (p_0+1)^{1/252})$ *has Hausdorff dimension* 1.

Proof. Those certified comparisons place the seed cell inside the region with first four prime outputs $(3, 5, 7, 29)$. Theorem 16.4 applies. □

The full admissible set in this corollary is much larger, in the metric sense, than the particular compact set obtained by retaining only the first two admissible primes at each stage. The latter set's upper box dimension zero is entirely compatible with Corollary 16.5.

17 A metric obstruction to uniform radical approximation

The very rapid approximation used to prove whole-family algebraic independence in the explicit prime-tree construction imposes a geometric cost. The following elementary estimate identifies that cost without making any assumption about the independence proof itself.

Theorem 17.1 (Uniform prime-root approximation forces zero dimension)**.** *Let* $K \subset [1, B]$ *be nonempty and bounded, with* $B > 1$. *Suppose* $E_n > 0$, $0 < \varepsilon_n \to 0$, *and for every* $x \in K$ *and every* n *there is a prime* $p = p(x, n)$ *such that*

$$|x - p^{1/E_n}| \leq \varepsilon_n.$$

If

$$\frac{E_n}{\log(1/\varepsilon_n)} \longrightarrow 0, \tag{17.1}$$

then $\dim_H K = 0$ *and the lower box dimension of* K *is zero. More strongly,* $\dim_H(K^r) = 0$ *for every positive integer* r.

Proof. For sufficiently large n, every approximating radical lies below $B+1$, hence $p \leq (B+1)^{E_n}$. It is unnecessary to use a prime-counting estimate: there are at most $(B+1)^{E_n}$ possible positive integer radicands. Thus K is covered by at most $(B+1)^{E_n}$ intervals of length $2\varepsilon_n$. For every fixed $s > 0$, the sum of the sth powers of their lengths is at most

$$(B+1)^{E_n}(2\varepsilon_n)^s,$$

whose logarithm tends to $-\infty$ by (17.1). This proves $\dim_H K = 0$. The same cover gives

$$\liminf_{\rho \downarrow 0} \frac{\log N(K, \rho)}{\log(1/\rho)} = 0,$$

by taking $\rho = 2\varepsilon_n$. Finally, taking products gives a cover of K^r by at most $(B+1)^{rE_n}$ sets of diameter $2\sqrt{r}\,\varepsilon_n$. The identical estimate proves the last claim. □

The theorem remains valid if "prime" is replaced by "positive integer." It does not assert that packing dimension or upper box dimension is zero: a small number of covering scales suffices for the conclusions proved here, whereas those other dimensions require further control. In particular, a full-Hausdorff-dimension subset of a prime-admissible set cannot retain the uniform approximation regime (17.1). For compact algebraically independent families, Corollary 18.3 gives the stronger obstruction that positive Hausdorff dimension is impossible regardless of the chosen approximation regime.

18 Algebraic consequences of the dimension contrast

The dimension conclusion has an algebraic consequence substantially stronger than uncountability. We use the following result of Edgar and Miller.

Theorem 18.1 (Edgar–Miller)**.** *Let $F \subseteq \mathbb{R}$ be an analytic subring. Either $F = \mathbb{R}$, or $\dim_H(F^m) = 0$ for every positive integer m.*

The analytic version follows from [14, Theorem 1 and the subsequent remarks]. Here "analytic" has its descriptive-set-theoretic meaning, namely a continuous image of a Polish space, not a regularity condition on functions.

Corollary 18.2 (The full spectrum generates the real field as a ring)**.** *Let $E_1 > 0$ and $E_{k+1}/E_k \geq 2$. Write*

$$\mathcal{P}(E) = \{x > 1 : \lfloor x^{E_k} \rfloor \text{ is prime for every } k \geq 1\}.$$

Then the unital ring generated by $\mathcal{P}(E)$ is all of $\mathbb{R}$:

$$\boxed{\mathbb{Z}[\mathcal{P}(E)] = \mathbb{R}.}$$

Equivalently, every real number is an integer polynomial in finitely many members of one and the same prime-representing spectrum.

Proof. For each k, the primality condition is a countable union of half-open root intervals. Therefore $\mathcal{P}(E)$ is Borel. Its generated ring is a countable union of continuous polynomial images of finite powers of $\mathcal{P}(E)$, hence is analytic. It has positive Hausdorff dimension by Corollary 14.5, because it contains $\mathcal{P}(E)$. Theorem 18.1 forces this ring to be $\mathbb{R}$. □

This is a representation theorem, with no asserted effective bound on the number or size of its parameters. It does not settle the arithmetic nature of a named real number represented in this way.

Corollary 18.3 (A sharp Hausdorff obstruction for independent families)**.** *If $K \subseteq \mathbb{R}$ is analytic and algebraically independent over $\mathbb{Q}$, then*

$$\dim_H(K^m) = 0 \qquad (m \geq 1).$$

Proof. The ring $F = \mathbb{Q}[K]$ is analytic, by the same polynomial-image argument. It is proper: if $\sqrt{2} = P(x_1, \dots, x_r)$ for a polynomial $P \in \mathbb{Q}[X_1, \dots, X_r]$ at distinct elements of K, algebraic independence would force $P^2 - 2$ to be the zero polynomial. This is impossible in the polynomial ring over $\mathbb{Q}$ (a nonconstant P has positive degree, and a constant rational P cannot square to 2). Thus Theorem 18.1 gives $\dim_H(F^m) = 0$, and $K^m \subseteq F^m$ completes the proof. □

Accordingly, a compact algebraically independent family cannot retain the positive Hausdorff dimension of its full prime-representing spectrum. This explains why the zero Hausdorff dimension of the compact independent sets constructed below is unavoidable. The assertion concerns Hausdorff dimension and its finite Cartesian powers; it imposes no corresponding upper box dimension bound by itself.

19 Separation governed by the exponent support

The degree of the ambient radical field is not always the relevant exponent in its norm estimate. A polynomial whose exponent vectors lie in a low-dimensional affine space has many conjugates of the same absolute value. Retaining that multiplicity gives the following refinement.

For a nonzero polynomial $P = \sum_{\nu \in S} a_\nu X^\nu$, its *support rank* is

$$\rho(P) = \dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}} \{\nu - \mu : \nu, \mu \in S\}.$$

It is the dimension of the Newton polytope, not the number of variables or the number of terms. Multiplication by a monomial does not change it. For $T \geq 1$ put

$$B_T(P) = \max\left\{2, \sum_{\nu} |a_\nu| T^{|\nu|}\right\}, \qquad L_T(P) = \sum_{|\nu| \geq 1} |a_\nu|\, |\nu| T^{|\nu|-1}.$$

The latter is a Lipschitz constant on $[0, T]^r$ in the supremum norm.

Theorem 19.1 (The support-lattice norm bound)**.** *Let $p_1, \dots, p_r$ be distinct primes, let $E \geq 2$ be an integer, and put $b_i = p_i^{1/E} \leq T$, using positive real roots. Let $0 \neq P \in \mathbb{Z}[X_1, \dots, X_r]$ satisfy $\deg_{X_i} P < E$ for every i. Fix $\nu_0 \in \operatorname{supp} P$, and let $d_1, \dots, d_\ell$ be the nonzero Smith invariant factors of the integer matrix with rows $\nu - \nu_0$, where $\nu \in \operatorname{supp} P$ and $\ell = \rho(P)$. With an empty product interpreted as 1, define*

$$M_E(P) = \prod_{j=1}^{\ell} \frac{E}{\gcd(E, d_j)}.$$

Then

$$|P(b_1, \dots, b_r)| \geq B_T(P)^{1 - M_E(P)} \geq B_T(P)^{1 - E^{\rho(P)}} > 0. \tag{19.1}$$

Proof. The prime-radical basis theorem gives

$$[\mathbb{Q}(b_1, \dots, b_r) : \mathbb{Q}] = E^r$$

and shows that $P(b_1, \dots, b_r) \neq 0$. For completeness, the monomials $b_1^{j_1} \cdots b_r^{j_r}$, $0 \leq j_i < E$, have rational Eth powers. Two different monomials have irrational quotient, by valuations at the distinct primes p_i. The real-radical independence theorem of Carr and O'Sullivan [9, Theorem 1.1] makes these E^r monomials linearly independent. They also span the generated field.

Let $F = \mathbb{Q}(b_1, \dots, b_r)$, let ζ be a primitive Eth root of unity, and write $\beta = P(b_1, \dots, b_r)$. Its E^r embeddings are indexed by $t = (t_1, \dots, t_r) \in (\mathbb{Z}/E\mathbb{Z})^r$ and send b_i to $\zeta^{t_i} b_i$. Consider the subgroup

$$H_E = \{t : (\nu - \nu_0) \cdot t = 0 \pmod{E} \text{ for every } \nu \in \operatorname{supp} P\}.$$

Smith normal form gives

$$h_E := |H_E| = E^{r-\ell} \prod_{j=1}^{\ell} \gcd(E, d_j), \qquad E^r/h_E = M_E(P).$$

For every $t \in H_E$,

$$P(\zeta^{t_1} b_1, \dots, \zeta^{t_r} b_r) = \zeta^{\nu_0 \cdot t} \beta.$$

Thus at least h_E conjugates have absolute value $|\beta|$. All conjugates have absolute value at most $B_T(P)$. Since β is a nonzero algebraic integer, its norm is a nonzero integer, and hence

$$1 \le |N_{F/\mathbb{Q}}(\beta)| \le |\beta|^{h_E} B_T(P)^{E^r - h_E}.$$

Taking h_Eth roots proves the first inequality. The second follows from $M_E(P) \le E^\ell$. □

Corollary 19.2 (A finite nonvanishing certificate)**.** *Under the assumptions of Theorem 19.1, suppose $\alpha_i \in [0, T]$ and $|\alpha_i - b_i| \le \varepsilon$. If*

$$L_T(P)\varepsilon \le \tfrac{1}{2} B_T(P)^{1 - M_E(P)},$$

then

$$|P(\alpha_1, \dots, \alpha_r)| \ge \tfrac{1}{2} B_T(P)^{1 - M_E(P)} > 0.$$

The two real numbers $P(\alpha_1, \dots, \alpha_r)$ and $P(b_1, \dots, b_r)$ have the same sign.

Proof. Their difference has absolute value at most $L_T(P)\varepsilon$. Apply (19.1). □

Theorem 19.3 (Sparse-support transfer)**.** *Let $\alpha_1, \dots, \alpha_r > 0$ be distinct. Let E_n be positive integers tending to infinity, and let $0 < \varepsilon_n \to 0$. Suppose that, for all sufficiently large n, there are pairwise distinct primes $p_{1,n}, \dots, p_{r,n}$ such that*

$$\max_i |\alpha_i - p_{i,n}^{1/E_n}| \le \varepsilon_n.$$

For an integer $\ell \ge 1$, assume

$$\limsup_{n \to \infty} \frac{\log(1/\varepsilon_n)}{E_n^\ell} = \infty. \tag{19.2}$$

Then every nonzero polynomial $P \in \overline{\mathbb{Q}}[X_1, \dots, X_r]$ of support rank at most ℓ satisfies $P(\alpha_1, \dots, \alpha_r) \ne 0$. The same assertion holds for Laurent polynomials, with their support viewed as a subset of $\mathbb{Z}^r$.

Proof. First take an integer polynomial. The approximants and their limits lie in $[0, T]^r$ for some fixed T. The individual degree conditions hold eventually. Along a subsequence supplied by (19.2),

$$L_T(P)\varepsilon_n < \tfrac{1}{2} B_T(P)^{1 - E_n^\ell} \le \tfrac{1}{2} B_T(P)^{1 - M_{E_n}(P)}.$$

Corollary 19.2 proves the assertion.

For algebraic coefficients, choose a finite Galois extension $L/\mathbb{Q}$ containing them and form the nonzero polynomial

$$Q(X) = \prod_{\sigma \in \mathrm{Gal}(L/\mathbb{Q})} \sigma(P)(X) \in \mathbb{Q}[X].$$

If $S \subset \nu_0 + V$, with $\dim V = \rho(P)$, then the support of Q is contained in $[L : \mathbb{Q}]\nu_0 + V$. Therefore $\rho(Q) \le \rho(P)$. Clearing its rational coefficient denominators and applying the integer-polynomial case gives $Q(\alpha) \ne 0$, hence $P(\alpha) \ne 0$. Finally, clearing negative exponents by multiplication by a monomial does not change support rank and does not affect vanishing at positive coordinates. □

Remark 19.4 (What is saved). A homogeneous polynomial in r variables has support rank at most $r-1$, so its lower bound uses E^{r-1} in place of E^r. A polynomial such as $X_1^2X_2^3 - 7X_3^5$ has support rank one regardless of its three variables. The estimate for it uses E, not E^3. No estimate of this form asserts that the growth conditions below are necessary; they are sufficient conditions produced by the norm method.

20 Slower schedules and multiplicative transcendence

Here is a completely specified binary construction to which the sparse criterion applies. It can use any starting prime cell contained in $(1,2)$; no large-prime certificate is needed for its existence. Put $E_0 = 252$, $s_\varnothing = 2$, and choose integer multipliers R_n such that

$$3 \mid R_n, \qquad R_n \geq 3 \cdot 10^{15}, \qquad E_{n+1} = E_n R_n. \tag{20.1}$$

For a binary word w of length n, define s_{w0} and s_{w1} to be the first and second primes strictly above $s_w^{R_n}$, and set

$$I_w = [s_w^{1/E_n}, (s_w+1)^{1/E_n}].$$

Proposition 20.1 (Uniform root approximation). *The cells are strictly nested and their siblings are disjoint. Every binary sequence determines a unique limit, and the set K of these limits is compact, perfect, and consists of prime-representing constants at every exponent E_n. For $a = 2^{1/252} > 1$, every descendant α of a level-n cell satisfies*

$$0 < \alpha - s_w^{1/E_n} < \frac{12}{E_n R_n} a^{-E_n R_n/3}. \tag{20.2}$$

Distinct limits have distinct prime labels at all sufficiently late levels.

Proof. We use the explicit theorem that $(m^3, (m+1)^3)$ contains a prime for all integers $m \geq \exp(\exp(32.892))$, proved in Cully-Hugill [11, Section 4]. At a node with prime s, exponent E, and multiplier R, the integer $M = s^{R/3}$ satisfies $M \geq 2^{10^{15}}$. In particular $\log\log M \geq 15\log 10 + \log\log 2 > 32.892$. The first two primes above s^R are therefore below $(M+2)^3$. Since $R/3 \geq 2$,

$$(s+1)^{R/3} - s^{R/3} > 2.$$

The upper endpoints of both child cells lie strictly below the upper endpoint of the parent; their lower endpoints lie strictly above the parent's lower endpoint. Distinct odd prime labels give disjoint cells. Their diameters are at most $1/E_n \to 0$, proving the nested-limit assertions and the homeomorphism with binary sequence space.

All cells remain in $[a, 3^{1/252}] \subset (1,2)$. If $b = s^{1/E}$, a descendant belongs to one of the next two child cells, so concavity gives

$$0 < \alpha - b \leq b\big((1+2/M)^{3/(ER)} - 1\big) < \frac{6b}{ERM} < \frac{12}{ER} a^{-ER/3}.$$

Distinct branches separate at one level and cannot meet thereafter. □

Theorem 20.2 (The support-rank growth criterion). *For the preceding tree, let $\ell \geq 1$. If*

$$\limsup_{n\to\infty} \frac{R_n}{E_n^{\ell-1}} = \infty, \tag{20.3}$$

then no nonzero polynomial over $\overline{\mathbb{Q}}$ of support rank at most ℓ vanishes on any finite tuple of distinct members of K. In particular:

1. *If* $\limsup R_n = \infty$*, then for distinct* $\alpha_1, \dots, \alpha_r \in K$ *and* $(m_1, \dots, m_r) \in \mathbb{Z}^r \setminus \{0\}$*,*
$$\alpha_1^{m_1} \cdots \alpha_r^{m_r} \notin \overline{\mathbb{Q}}. \tag{20.4}$$
The same holds for nonzero rational exponent vectors.
2. *If* (20.3) *holds with* $\ell = r$*, every* r *distinct members of* K *are algebraically independent.*
3. *If* (20.3) *holds with* $\ell = r - 1$ *and* $r \geq 2$*, then for distinct* $\alpha_1, \dots, \alpha_r \in K$ *the* $r - 1$ *ratios* $\alpha_2/\alpha_1, \dots, \alpha_r/\alpha_1$ *are algebraically independent.*

Proof. Taking logarithms in (20.2) gives
$$\frac{\log(1/\varepsilon_n)}{E_n^\ell} \geq \frac{\log a}{3} \frac{R_n}{E_n^{\ell-1}} + \frac{\log(E_n R_n/12)}{E_n^\ell}.$$
Apply Theorem 19.3 to any fixed finite tuple. If the monomial in (i) equaled an algebraic number c, the nonzero Laurent polynomial $X^m - c$, of support rank one, would vanish. Rational exponents reduce to integer exponents by raising to a common positive denominator. Part (ii) uses the bound $\rho(P) \leq r$. For (iii), a nonzero polynomial relation among the ratios would homogenize to a nonzero homogeneous polynomial in r variables, whose support rank is at most $r - 1$. □

Corollary 20.3 (A factorial-scale example)**.** *Take* $N = 10^{15}$ *and*
$$R_n = 3(N + n), \qquad E_n = 252\, 3^n \frac{(N + n - 1)!}{(N - 1)!}.$$
Then K *is a compact perfect set for which every nontrivial finite monomial in distinct members is transcendental. In particular every member, every product of finitely many distinct members, and every quotient of two distinct members is transcendental. The set has upper box dimension zero.*

Proof. Here $R_n \to \infty$, so Theorem 20.2(i) applies. There are 2^n level-n cells of diameter at most E_n^{-1}, and $\log E_n/n \to \infty$ by the factorial formula. If $E_n^{-1} \leq \delta < E_{n-1}^{-1}$, the covering number satisfies
$$\frac{\log N_\delta(K)}{\log(1/\delta)} \leq \frac{n \log 2}{\log E_{n-1}} \longrightarrow 0.$$
□

Corollary 20.4 (A smaller full-independence schedule)**.** *With* $E_0 = 252$*, define*
$$R_n = 3 \left\lceil 10^{15} + \exp((\log E_n)^2) \right\rceil, \qquad E_{n+1} = E_n R_n.$$
The entire binary limit set is algebraically independent over $\overline{\mathbb{Q}}$ *and has upper box dimension zero.*

Proof. For every fixed ℓ, $R_n/E_n^{\ell-1} \to \infty$. Thus Theorem 20.2 excludes every nonzero polynomial relation. Also $E_{n+1} \geq E_n^2$ eventually, which gives the dimensional conclusion as above. □

Remark 20.5 (Comparison with factorial continuation). The last multiplier is asymptotic to $3\exp((\log E)^2)$, substantially smaller than $E!$. The general r-variable growth condition $R_n/E_n^{r-1} \to \infty$ is already present for the countable sub-boundary in Saito and Takeda [38, Theorem 1.3]. Here it applies to the entire specified binary tree; the support-rank refinement gives additional conclusions at weaker growth rates. Neither the factorial-scale example nor the sparse-support criterion alone proves full algebraic independence when R_n grows only linearly in n. For instance, a relation involving $X_1 + X_2 - 1$ has support rank two and is not excluded by the rank-one argument.

21 A consequence for a pre-existing prime-constant boundary

We record a direct consequence for the family studied by Saito and Takeda, rather than only for the new binary subfamilies. Let c_1 be a positive integer and let $c_k \geq 2$ be integers for $k \geq 2$. Put

$$C_k = c_1 c_2 \cdots c_k, \qquad W(c_k) = \{A > 1 : \lfloor A^{C_k} \rfloor \in \mathbb{P} \text{ for every } k \geq 1\}.$$

The left sub-boundary is

$$\widetilde{\partial}_L W(c_k) = \{\sup U \in \mathbb{R} : U \text{ is a connected component of } \mathbb{R} \setminus W(c_k)\}.$$

The right sub-boundary is defined by replacing $\sup U$ with $\inf U$; write it as $\widetilde{\partial}_R W(c_k)$ and set $\widetilde{\partial} W(c_k) = \widetilde{\partial}_L W(c_k) \cup \widetilde{\partial}_R W(c_k)$. Only finite endpoints are included. Under the hypotheses below, $W(c_k)$ is closed and perfect by Saito–Takeda [38, Theorem 1.2], and these endpoints belong to $W(c_k)$.

Theorem 21.1 (Multiplicative transcendence on the left sub-boundary)**.** *If* $\limsup_{k\to\infty} c_{k+1} = \infty$*, then, for every finite tuple of distinct* $A_1, \ldots, A_r \in \widetilde{\partial}_L W(c_k)$ *and every nonzero* $m \in \mathbb{Z}^r$*,*

$$A_1^{m_1} \cdots A_r^{m_r} \notin \overline{\mathbb{Q}}.$$

More generally, if

$$\limsup_{k\to\infty} \frac{c_{k+1}}{C_k^{\ell-1}} = \infty,$$

no nonzero algebraic-coefficient polynomial of support rank at most ℓ *vanishes at such a tuple.*

Proof. Set $\theta = 21/40$ and $I = \{k : c_{k+1} \geq 3\}$, which is infinite. We use the following precise published input, Saito–Takeda [38, Lemma 5.2(1)]: for each $A \in \widetilde{\partial}_L W(c_k)$, writing $p_k(A) = \lfloor A^{C_k} \rfloor$, one has for all sufficiently large $k \in I$

$$0 \leq A - p_k(A)^{1/C_k} \leq A\, p_1(A)^{(\theta-1)C_{k+1}/c_1}.$$

Its hypotheses are $c_1 > 0$, integral $c_{k+1} \geq 2$, and infinitely many $c_{k+1} \geq 3$. Our additional integrality of c_1 ensures that every radical exponent C_k is an integer.

For a fixed finite tuple the prime labels become pairwise distinct, since $A_i \neq A_j$ implies $|A_i^{C_k} - A_j^{C_k}| \to \infty$. With $T = 1 + \max_i A_i$ and

$$\lambda = \frac{1-\theta}{c_1} \min_i \log p_1(A_i) > 0,$$

all sufficiently large $k \in I$ satisfy

$$\max_i |A_i - p_k(A_i)^{1/C_k}| \leq T \exp(-\lambda C_{k+1}).$$

Along the subsequence where $c_{k+1}/C_k^{\ell-1}$ is unbounded,

$$\frac{\log(1/\varepsilon_k)}{C_k^{\ell}} \geq \lambda \frac{c_{k+1}}{C_k^{\ell-1}} - \frac{\log T}{C_k^{\ell}} \longrightarrow \infty.$$

This subsequence lies in I eventually. Theorem 19.3 proves the more general assertion. Its rank-one case applied to $X^m - a$, for algebraic a, proves the monomial statement. □

Corollary 21.2 (The factorial exponent family)**.** *Let*

$$W_! = \{A > 1 : \lfloor A^{k!} \rfloor \in \mathbb{P} \text{ for all } k \geq 1\}.$$

Every nontrivial finite monomial in distinct elements of $\widetilde{\partial}_L W_!$ *is transcendental. In particular, for distinct* $A, B \in \widetilde{\partial}_L W_!$*, the quotient* A/B *is transcendental.*

Proof. Take $c_k = k$, including $c_1 = 1$, in Theorem 21.1. □

Remark 21.3 (The precise comparison). Saito–Takeda [38, Proposition 8.2 and Corollary 8.3] prove rational linear independence of two appropriate boundary constants under $c_{k+1} \to \infty$. On the all-left-sub-boundary case, Corollary 21.2 strengthens irrationality of the quotient to its transcendence, and the theorem treats arbitrary finite monomials. Their proposition also allows one right-sub-boundary point. The general support-rank argument does not cover that extra case: its natural approximant is a root of $p_k + 1$, which need not be prime. The quotient case can nevertheless be recovered by the separate argument below. These are proved refinements of a published theorem, not a claim that the cited paper posed the refinements as open conjectures. We have not established priority for the general support-lattice norm observation.

Theorem 21.4 (Transcendental quotients with one arbitrary boundary point)**.** *Assume* $\limsup_{k\to\infty} c_{k+1} = \infty$. *Let*

$$A_1 \in \widetilde{\partial} W(c_k), \qquad A_2 \in \widetilde{\partial}_L W(c_k), \qquad A_1 < A_2,$$

where $\widetilde{\partial} W(c_k)$ *is the union of the left and right sub-boundaries. Then* A_2/A_1 *is transcendental.*

Proof. Let $E = C_k$, and write $p_i(k) = \lfloor A_i^{C_k} \rfloor$. If A_1 is a left sub-boundary point, put $q_1(k) = p_1(k)$; if it is a right sub-boundary point, put $q_1(k) = p_1(k) + 1$. Put $q_2(k) = p_2(k)$. The two parts of Saito–Takeda [38, Lemma 5.2] show that there are constants $K, \lambda > 0$, depending on the fixed pair, such that for all sufficiently large $k \in I = \{k : c_{k+1} \geq 3\}$,

$$|q_i(k)^{1/E} - A_i| \leq K \exp(-\lambda C_{k+1}) \qquad (i = 1, 2).$$

For the right endpoint, the precise bound used is

$$0 \leq (p_1(k) + 1)^{1/C_k} - A_1 \leq (p_1(1) + 1)^{1/c_1} A_1^{(\theta-1)C_{k+1}}, \qquad \theta = 21/40.$$

The left-endpoint bound was stated in the preceding proof.

For all sufficiently large k, the integer $q_2(k)$ is a prime greater than $q_1(k)$, because $A_2 > A_1$. The polynomial

$$F_k(X) = q_1(k)X^E - q_2(k)$$

is Eisenstein at $q_2(k)$ and is therefore irreducible over $\mathbb{Q}$. In particular, its positive root $z_k = (q_2(k)/q_1(k))^{1/E}$ has degree E. Both root approximants stay in a fixed compact subinterval of $(0, \infty)$, so the quotient map is Lipschitz there. Consequently

$$|z_k - A_2/A_1| \leq K' \exp(-\lambda C_{k+1}) \tag{21.1}$$

for a fixed constant $K' > 0$.

Suppose $z = A_2/A_1$ were algebraic, and choose a nonzero $Q \in \mathbb{Z}[X]$ with $Q(z) = 0$, of degree d. For $E > d$, irreducibility of F_k makes $\operatorname{Res}(F_k, Q)$ a nonzero integer. Choose a fixed $T > 1$ with $q_i(k)^{1/E} \leq T$ and $z_k \leq T$ eventually, and set $B = \max\{2, \sum_j |[X^j]Q| T^j\}$. All complex roots of F_k have modulus z_k, whence

$$1 \leq |\operatorname{Res}(F_k, Q)| = q_1(k)^d \prod_{\zeta^E=1} |Q(\zeta z_k)| \leq q_1(k)^d |Q(z_k)| B^{E-1}.$$

Since $q_1(k) \leq T^E$, this yields

$$|Q(z_k)| \geq T^{-dE} B^{1-E} \geq \exp(-HE)$$

for a fixed $H > 0$. On the other hand, (21.1) and a derivative bound for Q give $|Q(z_k)| \leq K'' \exp(-\lambda E c_{k+1})$. Choose $k \to \infty$ along a subsequence with $c_{k+1} \to \infty$. The last two estimates are incompatible. Therefore z is transcendental. □

Remark 21.5. For $c_k = k$, this gives transcendence of A_2/A_1 for the full pair configuration of Saito–Takeda's Proposition 8.2 and Corollary 8.3, including their allowed right sub-boundary point. The proof uses the published approximation estimates and a classical integer resultant; it does not assume algebraic independence of the pair.

22 Relative independence of complete exponential towers

The perfect prime set can be thinned without changing its exponent sequence. This permits relative independence over prescribed constants and, more strongly, independence of the logarithm and every finite exponential iterate of every retained point. The selection argument is a specialization of the classical perfect-set method of Mycielski [28]; we include the proof and its effective refinement because the retained prime identities and the coefficient-field issue are essential here.

For $x > 1$, define

$$T_{-1}(x) = \log x, \qquad T_0(x) = x, \qquad T_{j+1}(x) = \exp(T_j(x)) \quad (j \geq 0).$$

An indexed family is algebraically independent if every polynomial in finitely many distinct indexed entries, with coefficients in the base field, is nonzero at those entries. Thus the assertion below also excludes coincidences between entries with different indices.

Lemma 22.1 (Functional independence of exponential towers)**.** *For each $m \geq 0$, the real-analytic functions $T_{-1}, T_0, \ldots, T_m$ are algebraically independent over $\mathbb{C}$. More generally, for independent real variables $x_1, \ldots, x_r$, the functions $T_j(x_i)$, $1 \leq i \leq r$, $-1 \leq j \leq m$, are algebraically independent over $\mathbb{C}$.*

Proof. For a multi-index $a = (a_{-1}, a_0, \ldots, a_m)$ of nonnegative integers and $x > e$, write $M_a(x) = \prod_{j=-1}^{m} T_j(x)^{a_j}$. Then

$$\log M_a(x) = a_{-1} \log\log x + a_0 \log x + \sum_{j=1}^{m} a_j T_{j-1}(x).$$

If $a \neq b$, the largest index at which they differ determines the sign of $\log(M_a/M_b)$ for sufficiently large x, and its absolute value tends to infinity. Consequently, among the finitely many monomials of a nonzero polynomial, the lexicographically largest, reading indices from m down to -1, dominates all the others as $x \to \infty$. Its nonzero coefficient prevents cancellation. This proves the one-variable claim. For several variables, group a purported identity by the monomials in $T_j(x_1)$. Holding the other variables fixed, the one-variable result forces every coefficient function to vanish. Induction on r finishes the proof. □

Lemma 22.2 (Analytic avoidance on a perfect set)**.** *Let J be a connected open real interval and let $P \subset J$ be a nonempty compact perfect set. If $f : J^r \to \mathbb{C}$ is real analytic and is not identically zero, its zero set in P^r is closed and nowhere dense relative to P^r.*

Proof. Closedness follows from continuity. If the zero set had nonempty relative interior, f would vanish on a product $U_1 \times \cdots \times U_r$ of nonempty relatively open subsets of P. Each U_i has an accumulation point in J. Apply the one-variable analytic identity theorem successively to the variables. The resulting identity on J^r contradicts the hypothesis. □

Proposition 22.3 (Cantor fusion with retained constraints)**.** *Under the assumptions on J, P in Lemma 22.2, let $(f_\nu)_{\nu \geq 1}$ be a countable family of nonzero real-analytic functions, where f_ν has some finite arity r_ν. There is a compact perfect set $K \subset P$, homeomorphic to $2^{\mathbb{N}}$, such that*

$$f_\nu(x_1, \ldots, x_{r_\nu}) \neq 0$$

whenever the x_i are pairwise distinct points of K. Every property holding pointwise throughout P is retained on K.

Proof. We construct a binary family of nonempty relatively clopen subsets of P, with disjoint siblings, nested closures, and diameters at level n at most 2^{-n}. Such sets exist because a compact perfect subset of the line contains a Cantor subset; we may first replace P by that subset. Suppose that the level $n-1$ sets have been selected. Choose two prospective points in each parent, giving 2^n point variables in a product of relatively open perfect sets. Exclude all coincidences, and, for each $\nu \le n$, exclude the zeros of f_ν on every ordered choice of r_ν distinct point variables. There are only finitely many exclusions. Each is nowhere dense by Lemma 22.2, applied to the corresponding coordinate variables; the diagonal exclusions are nowhere dense because P has no isolated points. A surviving configuration therefore exists. Continuity permits pairwise disjoint, sufficiently small clopen neighborhoods of its points on whose corresponding products every required value remains nonzero. These are the new level sets. The intersection of the unions of the level sets is a compact Cantor set K. A finite tuple of distinct points eventually lies in distinct level sets. At any later stage that also handles f_ν, the relevant product has been protected from its zeros. This proves the claim. □

Theorem 22.4 (Prime constants independent over any countable field)**.** *Let $P \subset (1,\infty)$ be a nonempty compact perfect set and let $(C_n)_{n\ge 0}$ be positive integers such that $\lfloor x^{C_n} \rfloor$ is prime for every $x \in P$ and every n. For any countable subfield $F \subset \mathbb{C}$, there exists a compact perfect $K \subset P$ such that the entire indexed family*

$$\mathcal{T}(K) = \big(T_j(x)\big)_{(x,j)\in K\times\{-1,0,1,2,\dots\}} \tag{22.1}$$

is algebraically independent over F and over every algebraic extension of F inside $\mathbb{C}$. The original exponent sequence works unchanged for every base point $x \in K$. Any upper box dimension or common-prefix bound already proved for P is inherited by K.

Proof. For every r, m and every nonzero polynomial $Q \in F[Y_{i,j} : 1 \le i \le r, -1 \le j \le m]$, include the function

$$f_Q(x_1,\dots,x_r) = Q\big(T_j(x_i)\big)$$

in the avoidance list. This is a countable family, and Lemma 22.1 shows that no listed function is identically zero. Apply Proposition 22.3. Every finite selection of distinct pairs (x, j) embeds in such a rectangular array, so the resulting assertion is precisely algebraic independence of (22.1). Algebraic independence persists under algebraic extension of the coefficient field: a finite tuple transcendental over F has the same transcendence degree after an algebraic base extension. The remaining statements follow from $K \subset P$. □

22.1 Effectiveness without deciding relations among constants

A sequence $(c_i)_{i\ge 0}$ of complex numbers is *uniformly computable* if one algorithm, on input (i, N), returns a rational complex number within 2^{-N} of c_i. Proposition 20.1 with the explicit schedule in Corollary 20.3 supplies a prime Cantor host of upper box dimension zero. It has a computable binary presentation: cylinder endpoints can be computed, cylinders at each level are separated, their diameters have a computable bound tending to zero, and the eventually-zero branches give a uniformly computable dense set.

Theorem 22.5 (Effective relative fusion)**.** *Suppose P has the computable presentation just described and $F = \mathbb{Q}(c_0, c_1, \dots)$ for a uniformly computable sequence. Then the set K in Theorem 22.4 can be chosen with a computable homeomorphism*

$$\Phi : 2^{\mathbb{N}} \longrightarrow K$$

whose inverse is computable on K. Consequently

$$\deg_T \Phi(\omega) = \deg_T \omega \quad \textit{and} \quad \deg_T T_j(\Phi(\omega)) = \deg_T \omega \qquad (j \geq -1).$$

In particular, K has points of every Turing degree, has a dense set of computable points, and has a computable least point. These properties hold while (22.1) is algebraically independent over the algebraic closure of F. No algorithm deciding equality in F is assumed.

Proof. It suffices to enumerate the genuinely nonzero polynomials with coefficients in the ring $R = \mathbb{Z}[c_0, c_1, \ldots]$. Indeed, clearing denominators turns any relation with coefficients in F into a nonzero polynomial relation with coefficients in R. Enumerate finite arrays of distinct variable monomials with formal expressions for their coefficients; combine equal variable monomials before performing any test. The resulting coefficient values are uniformly computable. Nonzeroness of a computable complex number is semidecidable: a rational approximation with an error bound excluding zero is a certificate. Dovetail these tests for every coefficient of every formal polynomial, and emit a polynomial as soon as at least one coefficient is certified nonzero. Every nonzero polynomial eventually appears; a polynomial all of whose coefficients vanish never enters the list. Thus no decision concerning algebraic relations among the c_i is needed.

At a fusion stage, candidate configurations from the uniformly computable dense set of P can be enumerated. Nonzeroness of the finitely many required function values is again semidecidable, using rational interval evaluation of polynomials, logarithms, and finite exponential iterates. Dovetail the searches over all candidates. The successful configurations form a nonempty relatively open set by the proof of Proposition 22.3; density therefore guarantees termination. After finding one, effective continuity gives rational neighborhoods on which every required value has a certified positive lower bound in modulus. Search sufficiently deep original cylinders contained in these neighborhoods, and impose the desired diameter bound. This completes a computable fusion construction.

The successive selected cylinders compute $\Phi(\omega)$ from ω. Conversely, the positive separation between the two selected children lets a Cauchy name for the point determine each successive branch bit. Hence Φ and its inverse are computable. The maps T_j are computable; their inverse on the relevant range is the computable j-fold logarithm for $j \geq 0$, or the exponential for $j = -1$. The degree identities follow. Eventually-zero codes yield a dense computable subset. Ordering the two children from left to right makes the all-zero code the least point of K. □

Corollary 22.6 (Independence relative to named constants)**.** *The effective construction applies to*

$$F = \mathbb{Q}(e, \pi, \gamma, \delta), \qquad \delta = \int_0^\infty e^{-t} \log(1+t)\, dt,$$

and to any field generated by a uniformly computable sequence of named constants. For example, it produces a computable prime-representing constant α for which

$$\log \alpha,\ \alpha,\ e^{\alpha},\ e^{e^{\alpha}},\ e^{e^{e^{\alpha}}}, \ldots$$

are algebraically independent over $\overline{\mathbb{Q}(e, \pi, \gamma, \delta)}$.

Proof. The constants displayed are computable. For γ, the sequence $H_n - \log n$ has an explicit error bounded by $1/n$. For δ, use computable quadrature on a finite interval and the tail estimate

$$\int_A^\infty e^{-t} \log(1+t)\, dt \leq \int_A^\infty e^{-t}(1+t)\, dt = (A+2)e^{-A}.$$

Apply Theorem 22.5 to the all-zero code. □

Remark 22.7 (Scope of the relative assertion). The last corollary makes no assertion that e, π, γ, δ are algebraically independent among themselves, and proves neither the irrationality of γ nor Schanuel's conjecture. The new constants are selected to avoid relations over the field of the old ones. The prime-output condition is asserted for $x \in K$; it is not asserted for its logarithm or exponential iterates with the same exponents. Nor is the least point of K claimed to equal the unrestricted least prime-representing constant for the schedule.

Remark 22.8 (An unavoidable effectiveness boundary). For an arbitrary countable field, the existence conclusion remains valid, but the effective conclusion need not. If the field contains all computable real numbers, any set algebraically independent over it contains no computable real. This does not contradict Theorem 22.5: the computable reals do not admit a uniformly computable enumeration. In particular, when a prescribed field already contains the old computable minimum, a relatively independent subtree necessarily omits that point.

Finite nonvanishing certificates. The effective proof gives more than a nonzero limit statement. Once a listed polynomial and a tuple of separated branch cylinders have been handled, it records a positive rational ε such that $|Q(T_j(x_i))| \geq \varepsilon$ throughout that cylinder product. Given oracles for a finite tuple of distinct branches and a nonzero polynomial with the represented coefficients, one can reach such a certificate in finite time. This is an algorithmic termination guarantee, not a polynomial-time bound or a uniform closed-form estimate in degree and height.

Effective construction in pseudocode. The following specification uses certified rational interval arithmetic; all searches are dovetailed, so a nonterminating failed candidate cannot block the construction.

1. Enumerate formal coefficient expressions in $\mathbb{Z}[c_0, c_1, \ldots]$ and finite polynomial arrays. Emit an array only after one evaluated coefficient is certified nonzero.
2. At stage n, obtain the first n emitted arrays. In each selected parent cylinder, enumerate pairs of eventually-zero branch points. For each simultaneous candidate configuration, first certify pairwise distinctness, then test the required arrays on all injective choices of its 2^n point variables.
3. Retain the first configuration for which all tests certify a nonzero value. Compute positive rational lower bounds on the finitely many moduli and sufficiently small rational neighborhoods preserving half these bounds.
4. Replace each selected point by a descendant prime cylinder contained in its neighborhood, with diameter at most 2^{-n}. Order siblings from left to right and record all certified lower bounds.
5. Given a binary input sequence, follow its selected cylinders. Their unique limit is the output. Given a Cauchy name for an output, recover its bits from the separated sibling intervals.

22.2 Combining further analytic constraints

Proposition 22.9 (Fusion for closed nowhere-dense relations). *Let P be a compact Cantor set and, for each $\nu \geq 1$, let $Z_\nu \subset P^{r_\nu}$ be closed and nowhere dense. There is a compact Cantor set $K \subset P$ such that no ordered tuple of pairwise distinct points of K belongs to any Z_ν. If P has the computable presentation used in Theorem 22.5, and membership in $P^{r_\nu} \setminus Z_\nu$ is uniformly semidecidable from Cauchy names, with a successful test providing a rational open neighborhood*

contained in that complement, the fusion is effective and preserves Turing degrees as in that theorem.

Proof. At stage n, apply every relation Z_ν, $\nu \le n$, to every injective coordinate selection among the 2^n prospective children. The coordinate projections are open maps, so inverse images of closed nowhere-dense sets under these projections are nowhere dense. Together with the diagonals, finitely many such inverse images leave a nonempty open set of configurations in the prescribed parents. Choose a configuration there and sufficiently small clopen neighborhoods of its coordinates. The nested-set argument of Proposition 22.3 applies unchanged. In the effective case, enumerate configurations from the dense computable set and dovetail the complement tests. One terminates, and its certified neighborhoods allow computable selection of descendant cylinders. □

Corollary 22.10 (Local analytic tests)**.** *In Proposition 22.9, one may include relations of the form*

$$Z = P^r \cap B \cap \{x : f(x) = 0\},$$

where B is a closed rational box contained in an open box U, and f is analytic and not identically zero on U. Such a relation is closed and nowhere dense. If f is computable with effective local continuity bounds, its complement has the certificates required in the effective proposition. Countably many global tower conditions and countably many local conditions can therefore be enforced in one and the same subtree.

Proof. The zero set intersected with B is closed because B is compactly contained in the domain of continuity. Were its intersection with P^r to have relative interior, the successive identity-theorem argument in Lemma 22.2, now inside U, would force f to vanish throughout U. To certify that a point is outside Z, either certify that one coordinate is strictly outside the corresponding closed interval of B, or certify that the point lies in U and that f is nonzero there. Every point in the complement admits one of these certificates. Effective continuity gives a suitable open neighborhood. □

23 Exponential freeness of a prime-generated field

Independence of the individual towers can be supplemented by a field property involving exponentials of arbitrary rational functions of the prime constants. We keep two coefficient fields distinct: F_0 is the prescribed countable field over which the towers are independent, whereas $F = \overline{\mathbb{Q}}(K)$ is the rational-function field generated by the constructed prime constants and the algebraic numbers. The latter is not the algebraic closure of that field.

Theorem 23.1 (Relative Lindemann–Weierstrass property)**.** *Let $P \subset (1, \infty)$ be a nonempty compact perfect prime-coding set with a fixed exponent schedule, and let $F_0 \subset \mathbb{C}$ be any countable field. A compact perfect subset $K \subset P$ can be chosen so that both of the following hold:*

1. *The family $\{T_j(\alpha) : \alpha \in K,\ j \ge -1\}$ is algebraically independent over F_0, with the notation of Theorem 22.4.*
2. *Put $F = \overline{\mathbb{Q}}(K)$. For every finite family of pairwise distinct $u_1, \dots, u_m \in F$, the numbers $e^{u_1}, \dots, e^{u_m}$ are linearly independent over F.*

Every member retains the fixed prime-output schedule of P. Under the hypotheses of Theorem 22.5, K can be chosen computably homeomorphic to Cantor space, preserving all Turing degrees as in that theorem.

Proof. Add the following tests to the tower conditions in the fusion argument:

$$H(\mathbf{x}) = \sum_{j=1}^{m} A_j(\mathbf{x}) \exp(R_j(\mathbf{x})), \tag{23.1}$$

where $A_j, R_j \in \overline{\mathbb{Q}}(X_1, \dots, X_r)$, the A_j are not all zero, and the R_j are pairwise distinct as formal rational functions. This is a countable effectively enumerable family, because algebraic numbers admit exact effective representations and equality of rational functions over algebraic numbers is decidable. Each expression is analytic off its poles.

We first prove that H is not identically zero on any open real box avoiding its poles. A nonzero polynomial has no open zero set. Consequently, the box contains a rational point $\mathbf{a}$ at which at least one A_j is nonzero and every difference $R_j - R_k$ is nonzero. The exponents $R_j(\mathbf{a})$ are then distinct algebraic numbers, and the algebraic coefficients $A_j(\mathbf{a})$ are not all zero. The classical Lindemann–Weierstrass theorem gives $H(\mathbf{a}) \neq 0$; see, for example, the proof in Bézivin–Robba [5].

Here is a closed formulation of the bad relation. Let D be the product of the nonzero denominator polynomials in the test. In $(1, \infty)^r$ put

$$Z_H = \{D = 0\} \ \cup \ \{\mathbf{x} : D(\mathbf{x}) \neq 0, H(\mathbf{x}) = 0\}.$$

This set is closed: a limit point with $D \neq 0$ is handled by continuity, and every other limit point is included in $D = 0$. Its restriction to P^r is nowhere dense. First remove the polynomial zero set of D, whose restriction is nowhere dense by Lemma 22.2. On any remaining small open box, H is analytic and nonzero as a function, so the same iterated identity argument applies. For complex-valued H, use the real-analytic function $|H|^2$ with the same zero set. Also, the complement of Z_H is effectively open: at a point outside it, rational approximation eventually certifies both $D \neq 0$ and $H \neq 0$. Computable derivative bounds on a sufficiently small box then give a rational neighborhood certificate for these strict inequalities.

Apply Proposition 22.9 simultaneously to these relations and to all the tower relations over F_0. This gives (1) and the topological conclusions. In the effective case, the newly added tests have computable coefficients and semidecidable complements, so the same effective fusion applies.

To deduce (2), take pairwise distinct $u_j \in F$ and coefficients $a_j \in F$ which are not all zero. Only finitely many members $\boldsymbol{\alpha}$ of K occur in their rational-function expressions; if none occur, adjoin one unused member of K. Algebraic independence of K over $\mathbb{Q}$, hence over $\overline{\mathbb{Q}}$, makes evaluation at this tuple injective. Thus the u_j have pairwise distinct formal representatives $R_j(\mathbf{X})$ and the a_j have representatives $A_j(\mathbf{X})$ which are not all zero. A relation $\sum_j a_j e^{u_j} = 0$ would make one of the excluded tests (23.1) vanish at $\boldsymbol{\alpha}$, a contradiction. □

Corollary 23.2 (Exponentials of rational functions)**.** *If $u_1, \dots, u_m \in F$ are linearly independent over $\mathbb{Q}$, then $e^{u_1}, \dots, e^{u_m}$ are algebraically independent over F. In particular, for every $\alpha \in K$ and $d \geq 1$,*

$$e, \ e^{\alpha}, \ e^{\alpha^2}, \dots, e^{\alpha^d}, \ e^{1/\alpha}$$

are algebraically independent over F.

Proof. For distinct integer multi-indices ν, the exponents $\sum_j \nu_j u_j$ are pairwise distinct. Expanding a nonzero polynomial in the e^{u_j} would give a prohibited F-linear relation among exponentials of distinct members of F. For the displayed example, the exponents $1, \alpha, \alpha^2, \dots, \alpha^d, 1/\alpha$ are linearly independent over $\mathbb{Q}$ because α is transcendental. □

Part III
Modular return and the arithmetic of least constants

The return lemma first supplies the congruence mechanism. Local minimality then links prime intervals to Diophantine approximation. The ordinary Mills conclusions are explicitly conditional on algebraicity.

24 Arbitrary linear perturbations of polynomial exponential clocks

The following return principle handles every nonzero integer linear coefficient. It supplies a universal modular-return property, stronger than the divisibility return sufficient for several transcendence results. The threshold on the seed is essential. Throughout this section, $v_p(0) = \infty$.

Lemma 24.1 (Unit-slope residue lemma)**.** *Let $A \geq 2$ be an integer, $Q \in \mathbb{Z}[X]$, and $D \in \mathbb{Z}$. If $q \geq 1$ and $\gcd(D, q) = 1$, then $Q(A^j) + Dj$ attains every residue class modulo q at arbitrarily large nonnegative integer indices j.*

Proof. Use strong induction on q, the case $q = 1$ being immediate. For $q > 1$, write $q = uv$, where every prime divisor of u divides A and $\gcd(A, v) = 1$. The sequence A^j is eventually periodic modulo q with a period $t < q$: take $t = \mathrm{ord}_v(A)$ if $v > 1$, and $t = 1$ otherwise. The same period works for $Q(A^j)$. Set $g = \gcd(t, q) < q$. By induction, for any prescribed residue c there are arbitrarily large j for which

$$Q(A^j) + Dj \equiv c \pmod{g}.$$

Choose such j beyond the preperiod. Translation by th changes the value modulo q by Dth. Since $\gcd(Dt, q) = g$, a nonnegative h can be chosen so that $Q(A^{j+th}) + D(j + th) \equiv c \pmod{q}$. The resulting index is at least j, proving arbitrarily late occurrence. □

Theorem 24.2 (Eventual universal return)**.** *Let $a \geq 2$, $d \neq 0$, and e be integers, and let $P \in \mathbb{Z}[X]$. Define*

$$R(n) = P(a^n) + dn + e.$$

There is an integer m_0 such that for every $m \geq m_0$ and every $q \geq 1$, the congruence

$$R(n) \equiv R(m) \pmod{q} \tag{24.1}$$

has arbitrarily large solutions n. Its solution set is eventually periodic and has positive natural density.

Proof. If $|d| = 1$, Lemma 24.1 proves the assertion directly. Otherwise let S consist of all primes not exceeding the largest prime divisor of $|d|$. In particular, every prime divisor of d belongs to S, and every prime divisor of $p - 1$ belongs to S when $p \in S$. For each $\ell \in S$, put

$$B_\ell = \max\bigl(\{v_\ell(p - 1) : p \in S,\ p \nmid a\} \cup \{0\}\bigr).$$

Choose m_0 so that, whenever $m \geq m_0$ and $\ell \in S$ divides a,

$$v_\ell(P(a^m) - P(0)) \geq v_\ell(d) + B_\ell. \tag{24.2}$$

This is possible because a^m divides $P(a^m) - P(0)$. Indeed, the uniform choice

$$m_0 = \max\{1, \lceil 2\log_2 |d| \rceil\}$$

is sufficient: for $\ell \mid a$ the left side of (24.2) is at least m, while $v_\ell(d) + B_\ell \le 2\log_2 |d|$.

Fix $m \ge m_0$ and factor $q = q_S q_{\text{out}}$, with q_S supported on S and q_{out} coprime to every prime in S. For $\ell \in S$ let $t_\ell = v_\ell(q_S)$, and put

$$K_\ell = \begin{cases} \max(t_\ell, B_\ell), & \ell \nmid a, \\ \max(t_\ell - v_\ell(d), B_\ell, 0), & \ell \mid a. \end{cases} \qquad M = \prod_{\ell \in S} \ell^{K_\ell}.$$

For $\ell \nmid a$, impose $n_0 \equiv m \pmod{\ell^{K_\ell}}$. For $\ell \mid a$, impose

$$d(n_0 - m) \equiv P(a^m) - P(0) \pmod{\ell^{t_\ell}}, \qquad n_0 \equiv m \pmod{\ell^{B_\ell}}. \tag{24.3}$$

These two congruences are compatible by (24.2): after removing the ℓ-power from d, its remaining factor is a unit, and the required difference $n_0 - m$ is divisible by ℓ^{B_ℓ}. Thus each local system has a solution modulo ℓ^{K_ℓ}. The Chinese remainder theorem gives a positive integer n_0 satisfying all of them.

For every sufficiently large j, set $n = n_0 + Mj$. We claim that

$$R(n) \equiv R(m) \pmod{q_S}. \tag{24.4}$$

At a prime $\ell \mid a$, the exponential part satisfies $P(a^n) \equiv P(0) \pmod{\ell^{t_\ell}}$ for large n, and (24.3) gives the claim. At a prime $p \nmid a$ with $t_p > 0$, the order of a modulo p^{t_p} divides $(p-1)p^{t_p-1}$. Every prime factor of this order lies in S. For a factor $\ell \mid a$, its valuation in that order is at most B_ℓ, so the required divisibility of $n - m$ follows from the second congruence in (24.3). For a factor $\ell \nmid a$, it follows from $K_\ell \ge \max(t_\ell, B_\ell)$. Consequently $a^n \equiv a^m \pmod{p^{t_p}}$. Also $n \equiv m \pmod{p^{t_p}}$, proving (24.4).

On this arithmetic progression we have

$$R(n_0 + Mj) = Q(A^j) + Dj + (dn_0 + e), \qquad Q(X) = P(a^{n_0}X), \quad A = a^M, \quad D = dM.$$

All prime factors of D lie in S, so $\gcd(D, q_{\text{out}}) = 1$. Lemma 24.1 supplies arbitrarily large j for which the desired congruence holds modulo q_{out} as well. Together with (24.4), this proves (24.1). Finally, R is eventually periodic modulo q, since both a^n and dn are. An eventually periodic solution set with arbitrarily late members has positive natural density. □

Remark 24.3 (The seed threshold cannot simply be omitted). For $R(n) = 2^n + 6n$ we have $R(1) = 8$. For $n \ge 3$, equality to 8 modulo 8 forces $n \equiv 0 \pmod 4$, whereas equality modulo 3 forces n to be odd. Hence no sufficiently late index returns to the first value modulo 24.

Corollary 24.4 (Transcendental minima for tails)**.** *Suppose additionally that P has degree $r \ge 1$ and positive leading coefficient, and that $a^r > 40/19$. For every sufficiently large integer N, the minimum*

$$\xi_N = \min\{x > 1 : \lfloor x^{R(n)} \rfloor \text{ is prime for every } n \ge N\}$$

exists and is transcendental.

Proof. Choose $N \ge m_0$ so large that $R(n)$ is positive and $R(n+1)/R(n) \ge 2$ for $n \ge N$. Write $C_k = R(N + k - 1)$, $c_1 = C_1$, and $c_{k+1} = C_{k+1}/C_k$. Then $c_1 \ge 1$, $c_{k+1} \ge 2$, $C_k \in \mathbb{N}$, and

$$\lim_{k\to\infty} c_{k+1} = a^r > 40/19.$$

These are conditions (B1)–(B4) of Saito's Type-B theorem, [39, Theorem 2.3]. Choose a sufficiently small $\varepsilon > 0$. Its index set $I = \{k : c_{k+1} \geq 40/19 + \varepsilon\}$ is cofinite. For every $m, L \geq 1$, Theorem 24.2, with modulus LC_m, gives arbitrarily large k such that

$$C_k \equiv C_m \pmod{LC_m}.$$

One may therefore take $k > m$ with $k \in I$, proving (B6). Taking $L = 1$ also proves (B5), namely $C_m \mid C_k$ at a later index in I. The cited theorem gives existence and transcendence of ξ_N. □

Remark 24.5 (Scope of the corollary). This includes every sufficiently late tail of $ba^n + dn + e$ with $a \geq 3$, $b \geq 1$, $d \neq 0$, and arbitrary integer e. It also includes $4^n - 10 \cdot 2^n + 37n + 5$ and $27^n + 2 \cdot 9^n - 8 \cdot 3^n - 1000n + 7$. Deleting a finite prefix changes the defining minimum; no identification with the original full-prefix minimum is claimed. The case $d = 0$ is outside the return theorem, so ordinary Mills' constant is not resolved.

Corollary 24.6 (Composite windows for Pisot and Salem powers)**.** *Let P have positive leading coefficient and degree at least one. For every Pisot or Salem number α and every fixed integer $H \geq 0$, there is an arithmetic progression of indices on which, after finitely many initial indices are removed, all the integers*

$$\lfloor \alpha^{R(n)} \rfloor + h, \qquad |h| \leq H,$$

are composite. No condition $a^{\deg P} > 40/19$ is needed here.

Proof. Let D be the degree of α and $T(t) = \mathrm{Tr}_{\mathbb{Q}(\alpha)/\mathbb{Q}}(\alpha^t)$. This is an integer linear recurrence with nonzero trailing coefficient. There is an index s_0 such that T is periodic modulo every prime p from index s_0 onward. Indeed, the companion matrix is invertible modulo primes not dividing the trailing coefficient, giving pure periodicity. Only finitely many remaining primes exist, and one may maximize their finite preperiods. Equivalently, use [35, Lemma 2.1(b)].

Choose $m \geq m_0$ such that $R(m) \geq s_0$ and $T(R(m)) > H + D + 1$. For each integer j with $|j| \leq H + D$, choose a prime divisor p_j of $T(R(m)) + j$. Let L be a common multiple of the periods of T modulo these finitely many primes. By Theorem 24.2, an eventually periodic positive-density set of indices satisfies $R(n) \equiv R(m) \pmod{L}$. For all sufficiently large such indices,

$$T(R(n)) + j \equiv T(R(m)) + j \equiv 0 \pmod{p_j}.$$

Since $R(n) \to \infty$ and $T(t) \to \infty$, these integers eventually exceed their fixed divisors p_j, and hence are composite.

Every nonprincipal conjugate of a Pisot or Salem number has modulus at most one. Consequently

$$|T(t) - \alpha^t| \leq D - 1, \qquad |\lfloor \alpha^t \rfloor - T(t)| \leq D.$$

The required floor window is therefore contained in the trace window already treated. Finally, the eventually periodic return set contains all sufficiently late members of some arithmetic progression. □

Proposition 24.7 (Intrinsic non-reversibility)**.** *Under the hypotheses of Corollary 24.6, R is an inhomogeneous integer linear recurrence sequence, but no integer inhomogeneous recurrence representation for it has trailing coefficient* 1 *or* -1.

Proof. An integer polynomial in a^n plus a linear polynomial in n is a linear recurrence sequence. Suppose a monic integer polynomial Q satisfies $Q(E)R =$ constant, where E is the shift operator. For every nonzero coefficient of X^j in P with $j \geq 1$, linear independence of distinct exponential sequences gives $Q(a^j) = 0$. The coefficient of n also gives $dQ(1) = 0$. In particular, some a^j divides $Q(0)$, so $a \mid Q(0)$. The trailing coefficient of the recurrence is $-Q(0)$ and cannot equal ± 1. □

The last two results enlarge the family answering Saito's Problem 1.7 in [35]; they are not an additional numbered-problem resolution. The window radius H is fixed, and its arithmetic progression can depend on H and α.

25 Local Type B and factorial perturbations

Put $\theta = 21/40$. We use the short-interval prime theorem and the prime-chain extension lemma in Saito, arXiv:2508.16068v3, Theorem 1.6 and Lemma 4.2. We also use his arithmetic approximation lemma, Lemma 6.1: sufficiently strong exponential approximation of integer powers of an algebraic number by integers forces a Pisot power, with degree bounded by one plus the reciprocal approximation exponent. The arguments below specify the new localization step and its consequences; they do not replace those analytic and Diophantine inputs.

Write

$$W(C) = \{A > 1 : \lfloor A^{C_k} \rfloor \text{ is prime for every } k \geq 1\}, \qquad c_{k+1} = C_{k+1}/C_k.$$

Call $A \in W(C)$ a *local left minimum* if there is $\eta > 0$ such that $(A - \eta, A) \cap W(C) = \varnothing$.

Theorem 25.1 (Local Type B)**.** *Let (C_k) be positive integers satisfying $c_{k+1} \geq 2$ and*

$$r = \limsup_{k\to\infty} c_{k+1} > 40/19.$$

Choose $\varepsilon > 0$ so that

$$I = \{k : c_{k+1} \geq 40/19 + \varepsilon\}$$

is infinite. Suppose that

$$\text{for each } m \text{ there is } k > m,\ k \in I,\ \text{with } C_m \mid C_k. \tag{25.1}$$

If a local left minimum $A \in W(C)$ is algebraic, then $r < \infty$ and there is an integer $g \geq 1$ such that $\beta = A^g$ is a Pisot number of degree ℓ satisfying

$$3 \leq \ell \leq 1 + \left(\frac{19r}{40} - 1\right)^{-1},$$

and $g \mid C_k$ for all sufficiently large $k \in I$. In particular, if $r > 60/19$, every local left minimum is transcendental.

If, in addition,

$$\text{for each } m, L \geq 1 \text{ there is } k > m,\ k \in I,\ \text{with } C_k \equiv C_m \pmod{LC_m}, \tag{25.2}$$

then every local left minimum is transcendental for all $r > 40/19$.

Proof. Set $p_k = \lfloor A^{C_k} \rfloor$. First, (B5) excludes $A^{C_m} \in \mathbb{N}$ for every m: that integer would be the prime p_m, whereas a later proper multiple C_k would make $A^{C_k} = p_m^{C_k/C_m}$ composite.

We show explicitly why the global-minimum argument localizes. Consider the exceptional set

$$J = \left\{k : \|A^{C_k}\| \leq \frac{1}{c_{k+1} p_k^{c_{k+1}-1}}\right\}.$$

If A is algebraic, J is finite. Indeed, an infinite J gives

$$\|A^{C_k}\| \leq \frac{1}{2p_k} \leq A^{-C_k} \quad (k \in J \text{ sufficiently large}).$$

The arithmetic approximation lemma gives a Pisot power A^g of degree at most two, with $g \mid C_k$ for late $k \in J$. Degree one gives a composite integer power at late indices. In degree two the equality case of that lemma makes A^g a quadratic Pisot unit, so

$$\|A^{C_k}\| = A^{-C_k} > \frac{1}{2p_k}$$

for sufficiently large $k \in J$, again a contradiction. This is the argument behind Saito's Lemma 6.2 and uses no minimality.

Choose k_0 beyond J. Saito's elementary Lemma 5.1 then gives the nested inequalities

$$p_k^{c_{k+1}} \le p_{k+1} < (p_k+1)^{c_{k+1}} - 1 \qquad (k \ge k_0). \tag{25.3}$$

Suppose infinitely many $m \in I$ violate

$$p_{m+1} \le p_m^{c_{m+1}} + p_m^{\theta c_{m+1}}. \tag{25.4}$$

At each sufficiently large such m, the short-interval theorem and prime-chain extension lemma give a replacement prime sequence (q_k) with $q_k = p_k$ for $k \le m$,

$$p_m^{c_{m+1}} \le q_{m+1} \le p_m^{c_{m+1}} + p_m^{\theta c_{m+1}} < p_{m+1},$$

and nested prime intervals thereafter. Because $m \in I$, the replacement at level $m+1$ also lies strictly below $(p_m+1)^{c_{m+1}} - 1$ when m is large. The nested intervals define a limit ζ_m whose prime outputs agree with p_k for $k_0 \le k \le m$ and with q_k afterwards. Moreover,

$$p_m^{1/C_m} \le \zeta_m < A, \qquad \zeta_m \ge 2^{-1/C_m} A \longrightarrow A.$$

For each of the finitely many indices $k < k_0$, the number A^{C_k} lies strictly inside the interval $[p_k, p_k+1)$, since integer powers have been excluded. Continuity therefore ensures the same outputs at ζ_m for all sufficiently large m. Thus $\zeta_m \in W(C)$, and eventually $A - \eta < \zeta_m < A$, contradicting local left minimality. This proves (25.4) for all late $m \in I$.

Taking c_{k+1}-th roots and using concavity yields

$$0 < A^{C_k} - p_k \le \frac{2}{c_{k+1} p_k^{(1-\theta)c_{k+1}-1}} \qquad (k \in I \text{ sufficiently large}). \tag{25.5}$$

If $r < \infty$, the arithmetic approximation lemma applied to this estimate gives a Pisot power $\beta = A^g$ with

$$\ell \le 1 + ((1-\theta)r - 1)^{-1}, \qquad g \mid C_k \quad (k \in I \text{ sufficiently large}).$$

Here replacing p_k by A^{C_k} affects only a bounded factor, since the exponents $(1-\theta)c_{k+1} - 1$ are bounded. If $r = \infty$, fix $T > 1$ and restrict to infinitely many indices for which this exponent is at least T. Then (25.5) is at most $p_k^{-T} \le 2^T A^{-TC_k}$, so the approximation lemma applies with the fixed constant $K = 2^T$. It forces degree at most $1 + 1/T < 2$, hence degree one, which is impossible. Hence algebraicity requires finite r and degree at least two.

The small fractional part in (25.5) and the Pisot property imply

$$p_k = \mathrm{Tr}(\beta^{C_k/g}) \qquad (k \in I \text{ sufficiently large}). \tag{25.6}$$

For completeness, the other conjugate powers have sum tending to zero; since the fractional part tends to zero, the trace is the floor rather than the next integer.

If $\ell = 2$, write the other conjugate as γ. Equation (25.6) forces $\gamma < 0$ and C_k/g odd at all late $k \in I$. Choose such an m and then $k > m$ as in (B5). The quotient $q = C_k/C_m$ is an odd integer at least three. With $u = \beta^{C_m/g}$ and $v = \gamma^{C_m/g} < 0$,

$$p_k = u^q + v^q = (u+v)\sum_{j=0}^{q-1}(-1)^j u^{q-1-j}v^j.$$

The second factor is a rational algebraic integer, hence an integer, and all its terms are positive. It exceeds one, while $u + v = p_m$ is prime. This contradiction proves $\ell \geq 3$ and the stated degree bound.

Finally assume (B6). Choose a sufficiently large $m \in I$, put $K = \mathbb{Q}(\beta)$, and let $\alpha = \beta^{C_m/g}$. The trace sequence $T(n) = \mathrm{Tr}_{K/\mathbb{Q}}(\alpha^n)$ is an integer linear recurrence with characteristic polynomial $N_{K/\mathbb{Q}}(X - \alpha)$, whose constant coefficient, up to sign, is $N_{K/\mathbb{Q}}(\alpha)$. Since α is Pisot,

$$0 < |N_{K/\mathbb{Q}}(\alpha)| < T(1) = p_m$$

for large m. Its recurrence matrix is therefore invertible modulo the prime p_m, and $T(n)$ is purely periodic modulo p_m, with some period L. By (B6), choose $k > m$ in I with $C_k/C_m \equiv 1 \pmod{L}$. Equation (25.6) gives

$$p_k = T(C_k/C_m) \equiv T(1) = p_m \equiv 0 \pmod{p_m}.$$

As $p_k > p_m$, this is impossible. □

Corollary 25.2 (Least completions of prime prefixes)**.** *Under the hypotheses of Theorem 25.1, fix a finite sequence of prime outputs that has at least one completion in $W(C)$. Its least completion exists and is a local left minimum. Whenever either transcendence conclusion of that theorem applies, all these least completions are transcendental. They form a countable dense subset of $W(C)$.*

Proof. The finite constraints define an interval $[a, b)$ containing a point of $W(C)$. The infimum of $W(C) \cap [a, b)$ belongs to that set: take a sequence tending down to the infimum and use right continuity of each floor map. Call it A. Condition (B5) excludes $A = a$, because a is a lower endpoint p_j^{1/C_j} for some prescribed output. Hence $(a, A) \cap W(C) = \varnothing$ and A is a local left minimum. Countability follows from the countable collection of finite prime prefixes. For any $B \in W(C)$, the minima of its successive prefixes converge to B, since the width of its level-k prime interval is at most $1/C_k \to 0$. □

25.1 The original perturbed cubic families

The following examples were among the initial families considered in this project. They are recorded as consequences of the return criterion, rather than counted again as new first-time transcendence results.

Corollary 25.3 (Three full cubic perturbations)**.** *For each of the exponent sequences*

$$C_k = 3^k - k, \qquad C_k = 3^k - 2k, \qquad C_k = 3^k - 2k + 1 \qquad (k \geq 1),$$

every local left minimum of $W(C)$ is transcendental. In particular, all three full minimum constants are transcendental.

Proof. Each sequence is positive, has consecutive ratios at least two, and has limiting ratio three. Indeed, for $C_k = 3^k - dk + e$,

$$C_{k+1} - 2C_k = 3^k + dk - d - e > 0$$

for $(d,e) = (1,0),(2,0),(2,1)$ and every $k \geq 1$. The existence theorem therefore gives a nonempty prime-representing set. It remains to verify (25.2) from every seed m.

For $d = 1$, Lemma 24.1 says that $3^n - n$ attains every residue modulo any positive integer q at arbitrarily large indices. Apply this with $q = LC_m$ and the target residue C_m.

For $d = 2$, write $q = 2^u v$, where $u \geq 0$ and v is odd, and restrict to indices $n = m + 2^u j$. At all such indices,

$$3^n - 2n + e \equiv 3^m - 2m + e \pmod{2^u}.$$

For $u \geq 1$ the exponential congruence follows because the order of 3 modulo 2^u divides 2^u; for $u = 0$ there is no constraint. On this progression the value equals

$$3^m (3^{2^u})^j - 2^{u+1} j - 2m + e.$$

The coefficient -2^{u+1} is a unit modulo v, so Lemma 24.1 supplies arbitrarily large j giving the required residue modulo v as well. The Chinese remainder theorem then gives $C_n \equiv C_m \pmod{q}$. Taking $q = LC_m$ proves (25.2). Since the index set I is cofinite, Theorem 25.1 applies to all three families. □

25.2 Factorials plus polynomial drift

An integer polynomial is called *intersective* here if it has a root modulo every positive integer.

Theorem 25.4 (Factorial perturbations)**.** *Let $P \in \mathbb{Z}[X]$ be intersective. For all sufficiently large integers N, put*

$$W_N = \{A > 1 : \lfloor A^{n! + P(n)} \rfloor \text{ is prime for every } n \geq N\}.$$

Then W_N is nonempty, and every local left minimum of W_N, including its least element and the least completion of any feasible finite prime prefix, is transcendental.

In particular, without deleting any initial constraint,

$$\boxed{\xi(k! + k) \text{ is transcendental.}}$$

The same conclusion holds for $\xi(k! + k^2)$.

Proof. For $C_n = n! + P(n)$, factorial growth gives $C_n > 0$ eventually and

$$C_{n+1}/C_n \sim n + 1 \longrightarrow \infty.$$

Choose N so that all these ratios are at least two. This ensures nonemptiness by Saito's existence theorem. Given $m \geq N$, choose a root t of P modulo $q = C_m$ and take $n > m$ sufficiently large with $n \equiv t \pmod{q}$. Then $q \mid n!$ and $q \mid P(n)$, so $C_m \mid C_n$. Condition (B5) holds, and the unbounded-ratio conclusion of Theorem 25.1 proves the claim.

For $P(X) = X$, the polynomial has root zero modulo every modulus, and

$$C_{k+1} - 2C_k = (k-1)(k! - 1) \geq 0 \qquad (k \geq 1).$$

Thus $N = 1$ is allowed. For $P(X) = X^2$,

$$C_{k+1} - 2C_k = (k-1)k! - k^2 + 2k + 1 > 0.$$

The assertion is immediate for $k = 1, 2$; for $k \geq 3$ use $k! \geq k(k-1)$, which makes the right side at least $k(k-1)(k-2) + k + 1 > 0$. □

The least-constant assertions in this theorem are direct applications of Saito's existing Type B theorem. The additional local-minimum conclusion uses the localization above. No new claim about the usual Mills constant is implied.

Proposition 25.5 (Exact factorial criterion for (B5))**.** *Fix any integer polynomial P and choose N so that $C_n = n! + P(n) > 0$ and $C_{n+1}/C_n \geq 2$ for $n \geq N$. Condition (B5) for this tail is equivalent to the statement that P has a root modulo C_m for every $m \geq N$.*

Proof. The forward construction in the proof of Theorem 25.4 proves sufficiency. Conversely, iterate (B5) starting at any fixed m to obtain an unbounded chain of indices $m < n_1 < n_2 < \cdots$ with $C_m \mid C_{n_j}$. At a level $n_j \geq C_m$, the factorial is divisible by C_m, so $P(n_j) \equiv 0 \pmod{C_m}$. This proves necessity. □

This criterion separates (B5) from (B6). For $C_k = k! + k^2$, (B5) holds, but (B6) fails at $m = 1$, $L = 4$. Indeed, $C_1 = 2$, whereas the residues of $C_2, \ldots, C_7$ modulo eight are $6, 7, 0, 1, 4, 1$. For $k \geq 8$, one has $C_k \equiv k^2 \pmod 8 \in \{0, 1, 4\}$. No later exponent is congruent to C_1 modulo $LC_1 = 8$.

Every polynomial having an integer root is intersective, but the class is strictly larger. For example,

$$P(X) = (X^2 - 13)(X^2 - 17)(X^2 - 221)$$

has no integer root and is intersective. For an odd prime not dividing $13 \cdot 17$, either 13 or 17 is a quadratic residue, or both are nonresidues and 221 is a residue. A nonzero root of the corresponding quadratic lifts to every prime power. At 13 use $17 \equiv 4$; at 17 use $13 \equiv 8^2$; at 2 use that $17 \equiv 1 \pmod 8$ is a square in $\mathbb{Z}_2$. The Chinese remainder theorem combines these local roots. Thus Theorem 25.4 includes genuinely non-linear drifts with no rational root.

More generally, let $H_n \in \mathbb{Z}$ satisfy: for every $q \geq 1$, $q \mid H_n$ for every sufficiently large n. If $C_n = H_n + P(n)$ is eventually positive with successive ratios at least two and upper limiting ratio greater than $60/19$, the same proof applies to every intersective P. For instance, $H_n = (n!)^s$, with any positive integer s, is admissible after a sufficiently late start.

The criterion does not cover every polynomial drift. For $P = 1$, (B5) would yield an unbounded divisibility chain from a modulus $C_m > 1$, forcing $C_m \mid 1$ at a sufficiently late level. Hence (B5) fails. This is an obstruction to the present method, not a disproof of transcendence for $\xi(n! + 1)$. Likewise, arbitrary early constraints added to a good tail can create prime-radical boundary candidates $A = p^{1/E}$; they require a separate argument unless some tail exponent is a proper multiple of E.

25.3 A modest quotient restriction at bounded integer ratios

The following statement does not imply algebraic independence. It records exactly what the elementary integer comparison still yields when the ratios remain bounded.

Proposition 25.6 (Rational quotient height)**.** *Suppose all c_{k+1} are integers at least two and $r = \limsup c_{k+1} < \infty$, with $r \geq 3$. Set $\kappa = 19r/40 - 1$. If $1 < A < B$ are local left minima of $W(C)$ and $B/A = u/v$ is a reduced positive rational, then*

$$u \geq A^{\kappa}.$$

For the cubic clock, this gives $u \geq A^{17/40}$.

Proof. For integer ratios, the nesting inequalities (25.3) hold for every element of $W(C)$, without assuming algebraicity. The lower inequality follows because $p_k^{c_{k+1}}$ is an integer. The only possible obstruction to the strict upper inequality would be the prime $(p_k + 1)^{c_{k+1}} - 1$, which factors

nontrivially. Consequently the downward replacement argument establishing (25.4) applies directly to every local left minimum. It gives (25.5) for both A and B at all late indices with $c_{k+1} = r$. There are infinitely many such indices, since the ratios are integers with finite upper limit r.

If $u < A^{\kappa}$, choose $0 < \kappa' < \kappa$ with $u < A^{\kappa'}$. At those indices, for sufficiently large k,

$$0 < A^{C_k} - p_k \le A^{-\kappa' C_k}, \qquad 0 < B^{C_k} - q_k \le B^{-\kappa' C_k}.$$

Because $v^{C_k} B^{C_k} = u^{C_k} A^{C_k}$, the integer

$$v^{C_k} q_k - u^{C_k} p_k$$

has absolute value at most

$$(vB^{-\kappa'})^{C_k} + (uA^{-\kappa'})^{C_k} \longrightarrow 0.$$

Indeed, $vB^{-\kappa'} < uA^{-\kappa'} < 1$. Yet this integer is nonzero: for large k, p_k, q_k are distinct primes larger than u, v, so equality would force $p_k \mid q_k$. This contradiction proves the bound. □

Whenever $A < 2^{40/17}$, the bound $A^{17/40} < 2$ excludes no positive integer numerator greater than one. Thus this proposition is a quantitative restriction for larger branch minima, not a resolution of the principal quotient or transcendence problem.

26 Sharper Type A bounds and late-tail Type C transcendence

In this section $\xi(C_k)$ denotes the least $A > 1$ for which $\lfloor A^{C_k} \rfloor$ is prime for every indicated index. We use Theorems 2.1 and 2.3 and Lemma 9.1 of Saito, *Transcendency of variants of Mills' constant*, arXiv:2508.16068v3. These results extend the Pisot reduction in *Mills' constant is irrational*, arXiv:2404.19461v2. All theorem-number comparisons in this section refer specifically to those arXiv versions. The version of record of the first paper, published in the *Ramanujan Journal* on 10 August 2026, has different numbering; its publicly accessible preview does not establish whether the Type A bound was changed. No priority claim relative to that version is made. Let $\kappa = 1.324717957\ldots$ be the smallest Pisot number, namely the positive real root of $X^3 - X - 1$.

26.1 An exact floor inequality improves Type A

Theorem 26.1. *Let $b > 40/19$, $h \ge 0$, and $\alpha b - h \ge 1$. Put $C_k = \lfloor \alpha b^k - h \rfloor$ for $k \ge 1$. Then $\xi(C_k)$ exists and is irrational. More precisely, it is either transcendental, or some $\xi(C_k)^{C_m}$ is a Pisot number of degree*

$$\ell \le 1 + \left(\frac{19b}{40} - 1 \right)^{-1}.$$

In particular, $\xi(\lfloor b^k \rfloor)$ is irrational for every $b > 40/19$.

Proof. Set $x_k = \alpha b^k - h \ge 1$. Since $b > 2$ and $h \ge 0$,

$$C_{k+1} = \lfloor b x_k + (b-1)h \rfloor \ge \lfloor 2x_k \rfloor \ge 2 \lfloor x_k \rfloor = 2C_k.$$

Also $C_1 \ge 1$, the exponents are integers, and $C_{k+1}/C_k \to b > 40/19$. All hypotheses of Saito's Theorem 2.1 are therefore satisfied. □

This improves the sufficient lower bound $1 + \sqrt{2}$ displayed in Saito's arXiv v3 Theorem 1.9(A). With $h = 0$, it also reduces the amplitude condition in his Corollary 2.2 to $\alpha \ge 1/b$: the additional condition $\alpha \ge 1/[b(b-2)]$ is unnecessary. This is an elementary sharpening of an

existing theorem, not a new irrationality mechanism. Neither the endpoint $b = 40/19$ nor the remaining range $2 < b \le 40/19$ is settled by this argument.

For completeness, if an integer sequence satisfies $C_k = \alpha b^k + o(b^k)$ with $\alpha > 0$ and $b > 40/19$, then $C_{k+1}/C_k \to b$, so every sufficiently late tail has an irrational minimum by the same theorem. This statement does not assert anything about an earlier prefix whose consecutive ratios fail to be at least two.

26.2 A finite asymptotic gcd controls every sufficiently late tail

For a positive integer sequence write

$$\operatorname{agcd}(C_n) = \lim_{m\to\infty} \gcd(C_m, C_{m+1}, \ldots).$$

Theorem 26.2. *Let (C_n) be positive integers. Suppose that, after deletion of a finite prefix, the following hold:*

1. *$C_{n+1}/C_n \ge c$ for a fixed $c > 40/19$;*
2. *for every m there exists $k > m$ with $C_m \mid C_k$;*
3. *$D = \operatorname{agcd}(C_n) < \infty$.*

Then the minimum $\xi_K = \xi((C_n)_{n\ge K})$ is transcendental for every sufficiently large K.

More explicitly, choose $x_0 \ge 1$ such that every $x \ge x_0$ has a prime in $[x, x + cx^{1-1/c}]$, and put $M = 2x_0^{1/c} + 1$. Once the hypotheses hold on the tail, the sufficient condition

$$C_K > D\frac{\log M}{\log \kappa}$$

implies that ξ_K is transcendental.

Proof. Such an x_0 exists by the Baker–Harman–Pintz short-interval theorem, since $1 - 1/c > 21/40$. Bertrand's postulate supplies a fixed prime p with $x_0^{1/c} < p \le 2x_0^{1/c}$. This prime is independent of K. Saito's Lemma 9.1 applies to every tail: its first consecutive ratio is at least c, and consequently $p > x_0^{1/(C_{K+1}/C_K)}$. It gives a constant A_K representing primes on that tail with first output p. Hence

$$\xi_K \le A_K < (p+1)^{1/C_K} \le M^{1/C_K}. \tag{26.1}$$

Choose ε with $0 < \varepsilon < c - 40/19$. The index set I in Saito's Theorem 2.3 is then the entire set of tail indices. The recurring-divisibility hypothesis is exactly its condition (B5). If ξ_K were algebraic, that theorem would give a positive integer g such that $\beta = \xi_K^g$ is a Pisot number of degree at least three and $g \mid C_n$ for all sufficiently large n. Thus $g \mid D$. By (26.1),

$$\kappa \le \beta = \xi_K^g \le \xi_K^D \le M^{D/C_K} < \kappa,$$

a contradiction. Finally $C_K \to \infty$ geometrically, so the displayed sufficient condition holds for all sufficiently large K. □

The theorem gives a late-tail conclusion under a lower ratio $c > 40/19$, whereas Saito's Type C theorem assumes all ratios are at least three. It does not prove transcendence of the minimum with a fixed earlier prefix still attached.

Corollary 26.3. *Let $a \ge 3$, $b \ge 1$ and $e \ne 0$ be integers. For every sufficiently large K, the minimum associated with $(ba^n + e)_{n\ge K}$ is transcendental.*

Proof. Write $C_n = ba^n + e$. Positivity and a lower consecutive-ratio bound $c > 40/19$ hold eventually, since $C_{n+1}/C_n \to a$. Increase the starting index so that

$$v_p(ba^n) > v_p(e) \qquad (p \mid a)$$

for every subsequent n. It follows that $v_p(C_n) = v_p(e)$ for such primes.

Fix an index m after this point and split $C_m = q_a q_u$, where all prime factors of q_a divide a and $\gcd(q_u, a) = 1$. The stabilized valuations show that q_a divides every subsequent C_n. If $q_u > 1$, choose $T = \mathrm{ord}_{q_u}(a)$; otherwise take $T = 1$. Then

$$C_{m+T} = ba^{m+T} + e \equiv ba^m + e = C_m \equiv 0 \pmod{q_u}.$$

Thus $C_m \mid C_{m+T}$, establishing recurring divisibility. Finally,

$$C_{n+1} - aC_n = (1-a)e,$$

so $\mathrm{agcd}(C_n)$ divides $|(a-1)e|$ and is finite. Theorem 26.2 applies. □

The restriction $e \neq 0$ matters: for a pure geometric clock the asymptotic gcd is infinite. This corollary covers constant perturbations, rather than relying on a nonzero linear perturbation of the index.

Corollary 26.4. *Let $a \geq 4$, $b \geq 1$ and $e \neq 0$ be integers. If K satisfies*

$$ba^K + e \geq 1, \qquad (a-2)ba^K \geq e, \qquad v_p(ba^K) > v_p(e) \quad (p \mid a),$$

then $\xi((ba^n + e)_{n \geq K})$ is transcendental. In particular, if

$$\gcd(a, e) = 1, \qquad ba + e \geq 1, \qquad e \leq ba(a-2),$$

then the full minimum $\xi((ba^k + e)_{k \geq 1})$ is transcendental. Consequently $\xi(ra^k - 1)$ and $\xi(ra^k + 1)$ are transcendental for every integer $a \geq 4$ and $r \geq 1$.

Proof. The displayed conditions ensure positivity, consecutive ratios at least two, and recurring divisibility by the preceding proof. If the minimum were algebraic, Theorem 2.3 of Saito would give

$$3 \leq \ell \leq 1 + \left(\frac{19a}{40} - 1\right)^{-1} \leq 1 + \frac{10}{9} < 3,$$

a contradiction. For the full-clock assertion, $\gcd(a, e) = 1$ implies $\gcd(a, C_m) = 1$ for every m, so periodicity of powers of a modulo C_m gives recurring divisibility directly. □

Corollary 26.5. *Let $2 \leq a < b$ be coprime integers, and put $C_n = a^n + b^n$. Every sufficiently late tail minimum is transcendental. If $b \geq 4$, then the full minimum $\xi((a^n + b^n)_{n \geq 1})$ is transcendental.*

Proof. The consecutive ratio is a weighted average of a and b, so it is greater than two and tends to b. For every m, $\gcd(C_m, ab) = 1$. The recurrence

$$C_{n+2} = (a+b)C_{n+1} - abC_n$$

is therefore invertible modulo C_m. Its finite state sequence is purely periodic modulo C_m, giving $C_m \mid C_{m+T}$ for some $T \geq 1$.

Let $D_N = \gcd(C_N, C_{N+1}, \ldots)$. Since D_N is coprime to ab, the same recurrence propagates divisibility backwards and gives

$$D_N \mid \gcd(C_1, C_2) = \gcd(a+b, a^2 + b^2) \mid 2.$$

The last divisibility uses $\gcd(a+b, ab) = 1$. Hence the asymptotic gcd is finite, and Theorem 26.2 applies. When $b \geq 4$, the degree contradiction $3 \leq \ell \leq 1 + (19b/40 - 1)^{-1} < 3$ already applies to the full clock by Saito's Theorem 2.3. □

For example, every sufficiently late tail of $2^n + 3^n$ has a transcendental minimum. This is a sum of independent geometric components and does not rely on a polynomial perturbation of one geometric component.

26.3 An explicit parity refinement in the cubic-base example

The following is an earlier elementary refinement. The stronger complete finite-state analysis in Theorem 53.3 lowers the full-schedule cutoff to 1.624×10^{14} in general and 8.12×10^{13} for even r.

Saito computes

$$\operatorname{agcd}(r3^k - 1) = \begin{cases} 1, & r \text{ even}, \\ 2, & r \text{ odd}, \end{cases}$$

but uses the uniform upper bound two in the final estimate of Theorem 1.9(C). Retaining the first case gives the following numerical refinement.

Corollary 26.6. *Let $x_0 = \exp(3\exp(32.76))$. For every even integer r satisfying*

$$r > \frac{\log(2x_0^{1/3} + 1)}{3\log\kappa} + \frac{1}{3},$$

the number $\xi(r3^k - 1)$ is transcendental. In particular, every even integer $r \geq 2.002 \times 10^{14}$ satisfies this condition.

Proof. The Type C conclusion gives a cubic Pisot number ξ^g in the algebraic case, with $g \mid \operatorname{agcd}(r3^k - 1) = 1$. Thus $g = 1$, and the same upper bound used in the source becomes

$$\xi \leq (2x_0^{1/3} + 1)^{1/(3r-1)} < \kappa,$$

contradicting minimality of κ among Pisot numbers. The displayed real threshold is approximately $2.0014802709094633 \times 10^{14}$, below the safe rounded bound 2.002×10^{14}. □

This last result is a numerical corollary, not an unconditional resolution for all $r \geq 1$. In particular, this section does not prove transcendence of $\xi(3^k - 1)$. It does not remove the global prime-between-cubes hypothesis from Saito's Corollary 2.7.

27 Finite prefixes and nonlinear polynomial perturbations

27.1 Finite-prefix stability and its exact endpoint obstruction

A *left-gap point* of a set $W \subset (1, \infty)$ means a point $A \in W$ for which $W \cap (A - \delta, A) = \varnothing$ for some $\delta > 0$. This is the same notion as a local left minimum. We use the localized Type B theorem proved in Theorem 25.1; its proof includes the replacement-point argument and the exclusion of integral powers at schedule coordinates. Thus no separate localization hypothesis is needed in this section.

Theorem 27.1 (Finite-prefix dichotomy)**.** *Let W be a prime-representing set whose every left-gap point is transcendental. Given positive integers $E_1, \ldots, E_s$, put*

$$W^* = W \cap \{x > 1 : \lfloor x^{E_j} \rfloor \text{ is prime for } 1 \leq j \leq s\}.$$

Every algebraic left-gap point of W^* belongs to

$$\bigcup_{j=1}^{s} \{p^{1/E_j} : p \text{ prime}\}.$$

If, for each j, some exponent C_k defining W is an integer multiple of E_j with $C_k/E_j \geq 2$, then every left-gap point of W^* is transcendental. In particular, whenever W^* is nonempty, its minimum is transcendental under this divisibility condition.

Proof. Suppose $A \in W^*$ and A^{E_j} is nonintegral for every j. All the finitely many added floor conditions are then constant in a neighborhood of A. A left gap for W^* is therefore a left gap for W, proving transcendence. Otherwise $A^{E_j} = p$ for a prime p, because the added condition is satisfied, giving the stated exceptional form. If $C_k/E_j = t \geq 2$, this would give $A^{C_k} = p^t$, contradicting the prime condition in W. The minimum, when the intersection is nonempty, exists by the same right-continuity argument as [39, Lemma 4.1], and is a left-gap point. □

Apply this theorem to a sufficiently late tail of $R(n) = P(a^n) + dn + e$, with $d \neq 0$, positive leading coefficient of P, and $a^{\deg P} > 40/19$. Theorem 24.2 gives eventual universal return, which verifies all Type B hypotheses for the tail, so all its left-gap points are transcendental. The theorem retains an arbitrary finite prime prefix; the only possible algebraic obstruction is a prime radical arising exactly at a newly imposed floor boundary. This is stronger than a tail-minimum assertion, but is not an unconditional removal of every possible prefix obstruction.

A useful sufficient check for a prefix exponent $C_m = R(m) > 0$ is

$$v_p(C_m) \leq m v_p(a) \qquad (p \mid a). \tag{27.1}$$

Then a^m is already zero modulo the part of C_m supported on primes dividing a. On its coprime part, choose a multiplicative period of a, and choose $n \equiv m$ modulo both that period and C_m. Consequently $C_m \mid R(n)$ for arbitrarily large n, supplying the required tail multiple. Only finitely many early checks are needed, as the next theorem shows.

The check is not automatic. For $R(n) = 3^n + 9n + 6$, $R(1) = 18$, whereas $R(n) \equiv 6 \pmod 9$ for every $n \geq 2$. Thus 18 divides no later exponent. Without an additional argument excluding the relevant prime radicals from the full set, the preceding theorem does not settle that full-prefix minimum.

27.2 A general theorem for nonlinear drift

Theorem 27.2 (Polynomial drift and tail transcendence)**.** *Let $a \geq 2$ be an integer and $P, Q \in \mathbb{Z}[X]$. Suppose P has positive leading coefficient and degree $r \geq 1$, and put*

$$C_n = P(a^n) + Q(n), \qquad D(X) = P(0) + Q(X).$$

Assume D is not the zero polynomial and $L = a^r > 40/19$. For every sufficiently large N, the minimum

$$\xi_N = \min\{x > 1 : \lfloor x^{C_n} \rfloor \text{ is prime for all } n \geq N\}$$

exists and is transcendental. This includes every nonconstant integer polynomial Q.

If $L \geq 4$, every left-gap point of every sufficiently late tail is transcendental. If $L \geq 4$ and the full sequence additionally has positive integer exponents, all successive ratios at least two, and the earlier recurring-divisibility conditions, then every left-gap point of the full prime-representing set is transcendental, without deleting a prefix.

Proof. First we prove eventual (B5). Since D is a nonzero polynomial, $D(m) \neq 0$ for every sufficiently large m, and $\log |D(m)| = O(\log m)$. For each prime $p \mid a$, eventually

$$v_p(D(m)) < m v_p(a) \leq v_p(P(a^m) - P(0)).$$

Hence

$$v_p(C_m) = v_p(D(m)) < m v_p(a). \tag{27.2}$$

Write $C_m = uv$, where u is supported on the primes dividing a and $\gcd(a, v) = 1$. By (27.2), $u \mid a^m$. Choose a positive period t of a modulo v (take $t = 1$ if $v = 1$). For every positive integer j, put $n = m + j\,\mathrm{lcm}(C_m, t)$. Then $a^n \equiv a^m \pmod{C_m}$ and $Q(n) \equiv Q(m) \pmod{C_m}$. Therefore

$$C_n \equiv C_m \equiv 0 \pmod{C_m}.$$

This gives arbitrarily late multiples of every sufficiently late exponent, proving (B5), including membership in the cofinite set I of Saito's theorem.

Next we bound the asymptotic gcd. Let $J = \{j \geq 1 : [X^j]P \neq 0\}$, let $s = \deg D$, and let b be the leading coefficient of D. With E denoting the forward shift and $\Delta = E - 1$, we have the exact integer identity

$$\Delta^s \prod_{j \in J} (E - a^j) C_n = K, \qquad K = s!\, b \prod_{j \in J} (1 - a^j) \neq 0. \tag{27.3}$$

Indeed, the product annihilates every nonconstant exponential term, and leaves a degree-s polynomial with leading coefficient $b \prod_{j \in J}(1 - a^j)$; the finite difference produces the stated constant. Every tail gcd divides K. In particular, any integer g dividing every sufficiently late C_n satisfies $g \mid K$.

For a sufficiently late tail, positivity, $c_{k+1} \geq 2$, integrality, and $c_{k+1} \to L > 40/19$ verify (B1)–(B4). By (B5) and [39, Theorem 2.3], algebraicity of its minimum would produce a Pisot number $\beta = \xi_N^g$ of degree ℓ satisfying

$$3 \leq \ell \leq 1 + \left(\frac{19}{40} L - 1\right)^{-1}, \qquad g \mid K. \tag{27.4}$$

When $L \geq 4$, the upper bound is less than 3, a contradiction. The localized theorem gives exactly the same contradiction for every left-gap point, proving the stronger assertion in this case.

It remains to treat $L = 3$. We show uniformly that $\xi_N \to 1$. Choose $40/19 < c < 3$, so all sufficiently late ratios are at least c. Let $\theta = 21/40$. The Baker–Harman–Pintz input used in [39, Theorem 1.6] supplies a prime in $[x, x + x^\theta]$ for every sufficiently large x. Choose a fixed sufficiently large prime p. Uniformly for every $r \geq c$ and every $q \geq p$,

$$q^r + q^{\theta r} + 1 < (q + 1)^r,$$

since $(q + 1)^r - q^r \geq r q^{r-1}$ and $(1 - \theta)c - 1 > 0$. Starting with the same prime p at the first coordinate of any late tail, select the subsequent primes in these intervals. Nested prime-root intervals give a member of that tail set less than $(p + 1)^{1/C_N}$. Consequently

$$1 < \xi_N < (p + 1)^{1/C_N} \longrightarrow 1. \tag{27.5}$$

For $L = 3$, (27.4) forces $\ell = 3$. Every cubic Pisot number β satisfies $\beta > 5/4$: if f is its monic minimal polynomial and β_2, β_3 its other conjugates, then

$$1 \leq |f(1)| = (\beta - 1)|1 - \beta_2||1 - \beta_3| < 4(\beta - 1).$$

But (27.5) and $g \mid K$ imply

$$\beta = \xi_N^g < (p + 1)^{|K|/C_N} < 5/4$$

for every sufficiently large N, a contradiction. □

Remark 27.3 (What was extended). The proof does not assert universal modular return for nonlinear Q. It proves precisely the divisibility condition (B5) that is needed, bounds the possible Pisot-power exponent by a fixed integer, and then excludes the remaining cubic possibility for late minima. The condition $D \neq 0$ excludes a pure exponential polynomial with zero constant term. In that case the constant K in (27.3) vanishes, and this argument gives no bound for g. The case $a = 2$, $\deg P = 1$ has limiting ratio 2 and also lies outside the present theorem.

27.3 A full nonlinear example where (B6) fails

Proposition 27.4. *For the full schedule*

$$C_n = 8^n + 9n^2 \qquad (n \geq 1),$$

the least prime-representing constant and every left-gap point are transcendental. Nevertheless Saito's condition (B6) fails at $m = 2$.

Proof. The ratios tend to 8. Moreover, $8^n \geq (3/2)n^2$ for $n \geq 1$, so

$$C_{n+1} - 2C_n = 6 \cdot 8^n + 9(-n^2 + 2n + 1) > 0.$$

For every $n \geq 1$,

$$v_2(C_n) = 2v_2(n) < 3n,$$

because the two summands have different 2-adic valuations. Thus (27.1) holds at every index, and the preceding congruence construction proves (B5) for the full schedule. The Type-B degree bound is incompatible with degree at least three, so the localized theorem proves the first assertion.

For the second assertion, $C_2 = 100$. Suppose $n \geq 3$ and $C_n \equiv 100 \pmod{128 \cdot 65537}$. Put $j = 3n$. Since $j \geq 9$, the congruence modulo 128 yields $j^2 \equiv 100 \pmod{128}$, and hence

$$j \equiv 10 \text{ or } 22 \pmod{32}.$$

The prime 65537 satisfies $\mathrm{ord}_{65537}(2) = 32$. It follows that 2^j is congruent to 1024 or -64, respectively. The other congruence would therefore require

$$j^2 \equiv -924 \text{ or } 164 \pmod{65537}.$$

Both right-hand sides are quadratic nonresidues. Indeed, quadratic reciprocity gives

$$\left(\frac{-924}{65537}\right) = \left(\frac{3}{65537}\right)\left(\frac{7}{65537}\right)\left(\frac{11}{65537}\right) = (-1)^3 = -1,$$

while

$$\left(\frac{164}{65537}\right) = \left(\frac{41}{65537}\right) = \left(\frac{19}{41}\right) = \left(\frac{3}{19}\right) = -1.$$

Thus no such later n exists. Taking $L_0 = 32 \cdot 65537 = 2097184$ makes $L_0 C_2$ a multiple of $128 \cdot 65537$, so (B6) fails at $(m, L) = (2, L_0)$. □

This example disproves an unrestricted all-seed modular-return extension to nonlinear drift. It does not disprove eventual universal return for a fixed polynomial Q; that stronger assertion is not needed for Theorem 27.2 and remains unproved here.

28 Recalled cubic identities and the exact gap exponent

This section collects the cubic identities and gap asymptotics used in the arithmetic restrictions that follow. Saito's reduction implies that, if A is algebraic, some $\beta = A^{3^m}$ has conjugates

$$\beta,\ -\rho,\ \gamma, \qquad 0 < \sigma = |\gamma| < \rho < 1, \qquad \rho \leq \beta^{-17/40}.$$

The number β is a totally real cubic Pisot number. Set $M = 3^r$, $p_r = \lfloor \beta^M \rfloor$, $g_r = p_{r+1} - p_r^3$, and $N = |\,\mathrm{N}(\beta)|$.

Lemma 28.1 (Eventual least-prime choice)**.** *For the actual least Mills constant, each sufficiently late next output is the least prime strictly above the cube of the preceding output.*

Proof. The prime-gap bound gives a prime between q^3 and $q^3+O(q^{63/40}) < (q+1)^3-1$ for every sufficiently large prime q. Thus every sufficiently large prime starts an indefinitely extendible nested cubic prime chain. If a smaller prime than the chosen next output existed above a late tower cube, replace that output by the smaller prime and extend it. The nested intervals preserve every earlier output and give a smaller prime-representing constant, contradicting minimality. □

Proposition 28.2 (Previously established gap exponent)**.** *Under algebraicity,*

$$g_r \sim 3(\beta^2\rho)^{3^r}, \qquad \frac{\log g_r}{\log(p_r^3)} \longrightarrow \kappa = \frac{2+\log\rho/\log\beta}{3}, \qquad \frac{1}{2} < \kappa < \frac{21}{40}.$$

Moreover κ is transcendental.

Proof. The small-conjugate sum $-\rho^M+\gamma^M$ lies in $(-1,0)$ eventually, so $p_r = \mathrm{Tr}(\beta^M)$. For $X=\beta^M$, $Y=-\rho^M$, $Z=\gamma^M$, Newton's identity gives

$$g_r = -3(X+Y)(X+Z)(Y+Z) \sim 3\beta^{2M}\rho^M.$$

Since $N=\beta\rho\sigma$ is a positive integer and $\sigma<\rho$, one has $\beta\rho^2 > N \geq 1$, giving $\kappa > 1/2$. The conjugate upper bound gives $\kappa \leq 21/40$.

We show that $t=\log\rho/\log\beta$ is irrational. Otherwise $\beta^u\rho^v=1$ for some positive integers u,v. In an S_3 splitting field, interchange the first two conjugates in $\beta^u(-\rho)^v=(-1)^v$ and divide the identities. Distinct absolute values force $u=v$, hence $\beta\rho=1$, contradicting $N=\beta\rho\sigma<1$. In a cyclic cubic splitting field, put $l_i=\log|\beta_i|$ and apply the three cyclic embeddings. The resulting equations $ul_1+vl_2=ul_2+vl_3=ul_3+vl_1=0$ have determinant $u^3+v^3>0$, contradicting $l_1>0$. Thus t is irrational. If t were algebraic, Gelfond–Schneider would make β^t transcendental, contrary to $\beta^t=\rho$. Therefore t, and hence κ, is transcendental; equality at the rational upper endpoint is impossible. □

29 Frobenius restrictions and quantitative prime divisors in the cubic shadow

Throughout this section we assume that the least Mills constant A is algebraic. Theorem 1.7 of Saito, *Transcendency of variants of Mills' constant*, arXiv:2508.16068v3, https://arxiv.org/html/2508.16068v3, supplies a totally real cubic Pisot power $\beta=A^{3^m}$ with conjugates

$$\alpha_1=\beta, \qquad \alpha_2=-\rho, \qquad \alpha_3=\gamma, \qquad 0<\sigma=|\gamma|<\rho<1.$$

Let $f(X)=X^3-aX^2+bX-C$ be its minimal polynomial, let $\Delta=\mathrm{disc}(f)$, and put $N=|C|$. For $M_r=3^r$, write

$$p_r=\mathrm{Tr}(\beta^{M_r}), \qquad g_r=p_{r+1}-p_r^3, \qquad h_r=g_r/3.$$

The sum $(-\rho)^{3^r}+\gamma^{3^r}$ lies in $(-1,0)$ for all sufficiently large r, so $p_r=\lfloor\beta^{3^r}\rfloor$ is the Mills prime. Also $h_r>0$: the prime p_{r+1} is at least p_r^3 and cannot equal that composite cube. The classical cubic identities give

$$F_r := \frac{h_{r+1}}{h_r} = \prod_{i<j}(\alpha_i^{2M_r}-\alpha_i^{M_r}\alpha_j^{M_r}+\alpha_j^{2M_r}) \in \mathbb{N}.$$

Indeed, with $X_i = \alpha_i^{M_r}$, expansion gives

$$h_r = -(X_1 + X_2)(X_2 + X_3)(X_3 + X_1).$$

The right side is a symmetric algebraic integer, hence an integer. Factoring each $X_i^3 + X_j^3$ proves the formula for F_r, which is again a symmetric algebraic integer and is positive because the roots are real. The dominant terms give $h_r \sim B^{M_r}$ and $F_r \sim B^{2M_r}$, where $B = \beta^2\rho$. These identities supply the arithmetic input for the transition-factor restrictions below.

Theorem 29.1 (Frobenius and congruence sieve)**.** *Let ℓ be a prime with $\ell \nmid 6C\Delta$ and $\ell \mid F_r$. Then f is reducible modulo ℓ. More precisely,*

factor degrees of f $\pmod{\ell}$	*necessary congruence*
$(1,1,1)$	$\ell \equiv 1 \pmod{2 \cdot 3^{r+1}}$
$(1,2)$	$\ell \equiv -1 \pmod{2 \cdot 3^{r+1}}$.

In particular, $\ell \geq 2 \cdot 3^{r+1} - 1$. The assertion concerning F_r is valid for any irreducible cubic with nonzero constant term, without a Mills hypothesis.

Proof. Reduce the three distinct nonzero roots in a finite splitting field over $\mathbb{F}_\ell$. A vanishing factor in F_r means that, for some pair, R^{3^r} is a primitive sixth root of unity, where R is the ratio of the corresponding original roots. Consequently

$$\operatorname{ord}(R) = 2 \cdot 3^{r+1}.$$

If Frobenius permutes the three roots in a 3-cycle, orient the chosen pair cyclically. Then $R^{1+\ell+\ell^2} = 1$. The exponent is odd, which is impossible for an element of the displayed even order. If all roots lie in $\mathbb{F}_\ell$, the order divides $\ell - 1$, proving the first congruence.

In the remaining factorization type, let a be the fixed root and b, c the two roots exchanged by Frobenius. If the vanishing factor uses the ratio b/c, then $(b/c)^\ell = c/b$, so its order divides $\ell + 1$. If it uses $R = b/a$, its order divides $\ell^2 - 1$. Suppose that $\ell \equiv 1 \pmod 3$. Then the whole factor 3^{r+1} in its order divides $\ell - 1$, which is also even. Thus $R^{\ell-1} = 1$, so R is fixed by Frobenius and $b = c$, a contradiction. Therefore $\ell \equiv -1 \pmod 3$. All of the factor 3^{r+1} in the order must divide $\ell + 1$, and $\ell + 1$ is even. This proves the second congruence. □

This restriction is stronger than the previously obtained fixed quadratic norm condition: a cubic-inert prime is completely excluded from F_r, rather than merely subjected to an even-valuation condition.

Lemma 29.2 (Local multiplier valuations)**.** *For each fixed prime ℓ, extend the normalized valuation v_ℓ to a splitting field and order the root valuations as*

$$0 \leq u_\ell \leq v'_\ell,$$

where the three entries are $0, u_\ell, v'_\ell$. For all sufficiently large r,

$$v_\ell(F_r) = 2 \cdot 3^r u_\ell \quad (\ell \neq 3), \qquad v_3(F_r) = 2 \cdot 3^r u_3 + c_3,$$

where $c_3 \in \{0, 1, 2\}$ is fixed. Moreover $2u_\ell \leq v_\ell(C)$.

Proof. Integrality makes all root valuations nonnegative. Their minimum must be zero: otherwise their power sum p_r is divisible by ℓ for every sufficiently large r, whereas p_r is a prime tending to infinity. The sum of the three valuations is $v_\ell(C)$, giving $2u_\ell \leq u_\ell + v'_\ell = v_\ell(C)$.

For a pair of unequal root valuations, the corresponding quadratic factor has valuation exactly $2 \cdot 3^r$ times the smaller valuation. For a pair of equal valuations t, divide by the $2 \cdot 3^r$-th power of one root. The remaining factor is $\Phi_6(z^{3^r})$, where z is a local unit and $\Phi_6(X) = X^2 - X + 1$.

When $\ell \neq 3$, sufficiently large r kills the 3-primary part of the order of the residue of z. A root of Φ_6 has order 6 in characteristic at least 5, and order 3 in characteristic 2. Thus $\Phi_6(z^{3^r})$ is eventually a unit. When $\ell = 3$, its reduction is $(X+1)^2$. If the residue of z differs from -1, its 3^r-th power also differs from -1, so the factor is a unit. Otherwise $z^{3^r} \to -1$ in the local field, and $\Phi_6(z^{3^r}) \to 3$; its normalized valuation is eventually exactly 1. At most two of the three pair ratios can reduce to -1, since three cyclically oriented such ratios would have product -1 instead of 1.

Finally, the sum of the smaller valuations over the three pairs is u_ℓ. Summing the preceding formulas proves the lemma. □

Theorem 29.3 (A large allowed prime at every late transition)**.** *Let S be the primes dividing $6C\Delta$, and let $(n)_S$ denote the part of a positive integer n supported on S. For all sufficiently large r,*

$$(F_r)_S \leq 9N^{3^r}.$$

Every such F_r has a prime divisor outside S. Consequently,

$$P^+(F_r) \geq 2 \cdot 3^{r+1} - 1, \qquad P^+(g_r) \geq 2 \cdot 3^r - 1$$

for all sufficiently large r, where P^+ denotes the largest prime divisor.

Proof. Apply Lemma 29.2 simultaneously to the finite set S. Its valuation inequalities give

$$(F_r)_S \leq 3^2 \prod_{\ell \in S} \ell^{3^r v_\ell(C)} = 9N^{3^r}.$$

On the other hand,

$$\frac{B^2}{N} = \frac{\beta^4\rho^2}{\beta\rho\sigma} = \frac{\beta^3\rho}{\sigma} > \beta^3 > 1.$$

Since $F_r \sim B^{2\cdot 3^r}$, it follows that $F_r/(F_r)_S \to \infty$. Thus a prime divisor outside S exists at each sufficiently late stage, and Theorem 29.1 gives the first largest-prime bound. The second follows from $F_{r-1} \mid h_r \mid g_r$. □

Theorem 29.4 (Number and radical of gap prime divisors)**.** *Under the same hypothetical algebraic Mills shadow,*

$$\omega(g_r) \geq \frac{r}{2} - O(1), \qquad \log \operatorname{rad}(g_r) \geq \frac{\log 3}{4} r^2 - O(r).$$

Equivalently,

$$\omega(g_r) \geq \frac{\log\log g_r}{2\log 3} - O(1), \qquad \log \operatorname{rad}(g_r) \geq \frac{(\log\log g_r)^2}{4\log 3} - O(\log\log g_r).$$

Proof. First we give a direct finite-field proof that a prime $\ell \notin S$ can divide at most two distinct multipliers F_j. For each of the three original pair ratios, its order in a finite splitting field is fixed. If its factor vanishes in F_j, that order must be exactly $2 \cdot 3^{j+1}$, so each pair can account for at most one index. Moreover, among the 3-primary orders of the three ratios the largest occurs at least twice. Indeed, write the ratios as R, Q, RQ; in a cyclic 3-group the product of elements of unequal orders has the larger order. Consequently their 3-primary orders have at most two distinct values, and the prime can occur at at most two indices j.

Choose an index j_0 after which Theorem 29.3 applies, and choose one prime $q_j \notin S$ dividing F_j for each $j_0 \leq j < r$. Each chosen prime occurs at most twice and every one divides h_r, giving $\omega(g_r) \geq (r - j_0)/2$. Furthermore,

$$\prod_{j=j_0}^{r-1} q_j \leq \operatorname{rad}(g_r)^2, \qquad q_j \geq 2 \cdot 3^{j+1} - 1.$$

Taking logarithms and summing yields $\log \operatorname{rad}(g_r) \geq (\log 3) r^2/4 - O(r)$. The equivalent forms use $g_r \sim 3B^{3^r}$, hence $\log \log g_r = r \log 3 + O(1)$. □

Proposition 29.5 (A 3-adic embedding and cyclic cubic shadows)**.** *The polynomial f has a root in $\mathbb{Q}_3$, so the cubic field admits an embedding into $\mathbb{Q}_3$ and has a prime above 3 with ramification index and residue degree both equal to one. If $\mathbb{Q}(\beta)/\mathbb{Q}$ is Galois, then 3 splits completely in $\mathbb{Q}(\beta)$. In the Galois case every prime divisor of F_r outside S splits completely in the cubic field and satisfies $\ell \equiv 1 \pmod{2 \cdot 3^{r+1}}$.*

Proof. We first establish the elementary prime-tail rigidity needed here, rather than leave it as a reference to an earlier draft. Write $T_r = \operatorname{Tr}(\beta^{3^r})$. The integral companion matrix of f gives the standard trace congruence

$$T_{r+1} - T_r \in 3^{r+1}\mathbb{Z}.$$

One proof groups the terms in the matrix trace into cyclic rotation orbits of closed index words; the orbits of length 3^j contribute multiples of 3^j, and the shorter repeated words satisfy $u^{3^t} \equiv u^{3^{t-1}} \pmod{3^t}$ for integers u. Thus T_r converges to a limit $L \in \mathbb{Z}_3$ with $L - T_r \in 3^{r+1}\mathbb{Z}_3$.

For a sufficiently late prime $q = T_r$, exclude the fixed divisors of $3C\Delta$, and let $d \in \{1, 2, 3\}$ be the degree of the finite splitting field of $f \pmod q$. If $v_3(q^d - 1) \leq r$, cubing is periodic on all the powered roots $\alpha_i^{3^r}$: writing $q^d - 1 = 3^t u$ with $3 \nmid u$, any positive K with $3^K \equiv 1 \pmod u$ makes $q \mid T_{r+jK}$ for every $j \geq 0$. For large j this contradicts primality. Hence $v_3(q^d - 1) > r$. For $d = 1, 2$ this forces $q \equiv \pm 1 \pmod{3^{r+1}}$; for $d = 3$, lifting the exponent gives $q \equiv 1 \pmod{3^r}$. In all cases $T_r \equiv \varepsilon \pmod{3^r}$, with a fixed sign $\varepsilon \in \{1, -1\}$ because $T_r \equiv a \pmod 3$. Therefore $L = \varepsilon$. Using $T_0 = a$ and $T_1 = a^3 - 3ab + 3C$, the congruences modulo 3 and 9 give $a \equiv \varepsilon \pmod 3$ and $C \equiv ab \pmod 3$. Consequently

$$a \equiv \varepsilon \pmod 3, \qquad f(X) \equiv (X - \varepsilon)(X^2 + b) \pmod 3.$$

If $b \not\equiv 2 \pmod 3$, the root ε of the displayed residual polynomial is simple. If $b \equiv 2 \pmod 3$, the root $-\varepsilon$ is simple. Hensel's lemma therefore supplies a root of f in $\mathbb{Z}_3$. This gives the asserted embedding and degree-one completion. In a Galois extension all primes above 3 have the same local degree, so a degree-one completion implies complete splitting.

For a prime outside S, the Frobenius permutation in a cyclic cubic field is either the identity or a 3-cycle. The latter is excluded by Theorem 29.1; the remaining conclusion follows from the same theorem. □

These results do not prove that a new rational prime divisor appears at every stage: a prime may first divide one pair-factor and later a different pair-factor. They also do not contradict the known distribution of primes in residue classes or splitting classes. They are additional necessary conditions on a hypothetical algebraic Mills constant, not a proof of unconditional transcendence or a resolution of the full totally real cubic Pisot problem. A contradiction still requires new information about the least-prime choice or the entire trace sequence.

30 Local gap criteria and effective bounded-degree certificates

Saito's Theorem 1.8 in arXiv:2508.16068v3 already proves that the Density Hypothesis, and hence the Riemann Hypothesis, implies the transcendence of Mills' constant. The results below separate a much more localized sufficient condition from an effective finite test for algebraic candidates of bounded degree.

Write A for the least Mills constant and

$$P_k = \lfloor A^{3^k} \rfloor, \qquad X_k = P_k^3, \qquad G_k = P_{k+1} - P_k^3.$$

We use two previously established facts: eventually $P_{k+1} = \text{nextprime}(P_k^3)$; and if A is algebraic, then

$$\frac{\log G_k}{\log X_k} \longrightarrow \kappa, \qquad \frac{1}{2} < \kappa < \frac{21}{40}, \qquad \kappa \text{ is transcendental.}$$

These facts were proved in Section 28.

Corollary 30.1 (A localized subsequence criterion)**.** *Mills' constant is transcendental if*

$$\liminf_{k\to\infty} \frac{\log(\text{nextprime}(X_k) - X_k)}{\log X_k} \le \frac{1}{2}.$$

Equivalently, it suffices that for every $\varepsilon > 0$ there are infinitely many k for which $(X_k, X_k + X_k^{1/2+\varepsilon}]$ contains a prime. It also suffices that the sequence $\log G_k / \log X_k$ does not converge, or that it has an algebraic real accumulation point.

Proof. Under algebraicity the displayed sequence converges to the transcendental number $\kappa > 1/2$. Each of the stated alternatives contradicts this. The equivalence of the first two formulations is the definition of the lower limit and eventual least-prime choice. □

Only a single prime at each of infinitely many prescribed tower cubes is required. No density of primes in these intervals, and no assertion for all real starting points, is needed. Conversely, an almost-everywhere prime-gap theorem does not automatically control this prescribed sparse sequence. This criterion is a consequence of the exact gap exponent, not an unconditional proof of transcendence.

Theorem 30.2 (Degree, signature, and polynomial structure)**.** *If the least Mills constant A is algebraic of degree d, then*

$$d = 3^{t+1} \quad \text{for some } t \ge 0, \qquad \beta = A^{d/3}$$

is a totally real cubic Pisot number with a negative dominant small conjugate. If $f(X) = X^3 - aX^2 + bX - c$ is its minimal polynomial, then

$$f(X^{d/3}) = X^d - aX^{2d/3} + bX^{d/3} - c$$

is the minimal polynomial of A. In particular, the signature of $\mathbb{Q}(A)$ is $(3, (d-3)/2)$.

Proof. Saito's Theorem 1.7 supplies A^{3^m} with the stated cubic Pisot configuration. Let $K = \mathbb{Q}(A^{3^m})$, a totally real cubic field. As long as A^{3^u} is a cube in K and $u > 0$, its unique real positive cube root $A^{3^{u-1}}$ belongs to K. It remains a totally real cubic Pisot number: every embedding cubes to an embedding of the preceding Pisot number, and the degree is three since its cube already generates K. The negative dominant small conjugate is preserved under taking odd real roots.

This procedure stops either at $A \in K$, when $d = 3$, or at $\beta = A^{3^t} \in K$ with $t \ge 1$ and β not a cube in K. We claim that $X^{3^t} - \beta$ is irreducible over K. More generally, if b is not a cube in a characteristic-zero field, then $X^{3^v} - b$ is irreducible for every $v \ge 1$. For $v = 1$ this is the criterion for a cubic to have a root. Inductively, if θ is a root of the irreducible polynomial $X^{3^{v-1}} - b$, then

$$\text{Norm}_{K(\theta)/K}(\theta) = b,$$

because the degree is odd. If θ were a cube in $K(\theta)$, its norm would make b a cube in K, a contradiction. Thus $X^3 - \theta$ is irreducible over $K(\theta)$, proving the induction by the tower law.

Applying this to β gives $[\mathbb{Q}(A) : K] = 3^t$, hence $d = 3^{t+1}$ and $A^{d/3} = \beta$. Integrality follows from the integrality of the Pisot power. The monic polynomial $f(X^{d/3})$ annihilates A and has degree d, so it is its minimal polynomial. Each of the three nonzero real roots of f has exactly one real $(d/3)$rd root, because $d/3$ is odd. This proves the signature. □

Theorem 30.3 (An explicit bounded-degree onset)**.** *Fix an integer $D \geq 3$ and a rational number $U > 1$. Define*

$$G = 3^j = \max\{3^v : v \in \mathbb{Z}_{\geq 0},\ 3^v \leq D/3\}, \qquad T = U^G,$$

and

$$K(D,U) = j + \min\{r \in \mathbb{Z}_{\geq 0} : 3^r \geq 4(T+1)\}.$$

Suppose that A is algebraic, $[\mathbb{Q}(A) : \mathbb{Q}] \leq D$, and $A \leq U$. Then for every $k \geq K(D,U)$ the quantities

$$h_k = \frac{P_{k+1} - P_k^3}{3}, \qquad e_k = \mathrm{cres}_{P_k}(h_k)$$

are well defined integers and satisfy

$$\boxed{e_{k+1} = e_k^3.}$$

Here $\mathrm{cres}_p(h)$ is the unique representative of h modulo the odd integer p in $(-p/2, p/2)$.

Proof. By Theorem 30.2, the degree is $d = 3^{t+1}$ and A^{3^t} has the stated totally real cubic Pisot configuration. Since $3^t = d/3 \leq D/3$, we have $t \leq j$. Thus $\beta = A^G$ is an odd power of that cubic Pisot number. Its degree remains three: conjugate fibers under powering have equal size, while the dominant Pisot conjugate is the unique member of its fiber, forcing this size to be one. The real-root configuration and signs are preserved.

We may therefore label the conjugates of β as

$$\beta,\ -\rho,\ \gamma, \qquad 0 < |\gamma| < \rho < 1, \qquad \beta \leq T.$$

If $f \in \mathbb{Z}[X]$ is its monic irreducible cubic polynomial, then $f(-1)$ is a nonzero integer, so

$$1 \leq |f(-1)| = (\beta+1)(1-\rho)|1+\gamma| \leq 2(T+1)(1-\rho).$$

It follows that $\rho \leq 1 - 1/[2(T+1)]$.

For $k \geq K(D,U)$ put $n = 3^{k-j} \geq 4(T+1)$. Then $\rho^n \leq e^{-2} < 1/4$. Since n is odd and $|\gamma| < \rho$,

$$-\frac{1}{2} < -\rho^n + \gamma^n < 0.$$

The trace is an integer, and hence

$$P_k = \lfloor \beta^n \rfloor = \beta^n - \rho^n + \gamma^n = \mathrm{Tr}(\beta^n).$$

Let b_k be the sum of pairwise products of these three conjugate powers, and put $c_k = \mathrm{Norm}(\beta)^n$. Newton's identity now gives

$$P_{k+1} = P_k^3 - 3P_k b_k + 3c_k, \qquad h_k = c_k - P_k b_k.$$

Furthermore $P_k \geq \beta^n/2$, whence

$$\frac{|c_k|}{P_k} \leq 2(\rho|\gamma|)^n \leq 2\rho^{2n} < \frac{1}{2}.$$

Thus centered reduction recovers c_k exactly, namely $e_k = c_k$. Since $c_{k+1} = c_k^3$, the assertion follows. The primes P_k are odd at all these stages, since $P_{k+1} > P_k^3$ and $K(D,U) \geq 2$. □

Corollary 30.4 (Finite exclusion certificates)**.** *Suppose U is a certified upper bound for the actual least Mills constant. If three consecutive actual Mills floors P_k, P_{k+1}, P_{k+2} are certified at some $k \geq K(D,U)$ and either one of the required divisibilities by 3 fails or $e_{k+1} \neq e_k^3$, then A is not algebraic of degree at most D.*

Alternatively, if one seeks to exclude algebraic numbers of degree at most D and primitive minimal-polynomial coefficient height at most H, it suffices to use $U = H + 1$, by Cauchy's root bound.

This is an effective necessary-condition theorem. It does not assert that a required failure has been computed. A prime chain obtained by repeated greedy choices is not sufficient input unless its entries have also been certified as floors of the actual least Mills constant. No decimal estimate depending on RH is used.

Theorem 30.5 (An effective gap margin at bounded degree)**.** *Retain the algebraicity, degree, and upper-bound hypotheses and the notation of Theorem 30.3. Put*

$$\Delta = (T+3)^{-2}, \qquad \eta = \frac{\log(1+\Delta)}{6\log T} > 0, \qquad R_{\mathrm{gap}} = \max\left\{4(T+1), \frac{\log 2}{\log(1+\Delta)}\right\}.$$

For every $k \geq j$ with $3^{k-j} \geq R_{\mathrm{gap}}$, one has

$$\boxed{G_k \geq \frac{27}{32} X_k^{1/2+\eta}.}$$

In particular, one certified actual gap violating this inequality excludes every algebraic candidate of degree at most D bounded by U.

Proof. Write $\sigma = |\gamma|$ for the smaller conjugate modulus of β. First we show $\rho - \sigma \geq \Delta$. If $\gamma = -\sigma$, the positive integer discriminant of the irreducible cubic gives

$$1 \leq \sqrt{\mathrm{disc}(f)} = (\beta+\rho)(\beta+\sigma)(\rho-\sigma) \leq (T+1)^2(\rho-\sigma).$$

If $\gamma = \sigma$, set $a = \beta - \rho + \sigma = \mathrm{Tr}(\beta) \in \mathbb{Z}$. Since $f(a)$ is a nonzero integer,

$$1 \leq |f(a)| = (\rho-\sigma)|a+\rho|\,|a-\sigma| \leq (T+3)^2(\rho-\sigma).$$

Both cases imply the desired separation. Since $\sigma < 1$, it follows that $\rho/\sigma \geq 1+\Delta$. The absolute norm $N = \beta\rho\sigma$ is a positive integer, so

$$\beta\rho^2 = N\frac{\rho}{\sigma} \geq 1+\Delta.$$

Consequently the algebraic gap exponent obeys the uniform bound

$$\kappa - \frac{1}{2} = \frac{\log(\beta\rho^2)}{6\log\beta} \geq \frac{\log(1+\Delta)}{6\log T} = \eta.$$

Now put $n = 3^{k-j}$ and $X = \beta^n$, $Y = -\rho^n$, $Z = \gamma^n$. The threshold for n gives $\rho^n < 1/4$ and $(\sigma/\rho)^n \leq 1/2$. The preceding theorem identifies $P_k = X + Y + Z < X$ and the next floor with the cubic trace. Thus the exact identity

$$G_k = X^3 + Y^3 + Z^3 - (X+Y+Z)^3 = -3(X+Y)(X+Z)(Y+Z)$$

applies. We have $X + Y \geq 3X/4$, $X + Z \geq 3X/4$, and $-(Y+Z) \geq \rho^n/2$, regardless of the sign of γ. Therefore

$$G_k \geq \frac{27}{32}X^2\rho^n = \frac{27}{32}X^{3\kappa} \geq \frac{27}{32}(P_k^3)^{1/2+\eta},$$

as claimed. □

The explicit gap bound is conditional on bounded-degree algebraicity. It supplies a finite exclusion test, but does not assert that the actual Mills chain has been shown to violate that test.

31 Square clocks: a block criterion in every algebraic degree

Write

$$\mathcal{W}_2 = \{A > 1 : \lfloor A^{2^k} \rfloor \text{ is prime for every } k \geq 1\}, \qquad \xi_2 = \min \mathcal{W}_2.$$

Existence of this minimum is known; its irrationality is the explicit Question 1.6 of Saito, *Mills' constant is irrational*, arXiv:2404.19461v2. The results below do not answer that question unconditionally. They give an all-degree criterion based on long consecutive blocks of square gaps.

We use the following established inputs from Saito, *Transcendency of variants of Mills' constant*, arXiv:2508.16068v3: Lemma 6.1 (exponentially close algebraic powers yield a Pisot power), Lemma 5.8 (identification of the nearest integer with a trace), Lemma 5.10 (irrational arguments of nonreal Pisot conjugates), Lemma 5.12, attributed there to Smyth (equal moduli occur only for a complex conjugate pair), and Lemma 5.13 (a finite irrationality exponent for such an argument, from linear forms in logarithms). All references to numbering in this section concern those specific arXiv versions.

Lemma 31.1 (A doubling survivor calculation)**.** *Fix $1/5 < a < 1/4$ and put $I = [a, 1-a]$. If*

$$\{2^j x\} \in I \qquad (0 \leq j \leq L),$$

where $L \geq 1$ is an integer, then

$$\operatorname{dist}\big(\{x\}, \{1/3, 2/3\}\big) \leq 2^{-L}.$$

Consequently, the only points whose entire forward doubling orbit stays in I are $1/3$ and $2/3$.

Proof. A point of I whose next iterate also lies in I must lie in

$$I_0 = [a, (1-a)/2], \qquad I_1 = [(1+a)/2, 1-a].$$

Since $2a > (1-a)/2$, doubling modulo one cannot carry a point of I_0 back into I_0. By reflection about $1/2$, it cannot carry a point of I_1 back into I_1 either. Thus the first L binary digits of $\{x\}$ alternate. They agree with the first L digits of $1/3 = 0.010101\ldots_2$ or $2/3 = 0.101010\ldots_2$. The corresponding binary cylinder has length 2^{-L}, proving the bound. Letting L tend to infinity proves the last assertion. □

Theorem 31.2 (Consecutive square-gap blocks)**.** *Let $A \in \mathcal{W}_2$, and put*

$$p_k = \lfloor A^{2^k} \rfloor, \qquad d_k = p_{k+1} - p_k^2.$$

Suppose that, for some fixed $\delta > 0$, there are integers $m_j \to \infty$ and $L_j \geq 1$ such that

$$d_k \leq p_k^{1-\delta} \quad (m_j \leq k \leq m_j + L_j), \qquad \frac{L_j}{m_j} \longrightarrow \infty. \tag{31.1}$$

Then A is transcendental. In particular, the conclusion holds if $d_k \leq p_k^{1-\delta}$ for every sufficiently large k.

Proof. Call an index satisfying the gap inequality good. For a good index, write $A^{2^k} = p_k + u_k$ with $0 < u_k < 1$. The strict lower bound holds because an integral value at one index would make the next output a composite square. Since

$$(p_k + u_k)^2 < p_{k+1} + 1 = p_k^2 + d_k + 1,$$

we obtain

$$0 < u_k < \frac{d_k + 1}{2p_k} \ll p_k^{-\delta} \ll A^{-\delta 2^k}. \tag{31.2}$$

Here an infinite set of good indices necessarily has $\delta < 1$; alternatively one can replace δ by any smaller positive number below one. Assume for contradiction that A is algebraic. Apply Saito's Lemma 6.1 to the increasing sequence 2^k with k good. It gives a positive integer g, dividing every sufficiently late such 2^k, for which $\beta = A^g$ is a Pisot number. Hence $g = 2^h$ for an integer $h \geq 0$.

If β has degree one, it is an integer at least two, and $p_k = \beta^{2^{k-h}}$ is composite for all sufficiently large k. Thus its degree ℓ is at least two. Write its conjugates as $\beta = \beta_1, \beta_2, \ldots, \beta_\ell$ and put

$$S(n) = \sum_{i=2}^{\ell} \beta_i^n.$$

By (31.2) and Saito's Lemma 5.8, at every sufficiently large good index, with $n = 2^{k-h}$, one has

$$\mathrm{Tr}(\beta^n) = p_k, \qquad S(n) = -u_k < 0. \tag{31.3}$$

Smyth's conjugate-modulus lemma implies that the small conjugates of maximum modulus consist of either a single real conjugate or one nonreal conjugate pair. In the first case, the even powers of that real conjugate are positive and dominate the remaining terms, so $S(2^r) > 0$ for all sufficiently large r. This contradicts (31.3).

In the second case write the maximal pair as $z = \rho e^{2\pi i\phi}$ and $\overline{z}$, with $0 < \rho < 1$. All other small conjugates have modulus at most $\sigma < \rho$. If there are no others, the remainder below is zero. We have

$$S(n) = 2\rho^n \cos(2\pi n\phi) + O(\sigma^n).$$

It follows from (31.3) that, at good indices,

$$\cos(2\pi 2^{k-h}\phi) \leq O((\sigma/\rho)^{2^{k-h}}) = o(1).$$

Choose any fixed $1/5 < a < 1/4$. Outside $[a, 1-a]$ the cosine is bounded below by the positive number $\cos(2\pi a)$. Therefore every sufficiently late good index satisfies

$$\{2^{k-h}\phi\} \in [a, 1-a]. \tag{31.4}$$

Saito's Lemma 5.10 says that ϕ is irrational.

Applying Lemma 31.1 along a block in (31.1), and multiplying its conclusion by three, gives

$$\|3 \cdot 2^{m_j - h}\phi\| \leq 3 \cdot 2^{-L_j}.$$

Saito's Lemma 5.13 supplies constants $c > 0$ and $\lambda > 0$, depending on z, such that $\|q\phi\| \geq cq^{-\lambda}$ for every integer $q \geq 2$. Explicitly, choose $\phi \in (0, 1)$, put $w = z/|z|$, and take the fixed logarithms $\log w = 2\pi i\phi$ and $\log(-1) = \pi i$. For a nearest integer p to $q\phi$, the nonzero linear form

$$q \log w - 2p \log(-1) = 2\pi i(q\phi - p)$$

has integer coefficients of size $O(q)$. Baker's *Transcendental number theory* (1975), Theorem 3.1, as stated in Saito's Theorem 5.5, gives the asserted bound. Combining the two bounds yields $L_j \leq \lambda m_j + O(1)$, a contradiction.

For the special case of all sufficiently large indices being good, the logarithmic-forms bound is unnecessary: the entire tail orbit in (31.4) survives, so Lemma 31.1 makes $2^{k-h}\phi$ congruent to $1/3$ or $2/3$, contradicting irrationality directly. □

The use of the unique maximal conjugate or conjugate pair is the reason that Theorem 31.2 has no restriction such as $\delta > 1/3$ or $\delta > 1/2$. An isolated-index criterion and this block criterion have different hypotheses: an arbitrary infinite set of good indices need not contain the blocks required here. This distinction is essential.

Corollary 31.3 (A conditional answer for the square constant)**.** *Suppose that, for some fixed $0 < \theta < 1/2$ and $C > 0$, the least prime strictly exceeding every sufficiently large real x is at most $x + Cx^{\theta}$. Then ξ_2 is transcendental.*

Proof. For every sufficiently large prime p, put $q = \mathrm{nextprime}(p^2)$. The hypothesis gives

$$p^2 < q \le p^2 + Cp^{2\theta} < (p+1)^2 - 1.$$

Consequently, from any sufficiently large prime at any square-clock index one can continue indefinitely by this least-prime rule. The intervals

$$[p_k^{1/2^k}, (p_k+1)^{1/2^k}]$$

are nested; moreover, each child's upper endpoint is strictly below its parent's upper endpoint. Their intersection therefore gives a prime-representing constant with the prescribed initial prime, with no ambiguity at the half-open upper endpoints.

For the least square constant, its prime sequence must therefore obey

$$p_{k+1} = \mathrm{nextprime}(p_k^2)$$

at every sufficiently large index. Indeed, any smaller admissible next prime could be continued by the preceding construction, inside all earlier cylinders, to obtain a smaller element of $\mathcal{W}_2$. Choose $0 < \delta < 1 - 2\theta$. Then $d_k \le Cp_k^{2\theta} \le p_k^{1-\delta}$ eventually, and apply Theorem 31.2. □

This is a conditional result. The required prime-interval bound is not supplied by the usual RH estimate of order $\sqrt{x}\log x$. No unconditional irrationality or transcendence of ξ_2 is asserted here.

31.1 Arbitrarily sparse multiplier-four approximants

Proposition 31.4. *Let $T \subset \{2, 3, \ldots\}$ be any infinite set with no two consecutive elements. For $N \ge 1$, retain the exponent indices*

$$S_N = \mathbb{N} \setminus \{t \in T : t > N\},$$

list them in increasing order as $n_{N,1}, n_{N,2}, \ldots$, and put

$$\eta_N = \min\{A > 1 : \lfloor A^{2^{n_{N,j}}} \rfloor \text{ is prime for every } j \ge 1\}.$$

Then each η_N is transcendental and

$$\eta_N \le \eta_{N+1} \le \xi_2, \qquad \lim_{N\to\infty} \eta_N = \xi_2.$$

The set T may be chosen with arbitrarily large successive gaps.

Proof. The retained exponent clock has integer successive ratios two or four. Since the omitted indices are isolated and infinitely many remain after each finite cutoff, ratio four occurs infinitely often. Saito's arXiv:2404.19461v2, Theorem 1.5, therefore supplies existence and transcendence of each η_N. This is an existing transcendence theorem, not a new proof of that input.

The inclusions $S_N \subset S_{N+1} \subset \mathbb{N}$ imply the displayed monotonicity. Let $\zeta = \lim \eta_N \le \xi_2$. If the sequence is eventually equal to ζ, the union of the retained constraints gives $\zeta \in \mathcal{W}_2$. Otherwise $\eta_N < \zeta$ for every N. Fix k. If $\zeta^{2^k} = t$ is an integer, then $t \ge 2$ and, for every sufficiently large N,

$$\lfloor \eta_N^{2^{k+2}} \rfloor = t^4 - 1 = (t^2 - 1)(t^2 + 1),$$

which is composite, contradicting the retained constraint $k + 2 \in S_N$. Hence ζ^{2^k} is nonintegral. Continuity of the floor away from integers now implies $\lfloor \zeta^{2^k} \rfloor$ is prime. This holds for every k, so $\zeta \in \mathcal{W}_2$ and $\zeta = \xi_2$ by minimality. □

A limit of transcendental numbers need not be irrational, so this approximation statement does not settle Question 1.6. A multiplier three behaves differently: once it is inserted into a pure square clock, later exponents have an odd factor and are not a subsequence of $2, 4, 8, \ldots$. The set inclusions in this proof then do not apply without additional hypotheses.

31.2 What the Type D sign argument can and cannot give

For an eventual 2/3 multiplier word, write the tail as $3^{\nu_1}2\,3^{\nu_2}2\cdots$, with infinitely many twos. Under the additional integer-ratio hypotheses of Saito's arXiv:2508.16068v3, Theorem 2.8, including ratios at least two throughout, that theorem assumes

$$\limsup_{N\to\infty} \frac{\nu_{N+1}}{\sum_{j=1}^{N} \nu_j + (\log 2/\log 3)N} = \infty \tag{31.5}$$

and concludes transcendence of the minimum. Item 2.9 is a remark about independence of the chosen initial index, not a further theorem. If t_N is the position of the Nth two, condition (31.5) is equivalent to

$$\limsup_{N\to\infty} t_{N+1}/t_N = \infty.$$

Indeed, $\nu_{N+1} = t_{N+1} - t_N - 1$, while the denominator in (31.5) is comparable to t_N, with fixed positive comparison constants. This is a convenient reformulation of an existing theorem, not a weakening of its hypothesis.

There is a version-specific discrepancy worth recording accurately: the introductory Theorem 1.9(D) of arXiv v3 displays $\lfloor \log k \rfloor$, whereas its abstract and detailed proof use $\lfloor \sqrt{\log k} \rfloor$. The latter gives two-positions $\lceil e^{j^2} \rceil$ (up to initial indexing), with unbounded successive ratios; the former gives $t_N \sim e^N$ and does not satisfy (31.5). Accordingly the argument in that version establishes the square-root-logarithm example, and does not by itself establish the logarithm example.

One cannot extend that proof to every bounded pattern using only the sign of the leading conjugate pair. For example, an alternating two/three word makes the exponents at successive relevant three-steps differ by a factor of six. The sign requirement then has the form

$$\{6^j x\} \in J = [1/4, 3/4] \qquad (j \ge 0).$$

The two maps

$$f_2(x) = (x + 2)/6, \qquad f_3(x) = (x + 3)/6$$

send J into the disjoint subintervals $[3/8, 11/24]$ and $[13/24, 5/8]$ of J. Their attractor is an uncountable invariant set, of Hausdorff dimension $\log 2/\log 6$, all of whose forward multiplication-by-six orbits remain in J. In particular it contains irrational angles. Thus the sign condition alone permits many nonperiodic orbits. This does not claim that those angles are arguments of algebraic conjugates; it identifies precisely the extra arithmetic information missing from this attempted extension. No new unconditional resolution for the alternating Type D clock is claimed.

Part IV
Polynomial patterns and digital structure

These results arise from related questions in the source literature. They share floor-function and recurrence techniques, but each theorem states its own hypotheses and uses no unproved assertion about Mills constants.

32 Palindromic progressions and the base-two obstruction

Let P_b denote the positive integers whose standard base-b expansion is palindromic. Question 27 of Pongsriiam and Saito, *Palindromes and Antipalindromes in Short Intervals*, Journal of Integer Sequences **26** (2023), Article 23.9.5, p. 26, asks whether every finite coloring of P_b has a monochromatic nonconstant three-term arithmetic progression, and whether the stronger assertion holds for every subset of positive upper relative density. The source is [30]. Here an arithmetic progression is always required to have positive common difference.

Theorem 32.1. *Both assertions in Question 27 hold if and only if $b \geq 3$. More precisely, for $b \geq 3$, every $A \subseteq P_b$ with*

$$\limsup_{N\to\infty} \frac{|A \cap [1,N]|}{|P_b \cap [1,N]|} > 0$$

contains a b-term arithmetic progression. For $b = 2$, there is an explicit four-coloring of P_2 with no monochromatic three-term progression. There is even a three-term-progression-free subset of P_2 having lower relative density $1/7$.

Proof. First suppose $b \geq 3$. Write $P_{b,L}$ for the palindromes with exactly L digits. Their number is

$$a_L = (b-1)b^{\lceil L/2\rceil - 1}.$$

Put $Q_L = \sum_{j\leq L} a_j$. For $L \geq 2$ we have $a_L \leq ba_{L-1} \leq bQ_{L-1}$, and consequently $Q_L \leq (b+1)Q_{L-1}$. Let the upper relative density of A be $\delta > 0$. There are arbitrarily large N with $|A \cap [1,N]| \geq (\delta/2)|P_b \cap [1,N]|$. If $b^{L-1} \leq N < b^L$, extending to the end of the digit-length block gives

$$\frac{|A \cap [1, b^L - 1]|}{Q_L} \geq \frac{\delta}{2(b+1)}.$$

It follows, by summing exact-length block counts and discarding any fixed finite initial segment, that arbitrarily large L satisfy

$$\frac{|A \cap P_{b,L}|}{a_L} \geq \eta, \qquad \eta = \frac{\delta}{4(b+1)} > 0.$$

For each such L, fix a leading digit $a \in \{1, \dots, b-1\}$ for which the corresponding slice of A has density at least η among palindromes with that leading digit. For $L \geq 2$ this slice is parametrized by the cube $\{0, \dots, b-1\}^d$, where $d = \lceil L/2 \rceil - 1$, through an injective affine map

$$\Phi(v_1, \dots, v_d) = a(b^{L-1} + 1) + \sum_{j=1}^{d} w_j v_j, \qquad w_j > 0.$$

The paired digit weights are $b^i + b^{L-1-i}$; the central weight, when present, is $b^{(L-1)/2}$.

The density Hales–Jewett theorem implies that, once d is large enough, the preimage of this slice of A contains a combinatorial line. Thus there is a nonempty set of free coordinates I such

that replacing all coordinates in I by the same digit t, for $t = 0, \dots, b-1$, gives points of the preimage. Their images are

$$X,\ X + D, \dots, X + (b-1)D, \qquad D = \sum_{j \in I} w_j > 0.$$

This proves the density assertion. In any finite coloring, at least one color has positive upper relative density, proving the coloring assertion.

For the binary case we first observe that palindromes of one fixed length L contain no three-term progression. Indeed, suppose $x < y < z$ have that length and $x + z = 2y$. Let h be the highest binary position where x and z differ. Symmetry makes $j = L - 1 - h$ the lowest position where they differ, so $v_2(z - x) = j$. Since y lies between x and z, it shares their common prefix above position h. Since y is palindromic, it also shares their common suffix below position j. Therefore $v_2(y - x) \geq j$, contrary to $z - x = 2(y - x)$.

Color each $n \in P_2$ by the pair $(L \bmod 2, s)$, where L is its digit length and s its second leading digit; for $n = 1$ set $s = 0$. Suppose $x < y < z$ of one color satisfy $x + z = 2y$, and let L be the largest digit length. If only z has length L, the parity condition gives $y < 2^{L-2}$, whence $z = 2y - x < 2^{L-1}$, a contradiction. If y, z have length L but x does not, then $x < 2^{L-2}$. Put $B = 2^L$. When $s = 0$, both y, z lie in $[B/2, 3B/4)$, so $x = 2y - z > B/4$. When $s = 1$, they lie in $[3B/4, B)$, so $x > B/2$. Both conclusions are impossible. Finally, if all three lengths are equal, the preceding observation applies. Thus this four-coloring has no monochromatic three-term progression.

For an explicit density counterexample, take the color class C consisting of even-length binary palindromes with leading digits 11. At the end of length $2m$,

$$|C \cap [1, 2^{2m} - 1]| = 2^{m-1}, \qquad |P_2 \cap [1, 2^{2m} - 1]| = 2^{m+1} - 2.$$

Before the next C-block begins, the denominator has increased by 2^m from length $2m + 1$, and by 2^{m-1} from the leading-10 part of length $2m + 2$, while the numerator is unchanged. Thus the successive minima and maxima of the relative density tend respectively to

$$\frac{1}{7} \quad \text{and} \quad \frac{1}{4}.$$

Within a C-block the ratio increases, and between such blocks it decreases. This proves the claimed lower relative density and completes the proof. □

The combinatorial input is the density Hales–Jewett theorem of Furstenberg and Katznelson; an elementary proof is D. H. J. Polymath, *A new proof of the density Hales–Jewett theorem*, Annals of Mathematics **175** (2012), 1283–1327, [29]. This answers the literal Question 27 in every base. The distinction between binary and larger bases is essential.

33 Positive-density polynomial patterns in Piatetski–Shapiro sequences

Saito and Yoshida, *Distributions of finite sequences represented by polynomials in Piatetski–Shapiro sequences*, arXiv:2006.13930v2 [41], Question 6.1, ask for the order of magnitude of polynomial patterns whose indices are restricted to a positive-density subset of the integers. The following argument gives an affirmative answer. Throughout, an arithmetic progression has positive common difference, and a polynomial pattern of degree at most d means that all its successive differences of order $d + 1$ vanish.

Lemma 33.1 (Local polynomial approximation and counting)**.** *Fix integers $d \ge 0$, $k \ge d+2$, and real numbers $K \ge 0$ and $0 < \eta \le 1$. There are $c > 0$ and L_0 such that, for every integer $L \ge L_0$, the following holds. Let $B \subseteq \{1, \dots, L\}$ have $|B| \ge \eta L$, let $a(1), \dots, a(L)$ be integers, and suppose that a real polynomial P of degree at most d satisfies*

$$|a(t) - P(t)| \le K \qquad (1 \le t \le L).$$

Then at least cL^2 arithmetic progressions of length k contained in B have a-values forming a polynomial pattern of degree at most d. The constants are independent of the coefficients of P.

Proof. Partition $[-K, K]$ into $Q = O_{d,K}(1)$ disjoint sets, each contained in an interval of length $\varepsilon = 2^{-d-2}$. Color $t \in B$ by the set containing $e(t) = a(t) - P(t)$. A color class B' has at least $(\eta/Q)L$ elements. The usual supersaturation consequence of Szemerédi's theorem, obtained by Varnavides averaging, gives at least $c(k, \eta/Q)L^2$ arithmetic progressions of length k in B'.

For each such progression $t_j = u + jv$, the numbers $e(t_j)$ belong to one interval $[b, b+\varepsilon]$. Since differences of a constant vanish, for $0 \le j \le k-d-2$,

$$\left|\Delta^{d+1} a(t_j)\right| = \left|\Delta^{d+1} e(t_j)\right| \le 2^{d+1}\varepsilon = \frac{1}{2}.$$

Here Δ denotes difference in the progression index j; the polynomial contribution vanishes because $P(u+jv)$ has degree at most d. The left-hand difference is an integer, hence it equals zero. Newton interpolation now gives a polynomial of degree at most d representing all the $a(t_j)$. □

Theorem 33.2 (Affirmative answer to Question 6.1)**.** *Let $d \ge 1$, $d < \alpha < d+1$, and $k \ge d+2$. If $A \subseteq \mathbb{N}$ has positive lower natural density, define*

$$T_A(N) = \#\left\{(u,v) \in \mathbb{N}^2 : \begin{array}{l} u, u+v, \dots, u+(k-1)v \in A \cap [1,N], \\ \left(\lfloor (u+jv)^\alpha \rfloor\right)_{j=0}^{k-1} \text{ has degree at most } d \end{array}\right\}.$$

Then

$$T_A(N) \asymp_{\alpha,d,k,A} N^{\,2-\alpha/(d+1)}.$$

The upper bound holds without a density assumption. If A has only positive upper natural density, the corresponding lower bound holds along an unbounded sequence of values of N.

Proof. The upper bound is Saito–Yoshida's Theorem 1.3, since $T_A(N) \le T_{\mathbb{N}}(N)$. For the lower bound choose $\delta > 0$ such that $|A \cap [1,N]| \ge \delta N$ for every sufficiently large N. Discarding integers below $\delta N/2$ leaves at least $\delta N/2$ elements of A. Put

$$\beta = 1 - \frac{\alpha}{d+1} > 0, \qquad L = \lfloor N^\beta \rfloor.$$

Partition the remaining interval $[\delta N/2, N]$ into disjoint full integer blocks of length L, discarding the last incomplete block. Since $L = o(N)$, at least $\delta N/3$ elements of A remain for all large N. Call a block good if it contains at least $(\delta/6)L$ elements of A. Bad blocks contain in total at most $\delta N/6$ elements of A, so there are at least $\delta N/(6L)$ good blocks.

On a block beginning at $x \ge \delta N/2$, expand $(x+t)^\alpha$ to degree d at $t = 0$, writing the resulting polynomial as $P_x(t)$. For $0 \le t < L$, Taylor's theorem gives

$$|(x+t)^\alpha - P_x(t)| \le C_{\alpha,d,\delta} N^{\alpha-d-1} L^{d+1} \le C_{\alpha,d,\delta}.$$

Thus the integer sequence $\lfloor (x+t)^\alpha \rfloor$ is uniformly within $K = C_{\alpha,d,\delta} + 1$ of P_x. By Lemma 33.1, each good block contributes at least cL^2 desired progressions. Progressions counted in distinct blocks are distinct. Consequently,

$$T_A(N) \ge \frac{\delta N}{6L}\, cL^2 \gg NL \asymp N^{\,2-\alpha/(d+1)}.$$

The same proof applies on any subsequence with $|A \cap [1,N]| \geq \delta N$, proving the upper-density assertion. □

Corollary 33.3 (Simultaneous finite families)**.** *Fix finitely many real exponents $d_i < \alpha_i < d_i+1$, with integers $d_i \geq 1$, and let $k \geq 2 + \max_i d_i$. For a set A of positive lower natural density, there are at least*

$$cN^{\,1+\min_i(1-\alpha_i/(d_i+1))}$$

index progressions of length k in $A \cap [1,N]$ on which every sequence $\lfloor n^{\alpha_i} \rfloor$ forms a polynomial pattern of degree at most d_i.

Proof. Use the common block length $L = \lfloor N^{\min_i(1-\alpha_i/(d_i+1))} \rfloor$. Each Taylor remainder is bounded independently of N. Color a point by the tuple of residual bins, using bin width 2^{-d_i-2} in the ith coordinate, and repeat the proof of Lemma 33.1 and Theorem 33.2. This gives a lower bound; no matching upper bound for the simultaneous family is asserted. □

33.1 An older theorem answers the prime-index condition

In Saito–Yoshida, *Arithmetic progressions in the graphs of slightly curved sequences*, J. Integer Sequences **22** (2019), Article 19.2.1, Question 12 (arXiv:1807.06971v3 [40], Question 6.2), the authors ask whether the primes satisfy their condition (C). This follows directly from an earlier result of Tao and Ziegler, *Narrow progressions in the primes*, arXiv:1409.1327v2 [42], Theorem 5. It should therefore be recorded as a literature implication, not as a new theorem about prime progressions.

Here are the details. Fix an integer $k \geq 3$ and a real $\gamma > 0$. Tao–Ziegler gives infinitely many progressions $a, a+r, \ldots, a+(k-1)r$ consisting of primes, with $r \ll (\log a)^{L_k}$. Set

$$b = 1 + \min(\gamma/4, 1), \qquad h_\gamma(x) = x^b.$$

The functions h_γ and h'_γ strictly increase, and

$$\frac{h'_\gamma(x+1)^2}{h_\gamma(x)^\gamma} \asymp x^{\,2b-2-b\gamma} \longrightarrow 0.$$

For a sequence of such prime progressions with $a \to \infty$, put $x = (a-1)^{1/b}$. Then

$$h_\gamma(x+1) - h_\gamma(x) \sim b(a-1)^{1-1/b} \gg (\log a)^{L_k},$$

so the whole progression lies in $(h_\gamma(x), h_\gamma(x+1)]$. Passing to increasing starting points makes $x \to \infty$, and all requirements of condition (C) hold. Saito–Yoshida's Proposition 9 (arXiv Proposition 5.1) then gives the affirmative answer concerning the graphs at prime indices.

34 Optimal three-term waiting times

For $1 < \alpha < 2$ and an integer $r \geq 1$, put $b_n = \lfloor n^\alpha \rfloor$ and define

$$L_{\alpha,3,r}(x) = \min \left\{ y \geq 0 : \begin{array}{l} \text{there exists } n \in [x, x+y] \cap \mathbb{Z} \\ \text{with } b_n + b_{n+2r} = 2b_{n+r} \end{array} \right\}.$$

In Appendix A of arXiv:2006.13930v2, immediately before Proposition A.2, Saito and Yoshida conjecture the following upper bound. Their Proposition A.2 supplies a matching lower bound in the limsup sense, so the exponent obtained here is sharp.

Theorem 34.1. *For every $1 < \alpha < 2$ and every fixed integer $r \geq 1$,*

$$L_{\alpha,3,r}(x) = O_{\alpha,r}(x^{\,1-\alpha/2}).$$

Proof. Put $f(t) = t^\alpha$, $N = \lceil x \rceil$, and

$$\lambda = f''(N) = \alpha(\alpha-1)N^{\alpha-2}, \qquad M = \left\lceil \frac{32}{r\sqrt{\lambda}} \right\rceil.$$

Let $n_j = N + rj$ and $b_j = \lfloor f(n_j) \rfloor$ for $0 \leq j \leq M$. Since $\lambda \to 0$ and $rM = O(N^{1-\alpha/2}) = o(N)$, throughout $[N, N + rM]$ we have, for all sufficiently large N,

$$\frac{\lambda}{2} \leq f''(t) \leq \lambda.$$

The real increments $A_j = f(n_{j+1}) - f(n_j)$ increase strictly and satisfy

$$0 < A_{M-1} - A_0 \leq (M-1)r^2\lambda < 1$$

for large N. Indeed, $Mr^2\lambda = O(r\sqrt{\lambda} + r^2\lambda) \to 0$. It follows that the nondecreasing sequence $\lfloor A_j \rfloor$ has at most two constant runs. One run, say $j_0 \leq j < j_0 + J$, has $J \geq M/2$.

Suppose that no desired three-term progression begins at any n_j with $0 \leq j \leq M-2$. The integer increments $d_j = b_{j+1} - b_j$ then satisfy $d_j \neq d_{j+1}$. On the selected run, $\lfloor A_j \rfloor = q$ is constant and

$$d_j = \lfloor A_j + \{f(n_j)\} \rfloor \in \{q, q+1\}.$$

Thus these increments alternate, and their consecutive pair sums are all $2q+1$. In particular,

$$b_{j_0+2v} = b_{j_0} + v(2q+1) \qquad (0 \leq 2v \leq J).$$

Take $h = \lfloor J/4 \rfloor$. For sufficiently large N, $J \geq 8$ and $h \geq J/8$. Therefore the three values $b_{j_0}, b_{j_0+2h}, b_{j_0+4h}$ form an arithmetic progression. Set $t = n_{j_0}$ and $D = 2rh$. Cancellation of their integer parts gives

$$|f(t+2D) - 2f(t+D) + f(t)| < 2.$$

On the other hand, $D \geq rM/8$, and the integral formula for the second difference yields

$$\begin{aligned} f(t+2D) - 2f(t+D) + f(t) &= \int_0^D \int_0^D f''(t+u+v)\,du\,dv \\ &\geq \frac{\lambda}{2}D^2 \geq \frac{\lambda r^2 M^2}{128} \geq 8, \end{aligned}$$

a contradiction. Hence a three-term progression with the original fixed step r starts at some n_j with $j \leq M-2$, proving

$$L_{\alpha,3,r}(x) \leq 1 + rM \leq 1 + r + \frac{32}{\sqrt{\lambda}} = O_{\alpha,r}(x^{1-\alpha/2}). \qquad \square$$

35 Balanced equations in Piatetski–Shapiro sequences

Write $\mathrm{PS}(\alpha) = \{\lfloor n^\alpha \rfloor : n \in \mathbb{N}\}$. Saito's Theorem 2.2 in *Finiteness of solutions to linear Diophantine equations on Piatetski–Shapiro sequences* (arXiv:2306.17813v2 [32]) gives a general exceptional-set dimension bound $3r/(s-4)$ for equations $y = \sum_{i=1}^r a_i x_i$ with positive coefficients. The following argument gives the stronger bound $(r+1)/s$ when the coefficients sum to one. In particular, the three-term progression conclusion of his Theorem 1.3 extends from almost every $\alpha > 10$ to almost every $\alpha > 3$.

Theorem 35.1. *Let $r \geq 2$ be an integer, and fix positive real numbers $a_1, \dots, a_r$ with $\sum_i a_i = 1$. For $r+1 < s < t$, let $E_{\boldsymbol{a}}(s,t)$ be the set of $\alpha \in [s,t]$ for which*

$$y = \sum_{i=1}^{r} a_i x_i$$

has infinitely many nontrivial solutions in $\mathrm{PS}(\alpha)^{r+1}$. A solution is nontrivial if not all its coordinates coincide. Then

$$\dim_{\mathrm{H}} E_{\boldsymbol{a}}(s,t) \leq \frac{r+1}{s}.$$

Consequently, for almost every $\alpha > r+1$, the equation has only finitely many nontrivial solutions in $\mathrm{PS}(\alpha)$.

Proof. All implied constants may depend on s, t, r and the fixed coefficients. Writing $y = \lfloor n^\alpha \rfloor$ and $x_i = \lfloor n_i^\alpha \rfloor$, a solution necessarily satisfies

$$\left| \sum_i a_i n_i^\alpha - n^\alpha \right| \leq 2. \tag{35.1}$$

Indeed, the actual error is less than one, since the weights sum to one. Strict increase of the sequence of power floors implies that nontriviality is equivalent to $(n_1, \dots, n_r) \neq (n, \dots, n)$. Moreover, (35.1) implies $n_i \leq Cn$ uniformly: $a_i n_i^\alpha \leq n^\alpha + 2$. Thus there are only finitely many solutions with bounded n.

Put $D = \max_i |n_i - n|$. For each fixed n, group index vectors into dyadic shells $H \leq D < 2H$, where $H = 1, 2, 4, \dots$ and $H \leq Cn$. We first show that the number of vectors in a shell for which (35.1) holds at some $\alpha \in [s,t]$ is at most

$$CH^{r-1}(H^2/n + 1). \tag{35.2}$$

For $H \geq \varepsilon n$, with fixed small $\varepsilon > 0$, this follows from the trivial $O(n^r)$ bound. If $H < \varepsilon n$, choose ε small enough that all n_i belong to $[n/2, 3n/2]$. Set $d_i = n_i - n$. Uniform Taylor expansion gives

$$\sum_i a_i n_i^\alpha - n^\alpha = \alpha n^{\alpha-1} \sum_i a_i d_i + O(n^{\alpha-2} H^2).$$

Consequently every admissible vector satisfies

$$\left| \sum_i a_i d_i \right| \leq CH^2/n + Cn^{1-s}. \tag{35.3}$$

The right side is independent of α. After the first $r-1$ deviations have been fixed in $O(H^{r-1})$ ways, the last deviation lies in an interval containing $O(H^2/n + 1)$ integers, since $a_r > 0$ is fixed. This proves (35.2). No rationality assumption on the coefficients is needed.

We next prove that the exponent set associated with one admissible vector lies in an interval of length

$$Cn^{2-s}/H^2. \tag{35.4}$$

Put $q_i = n_i/n$ and $F(z) = \sum_i a_i q_i^z - 1$. Since the coefficients sum to one, $F(0) = 0$, and nontriviality gives $F''(z) > 0$. Hence

$$zF'(z) - F(z) = \int_0^z uF''(u)\, du. \tag{35.5}$$

For $0 < q \leq C$, define

$$J_s(q) = \int_0^s u(\log q)^2 q^u \, du = 1 + (s \log q - 1) q^s.$$

The quotient $J_s(q)/(q-1)^2$ extends continuously and positively at $q=0$ and $q=1$, with limits 1 and $s^2/2$, respectively. It is positive elsewhere by its integral representation. Compactness therefore gives $J_s(q) \ge c(q-1)^2$ uniformly for $0 < q \le C$. Choosing an index for which $|q_i - 1| = D/n$, we obtain from (35.5), for $s \le z \le t$,

$$zF'(z) - F(z) \ge cH^2/n^2.$$

Let $\delta = 2n^{-s}$ and $B = \{z \in [s,t] : |F(z)| \le \delta\}$. This set contains the exponent set in (35.1). On B,

$$F'(z) \ge (c/t)H^2/n^2 - \delta/s \ge c'H^2/n^2$$

for all sufficiently large n, since $H \ge 1$ and $s > 2$. The set B is an interval, possibly empty. Indeed, if $x < y$ lie in B, convexity gives $F'(z) \ge F'(x) > 0$ on $[x,y]$, so all intermediate values remain in $[-\delta,\delta]$. The mean value theorem now proves (35.4).

Fix $(r+1)/s < \sigma < 1$. The sum of the σ-powers of the covering interval lengths for a fixed n is, by (35.2) and (35.4), at most

$$Cn^{(2-s)\sigma}\left(n^{-1}\sum_{H} H^{r+1-2\sigma} + \sum_{H} H^{r-1-2\sigma}\right), \tag{35.6}$$

where both sums are over dyadic $H \le Cn$. The first term is $O(n^{r-s\sigma})$, whose sum over n converges because $s\sigma > r+1$.

For $r \ge 3$, the second term is $O(n^{r-1-s\sigma})$, also summable. For $r = 2$ and $\sigma < 1/2$, the same estimate applies. At $\sigma = 1/2$, it is $O(n^{(2-s)\sigma}\log n)$, and $\sigma > 3/s$ forces $s > 6$, ensuring convergence. For $r = 2$ and $\sigma > 1/2$, it is $O(n^{(2-s)\sigma})$, which is summable because

$$(s-2)\sigma > (s-2)\frac{3}{s} = 3 - \frac{6}{s} > 1.$$

Thus the full Hausdorff sum converges. The interval lengths tend uniformly to zero, and their tails cover every exponent admitting infinitely many nontrivial solutions. It follows that $\dim_{\mathrm{H}} E_{\boldsymbol{a}}(s,t) \le \sigma$. Letting $\sigma \downarrow (r+1)/s$ proves the dimension bound. A countable compact-interval cover of $(r+1,\infty)$ gives the final assertion. □

Corollary 35.2. *For almost every $\alpha > 3$, the set $\mathrm{PS}(\alpha)$ contains only finitely many nonconstant three-term arithmetic progressions. For $3 < s < t$, the set of exceptions in $[s,t]$ has Hausdorff dimension at most $3/s$.*

Proof. Apply Theorem 35.1 with $r = 2$ and $a_1 = a_2 = 1/2$. □

The theorem establishes an upper bound, not the proposed exact dimension. It does not prove infinitude for almost every $2 < \alpha < 3$, and it does not settle whether any exponent $\alpha > 2$ admits arbitrarily long arithmetic progressions. The identity $F(0) = 0$ distinguishes the balanced case from the remaining case of mixed positive coefficients with sum greater than one. The argument supplies the stated upper bound without any rationality assumption on the positive coefficients.

36 Exact rational-power orders for algebraic tower limits

Let $h(x)$ denote the limit of the tower $x, x^x, x^{x^x}, \dots$ on $[e^{-e}, e^{1/e}]$, and put

$$\operatorname{ord}(A) = \min\{m \ge 1 : A^m \in \mathbb{Q}\},$$

with value ∞ if the set is empty. Kobayashi, Saito, and Takeda, *Journal of Integer Sequences* **26** (2023), Article 23.3.3 [21], Theorem 4, define

$$\mathcal{E}(k)=\left\{\left(\frac{kt}{s}\right)^{s/(kt)} : \begin{array}{l} 1\le t\le \vartheta\log k,\quad kt/e\le s\le kte,\quad s,t\in\mathbb{N},\\ (kt)^{1/t},s^{1/t}\in\mathbb{N},\quad \gcd(kt,s)=1\end{array}\right\},\qquad \vartheta=(\log 2-e^{-1})^{-1}.$$

They prove that, for an algebraic base A of order $k\ge 2$, algebraicity of $h(A)$ implies $A\in\mathcal{E}(k)$. Their Remark 6 asks whether $\mathcal{E}(k)$ itself is the exact-order-k exceptional set. The literal set equality fails because some candidates have smaller order. This observation does not contradict Theorem 4.

Proposition 36.1 (A counterexample to the literal equality)**.** *For $k=4$, the number*

$$A=\left(\frac{4}{9}\right)^{9/4}=\left(\frac{2}{3}\right)^{9/2}$$

belongs to $\mathcal{E}(4)$, satisfies $h(A)=4/9$, and has $\operatorname{ord}(A)=2$.

Proof. Take $t=1$ and $s=9$ in the definition of $\mathcal{E}(4)$. The inequalities $4/e<9<4e$ and $\gcd(4,9)=1$ verify every condition. The square $A^2=(2/3)^9$ is rational, whereas A is irrational: the 2-adic valuation of A^2 is the odd integer 9, which a square of a rational number cannot have. Thus $\operatorname{ord}(A)=2$. The map $f(y)=y^{1/y}$ is strictly increasing on $[1/e,e]$, and the tower identity implies $h(f(q))=q$ for every $q\in[1/e,e]$; this is Lemma 14 of the cited paper. Applying it with $q=4/9$ gives $h(A)=4/9$. □

The same base is already identified in the paper's order-two discussion. Consequently the proposition is a modest correction to the literal formulation of Remark 6, not a new transcendence mechanism. The following valuation condition gives a finite explicit classification for every order, including orders that are not square-free.

For relatively prime positive integers u,v with $u\ne v$, let

$$G(u,v)=\gcd\{v_p(u),v_p(v) : p \text{ prime, nonzero entries only}\}.$$

Thus $G(u,v)$ is the largest integer $g\ge 1$ such that both u and v are gth powers of positive integers. The convention discards zero valuations; it allows $u=1$ or $v=1$.

Lemma 36.2 (Exact order from valuations)**.** *If a,b are relatively prime positive integers with $a\ne b$, then*

$$\operatorname{ord}\Big((a/b)^{b/a}\Big)=\frac{a}{\gcd(a,G(a,b))}.$$

Proof. The mth power is rational if and only if $a \mid mb\,v_p(a/b)$ for every prime p. Since $\gcd(a,b)=1$, these conditions are equivalent to $a/\gcd(a,G(a,b)) \mid m$. Taking the least positive m proves the formula. □

Theorem 36.3 (All-order exceptional-set classification)**.** *Fix an integer $k\ge 2$. The set of algebraic numbers $A\in[e^{-e},e^{1/e}]$ with* $\operatorname{ord}(A)=k$ *and algebraic $h(A)$ is exactly*

$$\left\{\left(\frac{u}{v}\right)^{v^t/k} : \begin{array}{l} t,u,v\in\mathbb{N},\quad 1\le t\le\vartheta\log k,\quad u^t=kt,\\ ue^{-1/t}\le v\le ue^{1/t},\quad \gcd(u,v)=1,\\ \gcd(k,G(u,v))=1\end{array}\right\}.$$

For each listed base, $h(A)=(u/v)^t$. Equivalently, one obtains the exact exceptional set by retaining from $\mathcal{E}(k)$ precisely the parameters satisfying

$$\gcd(kt,G(kt,s))=t.$$

Proof. First suppose that A has order k and algebraic tower limit. The cited Theorem 4 gives parameters t, s in $\mathcal{E}(k)$. Write $u = (kt)^{1/t}$ and $v = s^{1/t}$, so that $u^t = kt$, $\gcd(u, v) = 1$, and

$$A = \left(\frac{u}{v}\right)^{v^t/k}.$$

Because $k \mid u^t$, one has $\gcd(v, k) = 1$. Prime valuations now give

$$\operatorname{ord}(A) = \frac{k}{\gcd(k, G(u, v))}. \tag{36.1}$$

Indeed A^m is rational exactly when $k \mid mv^t v_p(u/v)$ for every prime p; the factor v^t is invertible modulo k. The asserted coprimality condition follows. The bounds on $s = v^t$ give the stated bounds on v.

Conversely, suppose the displayed conditions hold and set $q = (u/v)^t$. Then $q \in [1/e, e]$, and

$$q^{1/q} = \left(\frac{u}{v}\right)^{tv^t/u^t} = \left(\frac{u}{v}\right)^{v^t/k} = A.$$

The identity $h(q^{1/q}) = q$ gives an algebraic tower value. Equation (36.1) gives exact order k. Finally $G(kt, s) = tG(u, v)$, so $\gcd(kt, G(kt, s)) = t \gcd(k, G(u, v))$. This proves the equivalent formulation. □

The classification is an elementary refinement of the paper's finite candidate set, using its Theorem 4 and Lemma 14. It does not require new bounds for transcendental numbers or prime gaps. The refinement concerns exact rational-power order; it does not claim a new transcendence criterion for tower values.

37 A sparse zeta criterion for arbitrary real normal numbers

Kanado and Saito, *Normality of algebraic numbers and the Riemann zeta function*, arXiv:2412.02337v2, Corollary 1.2, give a normality criterion for positive algebraic irrational numbers using every integer $k \geq 0$. Their more general Theorem 2.2 permits the following strengthening. This is a corollary of their analytic theorem with an additional invariant-measure argument, rather than a resolution of a numbered open problem or of the normality of any specified algebraic irrational number. The primary source is [18].

For $b \geq 2$, $\alpha > 0$, and an integer $k \geq 1$, put

$$Z_{k,b,\alpha}(N) = \frac{1}{\log N} \sum_{1 \leq |n| \leq N} \zeta\left(-k + \frac{2\pi i n}{\log b}\right) \frac{\exp(2\pi i n \log \alpha / \log b)}{n^{k+1}}.$$

Theorem 37.1. *Let $b \geq 2$ be an integer, let $\alpha > 0$ be any real number, and let $R \subseteq \mathbb{N}$ be unbounded. Then α is normal in base b if and only if*

$$\lim_{N \to \infty} Z_{2r-1,b,\alpha}(N) = 0 \qquad \text{for every } r \in R.$$

In particular, algebraicity is unnecessary, the line $k = 0$ may be omitted, and any unbounded selection of the positive odd values of k suffices.

Proof. Let B_m be the Bernoulli polynomials. For $m \geq 2$ their periodic extensions $B_m(\{x\})$ are continuous on the circle $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. Theorem 2.2 of Kanado–Saito, with $d = 1/\log b$ and

$\theta = \log\alpha/\log b$, states that for every fixed integer $k \ge 1$,

$$\frac{(k+1)!}{(2\pi i)^{k+1}} \sum_{1\le|n|\le N} \zeta\Big(-k+\frac{2\pi i n}{\log b}\Big)\frac{\exp(2\pi i n\log\alpha/\log b)}{n^{k+1}}$$
$$= -\frac{1}{(\log b)^k}\sum_{0<h<\log_b N} B_{k+1}(\{\alpha b^h\}) + O_{k,b,\alpha}(\log\log N).$$

There is no algebraicity hypothesis in that theorem. After division by $\log N$, changes of the endpoint of the h-sum cost $o(1)$. Conversely, every large integer cutoff H can be reached by taking N near $b^{H+1/2}$. Hence the asserted zeta limits are equivalent to

$$\frac{1}{H}\sum_{h=1}^{H} B_{2r}(\{\alpha b^h\}) \longrightarrow 0 \qquad (r \in R).$$

Necessity follows immediately from normality, or equivalently from uniform distribution of $\{\alpha b^h\}$.

For sufficiency, let μ be any weak subsequential limit of the empirical probability measures

$$\mu_H = \frac{1}{H}\sum_{h=1}^{H}\delta_{\{\alpha b^h\}} \quad \text{on } \mathbb{T}.$$

Such limits exist by compactness. They are invariant under the continuous circle map $T_b(x) = bx \pmod 1$, because the difference between the averages of $f\circ T_b$ and f telescopes to two endpoint terms for every continuous f. Continuity of the even Bernoulli polynomials gives

$$\int B_{2r}(x)\,d\mu(x) = 0 \qquad (r \in R).$$

The absolutely and uniformly convergent Fourier expansion

$$B_{2r}(\{x\}) = \frac{(-1)^{r+1}2(2r)!}{(2\pi)^{2r}}\sum_{n=1}^{\infty}\frac{\cos(2\pi n x)}{n^{2r}}$$

therefore yields

$$\sum_{n=1}^{\infty}\frac{a_n}{n^{2r}} = 0, \qquad a_n = \int \cos(2\pi n x)\,d\mu(x), \quad |a_n| \le 1.$$

If some a_n were nonzero, choose the least such index m. Multiplying by m^{2r} and letting $r \to \infty$ through R would give $a_m = 0$: the remaining terms tend to zero by dominated convergence, being dominated for $r \ge 1$ by the summable sequence $(m/n)^2$, $n > m$. Thus every a_n vanishes.

Let $\check{\mu}$ be the image of μ under reflection $x \mapsto -x$, and let λ be Lebesgue probability measure on $\mathbb{T}$. The vanishing cosine coefficients and Fourier uniqueness imply

$$\mu + \check{\mu} = 2\lambda.$$

Both measures on the left are positive, so $\mu \le 2\lambda$; in particular μ has an L^1 density. The T_b-invariance gives, for every nonzero integer n,

$$\widehat{\mu}(n) = \widehat{\mu}(bn) = \widehat{\mu}(b^j n) \longrightarrow 0$$

by the Riemann–Lebesgue lemma. Thus $\mu = \lambda$. Every subsequential weak limit is therefore λ, proving uniform distribution and hence normality. □

Corollary 37.2. *For any real $a > 1$ and $\alpha > 0$, the sequence $\{\alpha a^h\}$ is uniformly distributed modulo one if and only if $Z_{k,a,\alpha}(N) \to 0$ for every integer $k \ge 1$, with the same definition using a in place of b. No Diophantine hypothesis on a or α is necessary.*

Proof. Use Kanado–Saito Theorem 2.2 as above. The polynomials $1, B_m$ for $m \geq 2$ span exactly the polynomials p with $p(0) = p(1)$. These are uniformly dense in continuous functions on the circle: approximate a continuous periodic function on $[0, 1]$ by a polynomial p, then subtract $x(p(1) - p(0))$ to match its endpoint values. Thus vanishing of all the Bernoulli averages with degree at least two is equivalent to uniform distribution. □

The sparse-odd reduction uses the integer-base circle invariance and is not asserted for noninteger a. These criteria do not establish the required zeta cancellation for $\sqrt{2}$, $(3/2)^h$, Mills' constant, or a chosen prime-representing constant. A full Hausdorff dimension statement for a set of such constants likewise does not by itself imply that it contains a normal number.

38 Realization of asymptotic and zeta dimension profiles

Question 1.4 of Saito's *New fractal dimensions and some applications to arithmetic patches*, arXiv:1802.08839v4 [34], p. 3, asks whether there are relations between asymptotic dimension and zeta dimensions. Appendix B of that paper already disproves either universal ordering between these invariants. We establish the following precise strengthening. It characterizes every possible finite numerical profile on integer lattices; it does not preclude relations under additional geometric or algebraic hypotheses.

Theorem 38.1 (Realization of all lattice dimension profiles)**.** *For every integer $d \geq 1$, every $m \in \{0, \dots, d\}$ and every pair $0 \leq a \leq b \leq d$, there is an infinite set $X \subseteq \mathbb{Z}^d$ such that*

$$\operatorname{asdim} X = m, \qquad \underline{\dim}_\zeta X = a, \qquad \overline{\dim}_\zeta X = b.$$

Consequently, these are exactly the possible triples on infinite subsets of $\mathbb{Z}^d$.

We use the Euclidean metric, or equivalently the sup metric, throughout this section. For $Y \subseteq \mathbb{Z}^d$, the zeta dimensions are its lower and upper counting exponents:

$$\underline{\dim}_\zeta Y = \liminf_{R\to\infty} \frac{\log \#(Y \cap [-R, R]^d)}{\log R}, \qquad \overline{\dim}_\zeta Y = \limsup_{R\to\infty} \frac{\log \#(Y \cap [-R, R]^d)}{\log R}.$$

Indeed, at every fixed covering scale a set of bounded diameter contains a bounded number of lattice points. The resulting covering and counting functions differ by a multiplicative constant depending only on that scale and d.

Lemma 38.2 (Sparse sequences with prescribed counting exponents)**.** *For $0 \leq \alpha \leq \beta \leq 1$, there is a strictly increasing sequence of positive integers (x_n) such that*

$$x_{n+1} - x_n \longrightarrow \infty, \qquad \liminf_{n\to\infty} \frac{\log n}{\log x_n} = \alpha, \qquad \limsup_{n\to\infty} \frac{\log n}{\log x_n} = \beta, \qquad \frac{\log x_{n+1}}{\log x_n} \longrightarrow 1.$$

Its range S has asymptotic dimension zero and lower and upper zeta dimensions α, β.

Proof. If $\alpha = \beta = t > 0$, take

$$x_n = \left\lceil n^{1/t} (\log n + 1)^2 \right\rceil.$$

If $\alpha = \beta = 0$, take

$$x_n = \left\lceil \exp\big((\log n)^2 + \log n\big) \right\rceil.$$

Both constructions have the stated properties after an irrelevant finite initial adjustment.

Suppose now that $\alpha < \beta$. For sufficiently large s, construct a nonnegative locally absolutely continuous function g satisfying $0 \leq g'(s) \leq s$ almost everywhere. Put

$$\phi(s) = s + 2\log(s + 1) + g(s), \qquad q(s) = \frac{s}{\phi(s)}.$$

We alternate phases on which $g' = s$ and $g' = 0$. Globally $g(s) \le C + s^2/2$. During a phase of the first type,

$$\phi(s) - s\phi'(s) = 2\log(s+1) - \frac{2s}{s+1} + g(s) - s^2 < 0$$

for all sufficiently large s. Thus q decreases, and tends to zero if that phase is continued indefinitely. During a phase of the second type the same numerator, with the term $-s^2$ omitted, is positive. Hence q increases and tends to one if that phase is continued indefinitely.

Choose an integer j_0 large enough that

$$\alpha_j = \max(\alpha, 1/j) < \beta_j = \min(\beta, 1 - 1/j) \quad (j \ge j_0).$$

Start with $g = 0$ at a sufficiently large value of s, so that $q > \beta_{j_0}$. Run a decreasing phase until $q = \alpha_j$, then an increasing phase until $q = \beta_j$, and repeat. Each stopping point is finite. The switching times tend to infinity: otherwise q, which is Lipschitz on each fixed compact interval, would undergo infinitely many oscillations of amplitudes bounded below by a positive constant in a finite interval. It follows that

$$\liminf_{s\to\infty} q(s) = \alpha, \qquad \limsup_{s\to\infty} q(s) = \beta.$$

Set $x_n = \lceil \exp(\phi(\log n)) \rceil$ for all sufficiently large n. At almost every sufficiently large real u,

$$\frac{d}{du}\exp(\phi(\log u)) = \frac{\exp(\phi(\log u))\phi'(\log u)}{u} \ge (\log u + 1)^2.$$

Integration on $[n, n+1]$ proves that the integer gaps tend to infinity. Moreover, $\phi'(s) \le s + 3$ almost everywhere, whence

$$\phi(\log(n+1)) - \phi(\log n) = O\left(\frac{\log n}{n}\right) = o(1).$$

Rounding changes the logarithms by $o(1)$. These estimates prove the adjacent-log ratio assertion. They also show that sampling $q(s)$ at $s = \log n$ preserves its lower and upper limits.

For $C_S(R) = \#(S \cap [1, R])$ and $x_n \le R < x_{n+1}$, the count equals $n + O(1)$. Bounding its logarithmic exponent between the values with denominators $\log x_n$ and $\log x_{n+1}$ proves the asserted counting exponents. Finally, at each fixed scale r, the r-chain components of S are singletons outside a finite initial subset. Their diameters are therefore uniformly bounded. Those components give a uniformly bounded, r-disjoint cover, proving $\operatorname{asdim} S = 0$. □

Proof of the profile theorem. Apply the lemma with $\alpha = a/d$ and $\beta = b/d$, and let $B = S^d \subseteq \mathbb{Z}^d$. In the sup metric,

$$\#(B \cap [0, R]^d) = C_S(R)^d,$$

so the two zeta dimensions of B are a, b. Products of the uniformly bounded r-chain-component partitions of S form a uniformly bounded r-disjoint cover of B. Thus $\operatorname{asdim} B = 0$.

For $m \ge 1$, define

$$A_m = \bigcup_{n\ge 1} \left(2^{2^n} e_1 + \{0, \dots, n\}^m \times \{0\}^{d-m}\right).$$

Both zeta dimensions of A_m are zero, since its counting function is

$$O((1 + \log\log R)^{m+1}).$$

By inclusion in an m-coordinate lattice, $\operatorname{asdim} A_m \le m$. Conversely, A_m contains isometric copies of arbitrarily large unit-mesh m-dimensional cubes. If its asymptotic dimension were

at most $m-1$, then, for every fixed $r>0$, these cubes would admit covers by m families of r-disjoint sets of a common diameter bound $D(r)$.

Translate and restrict to centered subcubes, and pass to a diagonal limit. To make this compactness step explicit, assign each lattice point a color in $\{1,\dots,m\}$ and an equivalence class indicating its covering set. On each fixed finite cube there are finitely many such color-and-equivalence patterns. In the diagonal limit, equivalence, diameter at most $D(r)$, and separation at least r of distinct classes of the same color all persist, because each is a finite constraint. The limit provides m families of uniformly bounded r-disjoint sets covering $\mathbb{Z}^m$. This contradicts $\operatorname{asdim}\mathbb{Z}^m=m$. Consequently $\operatorname{asdim}A_m=m$. For $m=0$, use $A_0=\{0\}$.

Finally let $X=A_m\cup B$. Monotonicity and the finite-union theorem for asymptotic dimension give $\operatorname{asdim}X=m$. Writing $C_Y(R)=\#(Y\cap[-R,R]^d)$, we have

$$C_B(R)\le C_X(R)\le C_B(R)+C_{A_m}(R).$$

Since $C_{A_m}(R)=R^{o(1)}$, adding it does not change either nonnegative logarithmic counting exponent of B. Hence the zeta dimensions of X are a,b.

Every nonempty subset of $\mathbb{Z}^d$ has integer asymptotic dimension between zero and d, and its lower and upper zeta dimensions satisfy $0\le a\le b\le d$. The construction attains every triple allowed by these intrinsic constraints. □

The asymptotic-dimension facts used above are standard; see Bell–Dranishnikov, *Asymptotic dimension in Będlewo*, arXiv:math/0507570 [3], Proposition 5, Proposition 6 and Corollary 9. The realization theorem supplies a complete answer to the numerical-profile interpretation of Saito's question on lattice subsets. The formulation concerns arbitrary lattice subsets and does not rule out relations on more restricted classes such as homogeneous spaces or finitely generated groups.

38.1 An existing-theorem consequence for avoidance dimensions

Question 6.1 of Saito's *New bounds for dimensions of a set uniformly avoiding multi-dimensional arithmetic progressions*, arXiv:1910.13071v1 [31], p. 14, asks for a specified upper bound on $D_S(k,\varepsilon)$. Its $k=3$ case follows from existing results, as follows. Write $T=\log(1/\varepsilon)$ and $M_1=\lceil 1/\varepsilon\rceil$. Saito's equation (2.1) gives

$$D_S(3,\varepsilon)\le D_A(3,\varepsilon)\le 1+\frac{\log(r_3(M_1)/M_1)}{2\log M_1}.$$

Kelley–Meka, *Strong Bounds for 3-Progressions*, arXiv:2302.05537v6 [19], Theorems 1.1–1.2, imply $r_3(N)/N\le\exp(-c(\log N)^{1/12})$ for a constant $c>0$ and large N. Therefore

$$D_A(3,\varepsilon)\le 1-c'T^{-11/12}.$$

The deficit requested in Saito's question, with $M=\lceil 1/(8\varepsilon)\rceil$, is

$$\frac{\log(\log M(\log\log M)^2)}{\log(1/\varepsilon)}=O\left(\frac{\log T}{T}\right).$$

Since $T^{1/12}/\log T\to\infty$, the displayed bound is stronger for all sufficiently small ε. This is a deduction from established additive-combinatorial results, not a new proof of Roth's theorem or a resolution for every k.

Part V
Growth constants, endpoint geometry, and finite prime covers

39 Polynomial recursions: radical rigidity and a coefficient obstruction

This section concerns an integer orbit

$$x_{n+1} = P(x_n), \qquad x_n \in \mathbb{Z}, \qquad x_n \longrightarrow +\infty, \tag{39.1}$$

where

$$P(X) = aX^d + bX^{d-1} + \cdots \in \mathbb{Q}[X], \qquad a > 0, \quad d \geq 2.$$

Put

$$r = a^{1/(d-1)}, \qquad A = r^{-1}, \qquad s = \frac{b}{da}, \qquad \alpha = \lim_{n\to\infty} x_n^{d^{-n}}.$$

The known polynomial-iteration asymptotic is

$$\alpha > 1, \qquad x_n = A\alpha^{d^n} - s + O(\alpha^{-d^n}). \tag{39.2}$$

See [44, Section 2] and [22, equation (1.2)]. It follows by centering and scaling the recurrence to $y_{n+1} = y_n^d + O(y_n^{d-2})$, then summing the logarithmic errors.

Status of the contribution. Dubickas already completely classified the algebraic possibilities under $r \in \mathbb{Q}$ in [12]. We do not count that classification as new. The results below are explicit consequences and refinements of the established approximation theorems, formulated for arbitrary positive rational leading coefficient. Their proofs are given here; no claim of bibliographic priority is made.

39.1 The approximation input

We use the following two consequences of the theorem of Kulkarni–Mavraki–Nguyen, *Trans. Amer. Math. Soc.* 371 (2019), 3787–3804. See [20, 10]; both consequences follow from [22, Theorem 1.2]; the second is the fixed-coefficient argument underlying his Theorem 2.2.

Lemma 39.1 (Two established approximation consequences)**.** *1. Let $B > 1$ be rational and $c \neq 0$ algebraic. If $\|cB^m\| < \theta^m$ for infinitely many positive integers m, where $0 < \theta < 1$ is fixed, then $B \in \mathbb{Z}$ and $c \in \mathbb{Q}$.*

2. Let $\lambda > 1$ be algebraic, let $q \neq 0$ be an algebraic integer, and let $\beta \in \mathbb{Q} \setminus \mathbb{Z}$. For every $0 < \theta < 1$, only finitely many positive integers m satisfy $\|q\lambda^m + \beta\| < \theta^m$.

Here $\|t\|$ denotes distance to the nearest integer.

Derivation from the stated approximation theorem. For (1), apply Kumar's Theorem 1.2 to its singleton tuple $\delta_1 = B$, with fixed coefficient c. Part (i) makes B an algebraic integer, hence an integer. Part (iv), because every Galois automorphism fixes B, gives $\sigma(cB^m) = cB^m$ for every σ at all sufficiently large successful indices. Consequently c is rational. Constant coefficient height satisfies the required sublinear height condition, for example with $f(m) = \sqrt{m} + 1 + h(c)$.

For (2), we spell out the argument from Theorem 1.2, so that only the stated algebraic-integrality hypothesis on q is needed. To recall why integrality of q is decisive, apply Theorem

1.2 to the nondegenerate pair $(\lambda, 1)$ with coefficients (q, β). Part (i) makes λ an algebraic integer. Part (iii) says that the sum of the distinct conjugates of $q\lambda^m$, together with β, is an integer at all sufficiently large successful indices. That conjugate sum is itself an integer because $q\lambda^m$ is an algebraic integer. Thus β would be integral, a contradiction. There is no collision between β and a conjugate of $q\lambda^m$ at large indices: an algebraic number with a rational conjugate is rational, and would then equal β, whereas $q\lambda^m$ grows in absolute value. □

39.2 Exact classification of rational-power growth constants

Theorem 39.2 (Radical rigidity)**.** *For an orbit* (39.1)*, the following are equivalent:*

1. $\alpha^h \in \mathbb{Q}$ *for some positive integer* h*;*
2. $$P(X) = a(X + s)^d - s, \qquad s = \frac{b}{da}. \tag{39.3}$$

In this case

$$\alpha = a^{1/(d-1)}|x_0 + s|, \qquad \alpha^{d-1} = a|x_0 + s|^{d-1} \in \mathbb{Q}. \tag{39.4}$$

Furthermore, α *is rational if and only if both* (39.3) *and* $a^{1/(d-1)} \in \mathbb{Q}$ *hold. Whenever* α *is rational it is an integer.*

Proof. Suppose $B = \alpha^h \in \mathbb{Q}$. The sequence d^n modulo h is eventually periodic. Therefore some integers $n_0 \geq 0$, $\ell \geq 1$ and $0 \leq e < h$ satisfy

$$d^{n_0+\ell j} = e + hm_j \qquad (j \geq 0),$$

where m_j are eventually positive and tend to infinity. Choose a positive integer D_0 such that $D_0 s \in \mathbb{Z}$. Equation (39.2) gives

$$\|D_0 A\alpha^e B^{m_j}\| = O(B^{-m_j}).$$

Choose $B^{-1} < \theta < 1$ and discard finitely many indices. Lemma 39.1(1) shows that B is an integer and $A\alpha^e$ is rational. Hence the numbers $A\alpha^e B^{m_j} - s$ have bounded denominators. Their distances from the integers $x_{n_0+\ell j}$ tend to zero, so discreteness gives

$$x_{n_0+\ell j} = A\alpha^{d^{n_0+\ell j}} - s \qquad \text{for all sufficiently large } j. \tag{39.5}$$

Set $D = d^\ell$ and $R = P^{\circ\ell}$. Comparing two consecutive terms of (39.5) gives

$$R(x_{n_0+\ell j}) = A^{1-D}(x_{n_0+\ell j} + s)^D - s.$$

These are infinitely many distinct arguments, since the orbit escapes. Polynomial identity therefore gives

$$R(X) = A^{1-D}(X + s)^D - s.$$

Normalize the original polynomial by the affine map $L(X) = r(X + s)$: $Q = L \circ P \circ L^{-1}$. Then Q is monic of degree d and $Q^{\circ\ell}(Y) = Y^D$. A polynomial commutes with its iterates, so

$$Q(Y^D) = Q(Y)^D. \tag{39.6}$$

If $Q \neq Y^d$, let cY^t be its highest nonzero term below Y^d. The highest nonleading term on the right of (39.6) has degree $dD - d + t$ and coefficient Dc, whereas on the left it has degree tD. Since $dD - d + t > tD$, this is impossible. Thus $Q = Y^d$, which is exactly (39.3).

Conversely, under (39.3), put $y_n = r(x_n + s)$. Then $y_{n+1} = y_n^d$ and $y_n = y_0^{d^n}$. Positive escape forces $|y_0| > 1$, and forces $y_0 > 0$ if d is odd. Taking the defining limit yields (39.4). Since $x_0 + s \in \mathbb{Q} \setminus \{0\}$, this also proves the criterion for rationality.

Finally, if $\alpha \in \mathbb{Q}$, apply Lemma 39.1(1) directly to $\|D_0 A\alpha^{d^n}\| = O(\alpha^{-d^n})$. It follows that α is an integer. □

Corollary 39.3 (An irrationality test without root normalization)**.** *If $a^{1/(d-1)} \notin \mathbb{Q}$, the growth constant of every escaping integer orbit of P is irrational. If P is not of the form (39.3), then no positive integer power of its growth constant is rational.*

The second statement excludes all rational radicals, but does not by itself prove transcendence: a quadratic Pisot unit associated with a Chebyshev recurrence is a familiar obstruction to such an inference.

39.3 An explicit coefficient test for transcendence

Write the positive leading coefficient as $a = u/v$ in lowest terms, where $u, v \in \mathbb{Z}_{>0}$, and define

$$M_d(a) = \prod_{p \mid u} p^{\lceil v_p(u)/(d-1) \rceil}. \tag{39.7}$$

The empty product is 1.

Theorem 39.4 (Denominator obstruction)**.** *If the growth constant α of (39.1) is algebraic, then*

$$M_d(a)\frac{b}{da} \in \mathbb{Z}. \tag{39.8}$$

Consequently, failure of (39.8) proves that α is transcendental. For $P \in \mathbb{Z}[X]$, the condition is

$$da \mid M_d(a)b. \tag{39.9}$$

In particular, $d \nmid b$ is sufficient for transcendence in this integer-coefficient case, although (39.9) can be strictly stronger.

Proof. Put $M = M_d(a)$. The number $q = MA$ is an algebraic integer, because

$$q^{d-1} = \frac{M^{d-1}v}{u} \in \mathbb{Z}.$$

Multiplying (39.2) by M gives

$$Mx_n = q\alpha^{d^n} - Ms + O(\alpha^{-d^n}).$$

If α is algebraic and Ms is a nonintegral rational number, Lemma 39.1(2), with $\beta = -Ms$ and exponents d^n, contradicts this approximation. This proves (39.8).

If $a \in \mathbb{Z}_{>0}$, then $M_d(a) \mid a$. Condition (39.9) therefore implies $d(a/M_d(a)) \mid b$, in particular $d \mid b$. □

Optimality within this scalar-integrality argument. For a positive integer T, the coefficient TA is an algebraic integer if and only if

$$(TA)^{d-1} = T^{d-1}v/u \in \mathbb{Z},$$

equivalently $M_d(a) \mid T$. Necessity uses that a rational algebraic integer is integral, and sufficiency uses the displayed monic equation. Thus (39.8) is the strongest condition obtainable by this method of multiplying the asymptotic by an integer and applying Lemma 39.1(2). This is an optimality statement about this particular argument, not about all possible transcendence criteria.

Corollary 39.5 (A family outside rational leading normalization)**.** *Let q be a positive integer that is not a square and let*

$$x_0 \geq 1, \qquad x_{n+1} = q^3 x_n^3 + 3x_n^2 + 1.$$

Then $\lim x_n^{3^{-n}}$ is transcendental. More generally, if $a \geq 1$ is an integer, $a/M_3(a) > 1$, and $b \in \mathbb{Z}$ is not divisible by $3a/M_3(a)$, every escaping integer orbit of $aX^3 + bX^2 + cX + e$ has transcendental growth constant.

Proof. For the displayed family, choose a prime p with $v_p(q)$ odd. Then

$$v_p(q^3/M_3(q^3)) = 3v_p(q) - \lceil 3v_p(q)/2 \rceil = \lfloor 3v_p(q)/2 \rfloor \geq 1.$$

In fact $q^3/M_3(q^3) > 1$, so $3q^3/M_3(q^3)$ does not divide the coefficient 3. The denominator obstruction applies. The integer orbit escapes because its recurrence is strictly larger than x_n for every $x_n \geq 1$. Its leading normalization $(q^3)^{1/2}$ is irrational, so Dubickas's rational-normalization hypothesis does not cover this example. The more general assertion is (39.9). □

For instance, the recurrence

$$x_0 = 1, \qquad x_{n+1} = 8x_n^3 + 3x_n^2 + 1$$

starts with $1, 12, 14257, \ldots$ and has a transcendental growth constant. Here $M_3(8) = 4$ and $M_3(8)b/(3 \cdot 8) = 1/2$; the weaker condition $3 \nmid b$ would miss it.

39.4 Periodic polynomial schedules

Corollary 39.6 (A periodic transfer)**.** *Suppose $P_0, \ldots, P_{t-1} \in \mathbb{Q}[X]$ have degrees $d_i \geq 2$ and positive leading coefficients. Let an integer sequence satisfy*

$$x_{n+1} = P_{n \bmod t}(x_n), \qquad x_n \longrightarrow +\infty.$$

Put $F = P_{t-1} \circ \cdots \circ P_0$, $D = \prod_i d_i$, and $\gamma = \lim_{n\to\infty} x_{tn}^{D^{-n}}$. Then Theorems 39.2 and 39.4 apply to γ with polynomial F. In particular, if the top two coefficients $F(X) = a_F X^D + b_F X^{D-1} + \cdots$ fail $M_D(a_F)b_F/(Da_F) \in \mathbb{Z}$, then γ is transcendental.

Proof. The subsequence $z_n = x_{tn}$ is an escaping integer orbit of the single polynomial F. Apply the preceding results. □

For an explicit alternating example, take $P_0(X) = 2X^2 + X$ and $P_1(X) = X^2 + 1$, with $x_0 \geq 1$. Their two-step polynomial is

$$F(X) = 4X^4 + 4X^3 + X^2 + 1.$$

Here $D = 4$, $a_F = 4$, $b_F = 4$, $M_4(4) = 2$, and $M_4(4)b_F/(4a_F) = 1/2$. Hence the two-step growth constant $\gamma = \lim x_{2n}^{4^{-n}}$ is transcendental. This transfer is a direct composition corollary, not a separate deep theorem.

Scope. These results classify rational radicals exactly and certify transcendence in explicit coefficient families. They neither classify all algebraic exceptions for arbitrary leading coefficient nor settle the transcendence of the ordinary Mills constant. A polynomial orbit is not automatically an orbit of consecutive least primes.

40 Exact modular return and persistent failure for nonlinear factorial clocks

Lemma 40.1 (The Type-B input and its local form)**.** *Let (C_n) be positive integers with $C_{n+1}/C_n \geq 2$ and $C_{n+1}/C_n \to \infty$. Suppose that every C_m divides arbitrarily late terms C_n. Then*

$$W(C) = \{A > 1 : \lfloor A^{C_n} \rfloor \text{ is prime for every } n \geq 1\}$$

has a minimum, and every left-gap point of $W(C)$ is transcendental. Here a left-gap point is an $A \in W(C)$ for which some interval $(A - \eta, A)$ is disjoint from $W(C)$.

Proof. Existence follows from Saito's Theorem 1.3 [39]. For the arithmetic assertion, apply the already proved local Type-B Theorem 25.1. The ratio limit is infinite, its index set I is cofinite, and the assumed arbitrarily late divisibility return supplies its condition (B5). A left-gap point is precisely a local left minimum in that theorem's terminology. Its unbounded-ratio conclusion therefore proves transcendence. □

The polynomial-exponential argument in the preceding work established eventual divisibility return, but did not establish universal return for nonlinear polynomial perturbations. Here we settle the corresponding return question exactly for a different class: a modularly vanishing sequence plus an integer polynomial. This yields full, unshifted transcendence examples for which universal return fails at arbitrarily late seeds. The distinction between the two return conditions is therefore structural, rather than merely an early-index obstruction.

We say that an integer sequence $H(n)$ is *modularly vanishing* if, for each positive integer q, one has $q \mid H(n)$ for all sufficiently large n. We call $F \in \mathbb{Z}[X]$ *intersective* when it has an integer root modulo every positive integer. Put

$$C_n = H(n) + Q(n), \qquad Q \in \mathbb{Z}[X],$$

and assume that the C_n are positive and strictly increasing. At a seed m, write $\mathcal{R}_m$ for the universal-return condition

$$\text{for every } L \geq 1 \text{ there is } n > m \text{ with } C_n \equiv C_m \pmod{LC_m}.$$

For schedules with ratios tending to infinity, the index set I in Saito's Type-B condition is cofinite; failures of $\mathcal{R}_m$ at arbitrarily late seeds therefore also refute eventual Type-B condition (B6).

Theorem 40.2 (An exact polynomial-image criterion)**.** *Under the preceding assumptions, the following conditions are equivalent for each fixed seed m:*

1. *$\mathcal{R}_m$ holds.*
2. *For every $q \geq 1$, there are arbitrarily large n such that $C_n \equiv C_m \pmod{q}$.*
3. *The polynomial $Q(X) - C_m$ is intersective.*

For every modulus q, the set of return indices has natural density

$$\frac{\rho_Q(C_m; q)}{q}, \qquad \rho_Q(c; q) = \#\{x \in \mathbb{Z}/q\mathbb{Z} : Q(x) \equiv c \pmod{q}\}.$$

Proof. Suppose (1) holds, and fix q and $N > m$. Choose a multiple L of q so large that

$$LC_m > \max_{m < n \leq N} (C_n - C_m).$$

Strict increase ensures that none of the finitely many indices $m < n \le N$ can return modulo LC_m. Condition (1) supplies an index $n > N$ returning modulo LC_m, hence also modulo q. This proves (2). Condition (2) immediately implies (1), by taking $q = LC_m$.

For a fixed q, modular vanishing gives $C_n \equiv Q(n) \pmod q$ for every sufficiently large n. The polynomial residue sequence $Q(n) \bmod q$ is periodic with period q. Consequently a return occurs at arbitrarily large indices exactly when $Q(X) - C_m$ has a root modulo q. This proves the equivalence with (3), and counting residue classes gives the asserted density. □

Corollary 40.3 (Power perturbations: exact local criterion)**.** *If $Q(X) = X^r$ with $r \ge 1$, then $\mathcal{R}_m$ holds if and only if C_m is a perfect rth power of a positive integer.*

Proof. If $C_m = b^r$, the residue class $n \equiv b \pmod q$ gives a root for every q. Conversely, if C_m is not an rth power, some prime p satisfies $r \nmid v_p(C_m)$. Writing $e = v_p(C_m)$, the congruence $x^r \equiv C_m \pmod{p^{e+1}}$ would force $rv_p(x) = e$, which is impossible. Apply Theorem 40.2. □

For completeness we establish an all-index growth inequality; this avoids silently deleting an initial segment of the following family.

Lemma 40.4 (A uniform factorial–power doubling inequality)**.** *For every pair of integers $n, r \ge 1$,*

$$(n+1)! + (n+1)^r \ge 2(n! + n^r).$$

Proof. It suffices to prove $2n^r - (n+1)^r \le (n-1)n!$. At $n = 1$ this is immediate. For $n = 2, 3, 4$ the left side is nonpositive once $r \ge 2, 3, 4$, respectively; for the remaining exponents its respective maxima are $1, 2, 7$, which are smaller than $(n-1)n!$.

Suppose $n \ge 5$ and the left side is positive. Set $u = \log n$, $t = \log(1 + 1/n)$, and $r_0 = \log 2/t$, so that $x = r_0 - r > 0$. The inequalities $1 - e^{-tx} \le tx$, $xe^{-ux} \le 1/(eu)$, $t < 1/n$, and $r_0 < (n+1)\log 2 < \frac{7}{10}(n+1)$ give

$$2n^r - (n+1)^r = 2n^{r_0}e^{-ux}(1 - e^{-tx}) < \frac{2n^{7(n+1)/10}}{en\log n}.$$

Also $n! \ge (n/e)^n$. The logarithm of the ratio between $(n-1)(n/e)^n$ and the last upper bound is

$$F(n) = \tfrac{3}{10}(n+1)\log n - n + \log(n-1) + \log\log n + 1 - \log 2.$$

This is positive for every real $n \ge 5$. Indeed, $F(5) = \frac{9}{5}\log 5 - 4 + \log 2 + \log\log 5 > 0.04$, and

$$F'(x) = \tfrac{3}{10}\log x - \tfrac{7}{10} + \frac{3}{10x} + \frac{1}{x-1} + \frac{1}{x\log x} > 0 \quad (x \ge 5).$$

For the last inequality one may split into $[5, 8]$, $[8, 11]$, and $[11, \infty)$: on the first interval use $\log 5 > 1.6$ and $8\log 8 < 17$; on the second use $\frac{3}{10}\log 8 > 0.62$ and $1/(x-1) \ge 0.1$; on the third the first two terms are already positive. Thus $(n-1)n!$ exceeds the required upper bound. □

Theorem 40.5 (Full transcendence despite infinitely many return failures)**.** *Let $s, r \ge 1$ be integers and put*

$$C_n = (n!)^s + n^r \qquad (n \ge 1).$$

The least prime-representing constant for the full schedule exists and is transcendental. More generally, every left-gap point of its prime-representing set is transcendental.

If $r > s$, universal return fails at every prime seed $m = p$. More precisely,

$$C_n \not\equiv C_p \pmod{p^{s+1}} \qquad (n > p),$$

so Saito's condition (B6) fails with $m = p$ and $L = p$. These failures occur arbitrarily late, while the divisibility condition (B5) holds at every seed.

Proof. The preceding lemma gives $C_{n+1} \geq 2C_n$ also for general s, because

$$((n+1)^s - 2)(n!)^s \geq (n-1)n!$$

for $n \geq 2$ and $s \geq 1$; the case $n = 1$ is direct. Moreover $C_{n+1}/C_n \sim (n+1)^s \to \infty$. For any seed m, take an arbitrarily large multiple n of C_m. Then $C_m \mid n^r$ and, once $n \geq C_m$, also $C_m \mid n!$. Thus $C_m \mid C_n$, proving (B5). The other hypotheses of Lemma 40.1 follow from positivity, integrality, the doubling inequality, and the unbounded ratios. That lemma proves existence and both transcendence assertions.

Now suppose $r > s$ and let p be prime. Since $v_p(p!) = 1$,

$$v_p(C_p) = v_p\big((p!)^s + p^r\big) = s.$$

For $n > p$ with $p \nmid n$, the factorial term is divisible by p whereas n^r is not, so $v_p(C_n) = 0$. If instead $p \mid n$, then $n \geq 2p$ and $v_p(n!) \geq 2$. Both terms defining C_n are then divisible by $p^{\min(2s,r)}$, and $\min(2s, r) > s$. Neither case can give $C_n \equiv C_p \pmod{p^{s+1}}$. Finally $p^{s+1} \mid pC_p$, so return modulo pC_p is impossible as well. Infinitely many prime seeds prove the last assertion. □

Remark 40.6 (Scope and novelty). The transcendence conclusion is an explicit full-schedule consequence of Saito's existing Type-B criterion; it is not a new general transcendence method. The additional results are the exact polynomial-image return criterion and the persistent separation between (B5) and (B6), including an infinite family with completely explicit obstructing seeds and moduli. In particular $C_n = n! + n^2$ has a transcendental minimum although (B6) fails at every prime seed. These arguments do not settle eventual universal return for $P(a^n) + Q(n)$ in general, nor do they prove full-schedule transcendence for $3^n + n^2$.

41 Ceiling prime constants: endpoints and right extrema

For an integer $c \geq 3$, put

$$\mathcal{V}_c = \{B > 1 : \lceil B^{c^k} \rceil \text{ is prime for every } k \geq 1\}.$$

Tóth proves that this set is nonempty and constructs constants by descending from a prime p to a prime below p^c; see *A Variation on Mills-Like Prime-Representing Functions*, J. Integer Sequences **20** (2017), Article 17.9.8 [43]. The following results address an endpoint distinction that prevents automatic transfer of results about the smallest floor constant. The arithmetic inputs used below are the prime-gap theorem with exponent $\theta = 21/40$, the Corvaja–Zannier–Dubickas approximation criterion, and the Pisot conjugate-sum bound, in the precise forms recalled in Saito, *Mills' constant is irrational*, arXiv:2404.19461v2, Theorems 2.1 and 2.6 and Lemma 2.7 [36]. These are deductions from existing machinery; no literature priority claim is made for the ceiling formulation.

41.1 The infimum need not be a prime-representing constant

Proposition 41.1 (Endpoint alternatives). *Let $L = \inf \mathcal{V}_c$. Then the following statements hold.*

1. *If $L > 1$ and L^{c^k} is nonintegral for every $k \geq 1$, then $L \in \mathcal{V}_c$.*
2. *If $L > 1$ and $L \notin \mathcal{V}_c$, there are $k \geq 1$ and an even integer $t \geq 2$ such that $L^{c^k} = t$ and*

$$t^{c^j} + 1 \text{ is prime for every } j \geq 0.$$

In particular, this alternative is impossible unless c is a power of 2.

3. If $\mathcal{V}_c \cap (1,2] \neq \varnothing$, then $L = 1$ and $\mathcal{V}_c$ has no smallest element.

Proof. Choose $B_i \in \mathcal{V}_c$ tending to L from above (or equal to L when it already belongs to the set). At a noninteger value L^{c^k}, the ceiling is locally constant, proving the first assertion. An element $B \in \mathcal{V}_c$ cannot have an integer power B^{c^k}: that integer would be prime, whereas its cth power at the next index would be composite. Consequently nonattainment at $L > 1$ requires an integer boundary $t = L^{c^k}$. Since $B_i > L$, right limits of ceilings give $t^{c^j} + 1$ prime for every fixed $j \geq 0$. Already $t + 1$ prime forces t even. Write $c = 2^a u$ with u odd. If $u > 1$, then $(t^{2^a})^u + 1$ has the proper factor $t^{2^a} + 1$, contradicting the condition at $j = 1$. This proves the second assertion.

For the last assertion, take $1 < B \leq 2$ in $\mathcal{V}_c$. Then $B^{1/c} \in \mathcal{V}_c$: its first output is $\lceil B \rceil = 2$, and all subsequent outputs are those of B shifted by one index. Iteration produces $B^{c^{-r}} \downarrow 1$. The number 1 itself is excluded and does not have prime outputs. □

Corollary 41.2 (Unconditional absence of a minimum for infinitely many bases)**.** *For every sufficiently large integer m, the nonempty set $\mathcal{V}_{3^m}$ has infimum 1 and no smallest element.*

Proof. Choose any $B \in \mathcal{V}_3$, whose existence is Tóth's theorem. For each $m \geq 1$, put $A_m = B^{3^{1-m}}$. Then

$$A_m^{(3^m)^k} = B^{3^{m(k-1)+1}}$$

has prime ceiling for every $k \geq 1$. For sufficiently large m, $1 < A_m \leq 2$, so Proposition 41.1 applies. □

Remark 41.3. The criterion involving $t^{c^j} + 1$ is necessary, not sufficient, for a missing endpoint; no assertion about an infinite family of generalized Fermat primes is used. Similarly, the displayed numerical branch beginning with $2,7,337$ in Tóth's paper is not being used here to certify unconditionally that $\mathcal{V}_3 \cap (1,2]$ is nonempty. That would require an infinite extension argument for that particular finite prefix.

41.2 Rightmost completions and their arithmetic

A point $B \in \mathcal{V}_c$ is a *local right maximum* if $(B, B+\varepsilon) \cap \mathcal{V}_c = \varnothing$ for some $\varepsilon > 0$.

Lemma 41.4 (Attainment and local extension)**.** *Every nonempty bounded subset of $\mathcal{V}_c$ has its supremum in $\mathcal{V}_c$. Every feasible finite prime prefix has a greatest completion, which is a local right maximum. For every sufficiently large prime p, the one-term prefix $\lceil B^c \rceil = p$ is feasible.*

Proof. If $S \subset \mathcal{V}_c$ is nonempty and bounded, choose elements increasing to $b = \sup S > 1$. Left continuity of the ceiling gives $\lceil b^{c^k} \rceil$ prime for each fixed k. Thus $b \in \mathcal{V}_c$. The same proof preserves any prescribed finite prefix. Its completion set is bounded, and its greatest member has no integer power by the argument in Proposition 41.1. All finitely many prefix conditions therefore remain valid in a neighborhood of that member, proving local right maximality.

The prime-gap theorem implies, with $\theta = 21/40$, that the largest prime $q < p^c$ satisfies

$$p^c - K p^{c\theta} \leq q < p^c$$

for all sufficiently large primes p, with a fixed constant K. Since $c\theta < c - 1$ for $c \geq 3$, this gives $(p-1)^c + 1 < q < p^c$, and also $q > p$. Iteration yields primes p_n with

$$(p_n - 1)^c + 1 < p_{n+1} < p_n^c.$$

The intervals $((p_n - 1)^{1/c^n}, p_n^{1/c^n}]$ are nested, their left endpoints strictly increase, and their lengths tend to zero. Their unique limiting point lies strictly above every left endpoint and at or below every right endpoint, and therefore has the required prime ceilings. □

Corollary 41.5 (No global minimum for all sufficiently large bases)**.** *There is an integer c_0 such that $\inf \mathcal{V}_c = 1$ and $\mathcal{V}_c$ has no smallest element for every integer $c \geq c_0$.*

Proof. The large-prime threshold in the extension part of Lemma 41.4 can be chosen uniformly for all integer $c \geq 3$. Indeed the backward prime-gap bound has constants independent of c, and

$$p^c - (p-1)^c \geq p^{c-1}, \qquad \frac{p^{c-1}}{p^{c\theta}} = p^{c(1-\theta)-1} \geq p^{17/40}.$$

Thus one fixed sufficiently large prime P can serve as the first seed for every $c \geq 3$. The resulting constant satisfies $1 < B \leq P^{1/c}$. For $c \geq \max\{3, \lceil \log_2 P \rceil\}$ this lies in $(1, 2]$, and Proposition 41.1 applies. □

Theorem 41.6 (Arithmetic of ceiling right maxima)**.** *Let $c \geq 3$ be an integer and let B be a local right maximum of $\mathcal{V}_c$. Then B is irrational. More precisely:*

1. *If $c \geq 5$, then B is transcendental.*
2. *If $c = 4$, then either B is transcendental or B^{4^m} is a quadratic Pisot number for some $m \geq 1$.*
3. *If $c = 3$, then either B is transcendental or B^{3^m} is a cubic Pisot number for some $m \geq 1$.*

In particular, these conclusions apply to the greatest completion of every feasible finite prefix, and such completions exist unconditionally for every sufficiently large first prime.

Proof. Put $p_k = \lceil B^{c^k} \rceil$. We first prove that eventually

$$p_{k+1} = \max\{q < p_k^c : q \text{ prime}\}. \tag{41.1}$$

Indeed, let q denote the maximum on the right. Necessarily $p_{k+1} \leq q$, since p_k^c is composite. If the inequality is strict, start the extension in Lemma 41.4 with q at index $k+1$. Its resulting constant D satisfies

$$D > (q-1)^{1/c^{k+1}} \geq p_{k+1}^{1/c^{k+1}} > B,$$

while its level-k power lies in $(p_k - 1, p_k)$. This preserves all earlier prime outputs: the boundaries of a level-j ceiling cell, raised to the integer power c^{k-j}, are integers, so no earlier boundary lies in the interior of the unit interval $(p_k - 1, p_k)$. Both D and B lie in the level-k cell, whose width tends to zero. For arbitrarily large k such a $D > B$ would contradict local right maximality. This proves (41.1).

Set

$$\delta = c(1-\theta) - 1 = \frac{19c}{40} - 1 > 0.$$

The prime-gap estimate and $B^{c^{k+1}} > p_{k+1} - 1$ imply

$$0 < p_k - B^{c^k} \ll p_k^{-\delta} \ll B^{-\delta c^k}. \tag{41.2}$$

The estimate follows directly by applying the mean value theorem to $(p_k^c - K p_k^{c\theta} - 1)^{1/c}$; all quantities are positive for sufficiently large k.

Suppose that B is algebraic. The approximation criterion of Corvaja–Zannier and Dubickas, in Saito's Theorem 2.6 with $q = 1$ and exponents c^k, implies that $\beta = B^{c^m}$ is a Pisot number for some $m \geq 1$. Its degree ℓ is at least two, since an integer power of B is impossible. Write its other conjugates as $\beta_2, \ldots, \beta_\ell$ and put $\rho = \max_{i \geq 2} |\beta_i| < 1$. For $n = c^{k-m}$, the integer trace is eventually the unique nearest integer to β^n, hence equals p_k. Therefore

$$0 < \sum_{i=2}^{\ell} \beta_i^n = p_k - \beta^n \ll \beta^{-\delta n}.$$

The Pisot conjugate-sum lower bound gives, for some $\lambda > 0$, $\rho^n n^{-\lambda} \ll \beta^{-\delta n}$ along these exponents. Taking nth roots yields $\rho \le \beta^{-\delta}$. Since the nonzero integer norm has absolute value at least one,

$$1 \le |\operatorname{Norm}(\beta)| \le \beta\rho^{\ell-1} \le \beta^{1-\delta(\ell-1)}, \qquad \ell \le 1 + \frac{1}{\delta}.$$

For $c \ge 5$ this upper bound is strictly less than two, a contradiction. For $c = 4$ it allows only $\ell = 2$, and for $c = 3$ it allows only $\ell = 2$ or 3.

It remains to eliminate $\ell = 2$ when $c = 3$. If the conjugate of β is γ and $N = \beta\gamma \in \mathbb{Z}$, then for $n = 3^{k-m}$ and all sufficiently large k,

$$p_{k+1} = \beta^{3n} + \gamma^{3n} = p_k(p_k^2 - 3N^n).$$

This contradicts primality because $p_{k+1} > p_k > 1$. Finally, a rational B would have rational Pisot power β, which must be an integer. Thus all cases are irrational. □

Corollary 41.7 (Dense canonical transcendental completions)**.** *For every integer $c \ge 5$, the greatest completions of feasible finite prime prefixes form a countable dense subset of $\mathcal{V}_c$ consisting entirely of transcendental numbers.*

Proof. There are countably many finite prime prefixes. Given $B \in \mathcal{V}_c$, let D_n be the greatest completion of its first n outputs. Both B and D_n lie in the same level-n cell, whose width is at most c^{-n}, so $D_n \to B$. Theorem 41.6 supplies transcendence. □

Remark 41.8 (The sign cannot be ignored)*.* For a ceiling right maximum the Pisot conjugate sum is positive. For a floor left minimum it is negative. This is why the quadratic possibility at $c = 4$ is not eliminated by the usual even-power sign argument. The theorem does not assert that every ceiling constant is transcendental, does not assert the existence of a global least ceiling constant, and does not prove the transcendence of Tóth's displayed cubic numerical branch.

42 Sharp finite prime covers for algebraic power floors

We classify the fixed finite sets of primes that can cover all power floors of a suitable power of a Salem number. This sharpens existence questions about composite power floors studied in [13, 35, 16], and we contrast the result with Pisot units. These are classifications of *finite prime covers*; they do not classify every possible mechanism of eventual compositeness. No claim of bibliographic priority is made.

For a finite nonempty set of primes S, put $M_S = \prod_{p \in S} p$. A sequence of positive integers has an S-cover if every term has a proper prime divisor belonging to S.

42.1 The distribution input

Lemma 42.1 (Independent Salem angles)**.** *Let τ be a Salem number of degree $d = 2s + 2 \ge 4$, with conjugates $\tau, \tau^{-1}, z_1, z_1^{-1}, \dots, z_s, z_s^{-1}$, where $z_j = e^{2\pi i\theta_j}$ and $|z_j| = 1$. Then $1, \theta_1, \dots, \theta_s$ are linearly independent over $\mathbb{Q}$. Consequently, for all integers $a \ge 1$ and b, the vectors $((an + b)\theta_1, \dots, (an + b)\theta_s)$ are equidistributed modulo one.*

Proof. More strongly, suppose $\prod_j z_j^{m_j} = \omega$ for integers m_j and a root of unity ω. A rational linear relation among $1, \theta_1, \dots, \theta_s$ is a special case of such a relation. For each fixed j, choose an automorphism of the splitting field that sends z_j to τ. No other chosen z_k can map to either τ or τ^{-1}: the latter possibility would imply $z_k = z_j^{-1}$, which is not among the chosen representatives. Thus all the other images have modulus one, and the image of ω also has modulus one. Taking

absolute values gives $\tau^{m_j} = 1$, hence $m_j = 0$. The asserted equidistribution now follows from the Weyl criterion: every nontrivial integral character has irrational frequency, and its exponential sum is a bounded geometric sum divided by its length. □

Write

$$T_n = \mathrm{Tr}(\tau^n), \qquad E_n = \tau^{-n} + 2\sum_{j=1}^{s} \cos(2\pi n\theta_j).$$

The exact floor identity is

$$\lfloor \tau^n \rfloor = T_n - \lceil E_n \rceil. \tag{42.1}$$

We record the strict correction window, including its proof, because its endpoint is essential to all-index constructions.

Lemma 42.2 (Strict Salem window)**.** *For every $n \geq 1$,*

$$-(d-2) < E_n < d-2, \qquad d - \lceil E_n \rceil \in I_d := \{2, 3, \ldots, 2d-3\}.$$

For every $c \in \{-d+3, \ldots, d-2\}$, the set of n on any fixed positive-step arithmetic progression for which $\lceil E_n \rceil = c$ has a positive relative natural density w_c, independent of the progression.

Proof. Put $u = \tau^{-n} \in (0,1)$ and $x_j = |1 - z_j^n|^2 > 0$. The nonzero integer norm of $\tau^n - 1$ satisfies

$$1 \leq |\mathrm{N}(\tau^n - 1)| = \frac{(1-u)^2}{u} \prod_{j=1}^{s} x_j,$$

so $\prod_j x_j > u$. If $\sum_j x_j \leq u$, the arithmetic–geometric mean inequality gives $\prod_j x_j \leq (u/s)^s \leq u$, a contradiction. Hence $E_n = u + 2s - \sum_j x_j < 2s$; the lower bound follows from $E_n \geq u - 2s > -2s$.

For the density assertion, let $Y(t) = 2\sum_{j=1}^{s} \cos(2\pi t_j)$ on $[0,1)^s$ with Lebesgue probability measure. Its range is $[-2s, 2s]$. Every open interval $(c-1, c)$ under consideration meets the interior of that range, and its inverse image contains a nonempty open set. Its measure w_c is therefore positive. Each level set of Y has measure zero: after fixing $t_2, \ldots, t_s$, the equation in t_1 has at most two solutions, and Fubini's theorem applies. Equidistribution from Lemma 42.1, together with $\tau^{-n} \to 0$, therefore gives the claimed limiting frequencies. The vanishing perturbation causes no problem because the two boundary level sets have measure zero. □

42.2 An exact characterization of all finite covers

Theorem 42.3 (Salem cover criterion)**.** *Let τ be a Salem number of degree $d \geq 4$, let $J \subset \mathbb{Z}$ be finite and nonempty, and let S be a finite nonempty set of primes. The following conditions are equivalent.*

1. *There is an integer $h \geq 1$ such that, for every $n \geq 1$ and $j \in J$, the integer $\lfloor \tau^{hn} \rfloor + j$ has a proper prime divisor belonging to S.*
2. *The same assertion holds for all sufficiently large n, for some $h \geq 1$.*
3. *For every $r \in I_d$ and $j \in J$,*

$$\gcd(M_S, r + j) > 1. \tag{42.2}$$

If condition (3) fails, then for every $h \geq 1$ there is a $j \in J$ for which the indices n satisfying

$$\gcd(M_S, \lfloor \tau^{hn} \rfloor + j) = 1$$

have positive lower natural density. If condition (3) holds, a suitable h is effectively computable.

Proof. Let C be an integral companion matrix for the minimal polynomial of τ. Since a Salem polynomial is reciprocal with constant term one, C is invertible modulo every positive integer, and $T_n = \mathrm{tr}(C^n)$.

Suppose that (42.2) fails at (r, j), and fix an arbitrary h. Choose $L \geq 1$ with $(C^h)^L \equiv I \pmod{M_S}$. For all $n = Lm$, therefore, $T_{hn} \equiv d \pmod{M_S}$. On a positive relative density w_{d-r} of these indices, $\lceil E_{hn} \rceil = d - r$. Formula (42.1) then yields

$$\lfloor \tau^{hn} \rfloor + j \equiv r + j \pmod{M_S},$$

which is coprime to M_S. The lower density in the full index sequence is at least $w_{d-r}/L > 0$. Thus condition (2) fails.

Conversely, suppose condition (3) holds. Choose h_0 such that $C^{h_0} \equiv I \pmod{M_S}$, and take a sufficiently large multiple h of h_0 that

$$\lfloor \tau^h \rfloor + \min J > \max S.$$

For every $n \geq 1$, the trace is congruent to d, while the strict window gives $r = d - \lceil E_{hn} \rceil \in I_d$. Condition (3) provides a prime $p \in S$ dividing $r + j$, hence dividing the actual integer $\lfloor \tau^{hn} \rfloor + j$. The size inequality makes that divisor proper. Finite matrix orders can be found by exact modular arithmetic. A rational lower bound $1 < u < \tau$, obtained by algebraic root isolation, makes the size step effective as well. The implication (1)⇒(2) is immediate. □

Corollary 42.4 (Sharp prime budget for consecutive blocks)**.** *Let $H \geq 0$. Among finite prime sets that cover all integers*

$$\lfloor \tau^{hn} \rfloor,\ \lfloor \tau^{hn} \rfloor + 1, \ldots, \lfloor \tau^{hn} \rfloor + H \qquad (n \geq 1)$$

for some positive integer h, the unique smallest set under inclusion is

$$S_{d,H} = \{p \text{ prime} : p \leq 2d - 3 + H\}.$$

In particular, the minimum number of covering primes is exactly $\pi(2d - 3 + H)$, and some effectively computable power of every Salem number starts an all-composite block of length $H + 1$ at every index.

Proof. The integers $r+j$, for $r \in I_d$ and $0 \leq j \leq H$, exhaust $\{2, \ldots, 2d-3+H\}$. Condition (42.2) forces S to contain each prime in that interval by taking $r + j$ equal to that prime. Conversely, $S_{d,H}$ contains a prime divisor of every integer in the interval. Apply Theorem 42.3. □

Corollary 42.5 (Exact shift obstruction)**.** *For a finite nonempty set $J \subset \mathbb{Z}$, an all-index cover by some finite prime set exists after taking a power of τ if and only if*

$$J \cap \{-1, -2, \ldots, -(2d - 2)\} = \varnothing.$$

If this intersection is nonempty, no finite prime set covers the corresponding shifted floors, even eventually, for any positive power of τ.

Proof. For a fixed j, the consecutive integer interval $I_d + j$ contains 1 or -1 exactly when $-(2d-2) \leq j \leq -1$. Such an integer violates (42.2) for every S. Outside that range all integers in $I_d + j$ have absolute value at least two. Taking the union of their prime divisors over the finite set J supplies a finite S satisfying the criterion. □

The negative assertion concerns *finite prime covers*, not the possibility that every term is composite with unbounded smallest prime factor. In particular, it does not assert infinitely many primes among $\lfloor \tau^{hn} \rfloor - 1$.

Corollary 42.6 (Simultaneous floor and ceiling)**.** *Some effectively computable power α of every Salem number has both $\lfloor \alpha^n \rfloor$ and $\lceil \alpha^n \rceil$ composite for every $n \geq 1$. The smallest possible finite prime cover for both sequences is $\{p : p \leq 2d - 3\}$.*

Proof. No positive power of a Salem number is an integer: conjugating it to a unit-circle conjugate would contradict its modulus. Thus ceiling is floor plus one. Apply Corollary 42.4 with $H = 1$; the extra endpoint $2d - 2$ is even and greater than two, so it introduces no additional prime. □

42.3 Exact frequencies in degree four

Proposition 42.7 (Quartic divisor frequencies)**.** *For a quartic Salem number, the correction values $c = -1, 0, 1, 2$ have respective frequencies*

$$(w_{-1}, w_0, w_1, w_2) = (1/3, 1/6, 1/6, 1/3)$$

on every fixed positive-step arithmetic progression. More generally, for every Salem number of any degree and every $h \geq 1$, the even values of $\lfloor \tau^{hn} \rfloor$ have natural density exactly $1/2$.

Proof. In degree four, calculate the measures of the four inverse images of

$$(-2, -1), \quad (-1, 0), \quad (0, 1), \quad (1, 2)$$

under $2\cos(2\pi t)$; the respective lengths are $1/3, 1/6, 1/6, 1/3$. In arbitrary degree the distribution of $Y = 2\sum \cos(2\pi t_j)$ is symmetric under $Y \mapsto -Y$, by translating every t_j by $1/2$. Away from the measure-zero integral levels, $\lceil -Y \rceil = 1 - \lceil Y \rceil$, so the parity of the correction is split equally. The trace modulo two is periodic. On each of its finitely many residue progressions the correction has that same limiting distribution, proving the parity statement. □

For a concrete example

$$P(X) = X^4 - X^3 - X^2 - X + 1, \qquad \alpha = \tau^{260},$$

the substitution $Y = X + X^{-1}$ gives $Y^2 - Y - 3$. Its roots are $(1 \pm \sqrt{13})/2$, one above 2 and one in $(-2, 2)$. The quartic is irreducible: over $\mathbb{Q}(\sqrt{13})$ the discriminant $Y^2 - 4 = (\sqrt{13} - 1)/2$ for the large root has norm -3 and is not a square. Hence the large real root τ is Salem. A companion matrix and its exact power modulo 30 are

$$C = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}, \qquad D = C^{260} \equiv \begin{pmatrix} 21 & 10 & 20 & 10 \\ 20 & 11 & 20 & 10 \\ 20 & 10 & 21 & 10 \\ 20 & 10 & 20 & 11 \end{pmatrix} \pmod{30}.$$

Direct multiplication gives $D^3 = I$ and

$$\operatorname{tr}(D) = \operatorname{tr}(D^2) = \operatorname{tr}(D^3) = 4 \pmod{30}.$$

Thus $\operatorname{Tr}(\alpha^n) \equiv 4 \pmod{30}$ for all $n \geq 1$. Hence the floors are congruent to $5, 4, 3, 2$ modulo 30 with frequencies $1/3, 1/6, 1/6, 1/3$, respectively. The disjoint divisor classes are therefore

covering prime	2	3	5
natural density	1/2	1/6	1/3.

The former three-prime certificate is now sharp: every finite prime cover for powers of any quartic Salem number must contain $2, 3, 5$. The same example also makes every ceiling composite. The divisors are proper because $\tau > 3/2$ and $\tau^{260} > 6$.

42.4 A sharp contrast with Pisot units

Theorem 42.8 (Finite covers for two Pisot classes)**.** *Let $\beta > 1$ be a Pisot unit, and let S be a finite nonempty set of primes, with product M_S.*

1. *If β is totally real of degree $d \geq 2$, an S-cover of all $\lfloor \beta^{hn} \rfloor$ exists for some $h \geq 1$ if and only if $\gcd(M_S, d-1) > 1$. Thus degree two admits no finite prime cover after any power; in every higher totally real degree one prime suffices.*
2. *In the same totally real situation, one S-cover of both floor and ceiling for some power exists if and only if*

$$\gcd(M_S, d-1) > 1 \quad \textit{and} \quad \gcd(M_S, d) > 1.$$

   For $d \geq 3$ the minimum number of primes is exactly two.
3. *If β has degree three and its two other conjugates are non-real, a finite set S covers all floors of some power if and only if $\{2, 3\} \subseteq S$. This same condition is necessary and sufficient to cover both floors and ceilings.*

All the sufficient constructions are effective. Replacing all indices by all sufficiently large indices leaves the characterizations unchanged.

Proof. The companion matrix C is invertible modulo every M_S because β is a unit. In the totally real case, take an even multiple h of its order modulo M_S, sufficiently large that

$$0 < E_{hn} := \sum_{i=2}^{d} \beta_i^{hn} < 1 \qquad (n \geq 1).$$

This is possible because every $|\beta_i| < 1$. The trace is congruent to d, the floor is congruent to $d-1$, and the ceiling is congruent to d. A further size enlargement makes the indicated covering primes proper divisors, proving sufficiency in (1) and (2).

Conversely, for any proposed power h, restrict n to even multiples of an order of C^h modulo M_S. The same two residue statements hold for all sufficiently large such n. If either required gcd is one, the corresponding integers have no divisor in S, proving necessity. Since d and $d-1$ are coprime, two distinct primes are necessary in (2), and they suffice when $d \geq 3$.

For (3), write the conjugates as $\beta, z, \bar{z}$, where $z = \rho e^{2\pi i\theta}$ and $0 < \rho < 1$. The angle θ is irrational. Indeed, otherwise some positive power satisfies $z^m = \bar{z}^m$. The degree of β^m divides the prime degree three, and the conjugate collision rules out degree three, so β^m is rational. It is then a rational algebraic-integer unit, hence 1 or -1, contradicting $\beta > 1$.

Now $E_n = 2\rho^n \cos(2\pi n\theta)$ has absolute value less than one for all large n. On every arithmetic progression its positive and negative signs each have relative density $1/2$. On a matrix-return progression the floors consequently take residues 2 and 3 modulo M_S, each with positive relative density. This forces both primes 2 and 3 into S. Conversely, choose a sufficiently large matrix-return power. Its floors have residue 2 or 3 modulo six, and its ceilings have residue 3 or 4. Both are covered by $2, 3$; the size condition again ensures proper divisibility. □

This classification answers why the different constructions need one, two, or three small primes. It does not contradict the existence of infinitely many composite quadratic Pisot floors: having infinitely many composite terms is much weaker than covering every term of a power subsequence by one fixed finite set of primes.

43 Local dimension and effective spectra with arbitrarily long square stretches

Put $E_1 > 0$ and $E_{n+1} = c_{n+1}E_n$, and write

$$W_E = \{A > 1 : \lfloor A^{E_n} \rfloor \in \mathbb{P} \text{ for every } n \geq 1\}.$$

In this section the multipliers c_{n+1} are integers at least 2, and $c_{n+1} \geq 3$ for infinitely many n. No upper bound on the multipliers, and no bound on the lengths of runs of 2's, is imposed. Saito–Takeda's theorem [38] proves that this set is nonempty, closed, and perfect. The results below strengthen this topological conclusion locally and identify an ordinary computability consequence. They do not apply to an eventually all-square schedule.

43.1 Two prime-interval inputs and a uniform seed lemma

We use the following two published estimates. There are positive constants b, d, D and sufficiently large thresholds such that

$$\#(\mathbb{P} \cap [x, x + x^{21/40}]) \geq b\frac{x^{21/40}}{\log x}, \tag{43.1}$$

$$\#\mathcal{B}(x, \gamma) \leq Dx^{2/3-\gamma} \qquad (1/2 \leq \gamma \leq 1), \tag{43.2}$$

where $\mathcal{B}(x, \gamma)$ is any disjoint family of intervals $[u, u + u^\gamma] \subset [x, 2x]$ that contain at most $du^\gamma/\log u$ primes. The first is the Baker–Harman–Pintz estimate, as stated in [33, Lemma 3.1]; see also [6]. The second is Matomäki's Lemma 9 [25]; its constants and threshold are uniform in $\gamma \in [1/2, 1]$.

Lemma 43.1 (A bounded-tail good seed)**.** *There are $Q_0 > 0$ and $a > 0$ with the following property. Let $2 \leq t_j \leq 4$ for every $j \geq 1$, and let $q \geq Q_0$ be prime. If*

$$\#(\mathbb{P} \cap [q^{t_1}, q^{t_1} + q^{t_1-1}]) \geq \frac{d}{2}\frac{q^{t_1-1}}{t_1 \log q},$$

then the set of $y \in [q^{1/2}, (q+1)^{1/2})$ satisfying

$$\lfloor y^2 \rfloor = q, \qquad \lfloor y^{2t_1 \cdots t_j} \rfloor \in \mathbb{P} \quad (j \geq 1)$$

has Hausdorff dimension at least $\left(1 + 4/(q \log q)\right)^{-1}$.

Proof. Call a prime v good for a next multiplier t if its corresponding interval has at least $(d/2)v^{t-1}/(t \log v)$ primes. Suppose v is good for $t \in [2, 4]$, and consider its prime children $w \in [v^t, v^t + v^{t-1}]$. For a following multiplier $s \in [2, 4]$, the intervals $[w^s, w^s + w^{s-1}]$ are disjoint, because distinct prime labels differ by at least 2, and lie in $[v^{ts}, 2v^{ts}]$ once v is large. An interval belonging to a child that is not good is exceptional in (43.2), with $\gamma = 1 - 1/s$. Thus the number of such children is at most

$$Dv^{ts(2/3-(1-1/s))} = Dv^{t(1-s/3)} \leq Dv^{t/3}.$$

Uniformly for $t, s \in [2, 4]$, this is smaller than $(d/4)v^{t-1}/(t \log v)$ for sufficiently large v, since $t - 1 - t/3 \geq 1/3$. Discarding at most a fixed number of endpoint children still leaves at least $av^{t-1}/(t \log v)$ good children, for a fixed $a > 0$. Their closed root cells can be taken strictly inside the parent's cell: $w > v^t$ and $w + 1 < (v + 1)^t$ after the endpoint discards.

Iterate the selection. The resulting tree satisfies conditions (C1)–(C4) of Saito's Proposition 5.1, with $R = 4$, $\theta = 1$, $L = 1$, and $Q = a$. That proposition gives the stated dimension bound. This is precisely the part of Saito's construction after the initial good prime has been found; it does not require that the given seed be selected from an interval $[M, 2M]$. □

Lemma 43.2 (A local reset at a multiplier at least three)**.** *There is P_0 such that the following holds. Start with a prime root cell $J = (p^{1/E}, (p+1)^{1/E})$, where $p \geq P_0$. Suppose the next exponent is rE with $r \geq 3$, and all following exponent ratios are at least 2. There is a subset of J satisfying all these future prime conditions, whose dimension is at least*

$$\left(1 + \frac{4}{q \log q}\right)^{-1} \quad \text{for some prime } q > p^3.$$

The threshold P_0 is independent of the future schedule and of E, r.

Proof. Write $r = s2^m$, where $m \geq 0$ is an integer and $3 \leq s < 6$; for example, take $m = \lfloor \log_2(r/3) \rfloor$. Replace the jump r by the factors $s, 2, \ldots, 2$. Refine every subsequent ratio $u \geq 2$ into equal factors $u^{1/\lfloor \log_2 u \rfloor} \in [2, 4)$. The refined schedule after the first factor s has all factors in $[2, 4]$.

By subdividing $[p^s, p^s + p^{s-1}]$ into intervals of the length in (43.1), that interval contains at least $b' p^{s-1}/(s \log p)$ primes, uniformly for $s \in [3, 6]$ and large p. If $t \in [2, 4]$ is the next refined factor, the intervals $[q^t, q^t + q^{t-1}]$ associated with these prime candidates are disjoint and lie in $[p^{st}, 2p^{st}]$. Formula (43.2) excludes at most $Dp^{s(1-t/3)} \leq Dp^{s/3}$ bad candidates. Since $s - 1 - s/3 \geq 1$, there is a good candidate $q > p^s \geq p^3$. Also $q + 1 < (p+1)^s$, so its root cell at exponent sE is strictly inside J. Apply Lemma 43.1, followed by the bi-Lipschitz change of variable $A = y^{2/(sE)}$. All original exponents are among the refined exponents, which proves the assertion. □

43.2 Full dimension in every relative open set

Theorem 43.3 (Full local dimension)**.** *Assume $c_{n+1} \in \mathbb{Z}$, $c_{n+1} \geq 2$ for every n, and $c_{n+1} \geq 3$ for infinitely many n. For every open interval U with $U \cap W_E \neq \varnothing$,*

$$\boxed{\dim_H(U \cap W_E) = 1.}$$

In particular, every feasible finite prime-output prefix that occurs in W_E has a full-dimensional set of completions.

Proof. Choose $A \in U \cap W_E$ and let $p_n = \lfloor A^{E_n} \rfloor$. For every $j < n$, the quotient $M = E_n/E_j$ is an integer. On the half-open cell

$$I_n = [p_n^{1/E_n}, (p_n + 1)^{1/E_n}),$$

the integer part at exponent E_j is constant: after taking the E_n-th power, its possible discontinuities are integers of the form h^M, and none lies strictly between the consecutive integers p_n and $p_n + 1$. Since $A \in I_n$, the entire cell satisfies all conditions through index n.

Its length tends to zero, and $p_n \to \infty$. Consequently, for every sufficiently large n such that $c_{n+1} \geq 3$, the cell I_n lies in U and $p_n \geq P_0$. Lemma 43.2 supplies a subset of $U \cap W_E$ with dimension at least $(1 + 4/(q_n \log q_n))^{-1}$, where $q_n > p_n^3$. Letting n run through these infinitely many indices gives dimension at least 1. The ambient upper bound is 1.

For the last assertion, note that every $A \in W_E$ is an interior point of each of its finite prime cells. Indeed, A^{E_j} cannot be an integer, since a later integer exponent ratio would make a prime output into a proper prime power. Its prescribed finite prefix therefore holds in an open neighborhood of A, to which the theorem applies. □

Remark 43.4 (What the integer assumption does). The nesting argument uses the integrality of E_n/E_j essentially. The local reset lemma itself allows real ratios, but it applies only when the entire seed cell already respects the earlier conditions. Global full dimension for arbitrary real ratios at least 2 does not, by itself, prove full dimension in every prescribed prefix. No such unrestricted local assertion is made here.

43.3 Computability of the minimum and of a dense spectrum

Lemma 43.5 (An integer-arithmetic continuation rule)**.** *There are fixed integers $K, P \geq 1$ such that, for every integer multiplier schedule at least 2, every prime $p \geq P$ immediately before a multiplier at least 3 has a computable continuation. The continuation is uniform in the computable schedule and the seed. At a multiplier at least 3, at least two child primes can be chosen, each with a computable continuation.*

Proof. Set

$$H(v) = \left\lceil \frac{v}{K\lceil \log_2 v \rceil} \right\rceil,$$

and say that a prime v is square-good if $[v^2, v^2 + v]$ contains at least $H(v)$ primes. Choose K so large that, for sufficiently large v,

$$H(v) \leq \frac{dv}{2 \log v}.$$

Failure of square-goodness then makes $[v^2, v^2 + v]$ exceptional in (43.2) with $\gamma = 1/2$.

For an integer $c \geq 3$, (43.1) implies that $[p^c, p^c + p^{c-1}]$ contains at least $b'' p^{c-1}/(c \log p)$ primes, uniformly in $c \geq 3$ for large p. Indeed $p^{c-1}/(p^c)^{21/40} = p^{19c/40-1}$ tends to infinity uniformly, and subdivision into short intervals proves the bound. If the following multiplier is 2, remove those candidate primes q that are not square-good. Their disjoint square intervals lie in $[p^{2c}, 2p^{2c}]$, so at most $Dp^{c/3}$ are removed. The difference is at least two, uniformly for $c \geq 3$, after increasing P.

At a multiplier $c = 2$, require that the current seed p is square-good. There are at least $H(p)$ prime children in $[p^2, p^2 + p]$; if the next multiplier is also 2, discard at most $Dp^{2/3}$ that are not square-good. Since $H(p) \gg p/\log p$, at least two remain for large p. If the next multiplier is at least 3, no discard is needed.

Thus at each step select the least available child, imposing square-goodness precisely when the following multiplier is 2. All powers and interval endpoints are integers; primality and the finite prime counts are decidable, and $\lceil \log_2 v \rceil$ is computable by integer comparison. The initial multiplier is at least 3, so no square-good assumption on the initial seed is necessary. The child root cells are strictly nested, and their lengths tend to zero with a computable modulus. Their unique limit gives the required computable real. Choosing the first or second available child at a step with multiplier at least 3 gives the final assertion. □

Theorem 43.6 (Computable least constants, despite arbitrarily long square stretches)**.** *Suppose $E_1 > 0$ is computable, (c_{n+1}) is a computable integer sequence with $c_{n+1} \geq 2$, and $c_{n+1} \geq 3$ for infinitely many n. Then the least element of W_E is a computable real. Moreover, computable elements are dense in W_E.*

Proof. Write $M_n = E_n/E_1 \in \mathbb{N}$. For an index n immediately before a multiplier at least 3, enumerate primes q in increasing order and test the finite conditions

$$\left\lfloor q^{1/(E_n/E_j)} \right\rfloor \in \mathbb{P} \qquad (1 \leq j < n).$$

These tests are exact integer-root and primality computations. The cell $[q^{1/E_n}, (q+1)^{1/E_n})$ satisfies the first n prime conditions exactly when they hold. At any feasible cell, $q \geq 2^{M_n}$, because the first prime output is at least 2.

Choose n with $2^{M_n} \geq P$. By Lemma 43.5, every feasible cell at this index has an infinite continuation. Such cells exist by the nonemptiness of W_E, so the enumeration stops. Let q_n be its first successful prime. The actual least constant ξ_E therefore lies in

$$q_n^{1/E_n} \leq \xi_E < (q_n + 1)^{1/E_n}.$$

The existence of the minimum follows also directly from closedness, nonemptiness, and the lower bound $A \geq 2^{1/E_1} > 1$. For $E_n \geq 1$, the mean value theorem gives

$$(q_n + 1)^{1/E_n} - q_n^{1/E_n} \leq E_n^{-1}.$$

Given a requested precision, search for a later index before a multiplier at least 3 with E_n sufficiently large, perform this finite search, and approximate the two computable endpoints. This produces a Cauchy name for ξ_E.

For density, choose $A \in W_E$ and an arbitrary neighborhood U of A. A sufficiently late cell immediately before a multiplier at least 3 is contained in U, respects every earlier condition, and has prime label at least P. Applying the computable continuation rule produces a computable member of $U \cap W_E$. □

Remark 43.7 (Computability is not a numerical evaluation). The integers K, P in the preceding proof exist by the published analytic estimates; they have not been numerically evaluated in this draft. Fixing such integers yields an ordinary Turing algorithm, and the theorem asserts computability in this standard existence sense. It supplies neither a practical running-time bound nor a new certified decimal digit of the ordinary least Mills constant. In particular, the potentially enormous finite search preceding the first usable branching level is not replaced by assuming that the familiar small greedy prime chain extends forever. This distinction is the reason that Elsholtz's explicit, practically evaluable variants [15] remain substantially stronger computational information for their specified schedules.

Corollary 43.8 (Every local piece contains every Turing degree)**.** *Under the hypotheses of Theorem 43.6, every open interval U meeting W_E contains a computably embedded binary Cantor set of elements of W_E. The embedding and its inverse on its image are computable. Hence $U \cap W_E$ contains a real of every Turing degree.*

Proof. Begin with a sufficiently late feasible seed cell inside U. At each subsequent multiplier at least 3, use a bit to select the first or the second permitted child from Lemma 43.5. At all intermediate steps take the least permitted child. There are infinitely many bit choices, all searches terminate, and the computable upper bounds on root-cell lengths show that the map from binary sequences to limits is computable. Distinct bits select separated prime cells, so the map is injective. Its inverse is computable by comparing a Cauchy name of the limit with the finitely many separated child cells at each bit level. The limit never belongs to an integer-power endpoint, because a later integer multiplier would give a composite prime power. Thus every such comparison terminates. The two computable reductions preserve Turing degrees. No positive Hausdorff dimension is asserted for this particular two-child Cantor subset. □

44 A complete bounded exclusion for cubic Mills shadows

The results of this section address a finite, explicitly bounded part of Saito's cubic Pisot obstruction. They do not exclude all cubic Pisot numbers and do not prove that Mills' constant is transcendental. The finite verification uses only small-prime modular arithmetic; it does not assume values of the actual Mills prime sequence.

Lemma 44.1 (An elementary lower bound)**.** *Every totally real cubic Pisot number exceeds* 2.

Proof. Let its conjugates be β, u, v, with $\beta > 1$ and $0 < |u|, |v| < 1$, and let f be its monic irreducible polynomial. Since $|\beta uv|$ is a nonzero integer, $|uv| \geq 1/\beta$. The nonzero integer $f(1)f(-1)$ gives

$$1 \leq |f(1)f(-1)| = (\beta^2 - 1)(1 - u^2)(1 - v^2) \leq (\beta^2 - 1)(1 - 1/\beta)^2.$$

Indeed, $(1-u^2)(1-v^2) \le (1-|uv|)^2 \le (1-1/\beta)^2$. For $1 < \beta \le 2$ the last expression is at most $3/4$, a contradiction. □

Lemma 44.2 (Exact coefficient catalogue)**.** *Fix an integer $B \ge 3$. The monic minimal polynomials of totally real cubic Pisot numbers $\beta < B$ whose small conjugate of larger modulus is negative are exactly the following two disjoint finite families.*

1. *Opposite-sign small conjugates:*

$$X^3 - aX^2 - dX + c, \qquad 2 \le a < B, \quad d, c \ge 1, \quad d + c \le a.$$

2. *Two negative small conjugates:*

$$X^3 - aX^2 - dX - c,$$

where $1 \le a < B$, $1 \le d < 2B$, $1 \le c < B$, and

$$d \le 2a + 2, \qquad d \le a + c, \qquad B^3 - aB^2 - dB - c > 0,$$
$$a^2d^2 + 4d^3 - 4a^3c - 27c^2 - 18adc > 0.$$

Proof. In the first case write the roots as $\beta, -\rho, \sigma$, with $0 < \sigma < \rho < 1$. Then $a = \beta - \rho + \sigma$, $d = \beta(\rho - \sigma) + \rho\sigma > 0$, and $c = \beta\rho\sigma > 0$. The inequality $f(-1) < 0$ is exactly $d + c < a + 1$, or $d + c \le a$. Conversely these integer conditions give one root in each of $(-1, 0), (0, 1), (a, a + 1)$ by direct sign evaluation. Since $f(a) = -ad + c < 0$, the positive large root exceeds the trace a; therefore the negative small root has larger modulus. Thus $\beta < B$.

In the second case write the roots as $\beta, -\rho, -\sigma$, with $0 < \sigma < \rho < 1$. The preceding lemma gives $a = \beta - \rho - \sigma > 0$. Furthermore

$$d = a(\rho + \sigma) + \rho^2 + \rho\sigma + \sigma^2 < 2a + 3.$$

The other coefficient bounds, the positive discriminant, and $f(B) > 0$ are necessary. Conversely, positive discriminant gives three distinct real roots. There is exactly one positive root by Descartes' rule, and the other two are negative. The negative critical point is $(a - \sqrt{a^2 + 3d})/3 > -1$ because $d \le 2a + 2$. Together with $f(-1) < 0$, this puts both negative roots in $(-1, 0)$. The large root exceeds 1 because the product of the three root moduli is the positive integer c; $f(B) > 0$ puts it below B. Finally, each polynomial in either family is irreducible: a rational root would be an integer, necessarily the large root, leaving a monic integral quadratic with a nonzero integral constant term of absolute value less than one. □

Theorem 44.3 (A finite cubic exclusion with a density bound)**.** *Let $\beta < 100$ be a totally real cubic Pisot number whose dominant small conjugate is negative. Then there are a prime $\ell \le 47$, an integer $j \ge 0$, and an integer $1 \le L \le 1148$ such that*

$$\ell \mid \lfloor \beta^{3^{j+tL}} \rfloor \qquad \text{for every sufficiently large } t.$$

Consequently, the set of composite indices in $(\lfloor \beta^{3^r} \rfloor)_{r \ge 0}$ has lower density at least $1/1148$.

Proof with a finite arithmetic certificate. Write the polynomial as $X^3 - aX^2 + bX - c$, and set

$$(T_r, B_r, E_r) = \left(\sum_i \alpha_i^{3^r}, \sum_{i<j} (\alpha_i\alpha_j)^{3^r}, c^{3^r} \right).$$

Newton's identities give the integral map

$$\Phi(T, B, E) = (T^3 - 3TB + 3E,\ B^3 - 3TBE + 3E^2,\ E^3).$$

For a fixed prime ℓ, its orbit modulo ℓ is eventually periodic. A state with trace coordinate zero on the cycle yields an infinite arithmetic progression of traces divisible by ℓ. Since the dominant small conjugate is negative, these traces agree with the desired floors for every sufficiently large r.

The exact coefficient catalogue with $B = 100$ contains 213569 polynomials: 161700 from the first family and 51869 from the second. Exhaustive iteration modulo the following primes gives the listed counts of polynomials certified for the first time:

ℓ	2	3	5	7	11	13	17	19
count	134584	27162	27780	14024	4856	2501	1163	544

ℓ	23	29	31	37	41	43	47
count	492	182	142	86	35	14	4.

There are no survivors. The largest cycle period used is 1148, and the largest preperiod is 3. A complete standard-library Python verifier appears in Appendix B; all coefficients, discriminants, and modular orbit operations are exact integers. An independent companion-matrix audit checks the characteristic polynomial at the two cycle endpoints and the zero trace at the chosen intermediate index. The resulting progression has density $1/L \geq 1/1148$. Its entries eventually exceed ℓ, so they are composite rather than merely divisible by ℓ. □

The degree and minimal-polynomial structure needed below were proved in Theorem 30.2. We now combine that structure with the bounded modular exclusion.

Corollary 44.4 (Height and degree consequences for Mills' constant)**.** *Suppose that the least Mills constant A is algebraic. In its minimal-degree cubic representation*

$$d = 3^{t+1}, \qquad \beta = A^{3^t}, \qquad \mathrm{MinPoly}_A(X) = X^d - aX^{2d/3} + bX^{d/3} - c,$$

one necessarily has $\beta > 100$ and $a \geq 99$. In particular, the coefficient height of A is at least 99, independently of its degree. If $U > 1$ is any independently certified upper bound on A, then its degree must additionally satisfy

$$U^{d/3} > 100.$$

Proof. Theorem 30.2 supplies the degree and cubic-shadow structure. The bounded exclusion rules out $\beta < 100$, and equality is impossible for a cubic irrational. Its trace is $a > \beta - 2 > 98$, hence $a \geq 99$. The final inequality follows from $\beta = A^{d/3} \leq U^{d/3}$. □

The height statement is a bounded exclusion valid at all degrees; it is not an assertion that a numerical value of the actual Mills constant has been certified. No such bound U is assumed here.

Example 44.5 (The hand-checkable range below four)**.** There are precisely four polynomials in the catalogue below 4:

$$X^3 - 2X^2 - X + 1, \quad X^3 - 3X^2 - X + 1, \quad X^3 - 3X^2 - X + 2, \quad X^3 - 3X^2 - 2X + 1.$$

For their trace sequences, respectively, the indices $r \equiv 0 \pmod 2$, all r, all $r \geq 1$, and $r \equiv 1 \pmod 2$ give divisibility by $2, 3, 2, 2$. These statements follow immediately by applying Φ modulo 2 or 3. This is an independently hand-checkable subcase of the finite theorem.

45 Norm divisors that force an entire composite tail

Proposition 45.1 (An exact norm-prime obstruction)**.** *Let $f(X) = X^3 - aX^2 + bX - c$ be the minimal polynomial of a cubic Pisot number whose dominant small conjugate is negative, and put $T_r = \mathrm{Tr}(\beta^{3^r})$. If a prime $\ell \mid c$ divides any T_j, then*

$$\ell \mid T_r \quad \text{for every } r \geq j,$$

and the floor sequence is eventually composite. In particular, $\gcd(a, c) > 1$ *is sufficient for eventual compositeness.*

More exactly, suppose $\ell \geq 5$, $\ell \mid c$, and $\ell \nmid b(a^2 - 4b)$. Let u, v be the nonzero distinct roots of $X^2 - aX + b$ in $\mathbb{F}_{\ell^2}$, and put $R = u/v$. There is an index j with $\ell \mid T_j$ if and only if

$$\mathrm{ord}(R) = 2 \cdot 3^s \quad \text{for some integer } s \geq 0.$$

In that case the first such index is s and every subsequent trace is divisible by ℓ.

Proof. Modulo a prime dividing c, Newton's recurrence reads $T_{r+1} = T_r(T_r^2 - 3B_r)$, proving propagation and the gcd criterion. Under the additional hypotheses the three residual roots are $0, u, v$, so $T_r = u^{3^r} + v^{3^r}$. Its vanishing is equivalent to $R^{3^r} = -1$, and in a finite field of odd characteristic this is equivalent to the stated order condition with $s \leq r$. □

Corollary 45.2 (An explicit infinite family)**.** *For every integer $a \geq 8$ with $a \equiv \pm 1 \pmod 5$, let β_a be the large root of*

$$X^3 - aX^2 - 3X + 5.$$

Then β_a is cubic Pisot and

$$5 \mid \lfloor \beta_a^{3^r} \rfloor \qquad (r \geq 1).$$

Thus every one of these terms is composite. This family is not obtained from the initial gcd obstruction: $\gcd(a, 5) = 1$.

Proof. The opposite-sign coefficient catalogue applies since $3 + 5 \leq a$; in this case floor and trace agree for every odd power. The norm is -5 and $T_1 = a^3 + 9a - 15 \equiv a(a^2 - 1) \equiv 0 \pmod 5$. The propagation proposition applies, and all the relevant floors exceed 5. □

The norm-prime criterion is an elementary application of finite-field orders and Newton identities. The bounded catalogue and its complete modular certification are the additional concrete computations here; no claim of priority over all published computations is made.

Part VI
Boundary independence and further arithmetic extensions

46 Relations among all left boundary constants under sparse growth bursts

Theorem 21.1 already establishes support-rank independence for the entire left sub-boundary under an unbounded-limsup hypothesis. Here we record explicit continuation and local approximation bounds and deduce homogeneous-ratio consequences and a sparse-burst example. The recalled boundary formulation is the same theorem, not a second extension from a selected binary family.

Let C_1 be a positive integer, let $C_{n+1} = c_{n+1}C_n$ with integers $c_{n+1} \geq 2$, and put

$$W(C) = \{A > 1 : \lfloor A^{C_n} \rfloor \in \mathbb{P} \text{ for all } n\}.$$

Write $L(C)$ for its local left minima: $A \in L(C)$ if some $\eta > 0$ satisfies $(A - \eta, A) \cap W(C) = \varnothing$. Equivalently, these are the right endpoints of complementary intervals lying immediately to their left, including the least element of $W(C)$. They form Saito–Takeda's left sub-boundary $\widetilde{\partial}_L W(C)$; the terminology refers to the side on which the gap lies. For a nonzero polynomial $F = \sum_\nu a_\nu X^\nu$, define

$$\rho(F) = \dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}}\{\nu - \mu : a_\nu a_\mu \neq 0\}.$$

This is the affine dimension of its exponent support.

Lemma 46.1 (Prime-root separation recalled)**.** *Fix $T > 1$ and a nonzero integer polynomial F of support rank ρ. For all sufficiently large integers E and all distinct primes $p_1, \dots, p_r$ with $p_i^{1/E} \leq T$,*

$$0 < |F(p_1^{1/E}, \dots, p_r^{1/E})| \geq B^{1-E^\rho}, \qquad B = \max\left(2, \sum_\nu |a_\nu| T^{|\nu|}\right).$$

Consequently, if distinct positive numbers A_i admit these prime-root approximations with errors ε_n at integers $E_n \to \infty$, and

$$\limsup_n \frac{\log(1/\varepsilon_n)}{E_n^\ell} = \infty,$$

then no nonzero polynomial over $\overline{\mathbb{Q}}$ of support rank at most ℓ vanishes at the tuple $(A_1, \dots, A_r)$.

Proof. The separation bound is Theorem 19.1; the algebraic-coefficient transfer, including Laurent polynomials, is Theorem 19.3 applied along the subsequence on which the displayed quotient tends to infinity. □

Lemma 46.2 (Continuation through arbitrary integer square stretches)**.** *There are fixed integers $K, P \geq 1$ with the following property. For a prime $p \geq P$, call p square-good when $[p^2, p^2 + p]$ contains at least*

$$H(p) = \left\lceil \frac{p}{K\lceil \log_2 p \rceil} \right\rceil$$

primes. Given any sequence of integer multipliers at least 2, a seed $p \geq P$ has an infinite prime continuation if either it is square-good or its next multiplier is at least 3. At a next multiplier $c \geq 3$, the first child can be chosen in $[p^c, p^c + p^{21c/40}]$. All root cells in the continuation can be chosen strictly nested.

Proof. Choose K so large that $H(q) \le dq/(2\log q)$ for all sufficiently large q, where d is Matomäki's exceptional-interval constant. The estimate gives only $O(X^{1/3})$ primes $q \in [X, 2X]$ that fail to be square-good: apply it to their disjoint intervals $[q^2, q^2+q]$ at squared scale X^2, splitting into a fixed number of dyadic intervals and discarding at most one boundary-crosser per split.

At a step $c \ge 3$, Baker–Harman–Pintz [6] supplies $\gg p^{21c/40}/(c\log p)$ prime candidates in $[p^c, p^c + p^{21c/40}]$. If the following multiplier is 2, discard candidates that are not square-good; at most $O(p^{c/3})$ are removed. Since $21/40 > 1/3$, at least two candidates remain uniformly for $c \ge 3$ once $p \ge P$. Uniformity follows, for example, because $p^{c(21/40-1/3)}/c$ is increasing for $c \ge 3$ when p is sufficiently large. If the following multiplier is at least 3, no goodness restriction is needed.

At a square step, the current seed is required to be square-good. Its interval $[p^2, p^2+p]$ supplies $H(p) \gg p/\log p$ candidates; when the following multiplier is also 2, discard at most $O(p^{2/3})$ poor candidates. Again at least two remain for $p \ge P$. This maintains exactly the goodness condition needed for the next step. Every retained prime satisfies $p^c < q$ and $q+1 < (p+1)^c$. The assertion for $c \ge 3$ uses $21c/40 < c-1$, uniformly with positive margin; for $c = 2$ it follows directly from $q \le p^2 + p$. Iteration gives strictly nested prime root cells of lengths tending to zero and hence the claimed continuation. □

Lemma 46.3 (Uniform approximation of every left-gap endpoint)**.** *Let $A \in L(C)$, and put $p_n = \lfloor A^{C_n} \rfloor$. At every sufficiently late index with $c = c_{n+1} \ge 3$,*

$$p_{n+1} \le p_n^c + p_n^{21c/40}, \qquad 0 < A - p_n^{1/C_n} \le \frac{2}{cC_n} p_n^{1/C_n - 19c/40}.$$

The index threshold may depend on A, but the estimates hold for all larger indices with $c \ge 3$, whether the multipliers are bounded or not.

Proof. Integer multipliers ensure that the half-open prime root cell $[p_n^{1/C_n}, (p_n+1)^{1/C_n})$ respects all previous outputs. They also exclude $A^{C_n} \in \mathbb{N}$, since a later integer multiplier would make a proper prime power composite.

Suppose that the first displayed inequality fails at arbitrarily large indices with $c \ge 3$. Put $p = p_n$. By Lemma 46.2, a prime $q \in [p^c, p^c + p^{21c/40}]$ has an infinite continuation respecting the entire remaining integer multiplier schedule. Each replacement prime lies below p_{n+1}, and its root cell lies strictly inside the parent cell, because $21c/40 < c-1$ for $c \ge 3$. Its continuation gives $A_n' < A$ in $W(C)$ with $|A_n' - A| \le 1/C_n$. Infinitely many such replacements contradict the left-gap condition. The first inequality follows.

Consequently

$$A^{cC_n} < p_{n+1} + 1 \le p_n^c + 2p_n^{21c/40}.$$

Concavity of the cth root gives $A^{C_n} - p_n \le (2/c)p_n^{1-19c/40}$. Applying the mean value theorem to the C_nth root yields the stated bound. The finitely many cases with $C_n = 1$ can be discarded. □

Theorem 46.4 (Left-boundary support theorem, recalled)**.** *Let $\ell \ge 1$ and assume*

$$\boxed{\limsup_{n\to\infty} \frac{c_{n+1}}{C_n^{\ell-1}} = \infty.}$$

For every finite tuple of distinct members $A_1, \dots, A_r$ of $L(C)$, and every nonzero polynomial $F \in \overline{\mathbb{Q}}[X_1, \dots, X_r]$ with $\rho(F) \le \ell$,

$$F(A_1, \dots, A_r) \ne 0.$$

The assertion also holds for Laurent polynomials.

Proof. Apply Theorem 21.1, using $L(C) = \widetilde{\partial}_L W(C)$; multiply by a monomial for Laurent polynomials. □

Corollary 46.5 (Transcendental multiplicative combinations)**.** *If* $\limsup_n c_{n+1} = \infty$*, then for distinct* $A_1, \dots, A_r \in L(C)$ *and nonzero* $(u_1, \dots, u_r) \in \mathbb{Q}^r$*,*

$$\boxed{A_1^{u_1} \cdots A_r^{u_r} \notin \overline{\mathbb{Q}}.}$$

In particular every quotient of two distinct members of $L(C)$ *is transcendental. This applies to the factorial clock* $C_n = n!$*.*

Proof. Clear the denominators of the rational exponents. If the monomial were algebraic, a Laurent polynomial of the form $X^m - b$ with algebraic b and support rank 1 would vanish. Apply Theorem 46.4 with $\ell = 1$. For $C_n = n!$ the multipliers are $c_{n+1} = n + 1$. □

Corollary 46.6 (Homogeneous relations and ratios)**.** *If* $r \geq 2$ *and* $\limsup_n c_{n+1}/C_n^{r-2} = \infty$*, then for any distinct* $A_1, \dots, A_r \in L(C)$ *the* $r - 1$ *ratios* $A_2/A_1, \dots, A_r/A_1$ *are algebraically independent. If instead* $\limsup_n c_{n+1}/C_n^{r-1} = \infty$*, then the* r *constants themselves are algebraically independent.*

Proof. A polynomial relation among the ratios homogenizes to a nonzero polynomial in the A_i, all of whose exponent vectors lie in a single affine hyperplane, so its support rank is at most $r - 1$. Apply Theorem 46.4 with $\ell = r - 1$. An arbitrary polynomial in r variables has support rank at most r, which proves the second assertion. □

Example 46.7 (Long square stretches with an independent entire left boundary)**.** Take $C_1 = 2$ and recursively define

$$c_{n+1} = \begin{cases} C_n^n, & n \text{ is a power of 2,} \\ 2, & \text{otherwise,} \end{cases} \qquad C_{n+1} = c_{n+1} C_n.$$

For every fixed ℓ, the limsup condition of Theorem 46.4 holds along the powers of 2. Therefore the entire set $L(C)$ is algebraically independent. It is countable and dense in $W(C)$, as the right endpoints of the complementary intervals of a closed nowhere-dense perfect set, together with its minimum. The square stretches are unbounded in length, and $c_{n+1} = 2$ infinitely often; the growth limit in Saito–Takeda's Theorem 1.3 therefore does not hold. This example illustrates the sparse-burst limsup theorem, which is stronger than that published growth-limit hypothesis.

Remark 46.8 (Scope of the improvement)*.* For the factorial clock, Saito–Takeda's Proposition 8.2 gives rational linear independence of a pair with the stated boundary orientations. Corollary 46.5 proves the stronger transcendence of every pairwise quotient when both endpoints are local left minima, and also handles all finite nontrivial multiplicative combinations. It does not prove algebraic independence of arbitrary factorial-clock boundary tuples: a general polynomial may have support rank greater than one. Nor does any statement here apply to all arbitrary admissible constants in $W(C)$. The short approximation is a consequence of the left-gap property.

47 Density-one answers to the cubic Pisot problem

Saito's Problem 1.1 in arXiv:2504.14968v1 asks whether, for every cubic Pisot number β, the numbers $\lfloor \beta^{3^n} \rfloor$ are composite infinitely often. The results below give uniform infinite-family and density statements for this exact exponent sequence. They do not settle the word "every" in that problem, or Question 1.7 of arXiv:2404.19461v2 about the least Mills constant.

For a totally real cubic Pisot number write

$$f(X) = X^3 - aX^2 + bX - c, \qquad T_r = \mathrm{Tr}(\beta^{3^r}).$$

The two small conjugates have unequal moduli: equality would make them opposites and hence make their remaining conjugate β an integer. Therefore there is a fixed $s \in \{0, 1\}$ such that

$$\lfloor \beta^{3^r} \rfloor = T_r - s$$

for all sufficiently large r. Here $s = 0$ when the dominant small conjugate is negative and $s = 1$ when it is positive.

Lemma 47.1 (A uniform small-prime return condition)**.** *Let $p \equiv 2 \pmod 3$ be a prime with $p \nmid c\,\mathrm{disc}(f)$. Suppose $a \equiv s \pmod p$ and that f either splits completely or is irreducible over $\mathbb{F}_p$. Then the sequence $\lfloor \beta^{3^r} \rfloor$ has a positive lower density of composite terms.*

Proof. The roots of f lie in $\mathbb{F}_{p^e}$ with $e = 1$ or 3. Since $p \equiv -1 \pmod 3$ and e is odd, $\gcd(3, p^e - 1) = 1$. Choose $L \geq 1$ with $3^L \equiv 1 \pmod{p^e - 1}$. All roots are nonzero, so their 3^{jL}th powers equal the original roots, for every $j \geq 0$. It follows that $T_{jL} \equiv a \equiv s \pmod p$. The corresponding floors eventually exceed p, and hence are composite. The progression has density $1/L$. □

Theorem 47.2 (Relative density one among totally real cubic Pisot numbers)**.** *Let $\mathcal{T}(B)$ be the finite set of totally real cubic Pisot numbers $\beta < B$. Let $\mathcal{E}(B)$ be the subset for which the composite indices of $\lfloor \beta^{3^r} \rfloor$ have lower density zero. Then*

$$\frac{|\mathcal{E}(B)|}{|\mathcal{T}(B)|} \longrightarrow 0 \qquad (B \longrightarrow \infty).$$

In particular, Saito's Problem 1.1 has an affirmative answer for a relative density-one subset of totally real cubic Pisot numbers under this ordering.

Proof. Fix $s \in \{0, 1\}$ and a prime $p \equiv 2 \pmod 3$ with $p > 6$. Choose ordered triples (x, y, z) of distinct nonzero elements of $\mathbb{F}_p$ with $x + y + z = s$. There are p^2 ordered choices before the restrictions; the three zero-coordinate conditions and three equality conditions exclude at most $6p$ choices. Thus at least $(p^2 - 6p)/6$ distinct monic cubics with trace s split into three distinct nonzero roots. Each unordered root set corresponds to a unique coefficient pair (b, c). Consequently a proportion at least

$$\delta_p = \frac{p^2 - 6p}{6p^3}$$

of all triples $(a, b, c) \pmod p$ meet Lemma 47.1 for this fixed s.

For any finite set S of such primes, the Chinese remainder theorem shows that the proportion of integer coefficient triples in a box of side lengths comparable to B which avoid every corresponding return condition tends, as $B \to \infty$, to at most $\prod_{p \in S}(1 - \delta_p)$. The classical divergence $\sum_{p \equiv 2 \ (3)} 1/p = \infty$ implies that these finite products tend to zero as S increases. The avoiding triples therefore have count $o(B^3)$. Applying this separately for $s = 0$ and $s = 1$ and taking their union preserves the $o(B^3)$ bound.

Every $\beta < B$ under consideration has

$$|a| < B + 2, \qquad |b| < 2B + 1, \qquad 0 < |c| < B,$$

so its polynomial belongs to such a coefficient box. Conversely, there are $\gg B^3$ distinct such Pisot numbers: for each integer $2 \leq a < B - 1$, all the polynomials

$$X^3 - aX^2 - dX + c, \qquad d, c \geq 1, \quad d + c \leq a,$$

have roots in $(-1,0),(0,1),(a,a+1)$, and their negative small root has greater modulus. These cubics are irreducible, since an integer large root would leave an integral quadratic constant term of nonzero modulus less than one. Their number is $\sum_{a<B-1} a(a-1)/2 \gg B^3$, and distinct irreducible polynomials give distinct Pisot numbers. Dividing the exceptional $o(B^3)$ bound by this lower bound proves the assertion. □

The argument is a finite-residue sieve followed by a density limit. It makes no distribution assertion about the individual lacunary angles of complex conjugates. The theorem is deliberately stated for totally real cubic Pisot numbers, including the entire real configuration relevant to an algebraic Mills shadow.

Theorem 47.3 (Density one at every fixed nonzero norm)**.** *Fix a nonzero integer c. Among totally real cubic Pisot numbers $\beta < B$ with norm c and negative dominant small conjugate, a relative proportion tending to one have a positive lower density of composite terms $\lfloor \beta^{3^r} \rfloor$. This applies separately to the unit norms $c = 1$ and $c = -1$.*

Proof. Take $p \equiv 2 \pmod 3$ with $p \nmid 6c$, and impose $a \equiv 0 \pmod p$. The residual discriminant is

$$\operatorname{disc}(f) \equiv -4b^3 - 27c^2 \pmod p.$$

Cubing permutes $\mathbb{F}_p$, so exactly $(p-1)/2$ choices of b make this discriminant a nonzero square. For a separable cubic this means that Frobenius is an even permutation of the roots; its factorization type is therefore $1+1+1$ or 3. Lemma 47.1 applies with $s = 0$. Thus the fraction of good coefficient pairs $(a,b) \pmod p$ is exactly $(p-1)/(2p^2)$. The same Chinese-remainder and divergent reciprocal-prime argument gives $o(B^2)$ exceptional pairs in a box of size $O(B)^2$.

There are $\gg_c B^2$ Pisot numbers in the stated family. If $c < 0$, write $q = -c > 0$ and use $X^3 - aX^2 - dX + q$ with $1 \le d \le a - q$. If $c > 0$, take sufficiently large $a < B - 2$ and integer $3a/4 < d < 7a/8$. For $g(X) = X^3 - aX^2 - dX - c$,

$$g(-1) < 0, \qquad g(-1/2) > 0, \qquad g(0) < 0.$$

Hence its two negative roots are in $(-1,-1/2)$ and $(-1/2,0)$; the positive root is in $(a,a+2) \subset (1,B)$. Irreducibility follows as above. Each construction supplies $\gg_c B^2$ distinct polynomials, which proves the relative-density claim. □

Theorem 47.4 (Density one within Saito's shrinking conjugate locus)**.** *Fix a nonzero integer c. Let $\mathcal{S}_c(B)$ consist of the totally real cubic Pisot numbers $\beta < B$ of norm c, with conjugates $\beta, -\rho, \gamma$ satisfying*

$$0 < \sigma = |\gamma| < \rho < 1, \qquad \rho \le \min(\sigma^{17/23}, \beta^{-17/40}).$$

These are precisely the conjugate inequalities in Saito's Theorem 1.7 of arXiv:2508.16068v3, restricted to the fixed norm. Then

$$|\mathcal{S}_c(B)| \gg_c B^{63/40},$$

and the proportion of members for which $\lfloor \beta^{3^r} \rfloor$ has a positive lower density of composite indices tends to one as $B \to \infty$.

Proof. Put $q = |c| \ge 1$ and $\alpha = 23/40$. Because $q = \beta\rho\sigma$, the bound $\rho \le \beta^{-17/40}$ already implies

$$\rho^{40} \le \beta^{-17} \le q^{17}\beta^{-17}, \quad \text{hence} \quad \rho \le \sigma^{17/23}.$$

For a polynomial $X^3 - aX^2 + bX - c$ in this locus, the trace and pairwise-product formulas give

$$0 < a < B, \qquad |b| \le 2\beta\rho + \rho\sigma \le 2B^{\alpha} + 1.$$

The lower bound on a follows from the elementary fact that every totally real cubic Pisot number exceeds 2; alternatively it is immediate for the large-β portion, and finitely many smaller polynomials do not affect the density conclusion.

The fixed-norm sieve in Theorem 47.3 works unchanged in the rectangle of side lengths B and $O(B^\alpha)$: for each fixed finite set of sieve primes, both side lengths tend to infinity, so Chinese-remainder residue proportions still have the stated limits. Increasing the fixed finite set afterward shows that the number of avoiding pairs in this rectangle is $o(B^{1+\alpha})$. It remains to match this by a lower bound for actual admissible polynomials, rather than for the ambient rectangle.

Take large integers a, and integers d in the interval

$$\frac{1}{4}a^\alpha \le d \le \frac{1}{2}a^\alpha, \qquad f_{a,d}(X) = X^3 - aX^2 - dX - c.$$

Uniformly in these choices, $f_{a,d}(a) < 0$ and $f_{a,d}(a+1) > 0$, so there is a root $\beta \in (a, a+1)$. Write $u = \beta - a$. The equation for this root gives

$$u = \frac{d}{\beta} + \frac{c}{\beta^2} = \frac{d}{a}\left(1 + O_c(a^{-1})\right).$$

The other two roots have sum $-u$ and product c/β, so their quadratic discriminant is $u^2 - 4c/\beta$. As $d^2/a \asymp a^{3/20} \to \infty$, it is positive for all sufficiently large a, for either sign of c. Calling the larger small-root modulus ρ and the smaller one σ, the quadratic formula and the norm identity give, uniformly,

$$\rho = \frac{d}{a}\left(1 + O_c(a^{-3/20})\right), \qquad \sigma = \frac{q}{d}\left(1 + O_c(a^{-3/20})\right).$$

In particular the dominant small root is negative, $\sigma/\rho \to 0$, and both small roots have modulus less than one. The cubic is irreducible by the integer-root argument used above. Moreover,

$$\rho\beta^{17/40} = \frac{d}{a^{23/40}}\left(1 + O_c(a^{-3/20})\right) \le \frac{1}{2} + o(1) < 1.$$

Thus all these polynomials belong to the required locus. Taking a between $B/2$ and $B-2$ gives $\gg_c B^{1+\alpha} = B^{63/40}$ distinct members. The exceptional $o(B^{63/40})$ count from the same shrinking rectangle now proves the relative-density assertion. □

This strengthens the density conclusion inside the actual shrinking conjugate region allowed by Saito's Mills reduction. It still does not identify the hypothetical shadow of the distinguished least Mills constant: a density-zero exceptional family has not been shown empty.

48 An exact valuation theorem in the surviving unit branch

The preceding density theorems leave a possible thin exceptional set. The next calculation restricts an important part of that set without assuming any prime-distribution conjecture.

Theorem 48.1 (Exact linear 3-adic depth)**.** *Let $\varepsilon \in \{1, -1\}$ and let*

$$f(X) = X^3 - aX^2 + bX + \varepsilon$$

be an irreducible cubic, with $a \equiv \varepsilon \pmod 3$ and $b \equiv -1 \pmod 3$. Write T_r and B_r for the trace and second elementary symmetric function of the 3^rth powers of its roots, and put

$$s = v_3(a + \varepsilon b).$$

Then s is a finite positive integer and

$$\boxed{v_3(T_r - \varepsilon) = r + s \qquad (r \ge s).}$$

Proof. The norm of the roots is $c = -\varepsilon$, which is unchanged by odd powering. Put

$$x_r = T_r - \varepsilon, \quad y_r = B_r + 1, \quad z_r = x_r + \varepsilon y_r = T_r + \varepsilon B_r, \quad w_r = x_r - \varepsilon y_r = T_r - \varepsilon B_r - 2\varepsilon.$$

Newton's identities, expanded at the fixed state $(T, B, c) = (\varepsilon, -1, -\varepsilon)$, give the exact factorizations

$$z_{r+1} = z_r(3 + 3\varepsilon w_r + x_r^2 - \varepsilon x_r y_r + y_r^2),$$
$$w_{r+1} = w_r(9 + 3\varepsilon w_r + x_r^2 + \varepsilon x_r y_r + y_r^2).$$

The residue conditions ensure $3 \mid x_r, y_r$ at every index. The first multiplier is $3 \pmod 9$ and the second is divisible by 9. Therefore

$$v_3(z_r) = r + s, \qquad v_3(w_r) \geq 2r + 1.$$

The initial $z_0 = a + \varepsilon b$ cannot vanish, because $f(-\varepsilon) = -a - \varepsilon b$ and f is irreducible. Finally, $2x_r = z_r + w_r$. For $r \geq s$ the first summand has strictly smaller valuation, proving the formula. □

Lemma 48.2 (Projective return, recalled with proof)**.** *Let $p \nmid c\,\mathrm{disc}(f)$ divide T_r, and let Y_1, Y_2, Y_3 be the residual roots raised to 3^r in a finite splitting field. If the subgroup generated by their ratios has order coprime to 3, then $p \mid T_{r+jL}$ for every $j \geq 0$ and some $L \geq 1$.*

Proof. Choose L with $3^L \equiv 1$ modulo the exponent of that subgroup. For every j, the three 3^{jL}th powers of the Y_i differ from the Y_i by a common nonzero scalar. Their sum remains zero. □

Theorem 48.3 (Complete splitting forced in a positive-sign unit branch)**.** *Let β be a totally real cubic Pisot number with negative dominant small conjugate and polynomial*

$$f(X) = X^3 - aX^2 + bX + 1, \qquad a \equiv 1 \pmod 3, \quad b \equiv -1 \pmod 3, \quad v_3(a + b) = 1.$$

If $\lfloor \beta^{3^r} \rfloor$ is eventually prime, then, for every sufficiently large r, f splits completely modulo that prime. More strongly, one sufficiently late prime trace at which f does not split completely forces a positive-density progression of composite floors.

Proof. For large r, the floor equals T_r and exceeds every fixed prime divisor of $3\,\mathrm{disc}(f)$. Let $p = T_r$ be prime. Theorem 48.1 gives $p \equiv 1 \pmod 3$ and $v_3(p - 1) = r + 1$. If f is irreducible modulo p, its roots have the form u, u^p, u^{p^2}. Their ratios have orders dividing $p^2 + p + 1$, whose 3-adic valuation is exactly one. Raising to 3^r with $r \geq 1$ kills this 3-primary part, so Lemma 48.2 gives a composite progression.

If the factorization type is $1 + 2$, write the roots as $u \in \mathbb{F}_p$ and $v, v^p \in \mathbb{F}_{p^2}$. Since $v_3(p^2 - 1) = r + 1$, the 3-primary components of their 3^rth powers are in μ_3. The last two components are equal, since $p \equiv 1 \pmod 3$. Their product with the first component is the 3-primary component of the norm -1, namely one. Thus all three components are equal, and every root ratio has order coprime to 3. Projective return again applies. An eventually prime tail excludes both nonsplit cases. □

Corollary 48.4 (A positive-sign discriminant-cycle obstruction)**.** *Under the hypotheses of Theorem 48.3, suppose that $\mathrm{disc}(f)$ is not a rational square. Let D be the fundamental discriminant of its quadratic resolvent field. Iterate the cubic Newton state $(T_r, B_r, -1)$ modulo $|D|$. If any state of its eventual cycle has*

$$\chi_D(T_r) \neq 1,$$

then the floor sequence has a positive lower density of composite terms. Here χ_D is the quadratic Dirichlet character.

Proof. A cycle state occurs on an infinite arithmetic progression of indices. If its character is zero, a fixed prime divisor of D divides every trace on that progression. If the character is -1, then either the progression is eventually composite, or it contains a sufficiently late prime trace. At such a prime, Frobenius has odd sign, so the factorization type is $1+2$. Theorem 48.3 gives a composite progression. □

The exact valuation theorem extends the earlier character sieve into a positive-sign, norm--1 branch; it does not assert that the cycle obstruction always occurs. A surviving orbit could have character 1 at every cycle state, and completely split prime traces are not ruled out. Chebotarev's theorem supplies many primes of other splitting types, but does not force one of those primes to divide this prescribed sparse trace orbit after its 3-primary transient. Proving that such an appropriate recurrent divisor always exists remains the precise missing step in this approach to Saito's full Problem 1.1.

49 Square clocks: rational-function avoidance and interior gap tests

Saito's Question 1.6 asks whether the least $A > 1$ for which every $\lfloor A^{2^k} \rfloor$ is prime is irrational. The results of this section do not resolve that question. They give unconditional auxiliary theorems that restrict a rational candidate and an additional transcendence test distinct from the previously considered small-gap condition. Their deep inputs are established approximation theorems, so these deductions are not presented as a new transcendence method or as priority claims.

Write, for any $A > 1$,

$$p_k = \lfloor A^{2^k} \rfloor, \qquad u_k = A^{2^k} - p_k \in [0,1), \qquad D_k = p_{k+1} - p_k^2.$$

The exact relation

$$D_k = 2p_k u_k + u_k^2 - u_{k+1} \tag{49.1}$$

will be used repeatedly. Primality is not needed for the first two theorems.

49.1 Rational candidates cannot obey a rational-function law

Theorem 49.1 (Finite rational-function coincidences)**.** *Let $A > 1$ be rational but not an integer. For every rational function $R \in \mathbb{Q}(X)$, there are only finitely many k at which*

$$p_{k+1} = R(p_k).$$

Consequently, for every fixed finite collection of rational functions, all sufficiently late consecutive pairs (p_k, p_{k+1}) avoid all their graphs. The same statement holds with 2^k replaced by m^k for any fixed integer $m \geq 2$.

Proof. Suppose there are infinitely many coincidences. Since $p_k \to \infty$ and $p_{k+1}/p_k^2 \to 1$, the rational function must have the expansion

$$R(X) = X^2 + bX + O(1) \qquad (X \to +\infty), \qquad b \in \mathbb{Q}.$$

Equation (49.1) then gives, at the successful indices,

$$u_k = b/2 + O(p_k^{-1}).$$

Choose a positive integer q with $qb/2 \in \mathbb{Z}$. We obtain

$$\|qA^{2^k}\| = O(A^{-2^k}). \tag{49.2}$$

The Corvaja–Zannier–Dubickas approximation criterion, in the form of Saito's Theorem 2.6 in arXiv:2404.19461v2, implies that some power A^{2^j} at a successful exponent is Pisot. A rational Pisot number is an integer, whereas no positive integer power of a nonintegral rational number is an integer. This is a contradiction. Equivalently, one can use the singleton rational-base consequence of the Kulkarni–Mavraki–Nguyen theorem.

For $m \geq 2$, the asymptotic ratio forces $R(X) = X^m + bX^{m-1} + O(X^{m-2})$. Expanding $(p_k + u_k)^m$ gives $u_k = b/m + O(p_k^{-1})$, and the same proof applies. The finite-collection assertion follows by taking a finite union of the exceptional index sets. □

Corollary 49.2 (A necessary irregularity for a rational square constant)**.** *If a rational $A > 1$ has prime outputs $p_k = \lfloor A^{2^k} \rfloor$ for every $k \geq 1$, its output sequence cannot satisfy even infinitely many instances of any one fixed rational-function recurrence. In particular it cannot eventually follow a finite menu of rational polynomial recurrences.*

Proof. An integral $A > 1$ has composite outputs and is impossible. Apply Theorem 49.1. □

49.2 Transcendence from concentration inside the square interval

Theorem 49.3 (Interior algebraic-slope criterion)**.** *Suppose that for some real algebraic number $0 < \lambda < 2$, some $\varepsilon > 0$, and some $C > 0$, infinitely many k satisfy*

$$|D_k - \lambda p_k| \leq C p_k^{1-\varepsilon}. \tag{49.3}$$

Then A is transcendental. In particular,

$$D_k = p_k + O(p_k^{1-\varepsilon}) \quad \text{on an infinite subsequence} \quad \Longrightarrow \quad A \text{ is transcendental.}$$

Proof. Put $\eta = \min\{\varepsilon, 1\} > 0$. From (49.1), since $u_k, u_{k+1} \in [0, 1)$, we obtain

$$u_k = \lambda/2 + O(p_k^{-\eta}).$$

Thus on the successful subsequence

$$\|A^{2^k} + 1 - \lambda/2\| = O(A^{-\eta 2^k}). \tag{49.4}$$

Assume that A is algebraic. If λ is irrational, then $1 - \lambda/2$ is an algebraic irrational in $(0, 1)$; Kumar's Theorem 2.1 in arXiv:2207.08614v3 excludes infinitely many such approximations, with singleton base A and fixed coefficient 1. If λ is rational, the shift is a nonintegral rational number; Kumar's Theorem 2.2 applies because the coefficient 1 is an algebraic integer. In both cases select $A^{-\eta} < \theta < 1$ to absorb the implied constant into the required bound θ^{2^k}. This contradicts (49.4) and proves transcendence. □

Corollary 49.4 (Finite interior schedules)**.** *Let $\Lambda \subset (0, 2)$ be a finite nonempty set of real algebraic numbers. If, for some $\varepsilon > 0$ and $C > 0$, infinitely many k satisfy*

$$\min_{\lambda \in \Lambda} |D_k - \lambda p_k| \leq C p_k^{1-\varepsilon},$$

then A is transcendental.

Proof. One member of the finite set is successful infinitely often. Apply Theorem 49.3. □

This criterion concerns concentration near an interior fraction of the interval between consecutive squares. It does not require $D_k = o(p_k)$, and therefore is distinct from the previous small-gap criterion. The power-saving rate in (49.3) is essential to this proof: convergence $D_k/p_k \to \lambda$ without a quantitative rate is not enough to invoke the approximation theorem.

49.3 The unique algebraic polynomial-law exception

For this classification we use Saito's Theorem 2.6 and Lemma 2.7 in arXiv:2404.19461v2: exponentially close algebraic powers produce a Pisot power, and the sum of the smaller conjugate powers has a lower bound at its largest conjugate modulus, up to a polynomial factor. The argument below supplies the required quadratic classification directly.

Theorem 49.5 (Classification under an eventual polynomial law)**.** *Suppose every $p_k = \lfloor A^{2^k} \rfloor$ is prime and, for some $P \in \mathbb{Q}[X]$, one has $p_{k+1} = P(p_k)$ for all sufficiently large k. Then exactly the following alternatives apply:*

1. *$P(X) \neq X^2 + 2X - 2$, and A is transcendental;*
2. *$P(X) = X^2 + 2X - 2$, and for every sufficiently large starting index K,*

$$A^{2^K} = \frac{p_K + 1 + \sqrt{(p_K+1)^2 - 4}}{2}. \tag{49.5}$$

In particular A is algebraic and irrational in this alternative.

Thus in every case A is irrational.

Proof. The asymptotic ratio $p_{k+1}/p_k^2 \to 1$ forces $P(X) = X^2 + bX + c$, with $b, c \in \mathbb{Q}$. By (49.1),

$$u_k = b/2 + O(p_k^{-1}),$$

so $0 \leq b \leq 2$. If $0 < b < 2$, the interior-slope theorem already makes A transcendental. Suppose instead that A is algebraic. Then $b \in \{0, 2\}$, and

$$\|A^{2^k}\| = O(A^{-2^k}).$$

Saito's Theorem 2.6 gives a Pisot number $\beta = A^{2^K}$ for some K. We may increase K freely, since a positive integer power of a Pisot number is again Pisot. If β is an integer, then later outputs are proper integer powers and are composite, a contradiction. Thus its degree d is at least two.

Let $\beta_2, \ldots, \beta_d$ be its other conjugates and put $\rho = \max_{i \geq 2} |\beta_i| < 1$. The trace of β^m is an integer and differs from β^m by a quantity tending to zero. At the exponents $m = 2^j$, this trace is therefore the nearest integer for all sufficiently large j, and the preceding estimate gives

$$\left| \sum_{i=2}^{d} \beta_i^{2^j} \right| = O(\beta^{-2^j}).$$

Saito's Lemma 2.7 supplies, for some fixed $v > 0$, the lower bound $\rho^{2^j}(2^j)^{-v}$. Taking 2^jth roots yields $\rho \leq \beta^{-1}$. The nonzero integral norm consequently satisfies

$$1 \leq |N_{\mathbb{Q}(\beta)/\mathbb{Q}}(\beta)| \leq \beta \rho^{d-1} \leq \beta^{2-d}.$$

It follows that $d = 2$ and that the norm is ± 1. After increasing K once, the conjugate is positive and equals β^{-1}. The squared Pisot number still has degree two: it cannot be rational, since its conjugate would then equal itself despite having modulus less than one.

The integer traces now satisfy

$$T_j = \beta^{2^j} + \beta^{-2^j} \in \mathbb{Z}, \qquad T_{j+1} = T_j^2 - 2.$$

Since $0 < \beta^{-2^j} < 1$, the exact floors are $p_{K+j} = T_j - 1$. Hence $p_{K+j+1} = p_{K+j}^2 + 2p_{K+j} - 2$. The labels tend to infinity, so equality at infinitely many distinct arguments forces the polynomial identity $P(X) = X^2 + 2X - 2$.

Conversely, suppose this polynomial law holds from K onward. Put $t = p_K + 1 \geq 3$ and $\alpha = (t + \sqrt{t^2 - 4})/2$. Then $\alpha > 1$, $\alpha + \alpha^{-1} = t$, and the same trace recurrence proves by induction that $p_{K+n} = \alpha^{2^n} + \alpha^{-2^n} - 1$. Taking growth limits gives (49.5). The discriminant $t^2 - 4$ is not a square for $t \geq 3$, so this α is irrational; A therefore cannot be rational. □

The exceptional polynomial has not been shown to possess an infinite all-prime orbit here. The theorem identifies the only possible algebraic law under the stated hypothesis; it does not establish the existence of that law for Saito's least square constant.

49.4 What still prevents a solution of Question 1.6

For a rational candidate $A = a/b > 1$ in lowest terms, $b > 1$, the elementary lower bound

$$\{A^{2^k}\} \geq b^{-2^k}$$

is too small to contradict the floor inequalities. Neither primality nor minimality has been proved here to supply the exponentially accurate translated approximations required above. Likewise, the Type A/B theorems in arXiv:2508.16068v3 require a ratio limsup greater than 40/19, while the square clock has ratio exactly 2. No argument in this section supplies that missing input. The unconditional irrationality of the least square prime constant remains unresolved by the present work.

50 Square clocks: a dimension gap and the remaining computability obstruction

The global dimension results for unbounded multipliers and iterated exponential clocks do not imply local dimension statements for a fixed finite prefix. The present question is a local strengthening of Saito's dimension theorem and Saito–Takeda's perfectness theorem: what happens when every sufficiently late multiplier is exactly 2? The extra hypothesis of infinitely many multipliers at least 3 in Saito–Takeda's Theorem 1.2 does not apply.

Fix $E_0 > 0$, put $E_n = E_0 2^n$ for $n \geq 0$, and define

$$W_2(E_0) = \{A > 1 : \lfloor A^{E_n} \rfloor \in \mathbb{P} \text{ for every } n \geq 0\}.$$

This set is nonempty, closed, and has Hausdorff dimension 1, by the published existence, closedness, and dimension theorems. None of those facts alone implies the full local dimension asserted below outside a small exceptional set.

A prime label p at exponent E has the half-open root cell $[p^{1/E}, (p+1)^{1/E})$. For an admissible square-clock constant all powers A^{E_n} are nonintegers: an integral value would have a composite square at the next index. Thus every admissible point lies in the interior of each of its cells. A valid later prime cell has closure strictly inside its parent: if q follows p, then $p^2 < q$ and $q + 1 < (p + 1)^2$, since both p^2 and $(p + 1)^2 - 1$ are composite. We use half-open cells for finite prefix conditions and closed, strictly nested descendant cells for limit constructions.

50.1 Good square seeds and a sparse exceptional set

Matomäki's prime-interval estimate [25, Lemma 9] gives constants $d, D > 0$ such that, for sufficiently large x, at most $Dx^{1/6}$ disjoint intervals $[u, u + u^{1/2}] \subset [x, 2x]$ contain at most $du^{1/2}/\log u$ primes. Choose a fixed positive integer K sufficiently large, and set

$$H(p) = \left\lceil \frac{p}{K\lceil \log_2 p \rceil} \right\rceil.$$

For sufficiently large p, this is at most $dp/(2\log p)$. Call a prime p *good* if $[p^2, p^2+p]$ contains at least $H(p)$ primes, and *poor* otherwise. A sufficiently large threshold P will be fixed throughout. Both predicates are decidable by finite integer arithmetic and primality tests; numerical values of the universal constants are not evaluated in this note. In every computability or semidecision statement below, fixed integers K, P with the stated properties are hardcoded into the algorithm. Their existence proves ordinary computability, even though this argument does not supply their numerical values or a practical digit-computation procedure.

Lemma 50.1 (Few poor prime seeds)**.** *As $X \to \infty$,*

$$\#\{p \in \mathbb{P} \cap [X, 2X] : p \text{ is poor}\} \ll X^{1/3}.$$

Proof. The intervals $[p^2, p^2+p]$ for distinct primes p are disjoint. For $p \in [X, 2X]$, they lie in $[X^2, 4X^2+2X]$. Cover this range by a fixed number of dyadic intervals. At each dyadic boundary at most one of the disjoint intervals straddles the boundary; discard these finitely many intervals. All remaining poor intervals are exceptional intervals in Matomäki's estimate, with the squared scale $x \asymp X^2$. The bound is consequently $O((X^2)^{1/6}) = O(X^{1/3})$. □

Lemma 50.2 (A good seed has full-dimensional continuations)**.** *For every good prime $p \geq P$ and every $E > 0$, the square-continuation set*

$$\{A \in (p^{1/E}, (p+1)^{1/E}) : \lfloor A^{E2^j} \rfloor \in \mathbb{P} \ (j \geq 0)\}$$

has Hausdorff dimension 1. If E is computable, it contains a computable element and a computably embedded binary Cantor set.

Proof. Among the at least $H(p)$ prime children $q \in [p^2, p^2+p]$, at most $Dp^{2/3}$ are poor, by the same exceptional interval argument at the scale p^4. Thus, for sufficiently large p, there remain at least $ap/\log p$ good children, with a fixed $a > 0$. Their root cells are strictly inside the parent cell: $q > p^2$ because p^2 is composite, and $q+1 \leq p^2+p+1 < (p+1)^2$. Iterating gives a tree of good primes satisfying the hypotheses of Saito's Proposition 5.1 [33] with $R = 2$ and $L = 1$.

At a descendant prime q, that proposition bounds the dimension of its continuation subtree below by $(1 + 2/(q\log q))^{-1}$, after the appropriate power change of variable. Every such subtree lies in the original seed cell, and descendant labels tend to infinity. Taking these lower bounds to 1 proves the claim. Choosing the least good child at each step gives a computable limit. Choosing the first or second good child according to a binary sequence gives the stated computable embedding; the separated child cells give a computable inverse on its image. □

Theorem 50.3 (An exceptional set of dimension at most one third)**.** *Let (F_n) be any real exponent sequence with $F_n \geq c2^n$ for a constant $c > 0$. The set*

$$\mathcal{E}_F = \{A > 1 : \lfloor A^{F_n} \rfloor \text{ is a poor prime for infinitely many } n\}$$

has Hausdorff dimension at most $1/3$. In particular, outside a set of dimension at most $1/3$, every sufficiently late prime output of a square clock is good.

Proof. Fix a compact interval $[a, b] \subset (1, \infty)$ and $s > 1/3$. For large n, the prime root cell corresponding to a label p that meets $[a, b]$ has diameter at most

$$\frac{B}{F_n p}$$

for a constant B depending only on b; this follows from the mean value theorem, since $p^{1/F_n} \leq b$. Such labels satisfy $p \geq a^{F_n} - 1$. From Lemma 50.1 and a dyadic sum,

$$\sum_{\substack{p \geq Y \\ p \text{ poor}}} p^{-s} \ll_s Y^{1/3-s}.$$

Therefore the sum of the sth powers of diameters of all poor-label cells meeting $[a, b]$ at level n is

$$\ll_{s,a,b} F_n^{-s} a^{F_n(1/3-s)}.$$

This is summable in n, because $F_n \geq c2^n$ and $s > 1/3$. For every N, the union of these covers over $n \geq N$ covers the limsup set. The total s-content tends to zero as $N \to \infty$, proving that its s-dimensional Hausdorff measure is zero. Let $s \downarrow 1/3$ and use a countable cover of $(1, \infty)$ by compact intervals. □

50.2 A local dimension gap and a canonical perfect part

Theorem 50.4 (No intermediate local dimensions)**.** *For every open interval $U \subset (1, \infty)$,*

$$\boxed{\dim_H(U \cap W_2(E_0)) \leq \tfrac{1}{3} \quad \text{or} \quad \dim_H(U \cap W_2(E_0)) = 1.}$$

More precisely, if $A \in W_2(E_0)$ has good prime outputs at infinitely many indices, then every neighborhood U of A satisfies $\dim_H(U \cap W_2(E_0)) = 1$.

Proof. The half-open root cell of an output at level n respects every earlier prime condition, because the ratios $E_n/E_j = 2^{n-j}$ are integers. The point A lies in its interior, and these cells shrink to A. If good outputs occur at arbitrarily late indices, a good seed cell is contained in any prescribed neighborhood of A, and Lemma 50.2 applies.

Suppose $\dim_H(U \cap W_2(E_0)) > 1/3$. By Theorem 50.3, the set $U \cap W_2(E_0)$ cannot be contained in the exceptional limsup set. It thus contains a point whose outputs are all good from some index onward. The preceding argument gives a full-dimensional subset inside U. □

Theorem 50.5 (A perfect part with full local dimension)**.** *Define*

$$R = \{A \in W_2(E_0) : \dim_H(U \cap W_2(E_0)) = 1 \text{ for every open } U \ni A\}.$$

Then R is a nonempty closed perfect set, every open interval meeting R intersects R in dimension 1, and

$$\dim_H(W_2(E_0) \setminus R) \leq \tfrac{1}{3}.$$

Every point of $W_2(E_0) \setminus R$ has only poor outputs from some index onward. If E_0 is computable, R has a dense sequence of computable elements, and every nonempty relative open subset of R contains every Turing degree.

Proof. The complement of R in $W_2(E_0)$ is relatively open. It is a union of countably many relative intersections with rational open intervals having dimension less than 1, hence at most $1/3$ by Theorem 50.4. Countable stability of dimension proves the bound. Since $W_2(E_0)$ is closed, R is closed; since the full set has dimension 1, R is nonempty and has dimension 1. If U meets R, then $U \cap W_2(E_0)$ has dimension 1. Removing the set of dimension at most $1/3$ leaves $\dim_H(U \cap R) = 1$. In particular R has no isolated point. If a point outside R had infinitely many good outputs, the preceding theorem would place it in R, which proves the assertion about eventual poor outputs.

Assume E_0 is computable. Enumerate all finite valid square-prime prefixes ending at a good prime at least P, and for each prefix choose the computable continuation that always takes the least good child. Every enumerated point belongs to R. Conversely every open interval meeting R has dimension 1 in $W_2(E_0)$, hence contains a point outside the exceptional set, and consequently contains a complete valid good seed cell. An enumerated continuation lies in this cell. The sequence is therefore dense in R. The binary construction in Lemma 50.2, inside the same cell, lies in R and preserves Turing degrees. □

Corollary 50.6 (A semidecision procedure for full-dimensional prefixes)**.** *Suppose E_0 is computable. Given a finite valid square-prime prefix, there is an algorithm that halts exactly when its completion set has Hausdorff dimension* 1*. It searches for a later valid descendant with a good prime label at least P. There is also an algorithm that halts exactly when the completion set is empty. Thus the only unresolved output of the combined search is a nonempty completion set of dimension at most* $1/3$*.*

Proof. Reachability of a good descendant is a search through a finitely branching, decidable tree of prime labels. If a good descendant is found, Lemma 50.2 proves dimension 1. If none exists, every infinite branch is eventually poor and its real limit belongs to the exceptional set, so the dimension is at most $1/3$. For emptiness, inspect successive finite levels of the prefix tree. An empty level certifies emptiness. Conversely, if all levels are nonempty, König's lemma gives an infinite branch, whose strictly nested prime root cells define a genuine continuation. Hence an empty completion set eventually has an empty finite level. □

50.3 Isolated points and the least square constant

Proposition 50.7 (Every isolated square constant is computable)**.** *If E_0 is computable, every isolated point of $W_2(E_0)$ is computable. Every such point has only poor prime outputs from some index onward.*

Proof. The eventual-poor assertion follows from Theorem 50.5. For computability, choose a finite prime prefix cell whose intersection with $W_2(E_0)$ is the isolated singleton. Its tree of finite extensions is decidable and finitely branching, with exactly one infinite path. For any requested level n, search for a later finite level at which all surviving nodes have the same ancestor at level n. Such a level exists: otherwise a second infinite path would exist by König's lemma. The common ancestor determines a root cell of length tending effectively to zero with n, which computes the singleton. This proves existence of an algorithm for each isolated point; it does not give an algorithm deciding which prefixes isolate a point. □

Proposition 50.8 (Where noncomputability of the least square constant could hide)**.** *Suppose E_0 is computable, and let $\xi_2 = \min W_2(E_0)$. If $\xi_2 \in R$, then ξ_2 is computable. Consequently, if ξ_2 is not computable, it must be nonisolated, must have eventually only poor outputs, and must have a neighborhood U for which*

$$0 \le \dim_H(U \cap W_2(E_0)) \le \tfrac{1}{3}.$$

No unconditional computability or irrationality claim for ξ_2 is made here.

Proof. The least lower endpoint ℓ_n among all finite valid prefix cells through level n is computable by enumerating final primes and testing all earlier outputs by exact integer roots. The numbers ℓ_n increase and are bounded above by ξ_2. Let $\ell = \lim \ell_n$. We verify $\ell \in W_2(E_0)$, including the possible endpoint issue. If $\ell^{E_j} = h \in \mathbb{N}$, then $h \ge 2$. If the approximants eventually equal ℓ, the later output h^2 is composite, contrary to their finite validity. Otherwise they approach ℓ strictly from below; at level $j+2$ their outputs are then eventually $h^4 - 1$, also composite. Thus ℓ^{E_j} is never an integer. Its floor is eventually the corresponding prime floor of ℓ_n, so every output at ℓ is prime. Hence $\ell = \xi_2$. This gives lower semicomputability.

The computable points enumerated densely in R in Theorem 50.5 all lie above ξ_2. If $\xi_2 \in R$, their infimum is ξ_2, giving upper semicomputability by a running minimum of rational upper approximations. The two semicomputabilities make ξ_2 computable. If it is not computable, it lies outside R, and Theorem 50.5 gives the eventual-poor and local dimension conclusions. Proposition 50.7 excludes isolation. □

Remark 50.9 (What remains unresolved). These arguments do not prove $R = W_2(E_0)$, do not rule out isolated square constants, and do not show that every feasible finite square prefix has an infinite continuation. The possible exceptional region is smaller in dimension, but can still contain distinguished minima. The passage from a sparse exceptional set of prime labels to the absence of an infinite exceptional chain is precisely the missing step. No new algebraic-independence assertion about the boundary elements in Saito–Takeda's Question 1.1 follows from this dimension argument.

Remark 50.10 (A general quantitative transfer). The exponent $1/3$ in the exceptional-dimension and local-gap conclusions comes only from the counting estimate in Lemma 50.1. If that estimate is replaced by $O_\varepsilon(X^{\tau+\varepsilon})$ for every $\varepsilon > 0$, with $\tau < 1/2$, the same proofs replace $1/3$ by τ. The restriction $\tau < 1/2$ ensures that among the $\gg p/\log p$ candidate children of a good seed, only $O_\varepsilon(p^{2\tau+2\varepsilon}) = o(p/\log p)$ are poor. This is a transfer principle, not a claim of a new prime-distribution estimate.

51 Logarithmic prime-free windows for non-reversible clocks

The previously established return theorem for an exponential polynomial plus a nonzero linear drift gives composite windows of every fixed radius. Here the radius grows logarithmically with the index. This is an extension of the quantitative conclusion of Saito's prime-free-window theorem to this particular non-reversible family; it is not a second resolution of his already treated Problem 1.7.

We use the following previously proved arithmetic input, with all its quantifiers stated explicitly. It is recalled rather than counted as a new result.

Lemma 51.1 (Universal return input). *Let $a \geq 2$, $b \neq 0$, $e \in \mathbb{Z}$, and $P \in \mathbb{Z}[X]$. For $R(n) = P(a^n) + bn + e$, there is m_0 such that, for every $m \geq m_0$ and every positive integer q, the congruence $R(n) \equiv R(m) \pmod q$ has arbitrarily large solutions n. One may take $m_0 = 1$ when $|b| = 1$ and $m_0 = \max(1, \lceil 2\log_2 |b| \rceil)$ otherwise.*

The input, including the displayed seed threshold, is proved in Theorem 24.2. The quantitative argument below does not require a constructive bound from that proof; eventual periodicity supplies the needed bound independently.

Lemma 51.2 (A uniform bound on a recurrent index). *Under the hypotheses of Lemma 51.1, fix $m \geq m_0$ and $q \geq 2$. There is a solution of $R(n) \equiv R(m) \pmod q$ satisfying*

$$m < n \leq \max\{m+1, \lceil \log_2 q \rceil\} + q^2.$$

Proof. Write $q = uv$, with every prime divisor of u dividing a and $\gcd(v, a) = 1$. For $n \geq \lceil \log_2 q \rceil$ one has $a^n \equiv 0$ modulo u, while a^n is periodic modulo v with period $t = \mathrm{ord}_v(a)$ (take $t = 1$ if $v = 1$). Thus $R(n)$ is periodic from that point with period $\mathrm{lcm}(q, t) \leq q\varphi(v) \leq q^2$. Universal return ensures that a solution occurs in the eventual cycle. Every interval of one full period within that cycle contains one. Apply this to the interval starting at $\max\{m+1, \lceil \log_2 q \rceil\}$. □

Theorem 51.3 (Logarithmic windows beyond reversibility). *Let α be a Pisot or Salem number of degree D. Let*

$$R(n) = P(a^n) + bn + e,$$

where $a \geq 2$, $b \neq 0$, $e \in \mathbb{Z}$, and $P \in \mathbb{Z}[X]$ has positive leading coefficient and positive degree. For every $\varepsilon > 0$, there are infinitely many indices n such that all integers in

$$\left[\lfloor \alpha^{R(n)} \rfloor - \left(\frac{1}{4D} - \varepsilon\right) \log n, \lfloor \alpha^{R(n)} \rfloor + \left(\frac{1}{4D} - \varepsilon\right) \log n \right]$$

are composite. Here log *denotes the natural logarithm; a nonpositive radius is interpreted trivially. The constant is independent of the degree and coefficients of P, and of the nonzero coefficient b.*

Proof. Put $T(k) = \mathrm{Tr}_{\mathbb{Q}(\alpha)/\mathbb{Q}}(\alpha^k)$. It is an integer linear recurrence of order D, and

$$|T(k) - \alpha^k| \le D - 1, \qquad |T(k) - \lfloor \alpha^k \rfloor| \le D. \tag{51.1}$$

There is a fixed integer s_0 such that, for every prime p, the sequence $T(k) \bmod p$ is periodic for $k \ge s_0$, with some period $\ell_p \le p^D$. Indeed, its state vector has at most p^D states. For primes not dividing the nonzero trailing coefficient its transition matrix is invertible, hence the sequence is purely periodic. There are only finitely many other primes, so their preperiods have a common bound. This is also Saito's recurrence-period lemma.

Let $m \to \infty$ through integers satisfying $m \ge m_0$ and $R(m) \ge s_0$. Write

$$t = T(R(m)), \qquad H = t - 2.$$

For all large m, t is a positive integer and $H \ge 0$. Every integer $t + j$, $|j| \le H$, lies in $[2, 2t - 2]$, so it has a prime divisor $p_j \le 2t - 2$. Let

$$L = \mathrm{lcm}\{\ell_p : p \le 2t - 2,\ p \text{ prime}\}.$$

By the prime number theorem in the form $\vartheta(x) = \sum_{p \le x} \log p = x + o(x)$,

$$\log L \le D\vartheta(2t - 2) = (2D + o(1))t. \tag{51.2}$$

For $L \ge 2$, Lemma 51.2 gives $n > m$ with $R(n) \equiv R(m) \pmod{L}$ and

$$n \le L^2 + \max\{m + 1, \lceil \log_2 L \rceil\}.$$

If $L = 1$, take $n = m + 1$, for which the same congruence is automatic. Since R is eventually increasing and $R(m + 1) - R(m) \to \infty$,

$$\frac{T(R(m + 1))}{T(R(m))} \longrightarrow \infty.$$

Thus for all sufficiently large m and every $|j| \le H$, $T(R(n)) + j > 2t - 2 \ge p_j$. On the other hand, periodicity gives

$$T(R(n)) + j \equiv T(R(m)) + j \equiv 0 \pmod{p_j}.$$

Each integer in the trace window $[T(R(n)) - H, T(R(n)) + H]$ is therefore composite. Equation (51.1) transfers this to a window of radius $H - D = t - D - 2$ around the required floor.

Finally $\log(m+1) = o(t)$, because $R(m)$ grows exponentially and $t = \alpha^{R(m)} + O(1)$. Combining the bound on n with (51.2) gives

$$\log n \le (4D + o(1))t.$$

Consequently $t - D - 2 \ge (1/(4D) - \varepsilon) \log n$ for all sufficiently large m. The selected indices are unbounded since $n > m$, and the proof is complete. □

For example, every Pisot or Salem number α of degree D has infinitely many prime-free windows of radius $(1/(4D) - \varepsilon) \log n$ around $\lfloor \alpha^{3^n + n} \rfloor$ and $\lfloor \alpha^{2^{5n} + 7n - 4} \rfloor$. The theorem does not include a pure geometric clock a^n: its nonzero linear coefficient is the hypothesis that makes universal return available. No improvement of Saito's constant in the reversible case is asserted.

52 A finite-cover classification for quadratic norm-one unit clocks

This section exhibits both fixed-divisor cascades and all-composite sequences for which every fixed finite set of prime divisors eventually fails. The factorization ingredient is classical and underlies the quadratic-unit results discussed by Cass and Saito. The exact gcd and cover classification are supplied explicitly here, with no priority claim.

Let $V_0(X) = 2$, $V_1(X) = X$, and $V_{r+2}(X) = XV_{r+1}(X) - V_r(X)$. These integer polynomials satisfy

$$V_r(z + z^{-1}) = z^r + z^{-r}, \qquad V_r(V_s(X)) = V_{rs}(X).$$

Their values at $X = 1$ are the six-periodic sequence

$$V_r(1) = 2\cos(r\pi/3) = 2, 1, -1, -2, -1, 1 \quad (r \equiv 0, 1, 2, 3, 4, 5 \pmod 6). \tag{52.1}$$

Theorem 52.1 (Exact pairwise gcd)**.** *Let $A \geq 3$ be an integer and let $\beta > 1$ be the root of $X^2 - AX + 1$. For $a \geq 2$ put $F_n = \lfloor \beta^{a^n} \rfloor$, $n \geq 1$. For every $m < n$,*

$$\gcd(F_m, F_n) = \gcd(F_m, c_a), \qquad c_a = \begin{cases} 0, & \gcd(a, 6) = 1, \\ 2, & 2 \mid a,\ 3 \nmid a, \\ 1, & 6 \mid a, \\ 3, & 3 \mid a,\ 2 \nmid a. \end{cases}$$

In particular a fixed finite set of primes covers every sufficiently late F_n if and only if at least one of the following holds:

1. $\gcd(a, 6) = 1$;
2. *a is even, $3 \nmid a$, and A is odd;*
3. *a is odd, $3 \mid a$, and $A \equiv 1 \pmod 3$.*

In each listed case one prime suffices. In every other case the F_n are pairwise coprime and their least prime factors tend to infinity.

Proof. The conjugate of β is $\beta^{-1} \in (0, 1)$, so $F_n = V_{a^n}(A) - 1$. Reducing the composition identity modulo F_m gives

$$F_n \equiv V_{a^{n-m}}(1) - 1 \pmod{F_m}.$$

Equation (52.1) proves the asserted gcd formula. If $\gcd(a, 6) = 1$, it shows $F_1 \mid F_n$ for every n, and any prime divisor of F_1 is proper at all sufficiently late indices.

If a is even and not divisible by three, then $V_{a^n}(A)$ is odd exactly when A is odd. For even A this follows directly from the recurrence modulo two. For odd A, that recurrence has period three with values $0, 1, 1$, and $3 \nmid a^n$. Thus in this case either every F_n is even or every F_n is odd; the gcd formula gives the second alternative in the statement, or pairwise coprimality, respectively.

If a is odd and divisible by three, reduction modulo three gives $V_{a^n}(A) \equiv A \pmod 3$. To see this, if $A \equiv 0$, the recurrence makes every odd-index value zero. If $A \equiv 1$, the six-periodic values in (52.1) have all odd-index values congruent to one modulo three. Finally $V_r(-X) = (-1)^r V_r(X)$ handles $A \equiv -1$. Thus the gcd formula gives a common divisor three exactly when $A \equiv 1 \pmod 3$, and otherwise pairwise coprimality. The case $6 \mid a$ is pairwise coprime immediately.

A pairwise coprime sequence of integers greater than one cannot have an eventual cover by finitely many primes. Each bounded prime occurs in at most one term, so its least prime factors tend to infinity. □

Proposition 52.2 (All composite, but no finite prime cover)**.** *Let $A \geq 4$ be even, let β be the root greater than one of $X^2 - AX + 1$, and let a be even, not divisible by three, and have a prime divisor $q \geq 5$. Then all the integers $\lfloor \beta^{a^n} \rfloor$, $n \geq 1$, are composite and pairwise coprime. In particular,*

$$\boxed{\lfloor (2+\sqrt{3})^{10^n} \rfloor \quad (n \geq 1)}$$

is an explicit all-composite sequence whose least prime factors tend to infinity. No fixed finite prime-cover argument can certify all its terms.

Proof. Pairwise coprimality follows from Theorem 52.1. Since $q \equiv \pm 1 \pmod 6$, the integer polynomial $V_q(X) - 1$ is divisible by $X - 1$. Write $a^n = qu$. Then

$$\lfloor \beta^{a^n} \rfloor = V_q(V_u(A)) - 1$$

is divisible by $V_u(A) - 1 \geq A - 1 \geq 3$. This divisor is smaller than the number itself, since $q > 1$ and $\beta^{qu} > \beta^u$ imply strict increase of the traces. The quotient is an integer greater than one, proving compositeness. The explicit example takes $A = 4, a = 10$. □

53 A smaller unconditional cutoff in Saito's Type C

All source theorem numbers in this section refer to Saito's *Transcendency of variants of Mills' constant*, arXiv:2508.16068v3. His Theorem 1.9(C), proved using Theorem 2.6, establishes transcendence of $\xi(r3^k - 1)$ for $r \geq 4.003 \times 10^{14}$. The discussion preceding Corollary 2.7 asks whether the threshold can be replaced by 1; that corollary obtains this conclusion assuming a prime in $[x, x + 3x^{2/3}]$ for every $x \geq 1$. We lower the unconditional threshold, but do not reach $r = 1$.

Write $A_r = \xi((r3^k - 1)_{k \geq 1})$ and

$$D_r = \begin{cases} 1, & r \text{ even}, \\ 2, & r \text{ odd}. \end{cases}$$

Saito's Type-B and Type-C reductions give the following facts: if A_r is algebraic, there is a divisor g of D_r for which $\beta = A_r^g$ is a cubic Pisot number, and

$$\lfloor A_r^{r3^k - 1} \rfloor = \mathrm{Tr}(\beta^{(r3^k-1)/g}) \quad \text{for all sufficiently large } k. \tag{53.1}$$

Here the trace equality is Proposition 3.1(iv); all sufficiently large indices belong to its set I because the ratios exceed 3. With

$$x_0 = \exp(3\exp(32.76)), \qquad M = 2x_0^{1/3} + 1,$$

Theorem 2.6 also gives

$$\beta \leq M^{g/(3r-1)} \leq M^{D_r/(3r-1)}. \tag{53.2}$$

The explicit prime-interval input here is exactly the one used by Saito, from Mossinghoff–Trudgian–Yang and Cully-Hugill; no stronger prime-gap assumption is introduced.

Lemma 53.1 (The complete cubic Pisot list below two)**.** *The cubic Pisot numbers below 2 are the large real roots of*

$$X^3 - X - 1, \quad X^3 - X^2 - X - 1, \quad X^3 - X^2 - 1, \quad X^3 - 2X^2 + X - 1.$$

If α is the root of the first polynomial, the root of the last polynomial is α^2.

Proof. First, a totally real cubic Pisot number β exceeds 2. If its other roots are u, v, then $|uv| \geq 1/\beta$, and its minimal polynomial satisfies

$$1 \leq |f(1)f(-1)| = (\beta^2 - 1)(1 - u^2)(1 - v^2) \leq (\beta^2 - 1)(1 - 1/\beta)^2.$$

The right side is at most $3/4$ when $\beta \leq 2$.

Thus a cubic Pisot number $1 < \beta < 2$ has a nonreal conjugate pair. Its positive integer norm is less than 2, hence equals 1. Write its polynomial as $X^3 - aX^2 + bX - 1$. The trace bounds give $a \in \{0, 1, 2, 3\}$. Since the polynomial is negative at 1 and positive at 2, its integer coefficients satisfy

$$2a - 3 \leq b \leq a - 1.$$

This leaves the four displayed polynomials and the two possibilities $X^3 - 2X - 1$ and $X^3 - 3X - 1$. The former is reducible, and the latter has three real roots and so cannot be Pisot below 2. Each displayed polynomial is irreducible, has one real root in $(1, 2)$, and has a nonreal pair with common modulus $\beta^{-1/2} < 1$. Finally direct substitution, using $\alpha^3 = \alpha + 1$, shows that α^2 is the root of the fourth polynomial. □

Lemma 53.2 (Finite trace certificates for the four small candidates)**.** *Set $R = 400300000000000$. For $1 \leq r \leq R$, none of the four cubic Pisot numbers $\beta < 2$ can satisfy (53.1) with $g \mid D_r$ and all outputs prime.*

Proof with exact finite-ring certificates. Write $f(X) = X^3 - aX^2 + bX - 1$. For a prime p, work in $\mathbb{F}_p[X]/(f)$, and let t be the multiplicative order of X. If $X^e = v_0 + v_1X + v_2X^2$, its trace modulo p is

$$\tau(e) = 3v_0 + av_1 + (a^2 - 2b)v_2.$$

The powers and traces are computed exactly by the update

$$(v_0, v_1, v_2) \longmapsto (v_2, v_0 - bv_2, v_1 + av_2) \pmod{p}, \qquad (v_0, v_1, v_2) = (1, 0, 0) \text{ initially.}$$

For $g = 1$, the exponent begins at $3r - 1$ and evolves by $e \mapsto 3e + 2 \pmod{t}$. A zero of τ on its eventual cycle gives infinitely many composite traces. The following finite certificates show that avoiding such a zero forces $r \equiv 0 \pmod{t}$ for every listed pair (p, t):

f	(p, t)
$X^3 - X - 1$	$(2, 7), (41, 1723), (131, 17293), (347, 173), (397, 52669)$
$X^3 - X^2 - X - 1$	$(5, 31), (31, 331), (59, 3541), (71, 5113), (89, 8011)$
$X^3 - 2X^2 + X - 1$	$(2, 7), (41, 1723), (71, 5113), (131, 17293),$ $(347, 173), (409, 55897)$

The respective least common multiples of these orders are

$$1900442940636401, \qquad 1488257702170843, \qquad 10312502210725960469,$$

all larger than R. Thus no positive $r \leq R$ can avoid every obstruction. For $X^3 - X^2 - 1$, modulo 2 the order is 7 and every eventual exponent cycle has a zero trace, for both $g = 1$ and $g = 2$.

For $g = 2$, necessarily r is odd; write $r = 2j + 1$. The initial exponent is $3j + 1$ and evolves by $e \mapsto 3e + 1$. For the first and second polynomials, modulo 7 the order is 48 and every eventual exponent cycle has a zero trace. For the fourth polynomial $\beta = \alpha^2$, hence $A_r = \alpha$; its traces are exactly those of the first polynomial in the already excluded case $g = 1$.

These certificates require no approximated roots or prime outputs. They are verified by enumerating the indicated finite exponent sets, marking each eventual cycle that contains a zero trace, and testing all possible initial residues. The complete standard-library verifier in

Appendix C reproduces the calculation; it may be copied verbatim into a Python file. A zero appearing on a cycle recurs infinitely often, and the corresponding positive traces tend to infinity, so they eventually exceed p. They therefore contradict the eventual prime trace identity (53.1). □

Theorem 53.3 (An unconditional Type-C cutoff improvement)**.** *The full minimum $\xi((r3^k-1)_{k\ge1})$ is transcendental in each of the following ranges:*

1. *every integer $r \ge 162400000000000$;*
2. *every even integer $r \ge 81200000000000$.*

More precisely, it suffices that

$$r > \frac{D_r \log(2x_0^{1/3}+1)}{3\log 2} + \frac{1}{3}, \qquad x_0 = \exp(3\exp(32.76)).$$

Proof. For $r \ge 4.003 \times 10^{14}$ this is Saito's existing theorem. For smaller r, algebraicity would produce a cubic Pisot number $\beta = A_r^g$ as above. Lemma 53.2 excludes $\beta < 2$, and $\beta = 2$ is impossible because its degree is three. Thus $\beta > 2$. On the other hand, the displayed sufficient condition and (53.2) give $\beta < 2$, a contradiction.

The rounded integer cutoffs have rigorous slack. For example,

$$e^{32.76} < 168845000000000, \qquad \log 2 > 693147/10^6, \qquad \log(2e^{e^{32.76}}+1) < e^{32.76}+1$$

give both cutoffs by rational arithmetic. The first two inequalities are certified by, respectively, a Taylor sum with a geometric upper bound on its tail and the positive series $\log 2 = 2\sum_{j\ge0}((2j+1)3^{2j+1})^{-1}$; these rational checks are included in the verifier. □

The gain is a factor of about 2.46 compared with using the smallest Pisot number as the lower bound (and the corresponding parity refinement). The threshold remains large. No conclusion for $r = 1$, or for all positive r, is claimed. In particular this does not replace the prime-gap hypothesis of Saito's Corollary 2.7.

54 Exact obstructions to the proposed universal-return routes

Proposition 54.1 (Every Type-C seed obstructs universal return)**.** *For $C_n = r3^n - 1$, every seed m fails condition (B6), even if the index-set restriction in that condition is discarded.*

Proof. For every $n > m$,

$$v_3(C_n - C_m) = v_3(r) + m.$$

But $3 \nmid C_m$. Taking $L = 3^{v_3(r)+m+1}$ makes $C_n \equiv C_m \pmod{LC_m}$ impossible. Thus deleting any finite prefix cannot repair this particular return condition. □

Proposition 54.2 (A coupled obstruction for $3^n + n^2$)**.** *For the full schedule $C_n = 3^n + n^2$, condition (B6) fails at the first seed. Specifically,*

$$C_n \not\equiv 4 \pmod{9\cdot 757} \qquad (n \ge 2).$$

Proof. For $n \ge 2$, return modulo 9 would give $n^2 \equiv 4 \pmod 9$, hence $n \equiv 2$ or $7 \pmod 9$. The prime 757 has $\mathrm{ord}_{757}(3) = 9$; indeed $3^9 \equiv 1$, $3^3 \equiv 27 \not\equiv 1$. Consequently return modulo 757 would require

$$n^2 \equiv 4 - 3^2 = -5 \quad \text{or} \quad n^2 \equiv 4 - 3^7 \equiv 88 \pmod{757}.$$

Both are quadratic nonresidues. Quadratic reciprocity gives $(-5/757) = (5/757) = (2/5) = -1$, while $(88/757) = (2/757)(11/757) = (-1)(9/11) = -1$. Thus no return is possible. Since $C_1 = 4$, the choice $L = 9 \cdot 757$ suffices to disprove (B6). □

This last proposition rules out a direct full-schedule application of the universal-return clause. It does not rule out eventual return after a sufficiently large prefix, does not disprove transcendence, and does not settle the full minimum for $3^n + n^2$. The previously proved sufficiently-late-tail transcendence remains the available positive conclusion for that example.

55 A density-one version of the prime–totient quotient conjecture

Let

$$T(N) = \sum_{p \le N} \varphi\Big(\Big\lfloor \frac{N}{p} \Big\rfloor\Big), \qquad \kappa = \frac{1}{\zeta(2)}.$$

Conjecture 10 in Saito–Suzuki–Takeda–Yoshida, *Some remarks on the $[x/n]$-sequence*, arXiv:2312.15642v1, asks whether $T(x) \sim \kappa x \log\log x$ pointwise as $x \to \infty$. The paper appeared in Research in Number Theory **11** (2025), Article 50; the conjecture number here refers specifically to the arXiv version. We prove the following density and mean statements. They do not establish the pointwise conjecture. An earlier announced pointwise formula and a difficulty with its proof are discussed below.

Theorem 55.1 (Bounded mean absolute error and density-one asymptotic)**.** *For every sufficiently large integer X,*

$$\boxed{\frac{1}{X} \sum_{X \le N < 2X} \left| \frac{T(N)}{N} - \kappa \log\log N \right| \ll 1.}$$

Consequently, for every fixed $\varepsilon > 0$,

$$\#\left\{ X \le N < 2X : \left| \frac{T(N)}{N \log\log N} - \kappa \right| > \varepsilon \right\} \ll_\varepsilon \frac{X}{\log\log X} = o(X).$$

In particular, the conjectured asymptotic holds on a subset of the positive integers of natural density one.

The proof uses only the identity $\varphi(n)/n = \sum_{d \mid n} \mu(d)/d$, the prime reciprocal estimate $\sum_{p \le z} p^{-1} = \log\log z + O(1)$, and the following elementary correlation calculation.

Lemma 55.2 (Correlation of quotient divisibility blocks)**.** *For positive integers p, q, d, e, put*

$$I_{p,d}(N) = \mathbf{1}_{d \mid \lfloor N/p \rfloor}, \qquad F_{p,d}(N) = I_{p,d}(N) - \frac{1}{d}, \qquad g = \gcd(pd, qe).$$

Then, for every interval of X consecutive integers,

$$\left| \frac{1}{X} \sum_{N} F_{p,d}(N) F_{q,e}(N) \right| \le \frac{g^2}{pqde} + O\Big(\frac{pqde}{X}\Big).$$

The implicit constant is absolute; $p = q$ is permitted.

Proof. The first indicator has period pd and selects the p residue classes $0, \dots, p-1$ modulo pd, so its mean is $1/d$. Similarly the other mean is $1/e$. Their common period is $L = pqde/g$. For each r mod g, let A_r count the integers in $[0, p)$ congruent to r mod g, and define B_r similarly with q. The Chinese remainder theorem shows that the joint indicator selects $\sum_{r \bmod g} A_r B_r$ classes

modulo L. Write $A_r = p/g + a_r$ and $B_r = q/g + b_r$. Then $|a_r|, |b_r| \le 1$ and $\sum a_r = \sum b_r = 0$, giving

$$\frac{1}{L}\sum_{r \bmod g} A_r B_r = \frac{1}{de} + \frac{1}{L}\sum_{r \bmod g} a_r b_r.$$

Thus the covariance over a full common period has absolute value at most $g/L = g^2/(pqde)$. The product $F_{p,d}F_{q,e}$ has absolute value at most one. Its average over an arbitrary interval differs from its periodic average by $O(L/X)$, proving the claim. □

Lemma 55.3 (A summable covariance majorant)**.** *Writing $\omega(n)$ for the number of distinct prime factors of n, with $\omega(1) = 0$, one has*

$$\sum_{m,n\ge 1} \frac{\omega(m)\omega(n)\gcd(m,n)^2}{m^2 n^2} < \infty.$$

Proof. Set $A_j = \sum_{n\ge 1} \omega(n)^j/n^2$ for $j = 1, 2$ and $A_0 = \zeta(2)$. These are finite since $\omega(n) \le \log_2 n$. Write $m = ga, n = gb$ with $g = \gcd(m,n)$, drop the restriction $\gcd(a,b) = 1$, and use $\omega(ga) \le \omega(g) + \omega(a)$. The sum in the statement is bounded by

$$\sum_{g,a,b\ge 1} \frac{(\omega(g)+\omega(a))(\omega(g)+\omega(b))}{g^2 a^2 b^2} = A_2 A_0^2 + 3A_0 A_1^2 < \infty.$$

□

Proof of Theorem 55.1. Write $W = \log\log X$, choose $D = \lfloor W \rfloor$ and $Y = \lfloor X^{1/4} \rfloor$, and put

$$f(n) = \frac{\varphi(n)}{n}, \quad f_D(n) = \sum_{\substack{d\mid n \\ d\le D}} \frac{\mu(d)}{d}, \quad \kappa_D = \sum_{d\le D} \frac{\mu(d)}{d^2}.$$

Since $0 < f(n) \le 1$,

$$\frac{T(N)}{N} = \sum_{p\le N} \frac{1}{p} f(\lfloor N/p \rfloor) + O(1).$$

The primes $Y < p \le N$ contribute $O(1)$ uniformly for $X \le N < 2X$, by the prime reciprocal estimate. We retain all primes $p \le Y$, including the small ones.

The discarded divisor tail has small mean. For every $p \le Y$, each value of $\lfloor N/p \rfloor$ is attained at most p times, and

$$\frac{1}{X}\#\{X \le N < 2X : d \mid \lfloor N/p \rfloor\} \le \frac{1}{d} + O(p/X).$$

It follows that

$$\begin{aligned}\frac{1}{X}\sum_{X\le N<2X}\sum_{p\le Y}\frac{1}{p}|f(\lfloor N/p\rfloor) - f_D(\lfloor N/p\rfloor)| \\ \ll \frac{1}{D}\sum_{p\le Y}\frac{1}{p} + \frac{\pi(Y)\log(2X)}{X} \ll 1.\end{aligned}$$

Here the exact divisor identity bounds the absolute tail by $\sum_{d>D, d\mid n} 1/d$, and $d \le 2X/p$ in this average.

Define the centered truncated sum

$$Z(N) = \sum_{p\le Y}\frac{1}{p}\sum_{d\le D}\frac{\mu(d)}{d}F_{p,d}(N).$$

Lemma 55.2, including its diagonal case, gives

$$\frac{1}{X}\sum_{X\le N<2X}|Z(N)|^2 \ll \sum_{\substack{p,q\le Y\\ p,q\text{ prime}}}\sum_{d,e\le D}\frac{\gcd(pd,qe)^2}{(pd)^2(qe)^2}+\frac{D^2\pi(Y)^2}{X}$$
$$\le \sum_{m,n\ge 1}\frac{\omega(m)\omega(n)\gcd(m,n)^2}{m^2n^2}+O(D^2Y^2/X)\ll 1.$$

For the second line, each integer $m=pd$ has at most $\omega(m)$ representations with p prime. Lemma 55.3 therefore applies. Cauchy–Schwarz bounds the mean of $|Z(N)|$ by $O(1)$. Finally,

$$\kappa_D=\kappa+O(1/D),\qquad \sum_{p\le Y}\frac{1}{p}=W+O(1),\qquad W/D=O(1).$$

Combining these estimates proves the asserted bounded mean absolute error. Replacing $\log\log X$ by $\log\log N$ changes the quantity by $O(1/\log X)$. Markov's inequality proves the displayed exceptional-set estimate. To obtain one density-one subset on which the relative error tends to zero, use the shrinking tolerance $(\log\log X)^{-1/2}$ on dyadic intervals. The relative exceptional proportions then tend to zero. □

Corollary 55.4 (Cesàro form of Conjecture 10)**.** *As $X\to\infty$ through the positive integers,*

$$\sum_{N\le X}T(N)=\frac{X^2\log\log X}{2\zeta(2)}+O(X^2).$$

Proof. Multiply the dyadic mean absolute bound by $2X$ on each interval $[X,2X)$, and sum over dyadic intervals up to the requested endpoint. The total error is $O(X^2)$. Partial summation gives $\sum_{3\le N\le X}N\log\log N=(X^2/2)\log\log X+O(X^2/\log X)$. □

Remark 55.5 (Earlier announcement and scope). Carella, *Euler Totient Function And The Largest Integer Function Over The Shifted Primes*, arXiv:2105.00790v2 (2021), Theorem 2.1, announces the stronger formula $T(x)=\kappa x\log\log x+c_1x+O(x/\log x)$. Its equation (20), however, uses an incorrect prime-counting estimate. Already at its fixed parameter $n=1$ that estimate would imply

$$\pi(\lfloor x\rfloor)-\pi(\lfloor x/2\rfloor)=\frac{1}{2}\,\mathrm{li}(x)+O(xe^{-c\sqrt{\log x}}).$$

The prime number theorem and the expansion of li give instead

$$\pi(\lfloor x\rfloor)-\pi(\lfloor x/2\rfloor)-\tfrac{1}{2}\,\mathrm{li}(x)=-\frac{(\log 2)x}{2\log^2 x}+O(x/\log^3 x),$$

contradicting that error bound. This identifies a failure of the supplied proof, not a disproof of its announced final asymptotic. The later Saito–Suzuki–Takeda–Yoshida preprint explicitly poses the leading asymptotic as Conjecture 10; it does not discuss Carella's manuscript. Our independent averaging argument does not claim bibliographic priority and does not establish the pointwise conjecture. Its bounds permit exceptional individual integers with error of order $N\log\log N$. The density and averaged conclusions concern the same conjecture and are not two separate resolved open problems.

55.1 Explaining the observed half-square inclusion

Conjecture 8.3 of Kanado–Saito, *A system of certain linear Diophantine equations on analogs of squares*, arXiv:2205.12226v4, asks for infinitely many $a, b, c \geq 2$ such that

$$a^2 + b^2 - 1, \quad a^2 + c^2 - 1, \quad b^2 + c^2 - 1, \quad a^2 + b^2 + c^2 - 2$$

are all squares. Let $V(X)$ denote its ordered solutions $a \leq b \leq c \leq X$. Immediately after the displayed list $V(46300)$, the authors state that they do not know the reason for the observed inclusion $V(46300) \subset U_{\leq 46300}(1/2)$. Here $U_{\leq X}(\alpha)$ consists of triples of indices at most X whose positive values $\lfloor \alpha a^2 \rfloor, \lfloor \alpha b^2 \rfloor, \lfloor \alpha c^2 \rfloor$, all three pair sums, and their total sum belong to $S(\alpha) = \{\lfloor \alpha n^2 \rfloor : n \geq \alpha^{-1/2}\}$. The observed inclusion has an exact explanation valid at every cutoff.

Theorem 55.6 (Universal transfer of near-brick solutions)**.** *For every $X > 0$ and every*

$$\alpha \in \left\{ \frac{1}{8}, \frac{2}{8}, \ldots, \frac{7}{8}, \frac{1}{3}, \frac{2}{3} \right\}, \qquad V(X) \subseteq U_{\leq X}(\alpha).$$

In particular, the authors' observed inclusion at $\alpha = 1/2$ holds for every X.

Proof. Let $(a, b, c) \in V(X)$, and write the four companion squares as

$$u_{ab}^2 = a^2 + b^2 - 1, \quad u_{ac}^2 = a^2 + c^2 - 1, \quad u_{bc}^2 = b^2 + c^2 - 1, \quad v^2 = a^2 + b^2 + c^2 - 2.$$

At most one of a, b, c is even: two even coordinates would make their companion square congruent to 3 (mod 4). Every odd square is 1 (mod 8). If there is an even coordinate, write its square residue modulo 8 as $e \in \{0, 4\}$. A pair of odd coordinates has companion-square residue 1; a pair containing the even coordinate has companion-square residue e. The total companion square likewise has residue 1 if all coordinates are odd, and residue e otherwise.

Write each coordinate square as $8h+r$. The exact square equations show that each companion's quotient h is the sum of the corresponding coordinate quotients: the residue corrections are $1 + 1 - 1 = 1$, $e + 1 - 1 = e$, $1 + 1 + 1 - 2 = 1$, and $e + 1 + 1 - 2 = e$. For $1 \leq p \leq 7$,

$$\left\lfloor \frac{p}{8}(8h + r) \right\rfloor = ph + \left\lfloor \frac{pr}{8} \right\rfloor.$$

The residue 1 contributes zero, while the sole possible residue e contributes the same term on both sides of every relevant sum identity. This proves all four identities for $\alpha = p/8$.

Modulo 3, squares have residues $0, 1$, and at most one of a, b, c is divisible by 3, since two such coordinates would give a companion square congruent to 2 (mod 3). The identical argument with modulus 3 and exceptional residue $e = 0$ proves the assertions for $p/3$, $p = 1, 2$.

Finally no coordinate can equal 2: for another coordinate $b \geq 2$, $b^2 + 3$ lies strictly between b^2 and $(b+1)^2$. Thus $a, b, c \geq 3$, so all their floor values are positive for the listed $\alpha \geq 1/8$. The companion floor values are their positive sums. All membership and positivity requirements in $U_{\leq X}$ are therefore satisfied. □

Remark 55.7. The transfer proves the universal inclusion behind an explicitly unexplained numerical observation in the source. It is an elementary congruence result; it does not prove that $V(X)$ grows without bound, does not resolve Conjecture 8.3, and does not answer Question 8.1 about the existence of an $\alpha \in (0, 1)$ with $T(\alpha)$ empty.

A useful exact search reduction is also available.

Proposition 55.8 (Finite fibers and an exact divisor reduction)**.** *Fix $a \geq 2$. All possible positive integers b satisfying $a^2 + b^2 - 1 = u^2$ are obtained by*

$$b = \frac{v - w}{2}, \qquad u = \frac{v + w}{2}, \qquad vw = a^2 - 1, \quad v > w > 0, \quad v \equiv w \pmod 2.$$

In particular $b \leq (a^2 - 2)/2$. Consequently every fixed smallest coordinate occurs in only finitely many solutions of Conjecture 8.3, and any infinite family would have all three coordinates tending to infinity along a subsequence.

Proof. Factor $(u - b)(u + b) = a^2 - 1$. Both factors are positive integers of the same parity, giving the formula and its converse. Since $v \leq a^2 - 1$ and $w \geq 1$, the bound follows. Apply the same argument to c. For $a \leq b \leq c$, the set of solutions with a bounded is therefore finite. □

56 What would complete the remaining transfers?

The results locate explicit missing steps rather than removing them by renaming auxiliary problems. For Mills' constant, the density sieve must be upgraded to a recurrent-divisor theorem for each individual permitted cubic orbit. For the least square constant, one must force an approximation or recurrence of the kind proved sufficient here; neither minimality nor primality has yet supplied it. For Type C, the improved cutoff still leaves small parameters. For the prime–totient sum, an average estimate cannot exclude large errors at a sparse set of integers. For near-Euler bricks, the exact rational transfer does not create an unbounded family of integer solutions.

The boundary theorem illustrates a successful auxiliary-problem route: first sharpen prime-root separation according to affine support rank, then prove simultaneous short approximations at every local left minimum, and finally transfer the separation to those limits. The cube sieve uses a different connection: finite-field return, Chinese-remainder residue counts, and a shrinking coefficient region. Both arguments give statements over infinite families, without replacing their remaining universal quantifiers by finite computations.

Part VII
Focused reductions for the three remaining problems

57 The nonreal cubic Pisot case

Problem 1.1 of [35] asks whether every cubic Pisot number β satisfies

$$\#\{r \geq 0 : \lfloor \beta^{3^r} \rfloor \text{ is composite}\} = \infty.$$

Changing the starting index does not change this assertion. The following result settles its nonreal-conjugate case. It uses no equidistribution assertion for the angles $3^r\theta$, either on the whole sequence or on an arithmetic progression.

Theorem 57.1 (Nonreal cubic conjugates)**.** *Let β be a cubic Pisot number whose two other conjugates are nonreal. Then $\lfloor \beta^{3^r} \rfloor$ is composite for infinitely many integers $r \geq 0$.*

We give all of the trace, sign, and return arguments needed for the proof. Write the minimal polynomial and conjugates as

$$f(X) = X^3 - aX^2 + bX - c \in \mathbb{Z}[X], \qquad \beta, \alpha, \overline{\alpha}, \qquad \alpha = \rho e^{i\theta}, \quad 0 < \rho < 1.$$

For $r \geq 0$ set

$$T_r = \beta^{3^r} + \alpha^{3^r} + \overline{\alpha}^{3^r}, \qquad B_r = (\beta\alpha)^{3^r} + (\beta\overline{\alpha})^{3^r} + (\alpha\overline{\alpha})^{3^r}, \qquad E_r = c^{3^r}.$$

These are integers. Newton's identities give

$$\begin{aligned} T_{r+1} &= T_r^3 - 3T_rB_r + 3E_r, \\ B_{r+1} &= B_r^3 - 3T_rB_rE_r + 3E_r^2, \\ E_{r+1} &= E_r^3. \end{aligned} \tag{57.1}$$

In particular $T_r \equiv a \pmod 3$ for every r.

Lemma 57.2 (Distinct powered conjugates)**.** *For a Pisot number γ and every positive integer m, $\mathbb{Q}(\gamma^m) = \mathbb{Q}(\gamma)$. Consequently the mth powers of its distinct conjugates are distinct. In the present setting, $\alpha/\overline{\alpha}$ is not a root of unity, and $\cos(3^r\theta) \neq 0$ for every $r \geq 0$.*

Proof. Any embedding of $\mathbb{Q}(\gamma)$ fixing γ^m sends γ to a conjugate of modulus $\gamma > 1$. The Pisot property makes γ the only such conjugate. Separability therefore gives $[\mathbb{Q}(\gamma) : \mathbb{Q}(\gamma^m)] = 1$. The distinctness assertion follows by restricting embeddings to these equal fields. If $\alpha/\overline{\alpha}$ were a root of unity, some powered conjugates would coincide. If $\cos(3^r\theta) = 0$, then $(\alpha/\overline{\alpha})^{3^r} = -1$, which again makes this ratio a root of unity. □

Lemma 57.3 (Trace congruence)**.** *For every $r \geq 0$,*

$$T_{r+1} - T_r \in 3^{r+1}\mathbb{Z}. \tag{57.2}$$

Proof. Let A be an integral companion matrix of f, so $T_r = \operatorname{Tr}(A^{3^r})$. In the matrix expansion of $\operatorname{Tr}(A^{3^j})$, group closed index words by cyclic rotation. A word of least period 3^i is the repetition of a primitive cyclic word of length 3^i, with an integral weight w, and its orbit contributes $3^i w^{3^{j-i}}$. Consequently

$$\operatorname{Tr}(A^{3^j}) = \sum_{i=0}^{j} 3^i \sum_{w \in \mathcal{V}_i} w^{3^{j-i}},$$

where $\mathcal{V}_i$ is the finite multiset of weights of primitive cyclic words of length 3^i. The new terms with $i = j$ are divisible by 3^j. For every integer w and $h \geq 1$,

$$w^{3^h} - w^{3^{h-1}} \in 3^h\mathbb{Z};$$

this follows from Fermat's congruence at $h = 1$ and induction using the factorization of a difference of cubes. Thus every older term in the difference of the expressions for j and $j-1$ is also divisible by 3^j. Take $j = r+1$. □

For all sufficiently large r, define

$$s_r = \mathbf{1}_{\{\cos(3^r\theta)>0\}}, \qquad q_r = \lfloor \beta^{3^r} \rfloor = T_r - s_r.$$

The formula for q_r follows from $T_r - \beta^{3^r} = 2\rho^{3^r}\cos(3^r\theta)$, whose absolute value is strictly between zero and one for all sufficiently large r.

Lemma 57.4 (Both signs recur)**.** *Both values* 0 *and* 1 *occur infinitely often in* (s_r).

Proof. If two consecutive cosines are positive, the identity $\cos(3x) = \cos x(4\cos^2 x - 3)$ gives $\cos x > \sqrt{3}/2$. If both are negative, it gives $\cos x < -\sqrt{3}/2$. An eventually constant sign would therefore keep every sufficiently late angle within $\pi/6$ of a fixed center, either 0 or π modulo 2π. Write the signed displacement from that center as $\delta_r \in (-\pi/6, \pi/6)$. Tripling the angle gives $\delta_{r+1} = 3\delta_r$: since $3\delta_r \in (-\pi/2, \pi/2)$, no change by a multiple of 2π is possible. Boundedness for all future r forces $\delta_r = 0$. It follows that $(\alpha/\overline{\alpha})^{3^r} = 1$, contradicting Lemma 57.2. □

Proof of Theorem 57.1. Suppose, for a contradiction, that every sufficiently late q_r is prime. The floors grow strictly eventually, so these primes eventually avoid any fixed finite set. In particular, they are eventually odd and different from 3. If $a \equiv 0 \pmod 3$, the infinitely many indices with $s_r = 0$ give multiples of 3. If $a \equiv 1 \pmod 3$, those with $s_r = 1$ do so. Lemma 57.4 therefore forces

$$a \equiv 2 \pmod 3. \tag{57.3}$$

Modulo 2, the Newton map (57.1) is

$$(T, B, E) \longmapsto (T + BT + E,\ B + EBT + E,\ E).$$

Its complete eight-state graph is

$$000 \mapsto 000, \quad 001 \mapsto 111, \quad 010 \mapsto 010, \quad 011 \leftrightarrow 101,$$
$$100 \mapsto 100, \quad 110 \mapsto 010, \quad 111 \mapsto 111.$$

Every cycle except $011 \leftrightarrow 101$ has constant trace parity. Since $q_r = T_r - s_r$ is odd eventually, such a cycle would make s_r eventually constant, contrary to Lemma 57.4. Thus eventual primality forces the exact sign alternation

$$s_{r+1} = 1 - s_r \qquad \text{for all sufficiently large } r. \tag{57.4}$$

Fix a sufficiently late index r such that the prime $q = q_r$ does not divide $3c\,\mathrm{disc}(f)$. Let $d \in \{1, 2, 3\}$ be the degree of the splitting field of f over $\mathbb{F}_q$; these are the only possibilities for a separable cubic. Suppose that $v_3(q^d - 1) \leq r$. Write

$$q^d - 1 = 3^t N, \qquad 3 \nmid N, \qquad t \leq r.$$

Choose a positive even integer K for which $3^K \equiv 1 \pmod N$; if $N = 1$, take $K = 2$. For every $j \geq 0$,

$$q^d - 1 \mid 3^r(3^{jK} - 1).$$

All three residual roots are nonzero, so their 3^rth powers return at indices $r + jK$. Their traces therefore return modulo q. Since K is even, (57.4) returns the shift as well. It follows that

$$q \mid T_{r+jK} - s_{r+jK} = q_{r+jK} \qquad (j \geq 0).$$

For sufficiently large j, this contradicts primality because $q_{r+jK} > q$. We have proved

$$v_3(q_r^d - 1) \geq r + 1.$$

For $d = 1$, this gives $q_r \equiv 1 \pmod{3^{r+1}}$. For $d = 2$, exactly one of $q_r - 1, q_r + 1$ is divisible by 3, so $q_r \equiv \pm 1 \pmod{3^{r+1}}$. For $d = 3$, it first forces $q_r \equiv 1 \pmod 3$, and

$$v_3(q_r^3 - 1) = v_3(q_r - 1) + 1$$

gives $q_r \equiv 1 \pmod{3^r}$. Uniformly, therefore,

$$q_r \equiv \varepsilon_r \pmod{3^r}, \qquad \varepsilon_r \in \{-1, 1\}.$$

Equation (57.3) fixes the sign: $\varepsilon_r = 2s_r - 1$. Hence

$$T_r \equiv 3s_r - 1 \pmod{3^r}.$$

By (57.4), the representatives on the right alternate between -1 and 2. Comparing indices r and $r + 1$ therefore gives

$$T_{r+1} - T_r \equiv \pm 3 \pmod{3^r}.$$

For $r \geq 2$ this contradicts (57.2). Thus infinitely many floors are composite, as asserted. □

Remark 57.5 (Scope of the angle argument). Lemma 57.4 concerns the full tripling orbit. Irrationality alone does not give both signs on each progression of indices. For example, choose a digit sequence $d_j \in \{1, 2\}$ that is not eventually periodic and put $t = \sum_{j\geq 1} d_j 9^{-j}$. Every shifted base-9 tail lies strictly between 1/8 and 1/4. Thus t is irrational, $\cos(2\pi 9^n t) > 0$, and $\cos(2\pi\, 3 \cdot 9^n t) < 0$ for every $n \geq 0$. The signs alternate forever. The proof above does not try to rule out this itinerary by dynamics: it rules it out for an eventually prime cubic trace sequence by the arithmetic trace congruence. No assertion that $e^{2\pi i t}$ is algebraic is made in this example.

58 The three remaining problems and their logical relations

This part concerns the irrationality of $\xi(2^k)$, the unconditional transcendence of the least Mills constant $A = \xi(3^k)$, and Saito's Problem 1.1 asking whether every cubic Pisot number β has infinitely many composite values $\lfloor \beta^{3^r} \rfloor$. The square problem is Question 1.6 of [36]; the cubic problem is posed in [35]. The nonreal cubic case is settled by Theorem 57.1. The results below give further exact restrictions in the remaining cases, without settling the three full assertions.

Proposition 58.1 (One composite term is the universal target)**.** *Let $\mathcal{C}$ be a class of cubic Pisot numbers closed under $\beta \mapsto \beta^{3^j}$ for every $j \geq 0$. The following assertions are equivalent:*

1. *every $\beta \in \mathcal{C}$ has at least one composite value $\lfloor \beta^{3^r} \rfloor$ with $r \geq 1$;*
2. *every $\beta \in \mathcal{C}$ has infinitely many such composite values.*

This applies both to the full cubic Pisot class and to the class $\mathcal{S}$ of totally real cubic Pisot numbers with conjugates $\beta, -\rho, \gamma$ satisfying

$$0 < \sigma = |\gamma| < \rho < 1, \qquad \rho \leq \min\{\sigma^{17/23}, \beta^{-17/40}\}. \tag{58.1}$$

Either universal assertion for $\mathcal{S}$ implies the unconditional transcendence of the least Mills constant.

Proof. Apply the first assertion to β^{3^j}. It gives a composite value at an index larger than j for the original β, for every j. The reverse implication is immediate. Lemma 57.2 preserves cubic degree. Odd powers preserve the signs, order of the small-root moduli, and both inequalities in (58.1), proving closure of the stated classes.

If the least Mills constant were algebraic, Saito's Theorem 1.7 in [39] would supply $\beta = A^{3^m} \in \mathcal{S}$. Every value $\lfloor \beta^{3^r} \rfloor = \lfloor A^{3^{m+r}} \rfloor$ would be prime, contradicting the first assertion. □

The implication does not identify every algebraic member of $W(3^k)$ with its minimum. An eventually all-prime cubic Pisot orbit, if one existed, would give an algebraic member after a power change; it would not determine the arithmetic nature of the least member.

Proposition 58.2 (The norm restriction and a nonunit example)**.** *For every $\beta \in \mathcal{S}$, the positive integer $q = |\operatorname{Norm}(\beta)|$ satisfies*

$$1 \le q < \beta^{3/20}.$$

The bound $\rho \le \sigma^{17/23}$ follows already from $\rho \le \beta^{-17/40}$ and $q \ge 1$. These restrictions do not force $q = 1$: the largest root of

$$f(X) = X^3 - 10007X^2 - 90X + 2$$

belongs to $\mathcal{S}$ and has norm -2.

Proof. The norm identity is $q = \beta\rho\sigma$. Since $\sigma < \rho$,

$$q < \beta\rho^2 \le \beta^{3/20}.$$

Furthermore $\rho^{40} \le \beta^{-17} \le q^{17}\beta^{-17} = \rho^{17}\sigma^{17}$, giving the claimed redundancy.

For the example, the signs at rational endpoints place its roots in

$$(10007, 10008), \qquad (-39/2000, -19/1000), \qquad (1/100, 11/1000).$$

For the two small-root intervals the exact endpoint values are, in the displayed order,

$$-\frac{401353319}{8000000000}, \quad \frac{97466141}{1000000000}, \quad \frac{99301}{1000000}, \quad -\frac{200845669}{1000000000}.$$

Also $f(10007) < 0 < f(10008)$. The three sign-changing intervals account for all roots and contain no integer; the monic cubic is therefore irreducible by the rational-root theorem. The dominant small conjugate is negative and the norm is -2. The exact integer inequality

$$39^{40}10008^{17} < 2000^{40}$$

implies $\rho^{40}\beta^{17} < 1$. The remaining conjugate inequality follows from the first part of the proof. □

Passing to β^{3^j} also raises q to q^{3^j}; it does not improve this logarithmic norm bound. A bound $\rho \le \beta^{-1/2}$ would instead give $1 \le q < \beta\rho^2 \le 1$. The exponent $17/40$ does not reach that contradiction, and the nonunit example shows why an argument restricted to units cannot exhaust the possible shadows.

59 A rational square minimum would force a prime deficit

Let $A = \xi(2^k)$ and put $p_n = \lfloor A^{2^n} \rfloor$ for $n \geq 1$. Call a prime q *viable* if there is a real number $B > 1$ such that $\lfloor B^{2^j} \rfloor$ is prime for every $j \geq 0$ and $\lfloor B \rfloor = q$. Write V for the set of viable primes. Equivalently, $q \in V$ if its finitely branching tree of prime children

$$p^2 < q' < (p+1)^2 - 1$$

has an infinite path. Strict nesting of the closed descendant root cells proves the equivalence.

Lemma 59.1 (The least viable child)**.** *At every index $n \geq 1$,*

$$p_{n+1} = \min\{q \in V : p_n^2 < q < (p_n+1)^2 - 1\}.$$

Proof. The actual tail shows $p_{n+1} \in V$. Neither endpoint is prime: p_n^2 is composite and $(p_n+1)^2 - 1 = p_n(p_n+2)$ is composite. If a smaller viable child q existed, choose its infinite continuation. After taking the 2^{n+1}th root, this continuation lies strictly inside the p_n parent cell and below the p_{n+1} cell. It respects every earlier prime condition. Indeed, an earlier integer-power boundary becomes an integer at level n, so it cannot cut the interior of a single level-n prime cell. The resulting admissible constant would be smaller than A, contradicting minimality. □

For the good-prime predicate used in the preceding square-clock section, fix sufficiently large integers K, P and set

$$H(p) = \left\lceil \frac{p}{K\lceil \log_2 p \rceil} \right\rceil.$$

A prime $p \geq P$ is good when $[p^2, p^2 + p]$ contains at least $H(p)$ primes, and poor otherwise. Lemmas 50.1 and 50.2, proved from Matomäki's prime-interval estimate, give the two inputs

$$\{p \geq P : p \text{ is good}\} \subset V, \qquad \#\{p \in [X, 2X] : p \text{ is a poor prime}\} \leq CX^{1/3} \tag{59.1}$$

for a fixed constant C and all sufficiently large X. Only existence of a continuation, rather than its dimension, is needed in the next theorem.

Theorem 59.2 (Necessary prime deficit for a rational minimum)**.** *If $A = \xi(2^k)$ is rational, then for every fixed $\epsilon \in (0,1)$ and all sufficiently large n,*

$$\pi(p_n^2 + p_n^{1-\epsilon}) - \pi(p_n^2) \leq C p_n^{2/3}, \tag{59.2}$$

where C is the poor-prime constant in (59.1). For $0 < \epsilon < 1/3$, this upper bound is $o(p_n^{1-\epsilon}/\log p_n)$.

Proof. An integer $A > 1$ cannot have prime square outputs, so a rational A would be nonintegral. Write $u_n = A^{2^n} - p_n \in (0,1)$. Mahler's approximation theorem, in the form of Theorem 2.5 of [36], states that for every $\eta > 0$,

$$\|A^m\| > \exp(-\eta m)$$

for all sufficiently large integers m. Choose $\eta = \epsilon \log(A)/2$ and use $p_n \sim A^{2^n}$. It follows that

$$u_n \geq \|A^{2^n}\| > p_n^{-\epsilon}$$

for all sufficiently large n.

The exact repeated-squaring relation is

$$p_{n+1} - p_n^2 = 2p_n u_n + u_n^2 - u_{n+1}.$$

Since $0 < u_{n+1} < 1$, it yields

$$p_{n+1} - p_n^2 > 2p_n^{1-\epsilon} - 1 > p_n^{1-\epsilon}$$

eventually. Every prime in $(p_n^2, p_n^2 + p_n^{1-\epsilon}]$ is therefore an earlier child of p_n. Lemma 59.1 says it is nonviable, and (59.1) says it is poor. The whole interval lies in $[p_n^2, 2p_n^2]$ for large n. Applying the poor-prime bound with $X = p_n^2$ proves (59.2). Finally,

$$\frac{p_n^{2/3}}{p_n^{1-\epsilon}/\log p_n} = p_n^{-1/3+\epsilon}\log p_n \longrightarrow 0$$

when $\epsilon < 1/3$. □

Corollary 59.3 (A viable-prime window sufficient for irrationality)**.** *Suppose that, for some fixed $\delta \in (0,1)$ and infinitely many indices of the actual least square-clock orbit, there is a viable prime*

$$p_n^2 < q \le p_n^2 + p_n^{1-\delta}.$$

Then $\xi(2^k)$ is irrational. In particular, it suffices that such a good prime occur infinitely often. Alternatively, for some fixed $\epsilon \in (0, 1/3)$, a prime count exceeding the right side of (59.2) infinitely often also suffices.

Proof. Minimality gives $p_{n+1} \le q$ at the indicated indices. The exact square relation then gives

$$2p_n u_n \le p_n^{1-\delta} + 1, \qquad u_n = O(p_n^{-\delta}).$$

This contradicts Mahler's lower bound, for example with exponent $\delta/2$, if A is nonintegral rational. The integer case is already excluded. The other assertions follow from (59.1) and Theorem 59.2. □

At the scale $x = p_n^2$, the deficit window has length $x^{(1-\epsilon)/2}$, below the exponent $1/2$. An averaged bound on exceptional intervals does not exclude this particular orbit: since $p_{n+1} > p_n^2$, it has at most one sufficiently large label in a dyadic interval. The theorem supplies a precise necessary deficit, while the viable-prime window in Corollary 59.3 remains an unproved input for the actual minimum.

60 Exact parity depths in a cubic-unit branch

In the unit branch whose norm equals the limiting trace sign, the exact 3-adic depth can depend on the parity of the index. The next theorem treats that branch and supplies a terminating integer procedure for identifying both depths.

We use the Baker–Brumer theorem in the following standard form: if the p-adic logarithms of nonzero algebraic numbers are linearly independent over $\mathbb{Q}$, they are linearly independent over $\overline{\mathbb{Q}}$. See Brumer [7] and the explicit formulation in Maksoud [24, Theorem 2.1].

Lemma 60.1 (Multiplicative independence of two conjugates)**.** *Let β be a totally real cubic Pisot unit. Any two distinct conjugates of β are multiplicatively independent modulo roots of unity.*

Proof. Write the three real conjugates as $\beta_0, \beta_1, \beta_2$, with $\beta_0 = \beta$, and put $\ell_i = \log|\beta_i|$. Then $\ell_0 > 0$, $\ell_1, \ell_2 < 0$, and $\ell_0 + \ell_1 + \ell_2 = 0$ because the norm has absolute value one. The transitive Galois group of an irreducible cubic contains a three-cycle on its roots. Applying that cycle twice to a multiplicative relation, and taking real absolute logarithms, shows that its exponent vector is orthogonal to all cyclic permutations of (ℓ_0, ℓ_1, ℓ_2). These permutations span the plane

$x_0 + x_1 + x_2 = 0$: the two nontrivial Fourier eigenvalues of the associated circulant matrix are nonzero, since a real triple with one positive and two negative entries cannot be constant. Consequently the exponent vector is a multiple of $(1, 1, 1)$. A relation involving only two roots has its remaining exponent zero and is therefore trivial. □

Theorem 60.2 (Exact parity-dependent 3-adic depth)**.** *Let β be a totally real cubic Pisot unit with minimal polynomial*

$$f(X) = X^3 - aX^2 + bX - \varepsilon, \qquad \varepsilon \in \{1, -1\}, \qquad a \equiv \varepsilon \pmod 3, \quad b \equiv 1 \pmod 3.$$

Let $T_r = \mathrm{Tr}(\beta^{3^r})$ for $r \geq 0$. There are finite positive integers s_0, s_1 such that

$$\boxed{v_3(T_r - \varepsilon) = r + s_j \quad \text{if } r \equiv j \pmod 2 \text{ and } r \geq s_j - 1.}$$

Moreover, the following statements hold.

1. *Both integers s_0, s_1 are computable by a terminating algorithm using only modular integer arithmetic.*
2. *One has*
$$s_0 = 1 \iff v_3(a - \varepsilon) = 1, \qquad s_1 = 1 \iff v_3(b - 1) = 1.$$
Here the valuation of zero is understood to be infinity.
3. *If $s = v_3(a - \varepsilon b)$, then s is finite and $\min(s_0, s_1) \leq s$. In particular, $s = 1$ forces at least one parity offset to equal one.*

Proof. Modulo 3 the polynomial is $(X - \varepsilon)(X^2 + 1)$ and is separable. Its 3-adic splitting field is the unramified quadratic extension $K = \mathbb{Q}_3(i)$, where $i^2 = -1$. Hensel lifting labels its roots as

$$\alpha = \varepsilon \exp(L), \qquad u = i \exp(M), \qquad v = -i \exp(N), \qquad L, M, N \in 3\mathcal{O}_K.$$

Here $\alpha, L \in \mathbb{Q}_3$, and the nontrivial automorphism of $K/\mathbb{Q}_3$ interchanges u, v and M, N, and sends i to $-i$. Since $\alpha uv = \varepsilon$, injectivity of the exponential on $3\mathcal{O}_K$ gives $L + M + N = 0$. For $m = 3^r$ we obtain

$$T_r = \varepsilon \exp(mL) + (-1)^r i\big(\exp(mM) - \exp(mN)\big).$$

Define

$$\lambda_0 = \varepsilon L + i(M - N), \qquad \lambda_1 = \varepsilon L - i(M - N).$$

Both belong to $3\mathbb{Z}_3$. The exponential series gives

$$T_r - \varepsilon = 3^r \lambda_{r \bmod 2} + R_r, \qquad v_3(R_r) \geq 2r + 2. \tag{60.1}$$

Indeed, for $k \geq 2$ and $Z \in 3\mathcal{O}_K$, the kth exponential term has valuation at least $k(r+1) - v_3(k!) \geq 2r + 2$.

It remains to exclude a vanishing linear coefficient. Fixing an embedding of the global splitting field into $\overline{\mathbb{Q}}_3$, the roots u, v are images of two distinct global conjugates. The logarithms M, N are linearly independent over $\mathbb{Q}$: an integral relation would, by exponentiating in $1 + 3\mathcal{O}_K$, make a product of powers of u, v a root of unity, contradicting Lemma 60.1. The Baker–Brumer theorem makes them linearly independent over $\overline{\mathbb{Q}}$. Since $L = -M - N$, either equality $\lambda_j = 0$ would give a nontrivial algebraic linear relation between M and N, with coefficients $-\varepsilon \pm i$ and $-\varepsilon \mp i$. Thus λ_0, λ_1 are nonzero. Put $s_j = v_3(\lambda_j) \geq 1$. When $r \geq s_j - 1$, we have $r + s_j < 2r + 2$; the asserted valuation formula follows from (60.1).

For effectiveness, let B_r be the second elementary symmetric function of the 3^rth powers of the roots. Starting from $(T_0, B_0) = (a, b)$, use the exact Newton recurrence

$$T_{r+1} = T_r^3 - 3T_r B_r + 3\varepsilon, \qquad B_{r+1} = B_r^3 - 3\varepsilon T_r B_r + 3. \tag{60.2}$$

For each parity, search for an r of that parity for which $T_r - \varepsilon$ is nonzero modulo 3^{2r+2}. For the first such certificate, if $v = v_3(T_r - \varepsilon) < 2r + 2$, equation (60.1) proves $s_j = v - r$. The nonvanishing of λ_j ensures that this search terminates for both parities. All recurrence calculations may be performed modulo 3^{2r+2} for the current test.

At $r = 0$, equation (60.1) gives $\lambda_0 \equiv a - \varepsilon \pmod 9$. Writing $x = a - \varepsilon$ and $y = b - 1$, both divisible by 3, the first recurrence gives $T_1 - \varepsilon = -3\varepsilon y + 3\varepsilon x^2 + x^3 - 3xy$. Consequently $\lambda_1 \equiv -\varepsilon(b - 1) \pmod 9$, which proves the two criteria for $s_j = 1$.

Finally set $x_r = T_r - \varepsilon$, $y_r = B_r - 1$, $z_r = x_r + \varepsilon y_r$, and $w_r = x_r - \varepsilon y_r = T_r - \varepsilon B_r$. The recurrence factors as

$$w_{r+1} = w_r(3 + 3\varepsilon z_r + x_r^2 + \varepsilon x_r y_r + y_r^2).$$

Since $3 \mid x_r, y_r$, its multiplier is $3 \pmod 9$. Also $w_0 = a - \varepsilon b \neq 0$, since otherwise $f(\varepsilon) = 0$. Thus $v_3(w_r) = r + s$ exactly. Expanding the inverse-root expression for B_r gives

$$w_r = 2\varepsilon 3^r L + O_3(3^{2r+2}).$$

It follows that $v_3(L) = s$. Since $\lambda_0 + \lambda_1 = 2\varepsilon L$, the two offsets cannot both exceed s. □

Example 60.3 (The offsets need not agree)**.** For

$$f(X) = X^3 - 13X^2 - 8X - 1$$

there is one root in each of $(-1, -1/4)$, $(-1/4, 0)$, and $(13, 14)$. The rational-root test proves irreducibility, so the large root is a totally real cubic Pisot unit with negative dominant small conjugate. Newton iteration gives

$$T_0 - 1 = 12, \qquad T_3 - 1 \equiv 2187 = 3^7 \pmod{3^8}.$$

These are exact certificates for $s_0 = 1$ and $s_1 = 4$. Therefore

$$v_3(T_r - 1) = \begin{cases} r + 1, & r \text{ even}, \\ r + 4, & r \text{ odd and } r \geq 3. \end{cases}$$

This example demonstrates the valuation phenomenon; it is not claimed to be a prime-generating Pisot number.

Corollary 60.4 (Parity-specific complete splitting)**.** *Assume the hypotheses of Theorem 60.2 with $\varepsilon = 1$, and suppose the dominant small conjugate is negative. Fix a parity j for which $s_j = 1$. If $\lfloor \beta^{3^r} \rfloor$ is eventually prime, then f splits completely modulo every sufficiently late such prime with $r \equiv j \pmod 2$. In particular this conclusion holds on the even indices when $a \not\equiv 1 \pmod 9$, and on the odd indices when $b \not\equiv 1 \pmod 9$.*

Proof. For all sufficiently large r, the negative dominant small conjugate gives $\lfloor \beta^{3^r} \rfloor = T_r$. Let $p = T_r$ be prime, large enough that $p \nmid 3\,\mathrm{disc}(f)$, with $r \equiv j \pmod 2$. Then $p \equiv 1 \pmod 3$ and $v_3(p - 1) = r + 1$.

If f is irreducible modulo p, the ratios of its three roots have orders dividing $p^2 + p + 1$, whose 3-adic valuation is one. After raising the roots to 3^r, their ratios therefore have orders coprime to 3. If the factorization type is $1 + 2$, the roots lie in $\mathbb{F}_{p^2}$, and their 3^rth powers have 3-primary components in μ_3, since $v_3(p^2 - 1) = r + 1$. The two conjugate quadratic roots have the same component because $p \equiv 1 \pmod 3$. Their product with the remaining component is one, by the norm condition. All three components are consequently equal, and their ratios again have orders coprime to 3.

In either nonsplit case, choose $L \geq 1$ such that 3^L is one modulo the exponent of the group generated by these ratios. Cubing L times returns all three powered roots up to a common nonzero scalar. Their vanishing sum modulo p therefore recurs at $r + L, r + 2L, \ldots$. The corresponding traces eventually exceed p, yielding a positive-density progression of composite floors. This contradicts an eventually prime tail. □

61 A completely split prime seed need not return

The splitting conditions above and their opposite-norm counterparts do not imply recurrence of the seed prime as a divisor. The following example lies in the precise shrinking-conjugate region and admits a finite exhaustive certificate of nonreturn.

For an irreducible polynomial

$$f(X) = X^3 - aX^2 + bX - c$$

with roots $\beta, \gamma_2, \gamma_3$, write

$$T_r = \beta^{3^r} + \gamma_2^{3^r} + \gamma_3^{3^r}, \qquad B_r = (\beta\gamma_2)^{3^r} + (\beta\gamma_3)^{3^r} + (\gamma_2\gamma_3)^{3^r}.$$

If β is a totally real cubic Pisot number and the small conjugate of largest absolute value is negative, then $\lfloor \beta^{3^r} \rfloor = T_r$ for all sufficiently large r. In the example below this holds for every $r \geq 0$.

Proposition 61.1 (A nonreturning prime seed in the narrowed cubic locus)**.** *Let β be the largest root of*

$$f(X) = X^3 - 211X^2 - 7X + 1.$$

Then β is an irreducible totally real cubic Pisot number, with conjugates $\beta, -\rho, \sigma$ satisfying

$$0 < \sigma < \rho < 1, \qquad \rho \leq \min\{\sigma^{17/23}, \beta^{-17/40}\}.$$

Its first two sparse floors are the primes

$$T_0 = 211, \qquad T_1 = p = 9\,398\,359.$$

Nevertheless,

$$p \nmid T_r \qquad \text{for every } r \geq 2.$$

Moreover f splits completely modulo p. The example satisfies $a \equiv 1$, $b \equiv -1 \pmod 3$ and $v_3(a+b) = 1$, as well as the odd-trace parity condition.

Proof. Sign changes give roots in $(-1, 0)$, $(0, 1)$ and $(211, 212)$. A rational root would be 1 or -1, neither of which is a root; hence the cubic is irreducible. From the sum and product of its roots,

$$\rho - \sigma = \beta - 211 > 0, \qquad \rho\sigma = \beta^{-1}.$$

Since $f(-1/10) < 0 < f(0)$, one has $\rho < 1/10$. The exact inequality $212^{17} < 10^{40}$ gives $1/10 < 212^{-17/40} < \beta^{-17/40}$. The norm identity then implies

$$\rho^{40} \leq \beta^{-17} = (\rho\sigma)^{17}, \qquad \rho \leq \sigma^{17/23}.$$

For every odd integer $N \geq 1$, the trace error is $-\rho^N + \sigma^N \in (-1, 0)$, proving equality of floors and traces.

Newton's identities give $T_1 = 211^3 + 3 \cdot 211 \cdot 7 - 3 = p$ and $B_1 = -7^3 - 3 \cdot 211 \cdot 7 + 3 = -4771$. The primality assertions are certified by trial division up to the respective integer square roots. Substitution modulo p certifies the three distinct residual roots

$$-1\,584\,786, \qquad -1\,942\,624, \qquad 3\,527\,621.$$

For norm -1, the exact modular Newton map is

$$\Phi(t, b) = (t^3 - 3tb - 3,\ b^3 + 3tb + 3) \pmod p.$$

Start at $S = (0, -4771 \bmod p)$, the state of index 1. Its next state is

$$S_1 = (9\,398\,356, 7\,830\,596).$$

The integer-only verifier in Appendix D checks

$$\Phi^{85712}(S_1) = S_1, \qquad \text{the first coordinate of } \Phi^j(S_1) \neq 0 \quad (0 \le j < 85712).$$

These checks are a finite exhaustive certificate of the asserted nondivisibility at every later index. They involve no numerical approximation and no unproved primality test. Finally, $a + b = 204$ has 3-adic valuation one; modulo 2 the state $(T_r, B_r, c) = (1, 1, 1)$ is fixed. □

This is not a counterexample to infinite compositeness. Indeed,

$$T_2 = 830\,149\,079\,789\,343\,894\,643 = 7 \cdot 31 \cdot 3\,825\,571\,796\,264\,257\,579.$$

It rules out deriving the desired theorem merely by asserting that an observed prime trace must recur as a divisor, even after imposing the known shrinking-conjugate, parity and 3-adic restrictions.

For a fixed prime not dividing the norm, the finite Newton state is eventually periodic. A zero trace in the transient part need not reappear on its eventual cycle. Proposition 61.1 therefore excludes a universal proof based solely on return of an arbitrarily chosen prime seed; it does not exclude a different recurrent divisor.

62 The norm cascade in the positive-dominant real branch

For the remaining real sign configuration, the actual floor is eventually one less than the trace. Let β be a totally real cubic Pisot number whose dominant small conjugate is positive, and write its conjugates as β, ρ, δ, where $0 < |\delta| < \rho < 1$. Put $M = 3^r$ and

$$\begin{aligned} T_r &= \beta^M + \rho^M + \delta^M, \\ B_r &= (\beta\rho)^M + (\beta\delta)^M + (\rho\delta)^M, \\ E_r &= (\beta\rho\delta)^M, \qquad F_r = T_r - 1. \end{aligned}$$

These are integers, and $F_r = \lfloor \beta^{3^r} \rfloor$ for all sufficiently large r.

Proposition 62.1 (A sufficient cascade condition and its eventual exclusion)**.** *If a prime $p = F_r$ divides $\mathrm{Norm}(\beta^{3^r} - 1)$, then $p \mid F_{r+j}$ for every $j \ge 0$. However, for every fixed β in this real sign configuration, this divisibility hypothesis fails at every sufficiently large prime seed F_r. More precisely,*

$$0 < B_r - E_r < F_r$$

for all sufficiently large r. Also

$$\gcd(F_r, F_{r+1}) = \gcd(F_r, 3(B_r - E_r)),$$

and a sufficiently large prime F_r never divides F_{r+1}.

Proof. The exact norm identity is

$$\mathrm{Norm}(\beta^M - 1) = E_r - B_r + T_r - 1 = E_r - B_r + F_r.$$

If $p = F_r$ divides this norm, then modulo p the polynomial whose roots are the three Mth powers becomes

$$X^3 - X^2 + B_r X - E_r = (X - 1)(X^2 + E_r).$$

Its roots in an algebraic closure of the residue field are $1, w, -w$, counted with multiplicity. Their trace after every odd power is 1. Since all powers 3^j are odd, $p \mid T_{r+j} - 1$.

On the other hand, an exact division gives

$$\frac{B_r - E_r}{(\beta\rho)^M} = 1 + \left(\frac{\delta}{\rho}\right)^M + \left(\frac{\delta}{\beta}\right)^M - \delta^M \longrightarrow 1.$$

Also $F_r/\beta^M \to 1$, and consequently

$$\frac{B_r - E_r}{F_r} \sim \rho^M \longrightarrow 0.$$

This proves the displayed strict inequalities. The positive integer $B_r - E_r$ is eventually smaller than the prime F_r, so it cannot be divisible by that prime; the norm identity excludes the cascade hypothesis.

Finally Newton's identity $T_{r+1} = T_r^3 - 3T_rB_r + 3E_r$ gives $F_{r+1} \equiv 3(E_r - B_r) \pmod{F_r}$, proving the gcd formula. The same asymptotic gives $0 < 3(B_r - E_r) < F_r$ eventually, so a prime F_r cannot divide its next floor. □

This proposition concerns the particular norm $\mathrm{Norm}(\beta^{3^r} - 1)$ and propagation of the entire prime seed. It leaves open whether another proper divisor or a later finite-field return supplies infinitely many composite terms.

63 Exact remaining requirements

The square-clock reduction asks for one viable prime in a sublinear window infinitely often along the actual least orbit, or for a violation of its necessary prime deficit. The cubic reduction asks for a composite term for every member of a power-closed real Pisot class; the nonreal case is already covered. For the least Mills constant, Saito's shrinking real class suffices. The parity-depth theorem and the seed-return counterexample distinguish an exact local restriction from a universal divisor assertion: complete splitting can occur, and a prime divisor can lie only in the transient portion of a modular orbit. These results leave the irrationality of $\xi(2^k)$, unconditional Mills transcendence, and the full cubic infinite-compositeness problem unresolved.

64 Scope of the consolidated conclusions

The least constant, an arbitrary prime-representing base, a selected branch, a local boundary point, and a minimum after deleting initial conditions remain different objects throughout. None of the results permits these distinctions to be dropped. The precise final status of the three principal arithmetic targets is recorded in Part VII. Saito's unconditional irrationality of Mills' constant and his conditional transcendence theorem are external inputs, not conclusions first established here.

The twin-output window equivalence and certified finite twin pairs do not prove that their windows are nonempty infinitely often. Extremely close approximations to a prime-generating base do not remove the precision or primality-certification costs of evaluating its large outputs. Failure of a fixed finite prime cover does not imply infinitely many prime values, as the pairwise-coprime composite examples demonstrate.

For the polynomial-pattern results, the balanced-equation bound does not determine the exact exceptional dimension. The waiting-time parameter is fixed. The normality criterion does not establish its analytic cancellation condition for a named algebraic irrational. Density-one and averaged asymptotics do not establish a pointwise conjecture. The general lattice dimension-profile result concerns arbitrary subsets and leaves room for additional restrictions on groups or homogeneous spaces.

Coefficient fields. In Theorem 23.1, the prescribed field F_0 may contain e, π, or other named constants. This does not put them into $F = \overline{\mathbb{Q}}(K)$. The corollary makes e transcendental over F and implies $\pi \notin F$: otherwise $e^{i\pi} + e^0 = 0$ contradicts the stated exponential independence. This last assertion is nonmembership, not transcendence over F. No general form of Schanuel's conjecture is asserted.

Provenance and assistance. This edition extends the earlier collected manuscript by incorporating the subsequent work on twelve classification and endpoint problems, the thirteen further research routes, and the final focused reductions. The counts in those working documents are not counts of solved named conjectures. Repeated front matter and bibliographic entries have been removed, and repeated arguments are cross-referenced where appropriate. The preparation used AI assistance and independent internal checks. The manuscript remains a proof draft requiring authorial and external review before any publication or priority claim.

Appendices

A Executable exact verification

The following is the complete verifier. Copy this single listing to `verify_certificates.py` and run it with Python 3.8 or later. It uses only the standard library, no internet access, and no probabilistic primality tests. The seed and all seven later primes are proved by the criteria in Section 8. The final blocks verify the finite floor table, the integer inequalities used for precision, and the 561 certified composite offsets. They do not attempt to construct the astronomical next greedy prime or print the asserted common decimal block.

The listing is embedded directly in this single LaTeX source. It has no external data dependencies.

```
"""Exact certificates for the sparse-prime-coding examples.
Python 3 standard library only. No probabilistic primality test is used.
Run: python verify_certificates.py
"""
from math import factorial, gcd, isqrt, prod

P0 = 513284270838218284273581257165 3
# n: (witness, complete factorization of n-1).
CERT = {
    3: (2, {2: 1}),
    5: (2, {2: 2}),
    7: (3, {2: 1, 3: 1}),
    11: (2, {2: 1, 5: 1}),
    13: (2, {2: 2, 3: 1}),
    19: (2, {2: 1, 3: 2}),
    23: (5, {2: 1, 11: 1}),
    29: (2, {2: 2, 7: 1}),
    37: (2, {2: 2, 3: 2}),
    61: (2, {2: 2, 3: 1, 5: 1}),
    131: (2, {2: 1, 5: 1, 13: 1}),
    139: (2, {2: 1, 3: 1, 23: 1}),
    149: (2, {2: 2, 37: 1}),
    157: (5, {2: 2, 3: 1, 13: 1}),
    181: (2, {2: 2, 3: 2, 5: 1}),
    241: (7, {2: 4, 3: 1, 5: 1}),
    557: (2, {2: 2, 139: 1}),
    3259: (3, {2: 1, 3: 2, 181: 1}),
    4457: (3, {2: 3, 557: 1}),
    12541: (14, {2: 2, 3: 1, 5: 1, 11: 1, 19: 1}),
    14149: (6, {2: 2, 3: 3, 131: 1}),
    56597: (2, {2: 2, 14149: 1}),
    59242103: (5, {2: 1, 61: 1, 149: 1, 3259: 1}),
    355452619: (2, {2: 1, 3: 1, 59242103: 1}),
    853482761: (3, {2: 3, 5: 1, 13: 1, 29: 1,
                    56597: 1}),
    411378690803: (2, {2: 1, 241: 1, 853482761: 1}),
    P0: (2, {2: 2, 157: 1, 4457: 1, 12541: 1,
             355452619: 1, 411378690803: 1}),
}
proved = {2}


def require(condition):
    if not condition:
        raise ArithmeticError("Certificate verification failed")


def prove_seed(n):
    if n in proved:
        return
    a, factors = CERT[n]
```

```
    for q in factors:
        prove_seed(q)
    require(prod(q ** e for q, e in factors.items()) == n - 1)
    require(pow(a, n - 1, n) == 1)
    for q in factors:
        require(gcd(pow(a, (n - 1) // q, n) - 1, n) == 1)
    proved.add(n)


def lucas_u(k, n, P, Q):
    """U_k(P,Q) modulo n, by exact binary doubling."""
    u, v = 0, 1
    for bit in bin(k)[2:]:
        u2 = u * (2 * v - P * u) % n
        v2 = (v * v - Q * u * u) % n
        if bit == "0":
            u, v = u2, v2
        else:
            u, v = v2, (P * v2 - Q * u2) % n
    return u


def check_cell(s, n):
    require(s ** 4 < n < n + 1 < (s + 1) ** 4)


def prove_plus(s, j, a):
    require(s in proved)
    n, F = s ** 4 + j * s ** 2 + 1, 2 * s ** 2
    require((n - 1) % F == 0 and F * F > n)
    require(pow(a, n - 1, n) == 1)
    for q in (2, s):
        require(gcd(pow(a, (n - 1) // q, n) - 1, n) == 1)
    check_cell(s, n)
    proved.add(n)
    return n


def prove_minus(s, j, P, Q):
    require(s in proved)
    n, F = s ** 4 + j * s ** 2 - 1, 2 * s ** 2
    D = P * P - 4 * Q
    require(gcd(n, 2 * Q * D) == 1)
    require((n + 1) % F == 0 and (F - 1) ** 2 > n)
    require(lucas_u(n + 1, n, P, Q) == 0)
    for q in (2, s):
        require(gcd(lucas_u((n + 1) // q, n, P, Q), n) == 1)
    check_cell(s, n)
    proved.add(n)
    return n


prove_seed(P0)
require(len(proved) == 28)
require(P0 == 29 ** 21 + 24)
# Each entry is a proper divisor of 29**21+i, for i=1,...,23.
DIVISORS = [2, 31, 2, 3, 2, 5, 2, 571, 2, 3, 2, 13,
            2, 17, 2, 3, 2, 72673, 2, 7, 2, 3, 2]
for i, d in enumerate(DIVISORS, 1):
    n = 29 ** 21 + i
    require(1 < d < n and n % d == 0)
print("Seed: 28 certificate nodes; all 23 intervening integers composite.")

p1 = prove_plus(P0, 33, 2)
p2 = prove_plus(p1, 765, 5)
p3 = prove_plus(p2, 4583, 13)
require([len(str(n)) for n in (p1, p2, p3)] == [123, 492, 1966])
print("Prime tower: 123, 492, 1966 digits; all certificates passed.")
```

```
t_plus = prove_plus(P0, 20303, 17)
t_minus = prove_minus(P0, 20303, 1, -1)
w_plus = prove_plus(t_plus, 447479, 7)
w_minus = prove_minus(t_plus, 447479, 1, -3)
for lo, hi, digits in ((t_minus, t_plus, 123),
                       (w_minus, w_plus, 492)):
    require(hi - lo == 2)
    require(len(str(lo)) == len(str(hi)) == digits)
print("Twin pairs: 123 and 492 digits; all certificates passed.")
print("All seven selected child cells are strictly nested.")


# All finite root comparisons use exact integer powers.
EXPONENTS = (4, 6, 7, 12)
ROWS = [
    (2, 4, (2, 2, 3, 8)),
    (11, 12, (2, 3, 4, 11)),
    (5, 7, (2, 3, 5, 15)),
    (17, 12, (2, 4, 5, 17)),
    (5, 6, (2, 5, 6, 25)),
    (7, 7, (3, 5, 7, 28)),
    (29, 12, (3, 5, 7, 29)),
]
for p, d, outputs in ROWS:
    for c, m in zip(EXPONENTS, outputs):
        require(m ** d <= p ** c < (m + 1) ** d)
for c, m in zip(EXPONENTS[:3], (3, 5, 7)):
    require(m ** 12 <= 29 ** c < 30 ** c < (m + 1) ** 12)
require(P0 + 1 < 2 * 29 ** 21 < 30 ** 21)
require(30 * 3 ** 12 < 4 ** 12)
require(29 * 10 ** 12 > 13 ** 12)
for c, q in zip(range(7, 15), (5, 5, 7, 7, 11, 11, 11, 13)):
    require(3 ** c <= q ** 6 < 4 ** c)
require(4 ** 15 > 2 ** 6 * 3 ** 15)
print("Finite minima and all displayed root comparisons passed.")

R = factorial(252)
E1 = 252 * R
require(2 * 10 ** 497 < R < 21 * 10 ** 496)
require(len(str(R)) == 498 and len(str(E1)) == 500)
require(E1 > 5 * 10 ** 499)
require(10 ** 30 < P0 < 10 ** 31)
# Bounds for the number of decimal digits of the next greedy prime.
require(30 * R > 6 * 10 ** 498)
require(31 * R + 2 < 7 * 10 ** 498)
require(252 * 2 ** 251 < 10 ** 79)
require(252 * 10 ** 495 + 79 < 10 ** 498)
require(252 ** 252 > 10 ** 605)
require(252 * 10 ** 601 + 79 < 10 ** 604)
print("Factorial sizes and decimal-boundary inequalities passed.")

def small_prime(n):
    return n >= 2 and all(n % d for d in range(2, isqrt(n) + 1))

small_primes = [q for q in range(2, 563) if small_prime(q)]
for q in small_primes:
    require(R % (q - 1) == 0)
    require(pow(P0, R, q) == 1)
for h in range(1, 562):
    q = next(q for q in small_primes if (h + 1) % q == 0)
    require((pow(P0, R, q) + h) % q == 0)
print("All 561 offsets in the composite block are certified.")
```

B Reproducing the finite cubic verification

The following Python 3 code uses only the standard library. It regenerates the entire finite family and finds periodic modular obstructions; no stored certificate file, floating-point root approximation, or external primality package is required. The reported count is the size of the exact coefficient catalogue proved in the main text; the verification uses its integer inequalities directly, rather than numerically approximating its roots.

```
"""Self-contained exact verifier: cubic Pisot shadows beta < 100.
Python 3 standard library only; no primality or real-root oracle is used.
"""
from collections import Counter

BOUND = 100
PRIMES = (2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47)
EXPECTED = {2:134584, 3:27162, 5:27780, 7:14024, 11:4856,
            13:2501, 17:1163, 19:544, 23:492, 29:182,
            31:142, 37:86, 41:35, 43:14, 47:4}

def polynomials(B):
    # Coefficients of X^3 - a X^2 + b X - c.
    for a in range(1, B):
        # Two negative small roots.
        for d in range(1, 2*B):
            for c in range(1, B):
                D = a*a*d*d + 4*d**3 - 4*a**3*c - 27*c*c - 18*a*d*c
                if (d <= 2*a+2 and d <= a+c and D > 0
                        and B**3 - a*B*B - d*B - c > 0):
                    yield (a, -d, c)
        # Opposite signs, negative root dominant in modulus.
        for d in range(1, a):
            for c in range(1, a-d+1):
                yield (a, -d, -c)

def step(s, p):
    a, b, c = s
    return ((a**3-3*a*b+3*c) % p,
            (b**3-3*a*b*c+3*c*c) % p, pow(c, 3, p))

def certificate(f, p):
    s = tuple(x % p for x in f)
    seen, states = {}, []
    while s not in seen:
        seen[s] = len(states)
        states.append(s)
        s = step(s, p)
    mu, period = seen[s], len(states)-seen[s]
    for j in range(mu, len(states)):
        if states[j][0] == 0:
            return mu, period, j
    return None

def run():
    remaining = list(polynomials(BOUND))
    assert len(remaining) == 213569
    assert len(set(remaining)) == len(remaining)
    counts = Counter()
    max_period = max_mu = 0
    for p in PRIMES:
        assert all(p % d for d in range(2, p) if d*d <= p)
        survivors = []
        for f in remaining:
            cert = certificate(f, p)
            if cert is None:
                survivors.append(f)
            else:
                mu, period, zero = cert
```

```
                counts[p] += 1
                max_period, max_mu = max(max_period, period), max(max_mu, mu)
        remaining = survivors
    assert not remaining
    assert dict(counts) == EXPECTED
    assert (max_period, max_mu) == (1148, 3)
    small = list(polynomials(4))
    assert small == [(2,-1,-1), (3,-1,-1), (3,-1,-2), (3,-2,-1)]
    print('PASS: 213569 polynomials; all certified by primes <= 47.')
    print('First-witness counts:', dict(counts))
    print('Maximum period =', max_period, '; maximum preperiod =', max_mu)

if __name__ == '__main__':
    run()
```

C Exact certificate for the Type-C improvement

The following Python 3 code uses only the standard library. It enumerates complete finite residue cycles, checks every survivor claimed in the proof, and verifies the rounded cutoffs with rational inequalities. The assertion that the four small Pisot candidates are exhaustive is proved analytically in the main text, not inferred from this computation.

```
"""Exact finite-field certificate for the strengthened Type-C cutoff.
Python standard library only. No numerical root or primality oracle.
"""
from math import lcm
from fractions import Fraction as Q

TABLE = {
 (0,-1,1): [(2,7),(41,1723),(131,17293),(347,173),(397,52669)],
 (1,-1,1): [(5,31),(31,331),(59,3541),(71,5113),(89,8011)],
 (2,1,1): [(2,7),(41,1723),(71,5113),(131,17293),(347,173),(409,55897)]
}
EXPECTED = [1900442940636401,1488257702170843,10312502210725960469]

def trace_period(f,p):
 a,b,c=f
 assert c==1
 assert all(p%d for d in range(2,p) if d*d<=p)
 v=(1,0,0); traces=[]
 while True:
  traces.append((3*v[0]+a*v[1]+(a*a-2*b)*v[2])%p)
  v=(c*v[2]%p,(v[0]-b*v[2])%p,(v[1]+a*v[2])%p)
  if v==(1,0,0):return traces

def surviving_parameters(f,p,g):
 tr=trace_period(f,p); t=len(tr); status={}
 for start in range(t):
  if start in status:continue
  z=start; seen={}; seq=[]
  while z not in seen and z not in status:
   seen[z]=len(seq);seq.append(z);z=(3*z+2//g)%t
  good=status[z] if z in status else all(tr[w]!=0 for w in seq[seen[z]:])
  for w in seq:status[w]=good
 # g=1: parameter j=r, initial exponent 3r-1.
 # g=2: parameter j=(r-1)/2, initial exponent 3j+1.
 good=[j for j in range(t) if status[(3*j-1 if g==1 else 3*j+1)%t]]
 return t,good

def verify():
 moduli=[]
 for f,rows in TABLE.items():
  for p,t in rows:
   assert surviving_parameters(f,p,1)==(t,[0]),(f,p)
  M=lcm(*(t for p,t in rows));moduli.append(M)
  assert M>400300000000000
  print('g=1:',f,'forces r divisible by',M)
 assert moduli==EXPECTED
 for f,p,g,t in [((0,-1,1),7,2,48),((1,-1,1),7,2,48),
                 ((1,0,1),2,1,7),((1,0,1),2,2,7)]:
  assert surviving_parameters(f,p,g)==(t,[]),(f,p,g)
  print('No surviving parameter:',f,'p=',p,'g=',g)
 # Rational enclosures: exp(32.76) and log(2).
 x=Q(819,25);term=Q(1);s=term
 for j in range(1,201):term*=x/j;s+=term
 first=term*x/201
 upper=s+first/(1-x/202)
```

```
 assert upper<168845000000000
 lower=2*sum((Q(1,(2*j+1)*3**(2*j+1)) for j in range(12)),Q(0))
 assert lower>Q(693147,1000000)
 for D,R in [(1,81200000000000),(2,162400000000000)]:
  assert D*(168845000000000+1)/(3*Q(693147,1000000))+Q(1,3)<R
 print('PASS: all exact certificates verified.')

if __name__=='__main__':verify()
```

D Exact certificate for the nonreturning prime seed

The following standalone Python program uses only the standard library. It proves primality by exhaustive trial division, checks the residual roots by substitution, and traverses every state in the exact modular cycle used in Proposition 61.1. Thus the nonreturn assertion concerns all subsequent indices, rather than a finite tested prefix. The final additional assertion checks the integer inequality in Proposition 58.2.

```
"""Exact certificate for a nonreturning prime trace in Saito's cubic locus.

This refutes a proposed divisor-return lemma; it does not refute Saito's
infinite-compositeness problem.  No floating-point calculations or packages.
"""
from math import isqrt

def prime(n):
    return n >= 2 and all(n % d for d in range(2, isqrt(n) + 1))

def phi(t, b, p):
    return ((t**3 - 3*t*b - 3) % p,
            (b**3 + 3*t*b + 3) % p)

a, d, p = 211, 7, 9_398_359
assert prime(a) and prime(p)
assert a**3 + 3*a*d - 3 == p
B1 = -d**3 - 3*a*d + 3
assert B1 == -4771

# Three distinct roots of f mod p, all certified by substitution.
roots = [-1_584_786, -1_942_624, 3_527_621]
assert len({r % p for r in roots}) == 3
assert all((r**3-a*r*r-d*r+1) % p == 0 for r in roots)

# An explicit period after the seed.  The full Newton state, not merely
# its trace, returns, so this is a finite proof about every later index.
L = 85_712
s = (0, B1 % p)
first = phi(*s, p)
s = first
cycle = set()
for _ in range(L):
    assert s[0] != 0
    assert s not in cycle
    cycle.add(s)
    s = phi(*s, p)
assert s == first
assert len(cycle) == L

# The example is not a counterexample to infinite compositeness.
T2 = p**3 - 3*p*B1 - 3
assert T2 == 7*31*3_825_571_796_264_257_579

# Rational sign bounds used to certify Saito's conjugate inequalities.
# f(-1/10)<0<f(0); beta<212; and 212^17<10^40.
assert -1 - 10*a + 100*d + 1000 < 0
assert -1-a+d+1 < 0 < 1
assert 1-a-d+1 < 0
assert -d*a+1 < 0 < (a+1)**3-a*(a+1)**2-d*(a+1)+1
assert 212**17 < 10**40

print("PASS: prime seed p occurs exactly once, at index 1;")
print("the 85,712-state certified tail has no zero trace modulo p.")

# Additional exact inequality for the nonunit example.
assert 39**40 * 10008**17 < 2000**40
```

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